Generative AI

Hands-on Workshop

From Zero to Production: Build your own GenAI solution

#**WISSEN**TEILEN powered by

Lars Röwekamp | **Tim** Wüllner | **open** knowledge **GmbH**

Lars Röwekamp

LinkedIn: **lars-roewekamp**

CIO New Technologies
OPEN KNOWLEDGE
(Architecture, Cloud, AI & ML)

Tim Wüllner

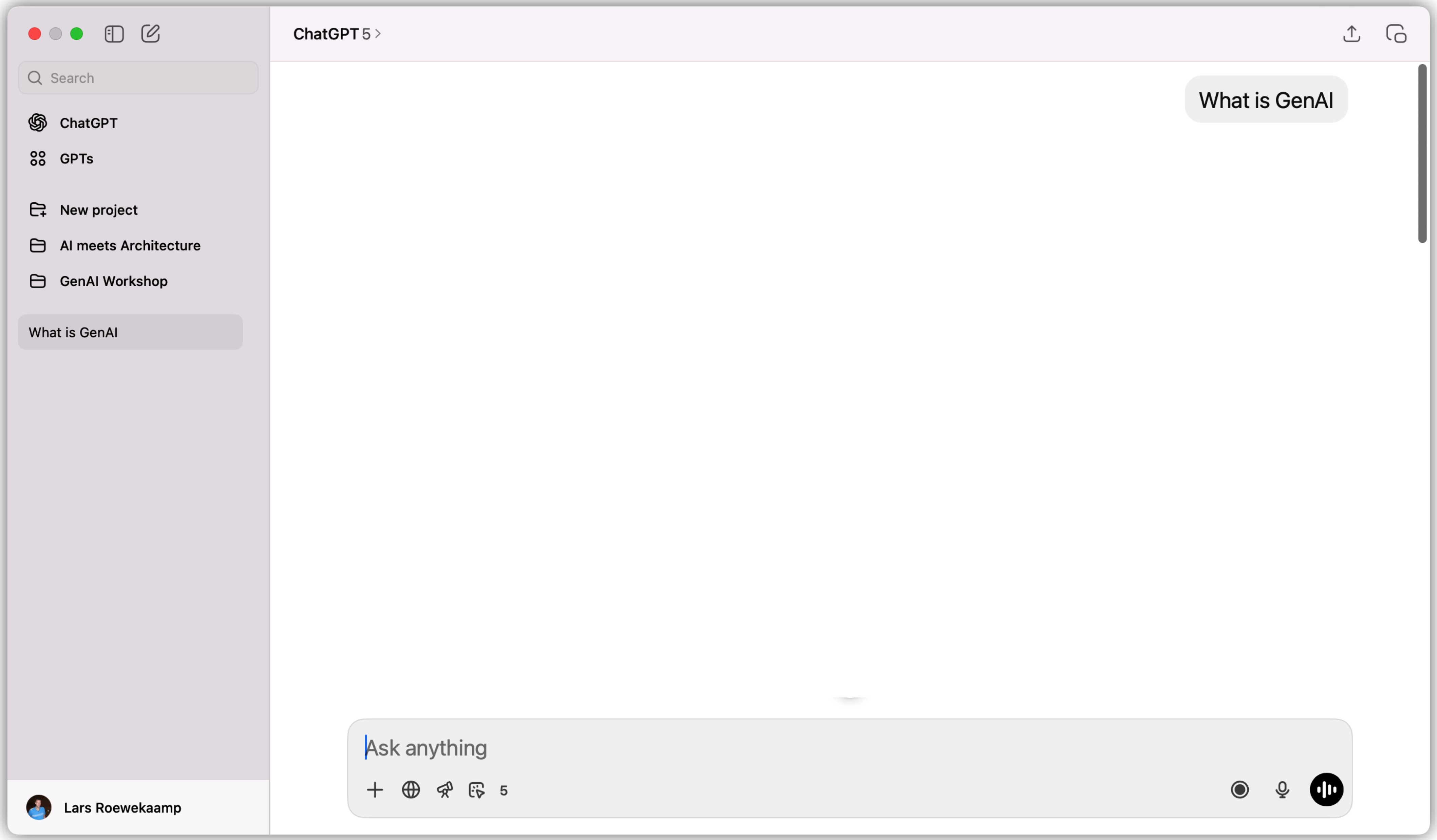
LinkedIn: **tim-wuellner**

Machine Learning Engineer
OPEN KNOWLEDGE

00

What‘s GenAI?

How does „Generative AI“ actually work?


ChatGPT 5
Search
ChatGPT
GPTs
New project
AI meets Architecture
GenAI Workshop
What is GenAI
Lars Roewekaamp
What is GenAI
Ask anything
5

How does a LLM generate a response?

The Idea:

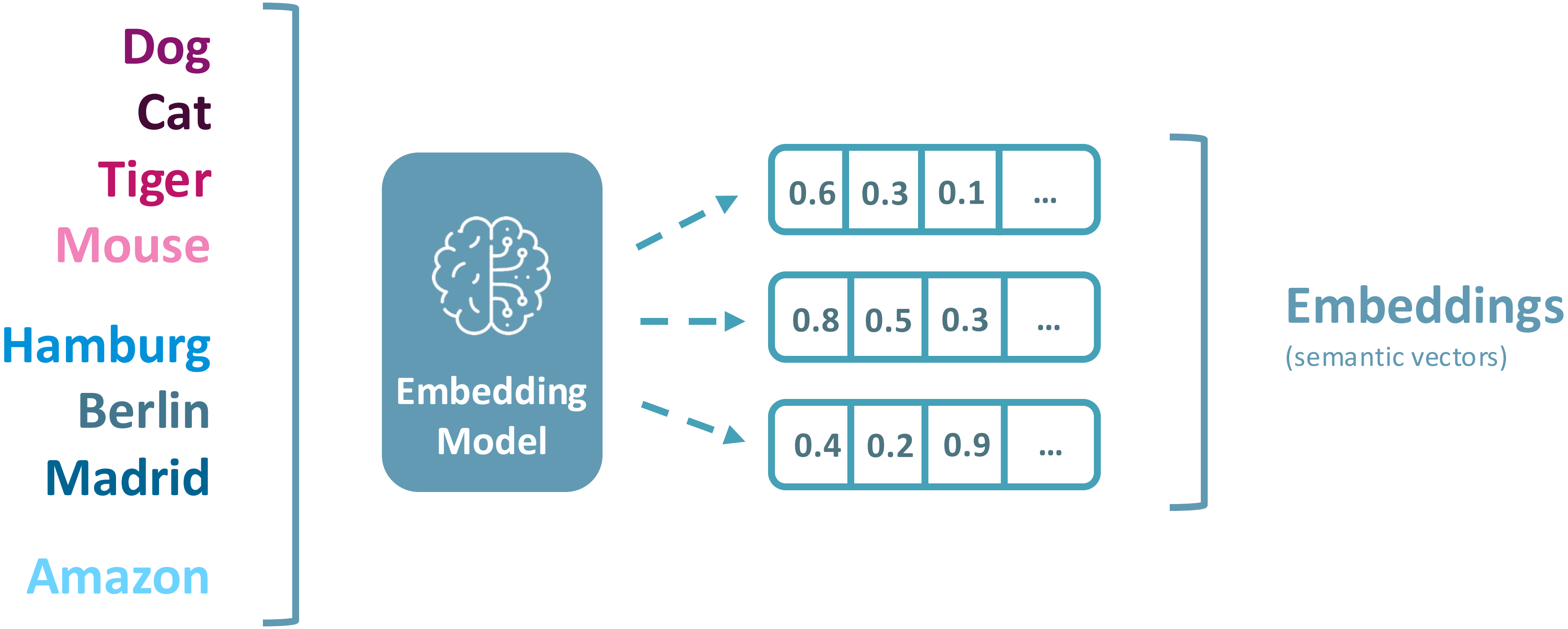
The question and answer are SEMANTICALLY very similar.

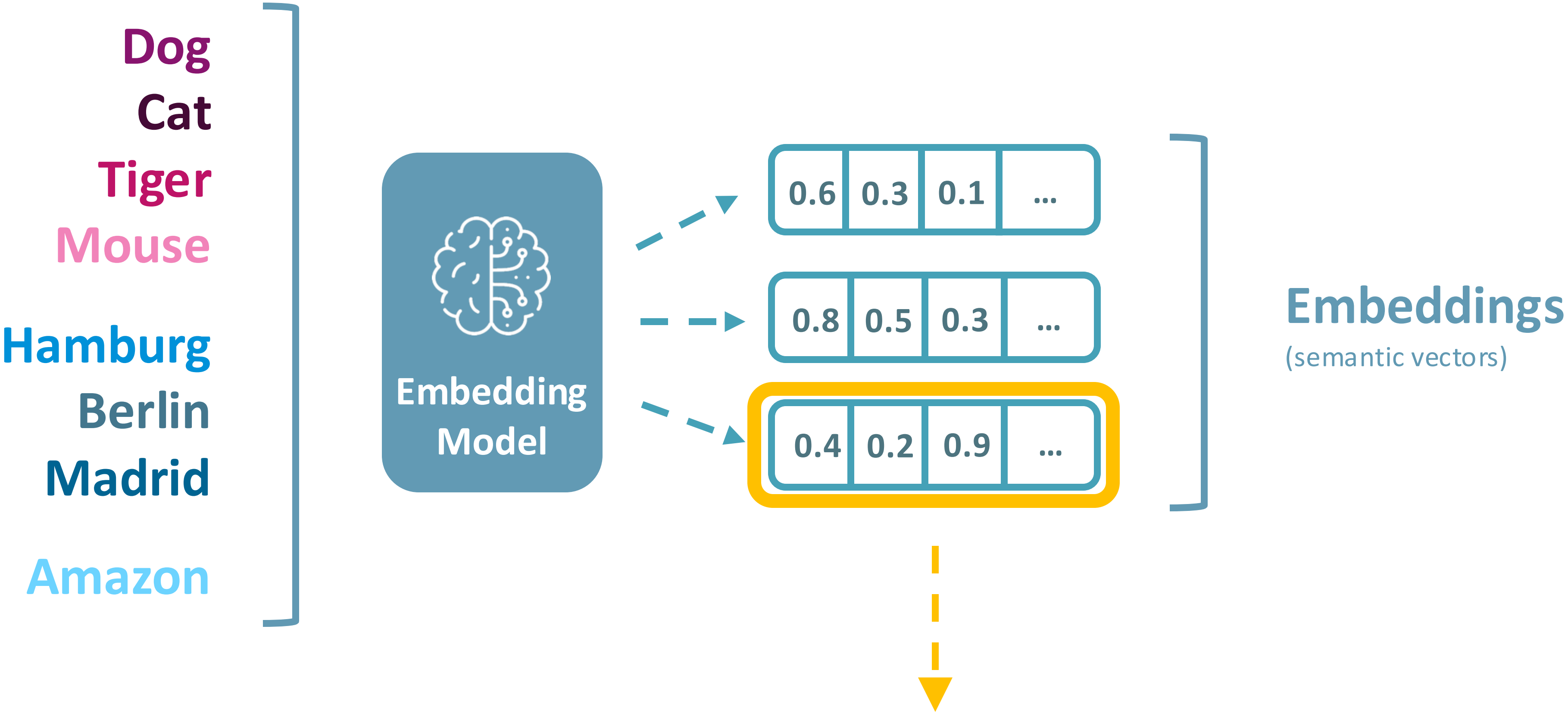
How does a LLM generate a response?

The Idea:

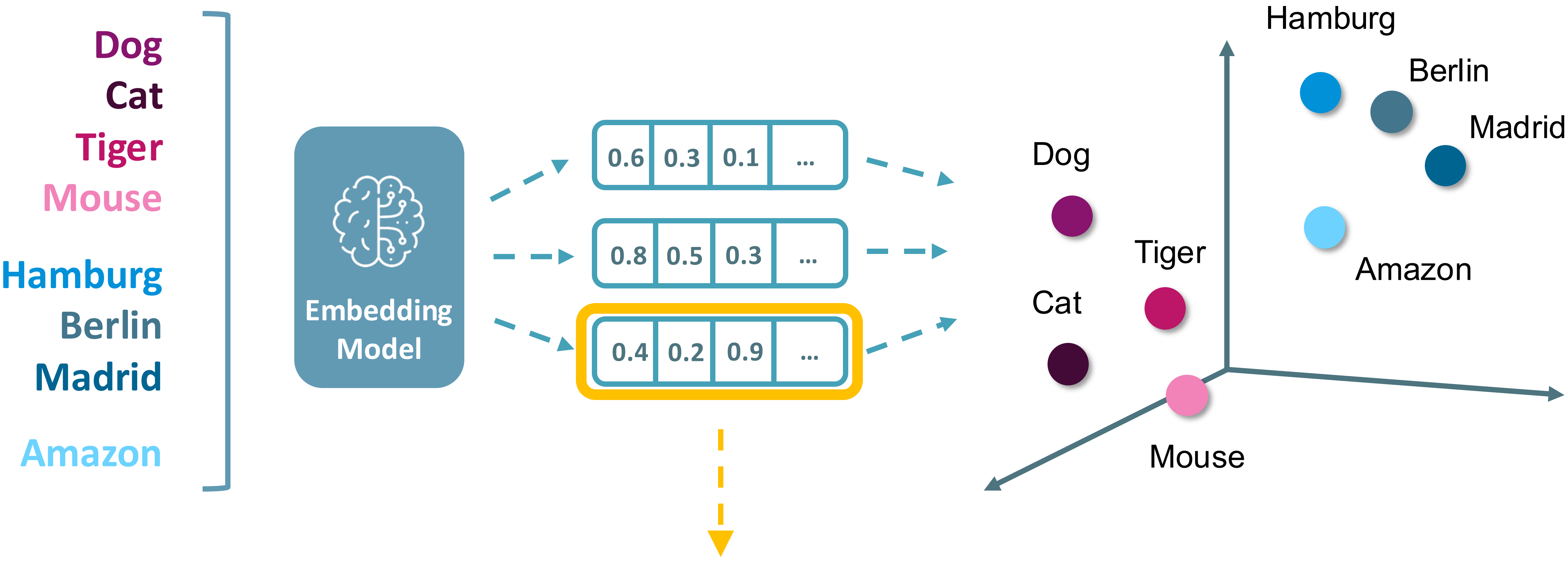
The question and answer are SEMANTICALLY very similar.

Sematic proximity is represented MATHEMATICALLY by the aid of vectors.

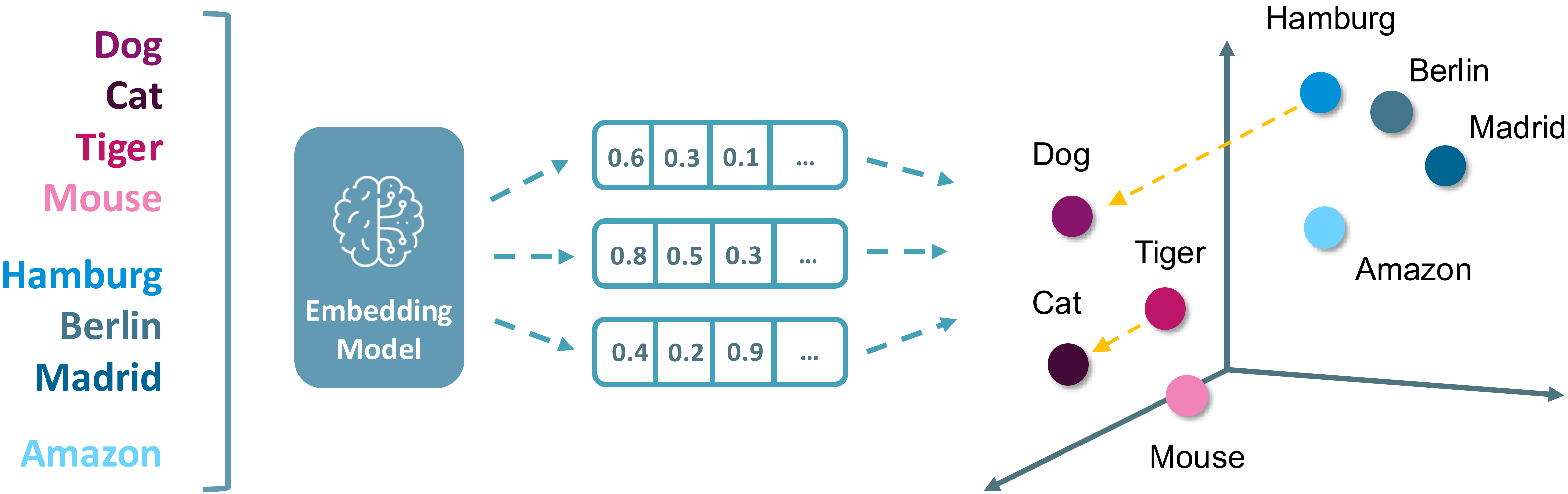

Dog
Cat
Tiger
Mouse
Hamburg
Berlin
Madrid
Amazon
Embedding Model
0.6 0.3 0.1 ...
0.8 0.5 0.3 ...
0.4 0.2 0.9 ...
Embeddings
(semantic vectors)



Berlin: City, Europe, Capital of Germany, ...



Berlin: City, Europe, Capital of Germany, …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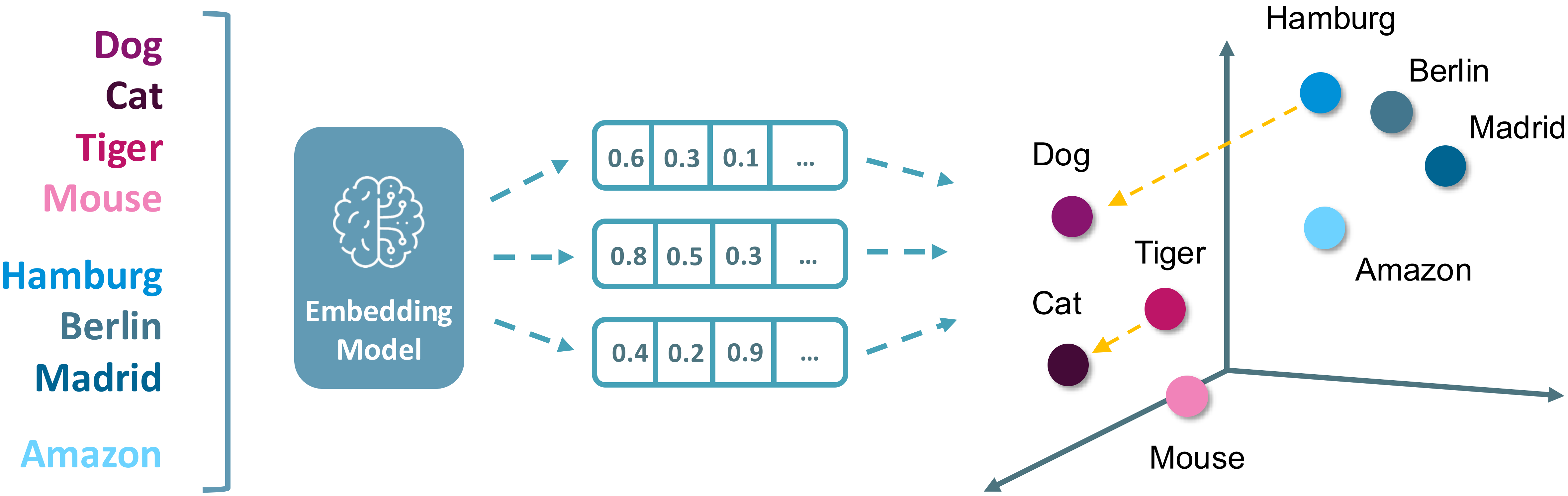
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Hamburg
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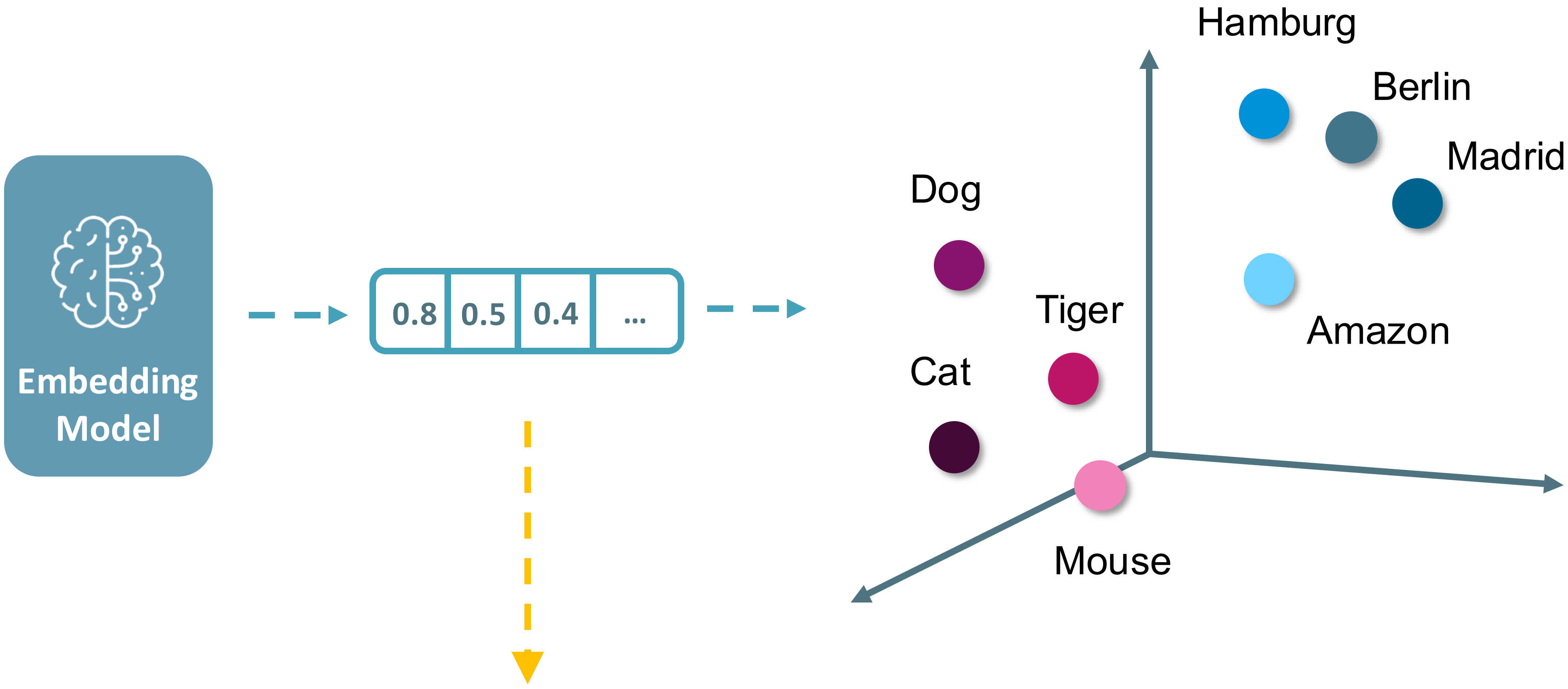
Amazon
Cat
Mouse

QUESTION:

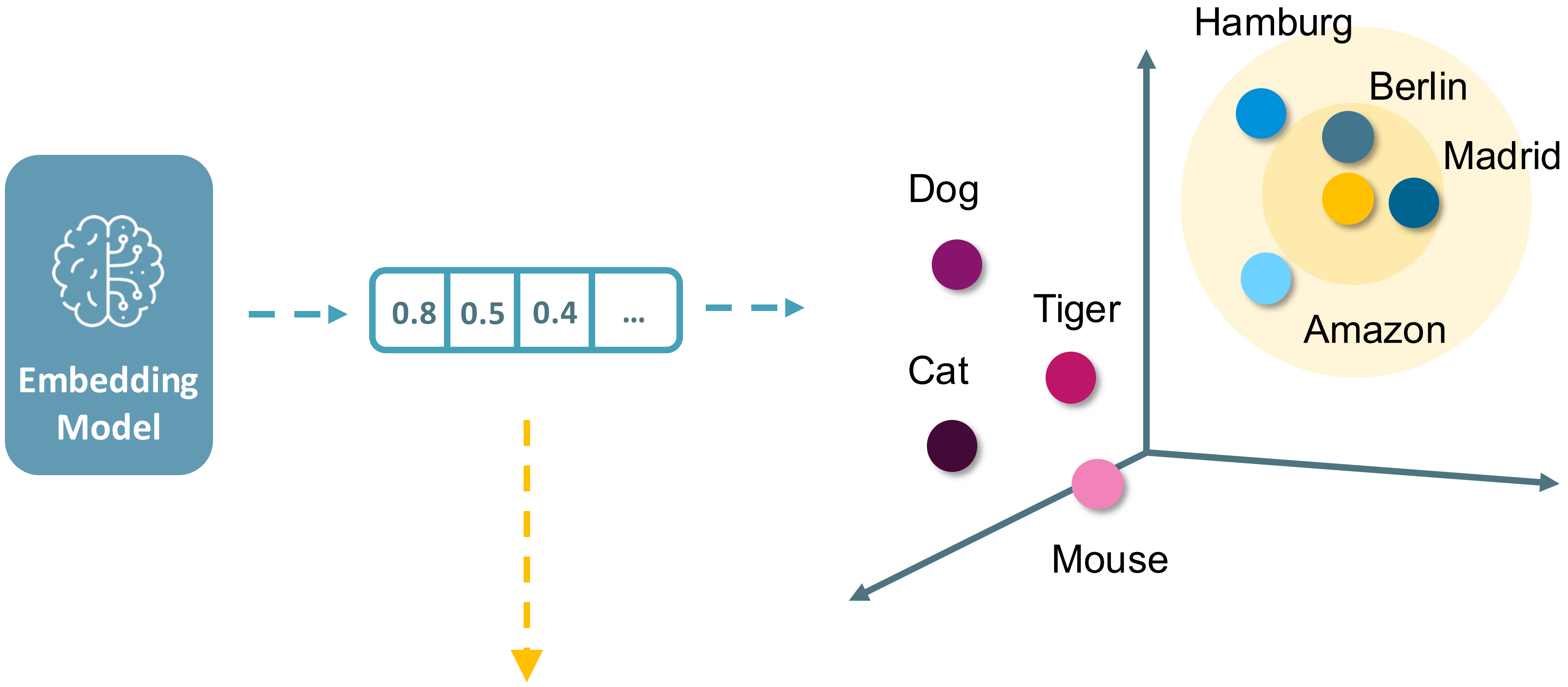
„Can you name a few European capitals?“



„Can you name ..?“: City, Europe, Capital of ...

QUESTION:

„Can you name a few European capitals?“



„Can you name ..?“: City, Europe, Capital of ...

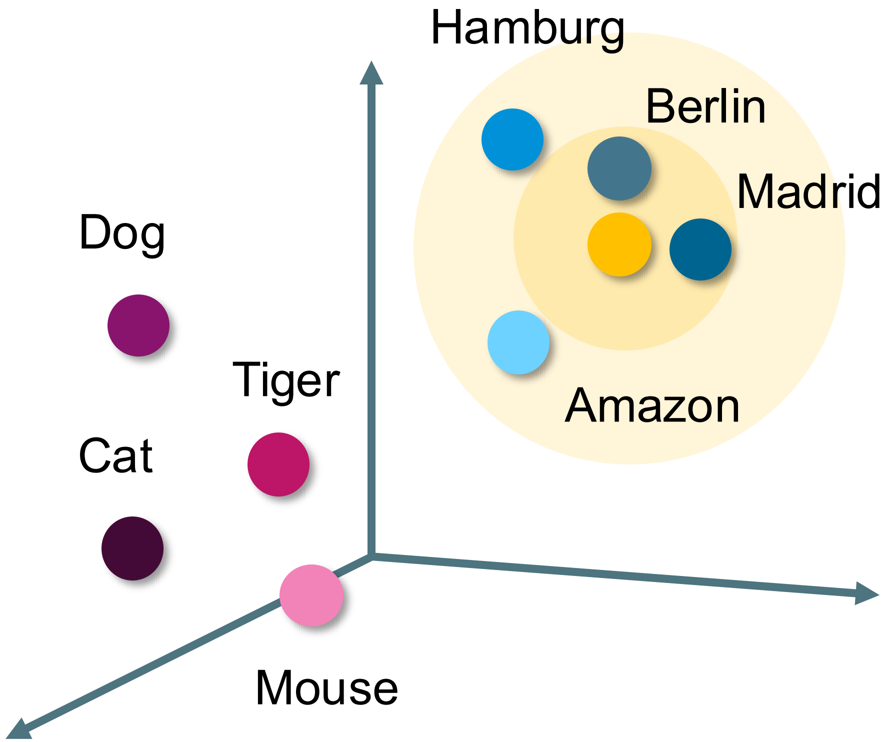
Our goal?

QUESTION:

„Can you name a few European capitals?“

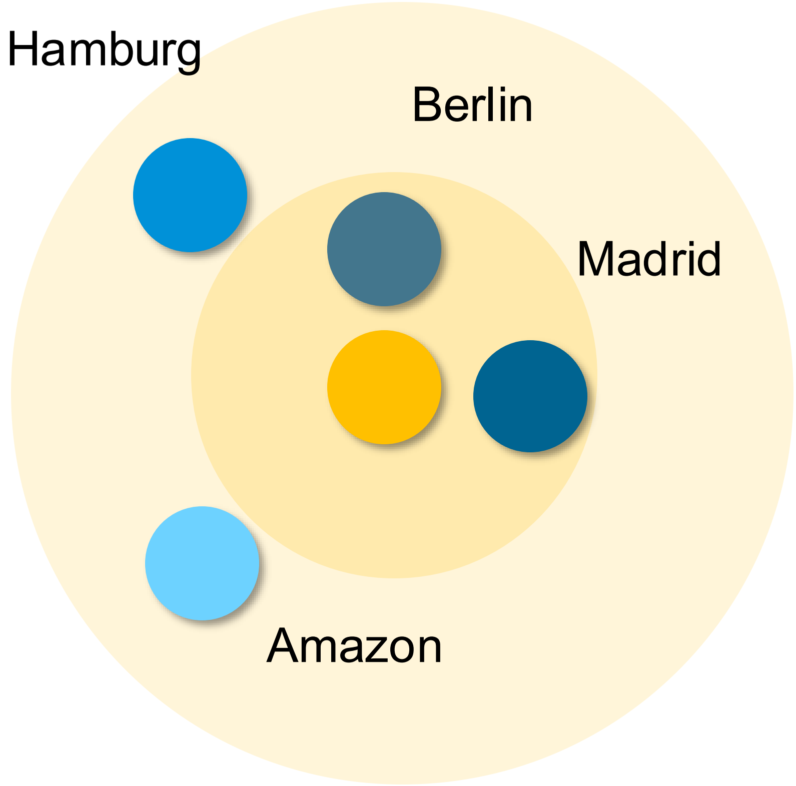
Embedding Model

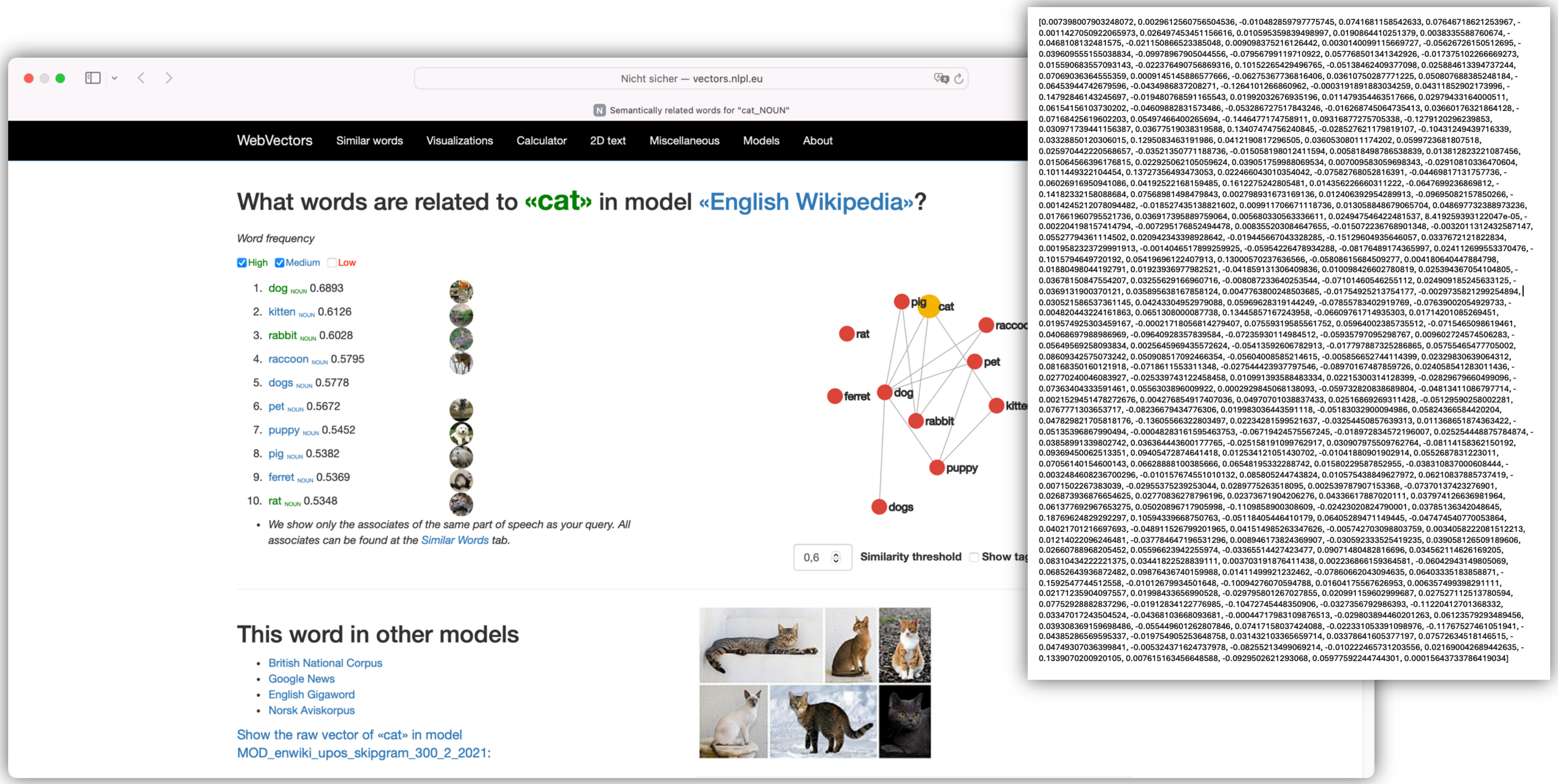
0.8	0.5	0.4	…



„Can you name ..?“: City, Europe, Capital of …

Our goal? Do everything possible to ensure that the ‚question' and the ‚answer' are as close together as possible!



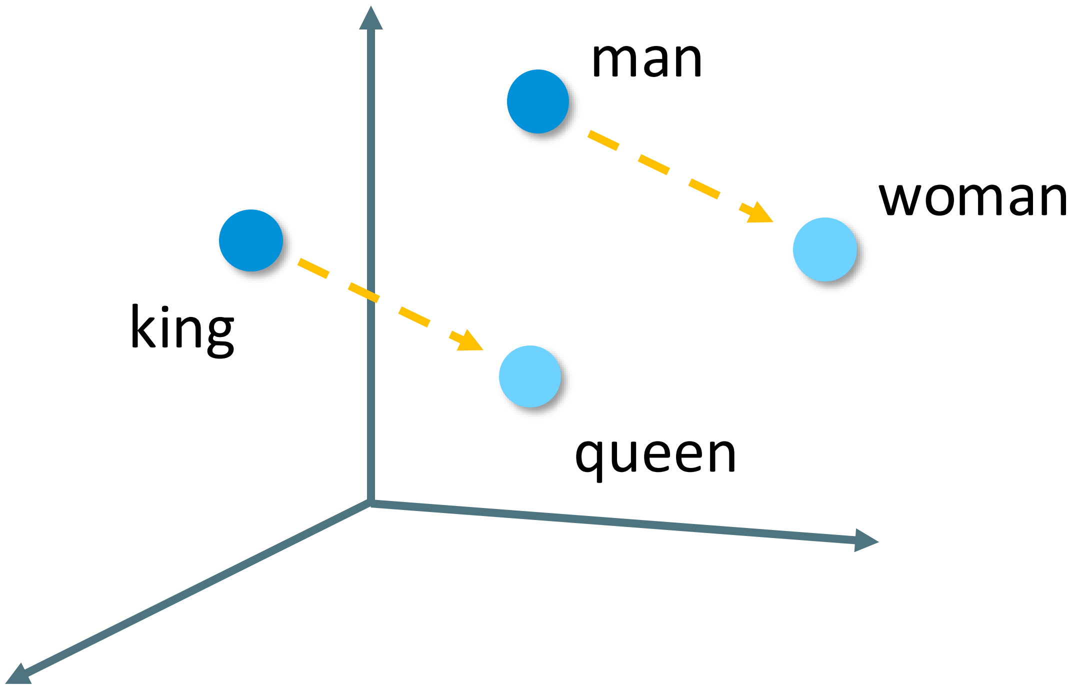


Source: http://vectors.nlpl.eu/explore/embeddings/en/MOD_enwiki_upos_skipgram_300_2_2021/cat_NOUN/

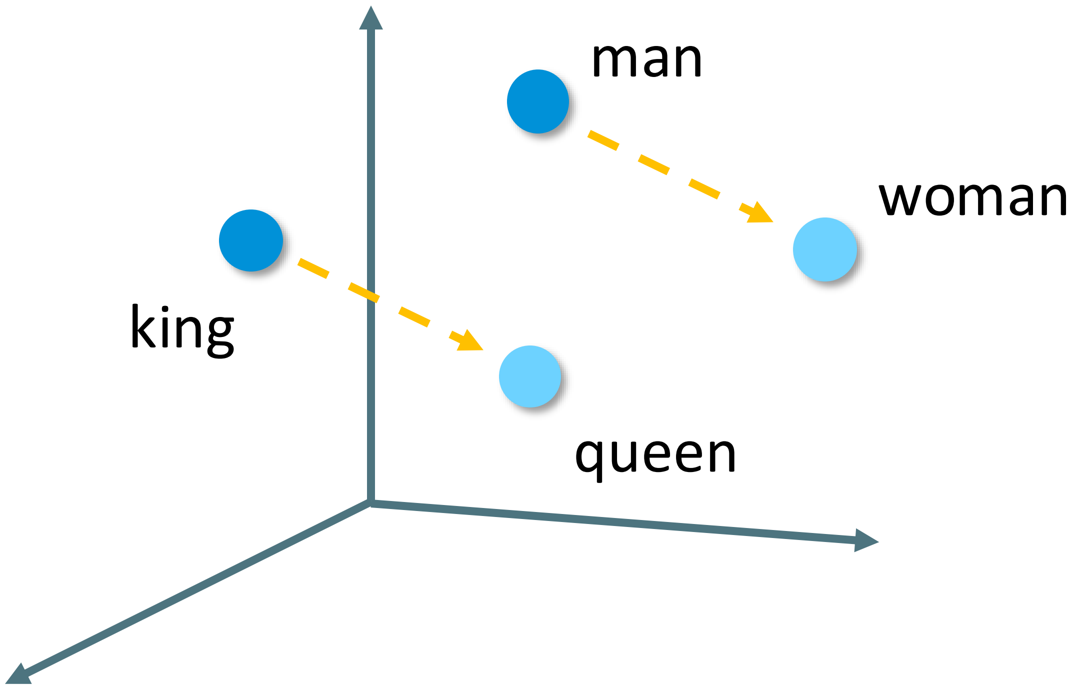
OFFENKUNDIGGUT

OPEN KNOWLEDGE

Okay, but what‘s so great about that?



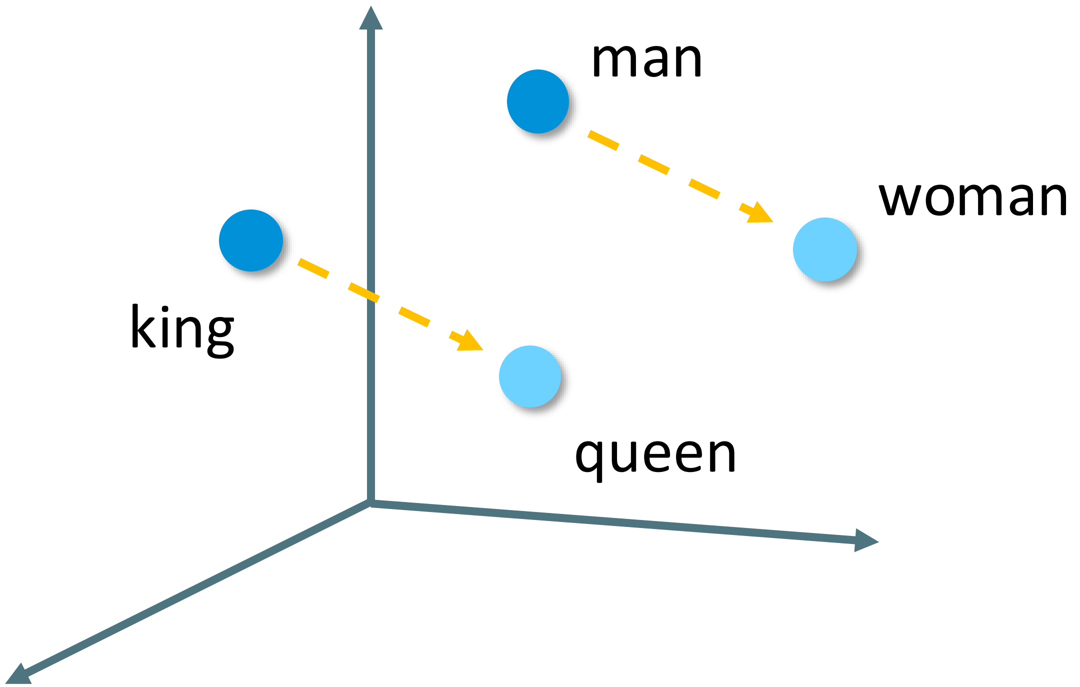
Okay, but what‘s so great about that?
We can calculate with it and ‚understand‘.



queen – woman + man = **king**

Okay, but what's so great about that?

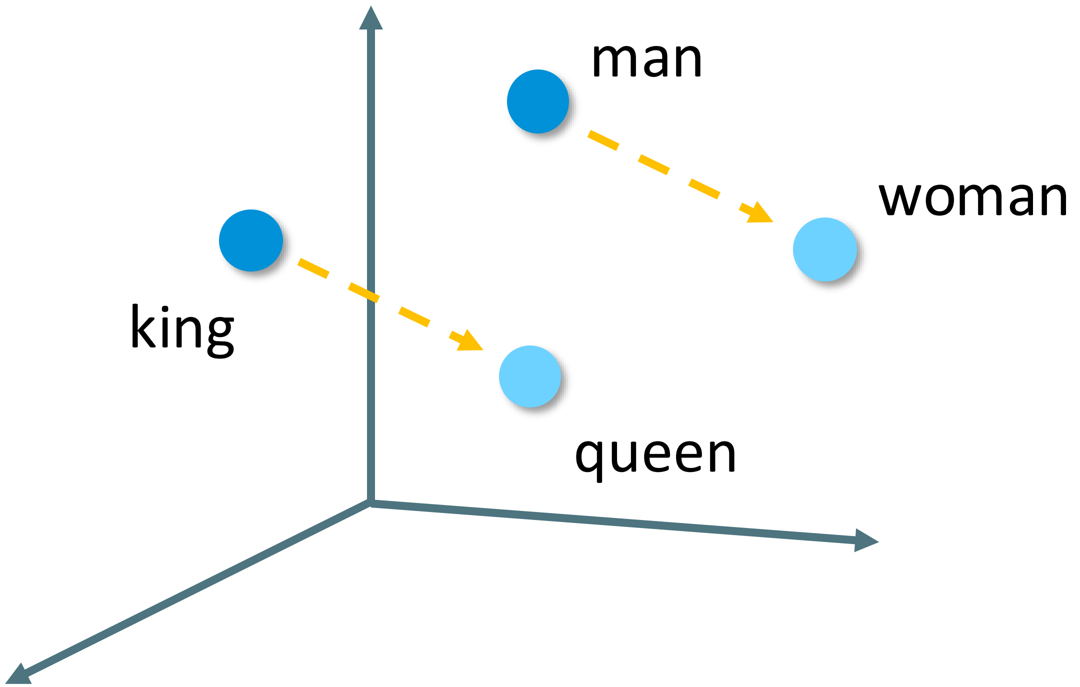
We can calculate with it and ‚understand'.



doctor – man + woman = ?

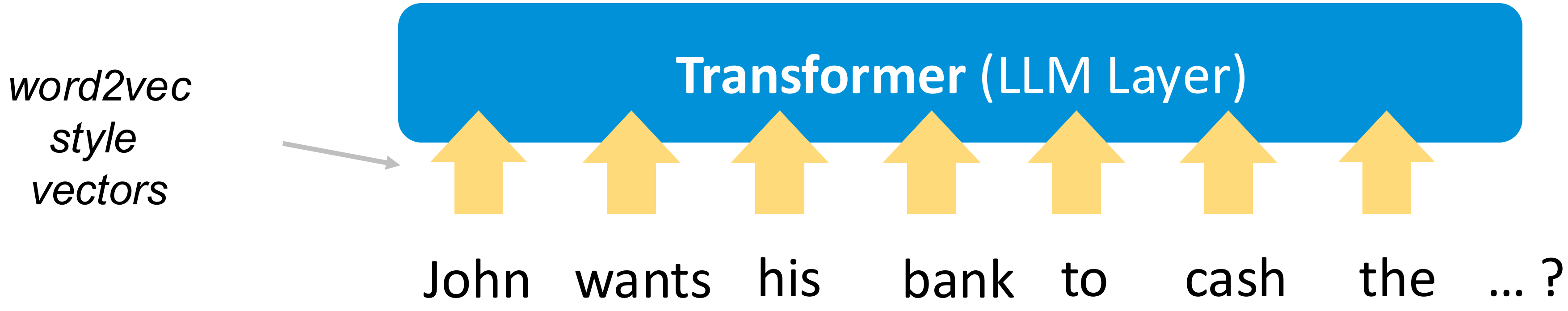
Okay, but what‘s so great about that?

We can calculate with it and ‚understand‘.

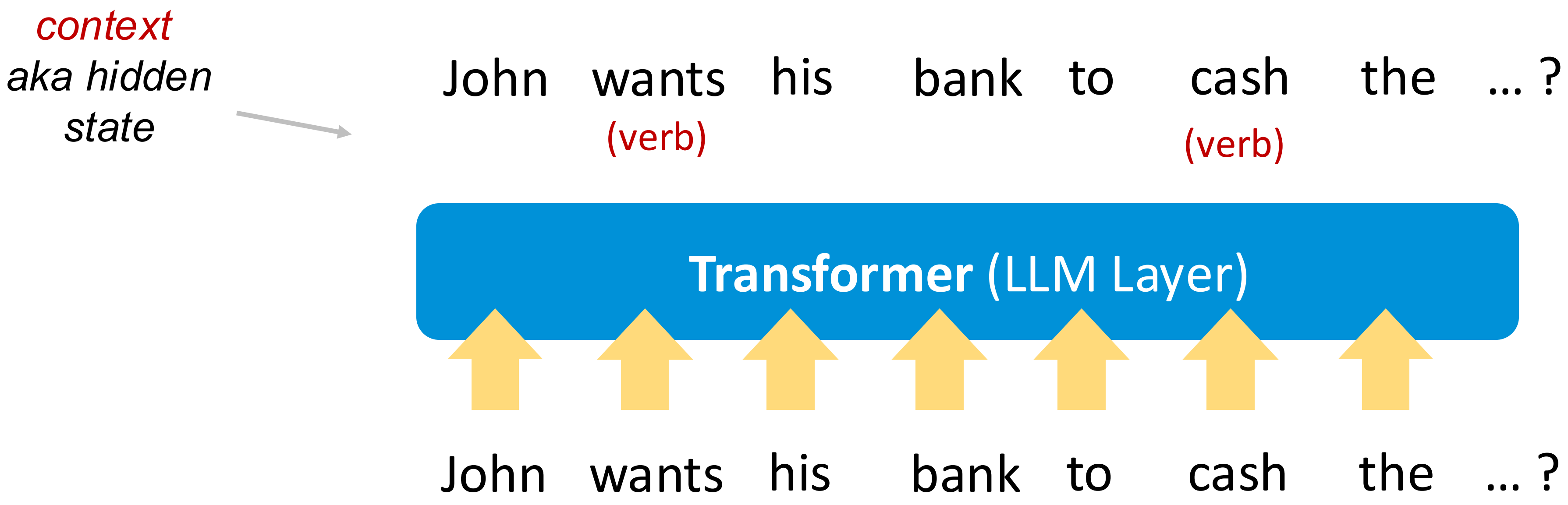


doctor – man + woman = **nurse**

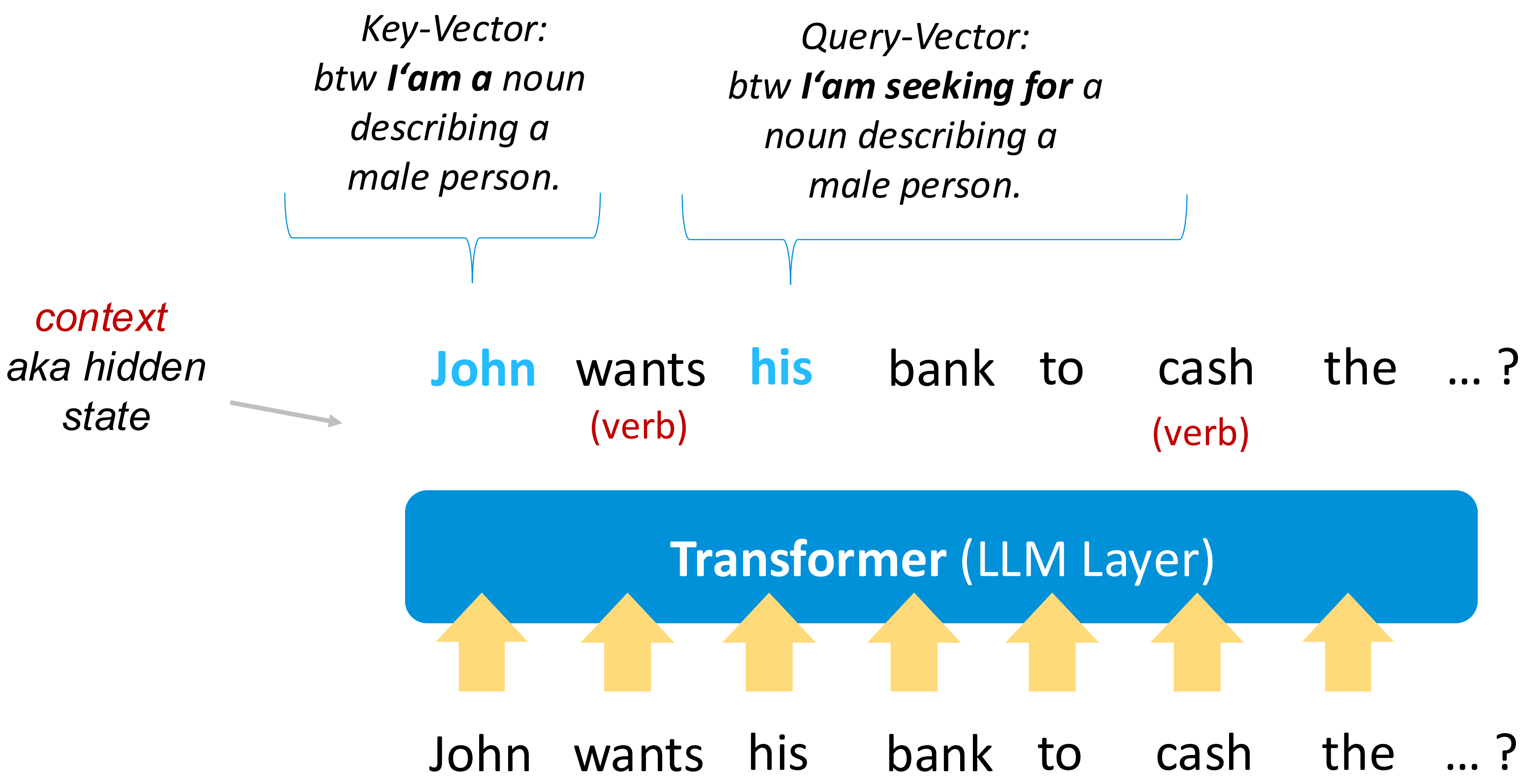
Generative AI under the hood



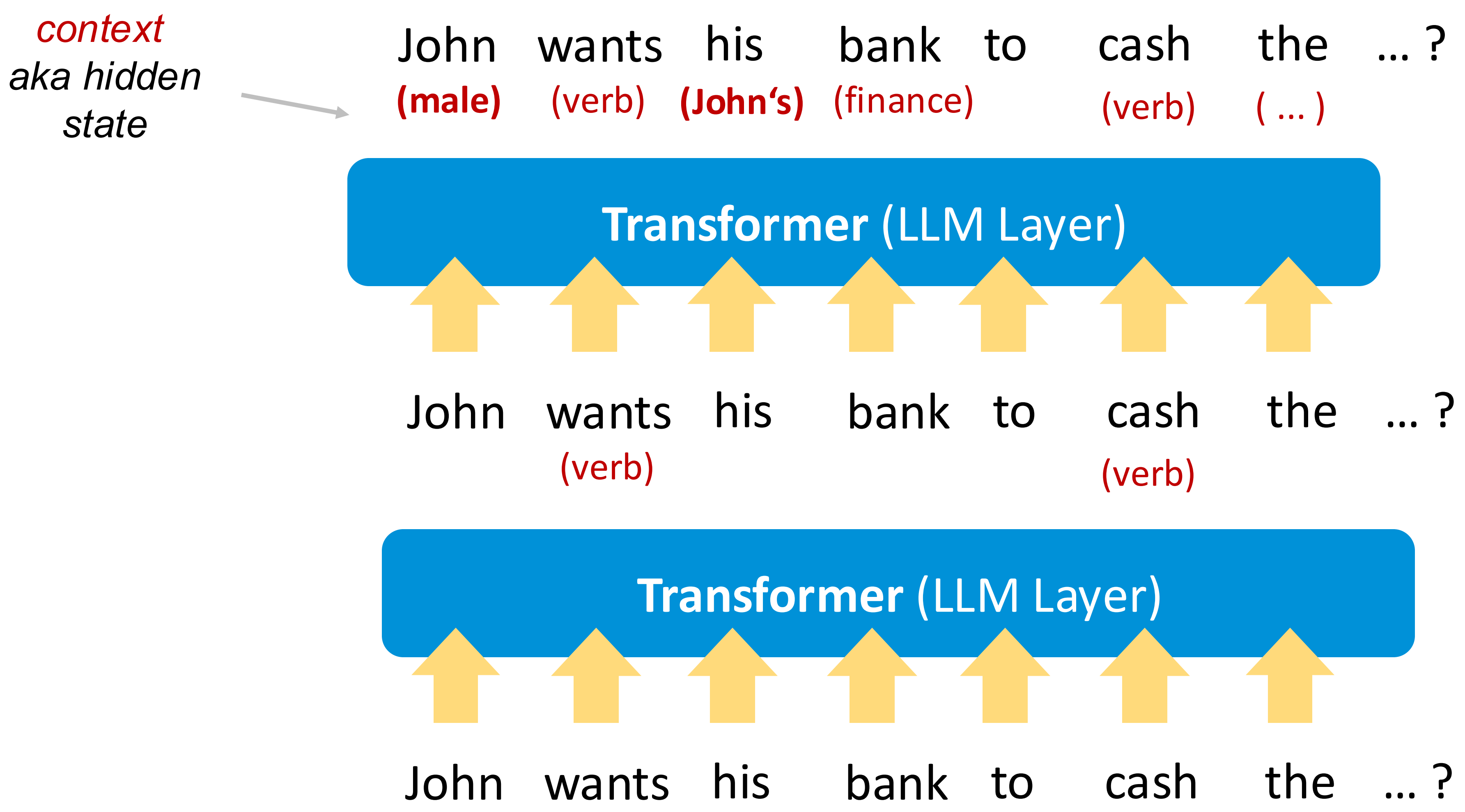
Generative AI under the hood



Generative AI under the hood



Generative AI under the hood



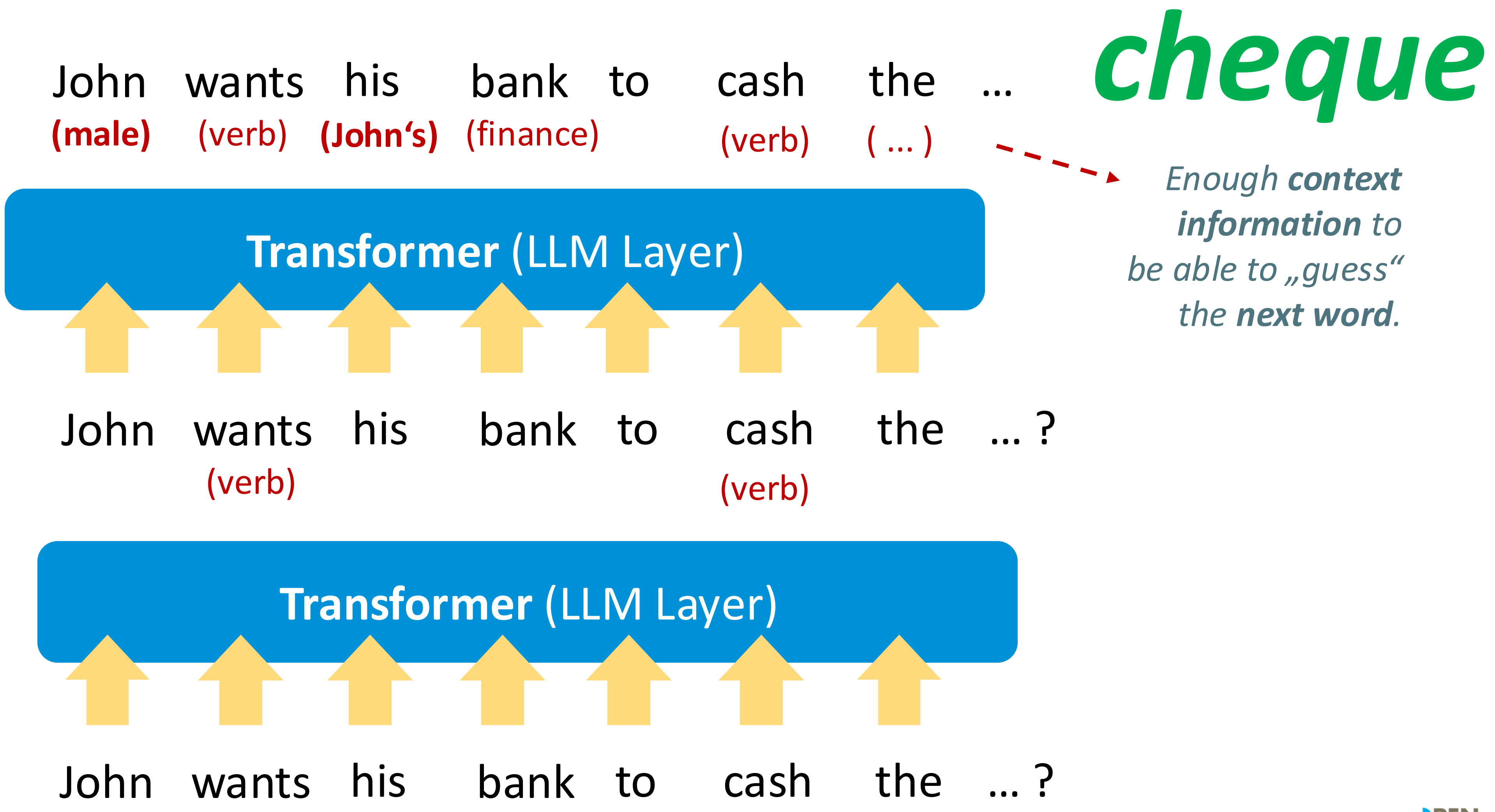
Generative AI under the hood

John wants his bank to cash the … ?

(main character, male, married to Cheryl, cousin of Donald, from Minnesota, currently in Boise, …)

John wants his bank to cash the … ?

Generative AI under the hood



Generative AI under the Hood

John wants his bank **to** …

Large Language Model

Next Token	Probability
increase	0.2
cash	**0.4**
close	0.1
…	…

John wants his bank **to cash the** …

Large Language Model

Next Token	Probability
account	0.15
cheque	**0.75**
money order	0.05
…	…

01

GenAI by Example?

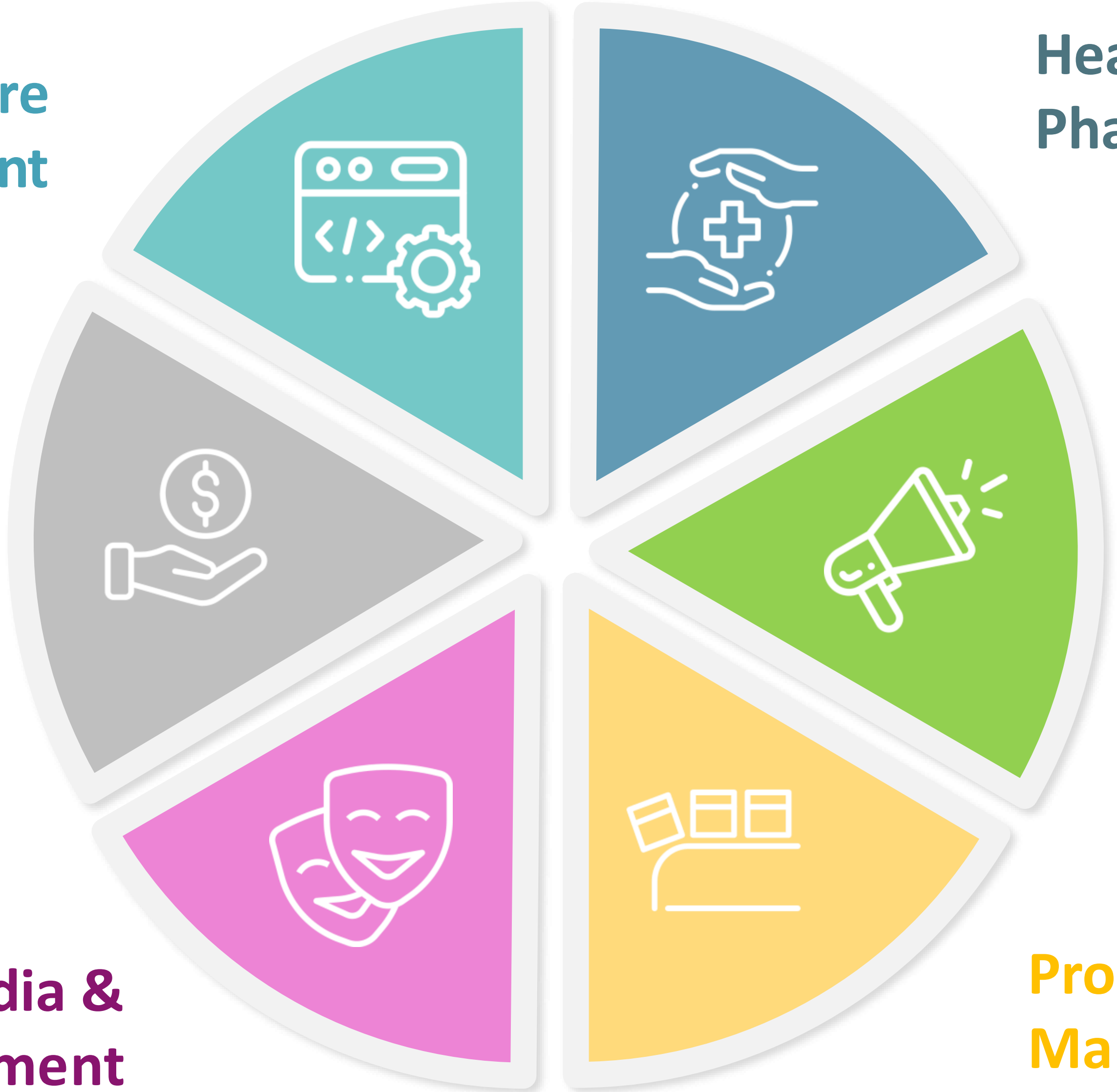
Where can I realistically use GenAI - and where not?

Generative AI by Example

Generative AI by Example

Software Development

Health Care & Pharmaceuticals



Financial Services

Advertisment & Marketing

Media & Entertainment

Production & Manufactoring

GenAI by Example

„Know Your Business!“

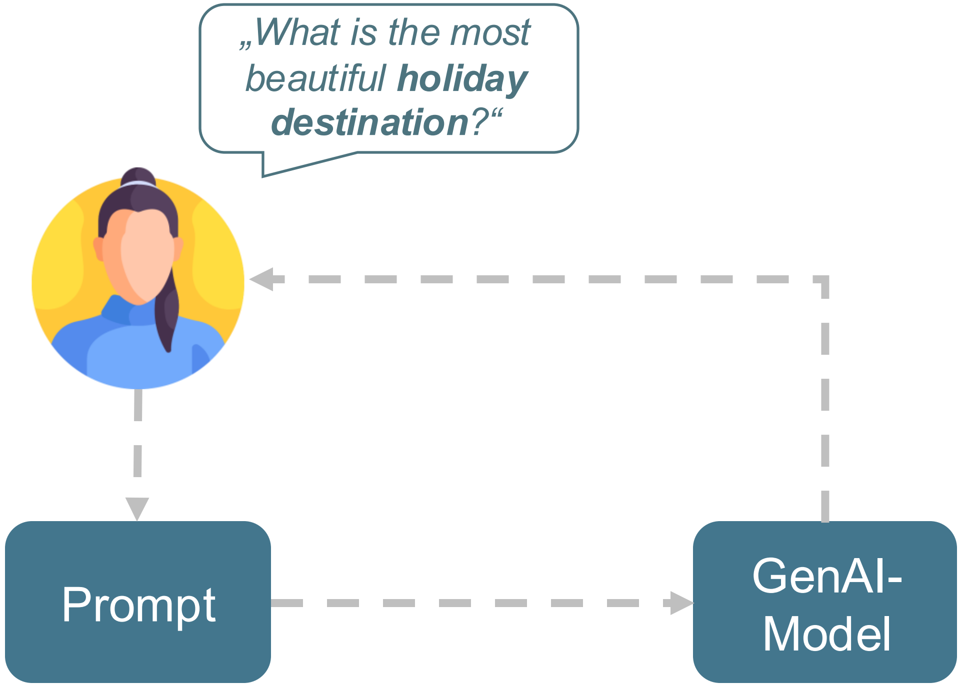
*if you want to succeed in the end.

02

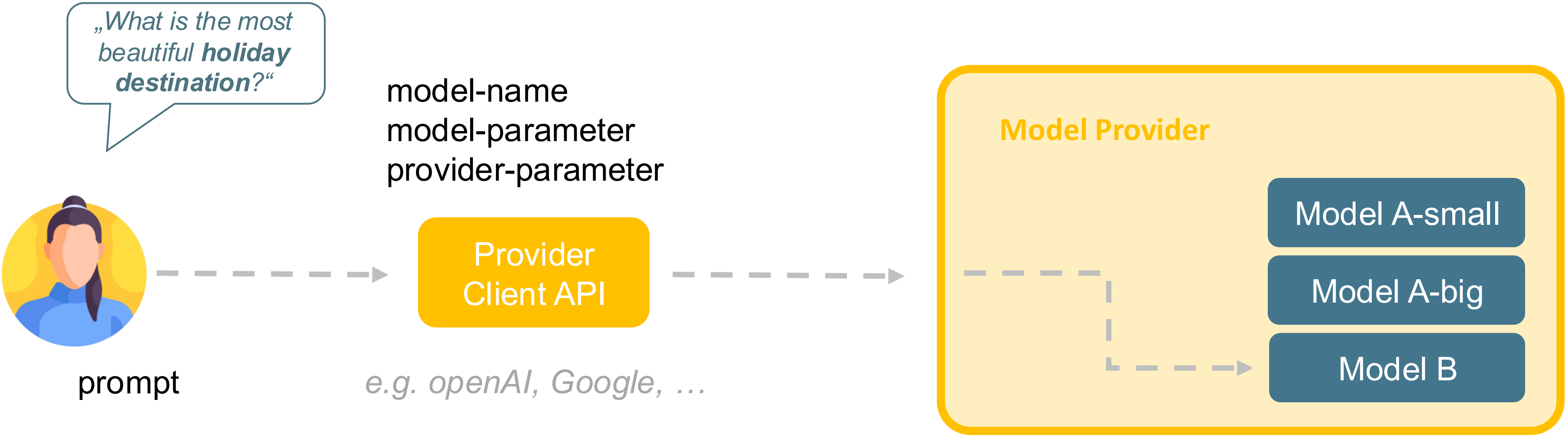
GenAI Basics

How to build a first, simple GenAI system?

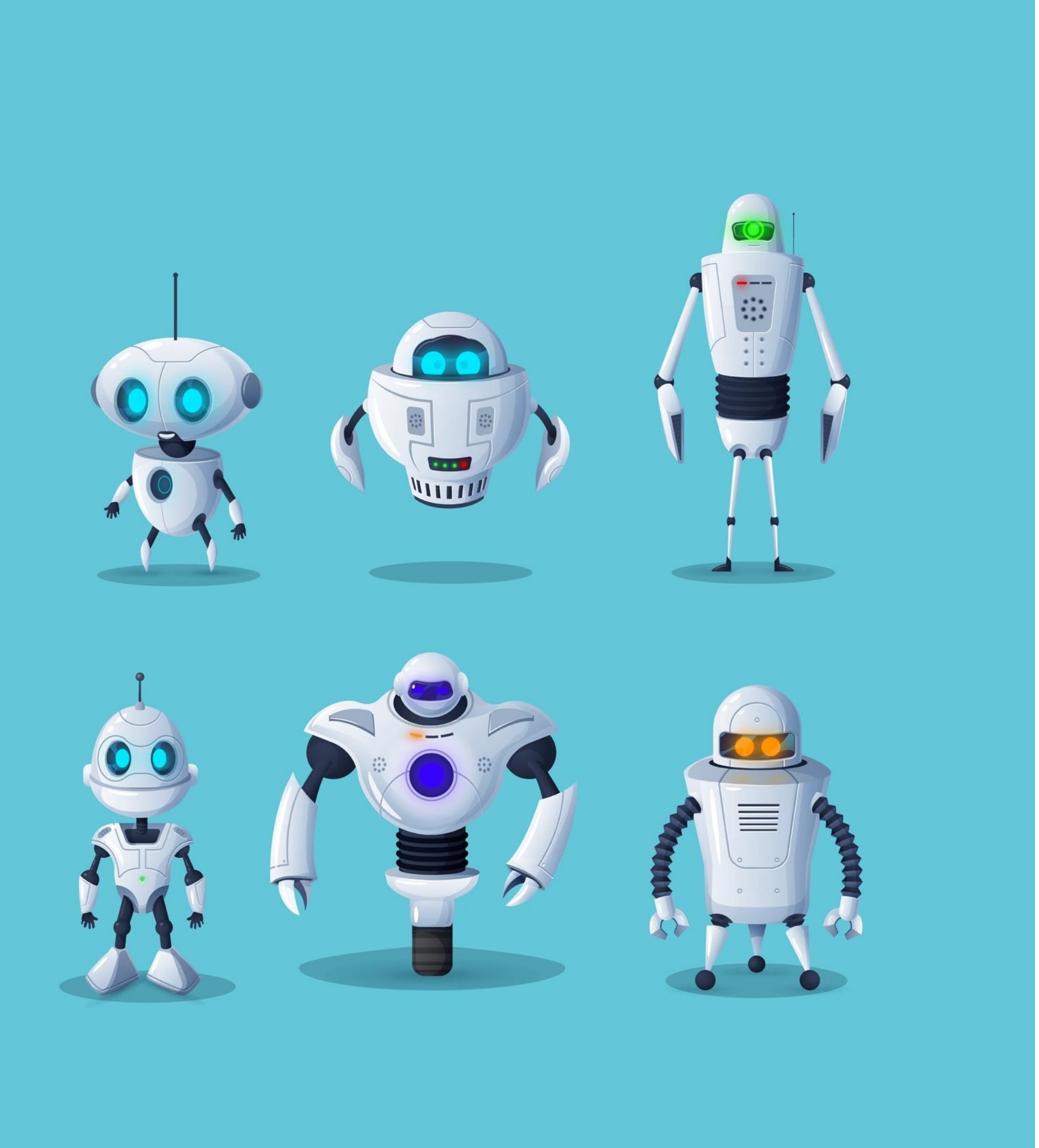
GenAI Basics



GenAI Basics Model Integration



Reminder:
00-hello-genai.ipynb



Hands-On

Hello IPYNB*, hello GenAI

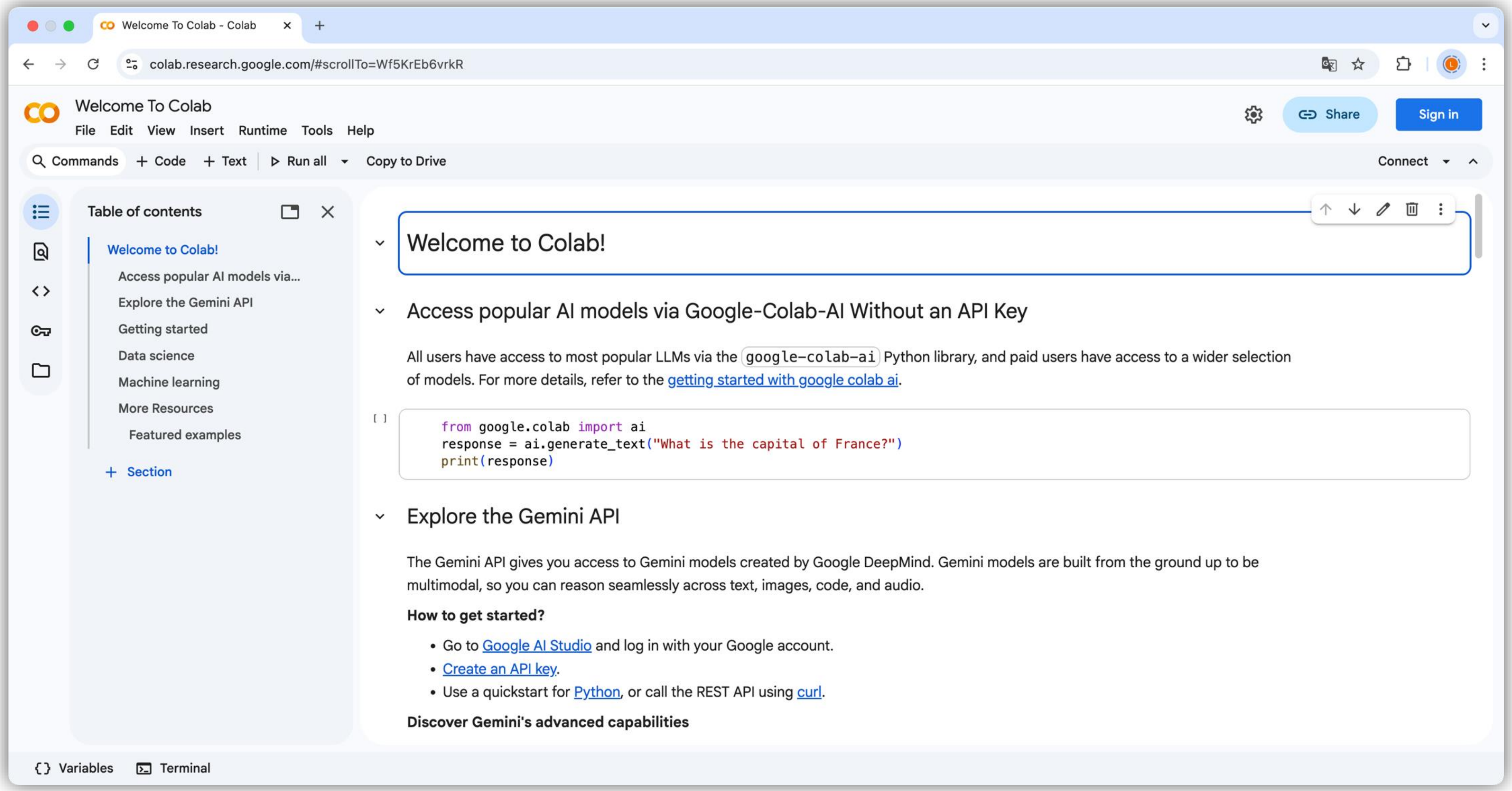
Create a Jupyter Notebook in the Google Colab environment.

Calling a GenAI model with a prompt via a provider-specific API.

*Python Notebook aka Jupyter Notebook

https://colab.research.google.com/

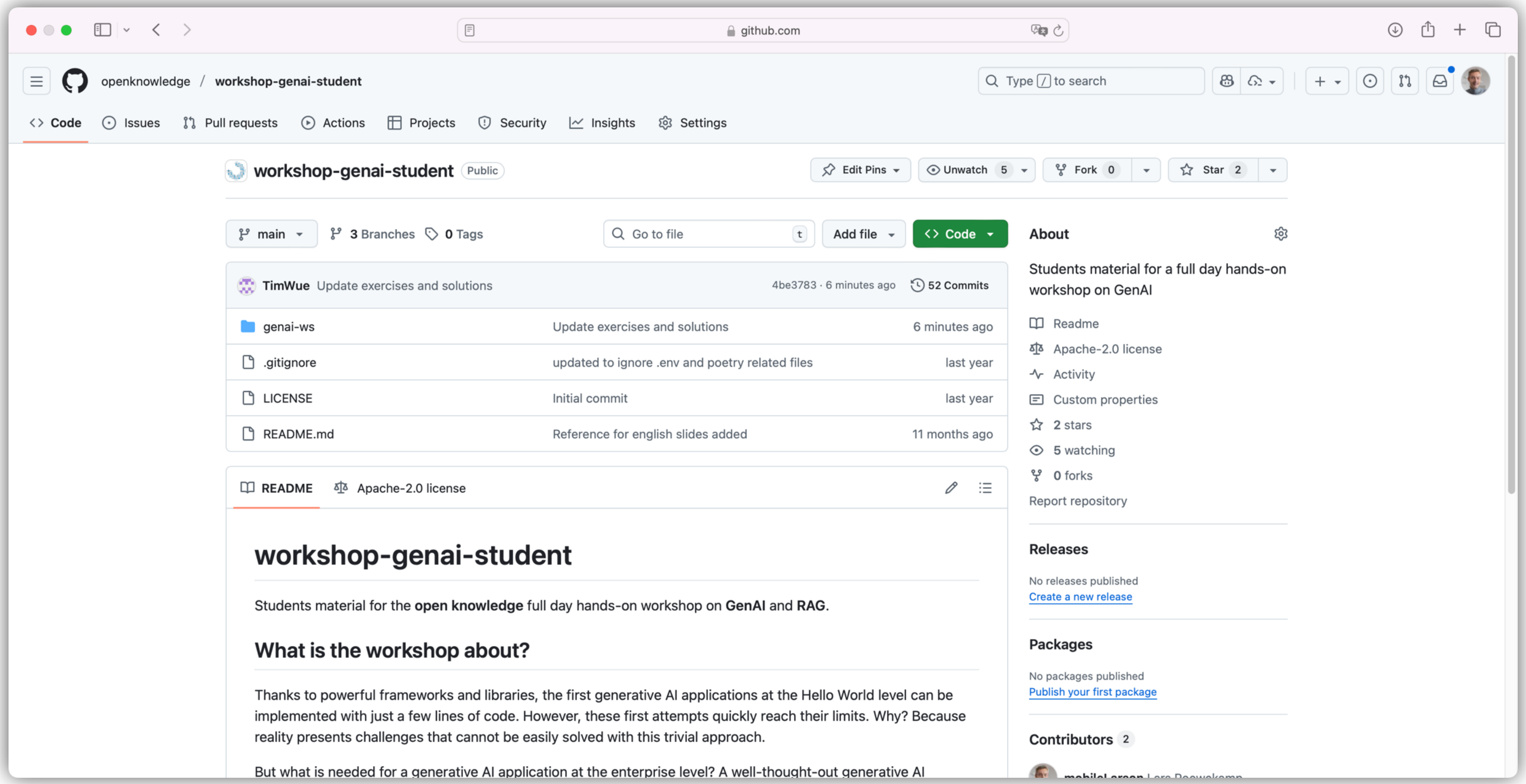
Hands-on Hello Colab



https://colab.research.google.com/

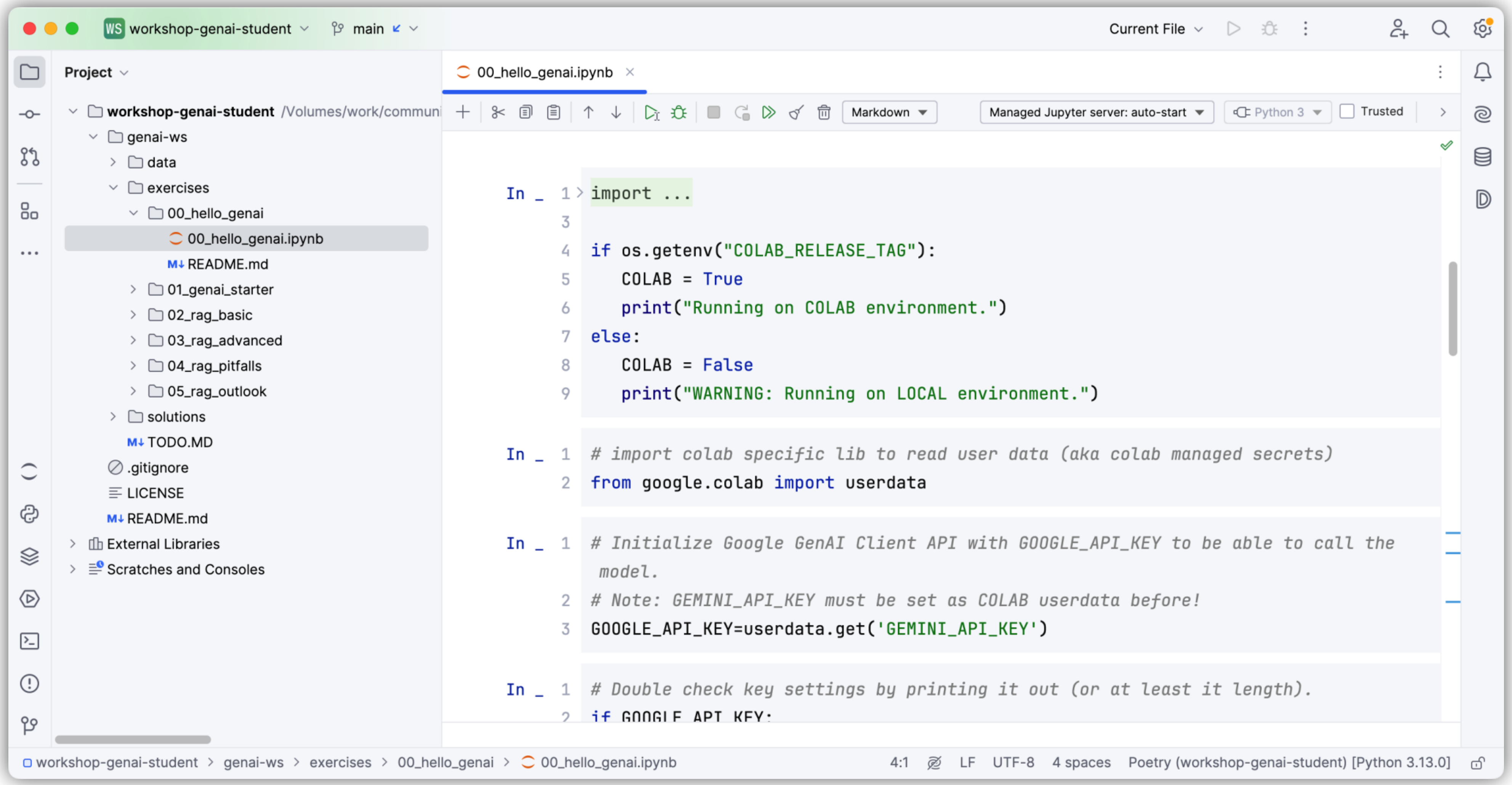
https://github.com/**openknowledge/workshop-genai-student**

Hands-on Hello GenAI World Jupyter Notebook



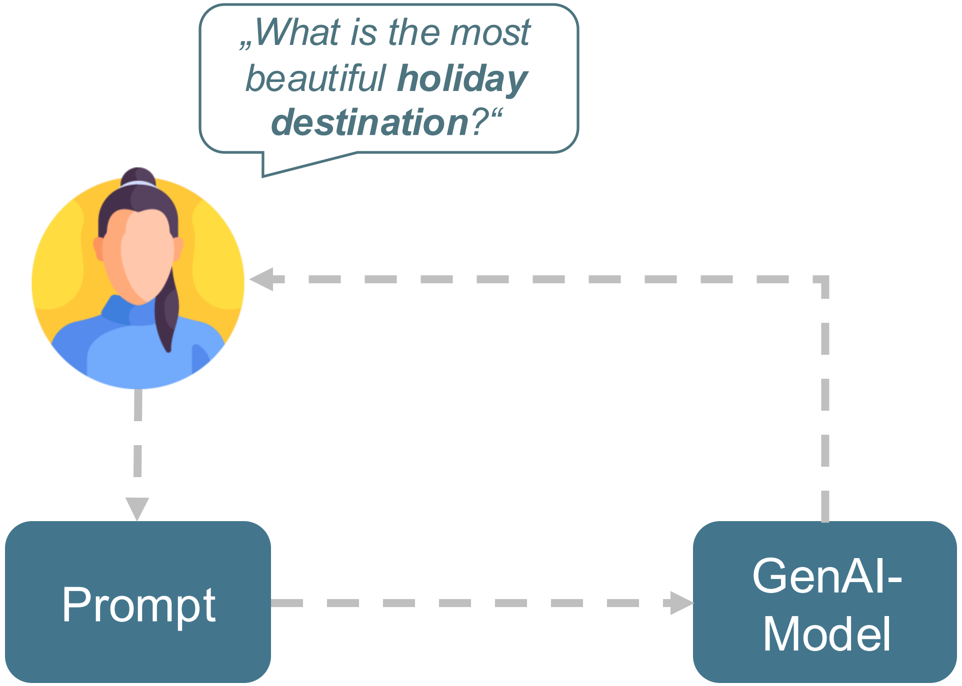
https://github.com/**openknowledge/workshop-genai-student**

Hands-on Hello GenAI World Jupyter Notebook

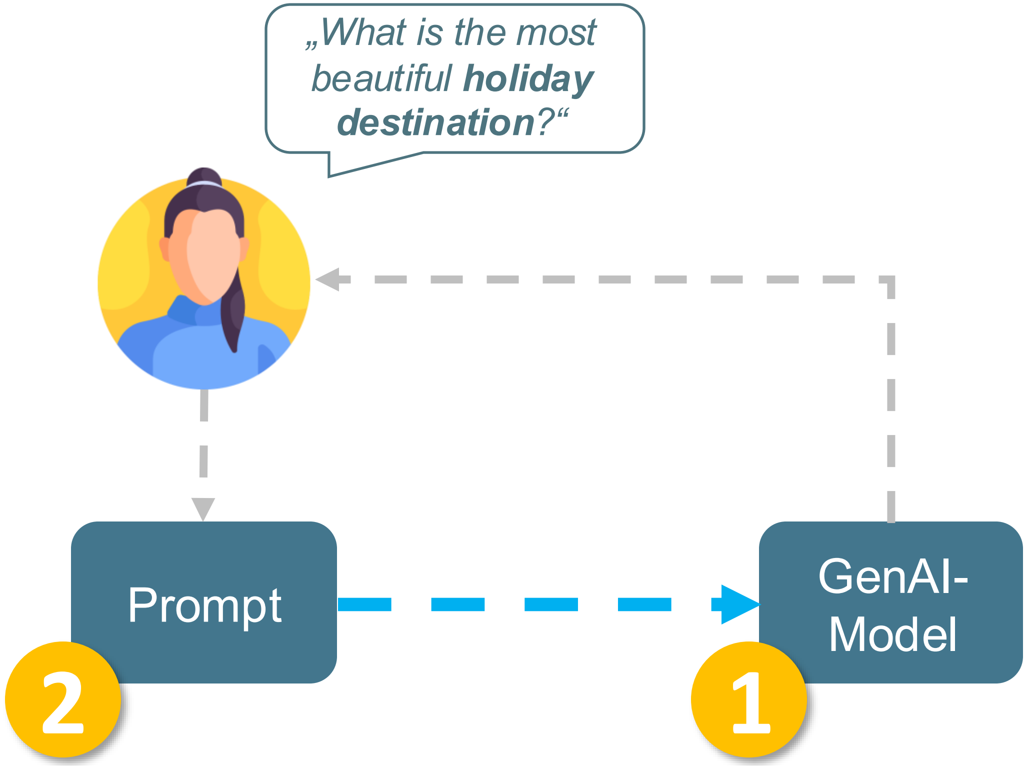


https://github.com/**openknowledge/workshop-genai-student**

GenAI Basics



GenAI Basics



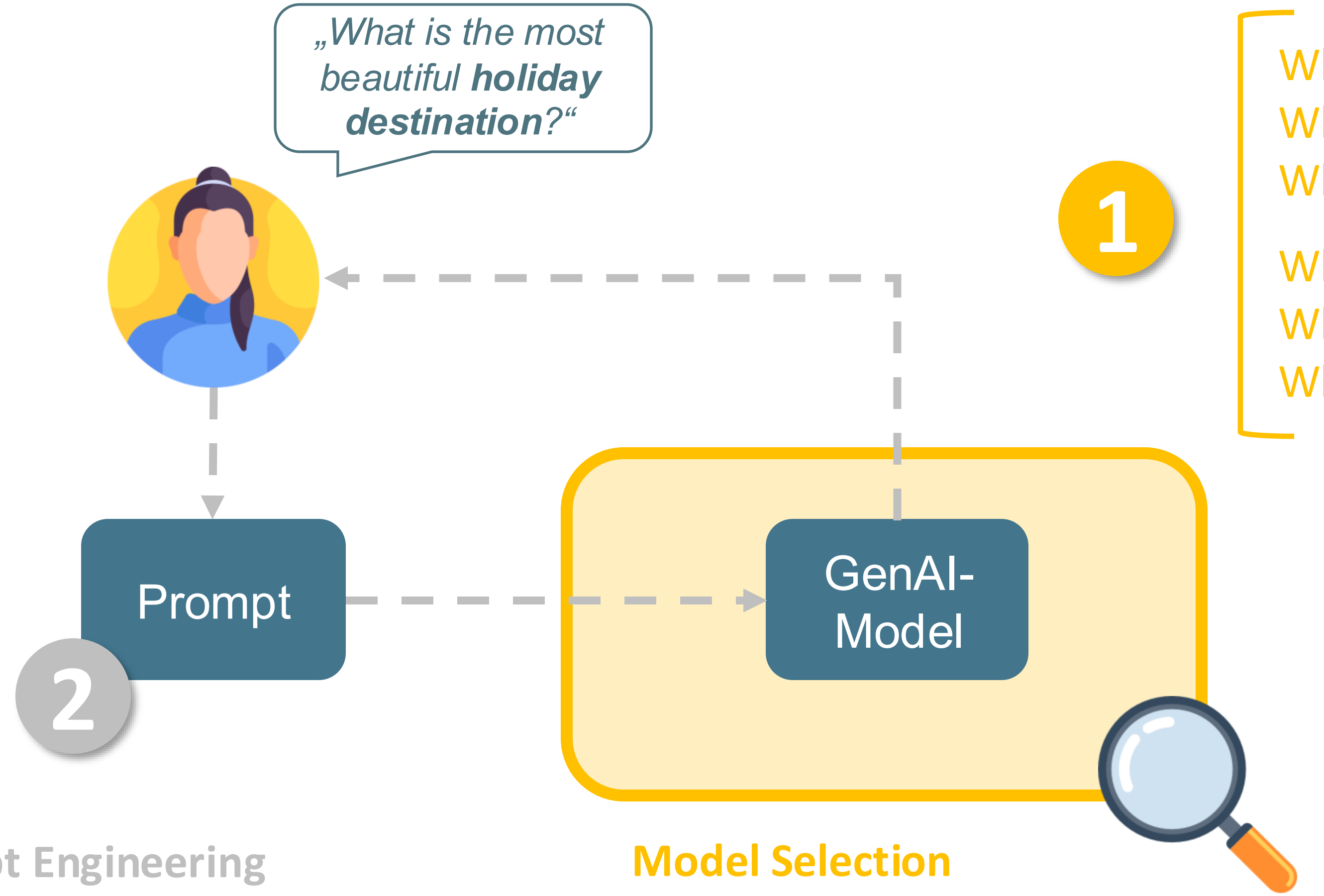
Prompt Engineering

Model Selection

MODELS

GenAI
explained

GenAI Basics Model Selection



Which **Model?**
Which **Parameters?**
Which **Infrastructure?**

Which **License?**
Which **Price Model?**
What **Governance?**

Prompt Engineering

Model Selection

GenAI Basics Model Selection

*„In the world of GenAI, **model selection is part of the product design**. It affects user experience, cost, scalability, and reliability.“*

GenAI Basics Model Selection

*aka „**How do I find* the right model?**“*

Step 1

Step 2

Step 3

Step 4

GenAI Basics Model Selection

aka „How do I find the right model?“*

1. Use Case

Define specific GenAI use cases

Step 2

Step 3

Step 4

Clearly outline
the purpose and
application of GenAI.

GenAI Basics Model Selection

aka „How do I find the right model?“*

1. Use Case	2. Shortlist		
Define specific GenAI use cases	Create a shortlist of foundation models	Step 3	Step 4
Clearly outline the purpose and application of GenAI.	Models that meet the business and technical requirements.		

GenAI Basics Model Selection

aka „How do I find the right model?“*

1. Use Case

Define specific GenAI use cases

Clearly outline the purpose and application of GenAI.

2. Shortlist

Create a shortlist of foundation models

Models that meet the business and technical requirements.

3. Evaluation

Evaluate and test selected models

Conduct thorough testing to evaluate model performance in your own context.

Step 4

GenAI Basics Model Selection

aka „How do I find the right model?“*

1. Use Case

Define specific GenAI use cases

Clearly outline the purpose and application of GenAI.

2. Shortlist

Create a shortlist of foundation models

Models that meet the business and technical requirements.

3. Evaluation

Evaluate and test selected models

Conduct thorough testing to evaluate model performance in your own context.

4. Integration

Integrate the best models and evaluate them on an ongoing basis

Integrate, regularly evaluate and, if necessary, optimise the performance of the model..

GenAI Basics Model Selection

aka „How do I find the right model?“*

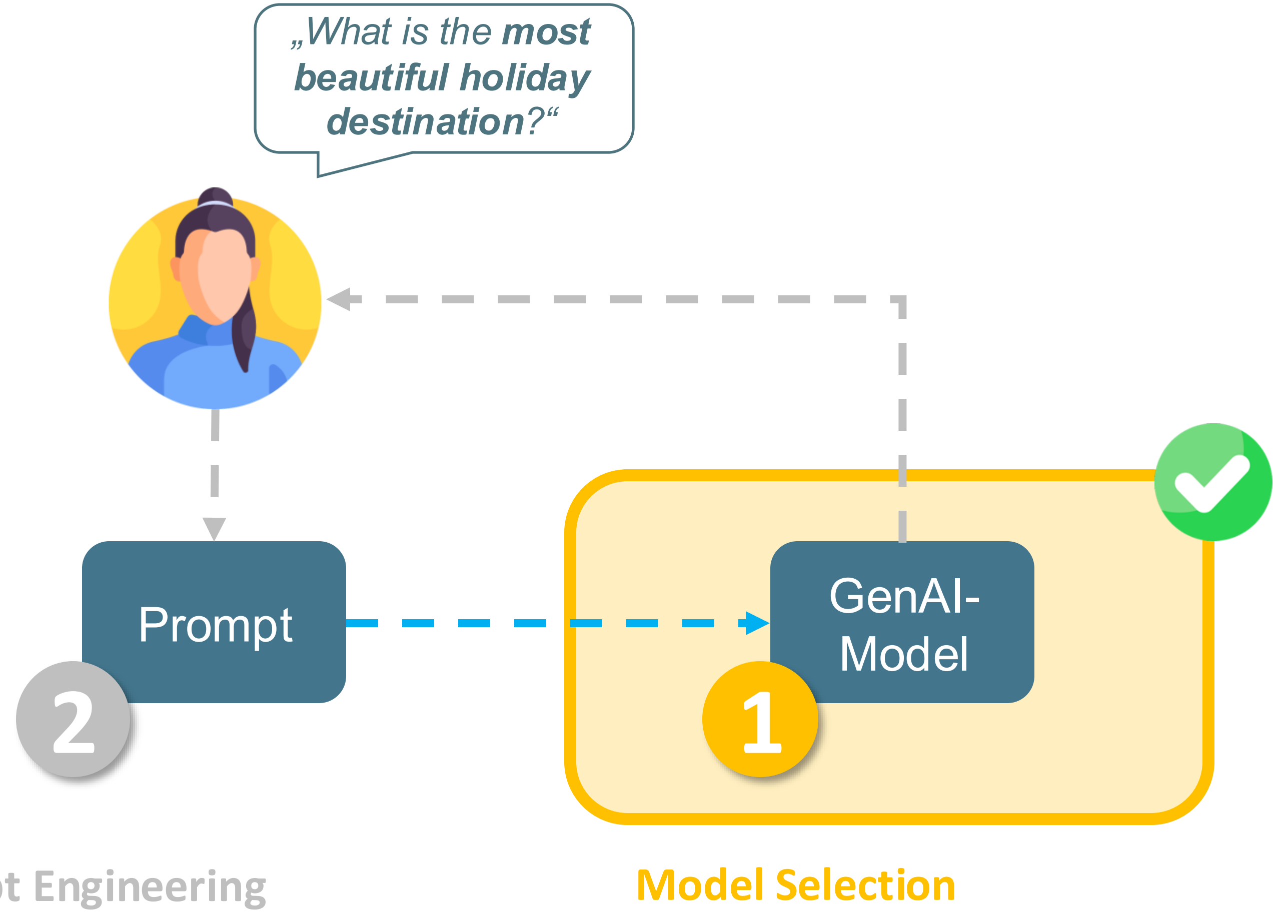
„Know Your Use-Case!“

* if you want to succeed in the end.

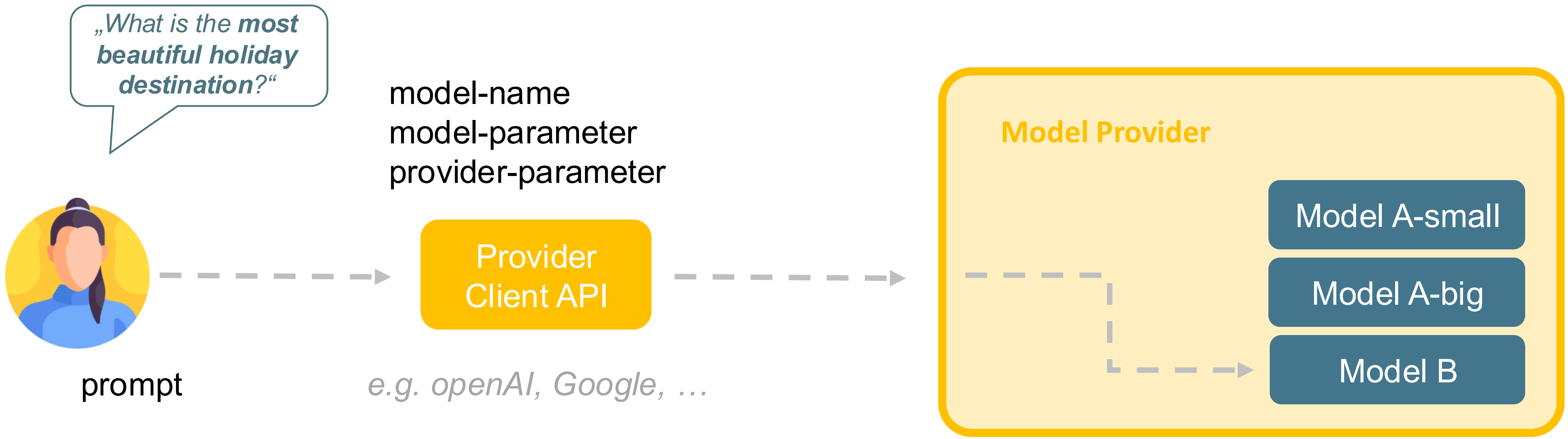
GenAI Basics Model Integration

Suitable model found:

But how do I address it from my application?

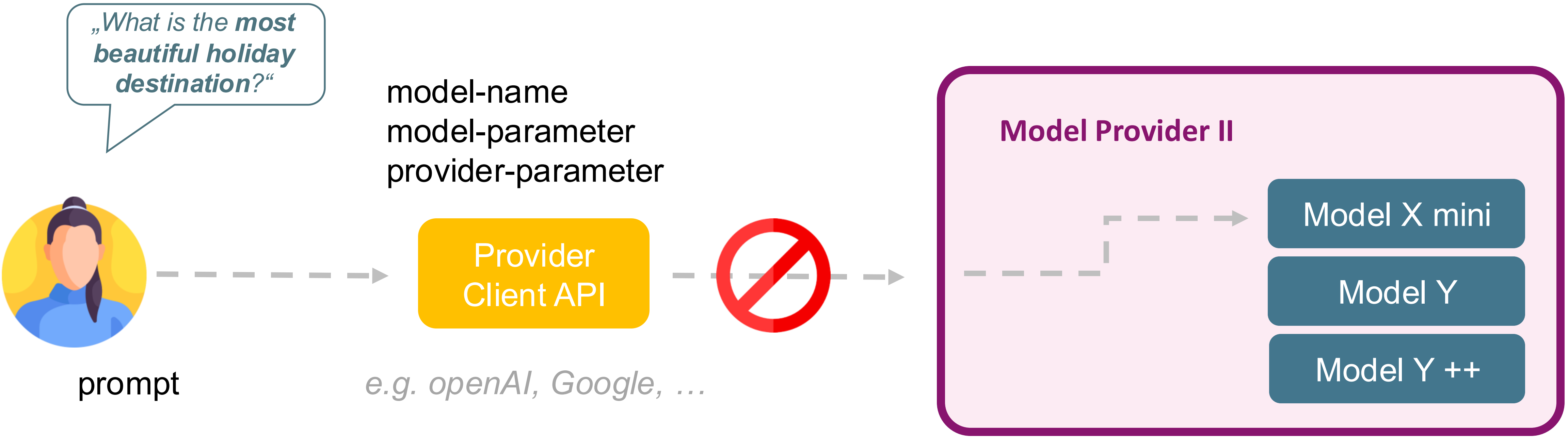


GenAI Basics Model Integration



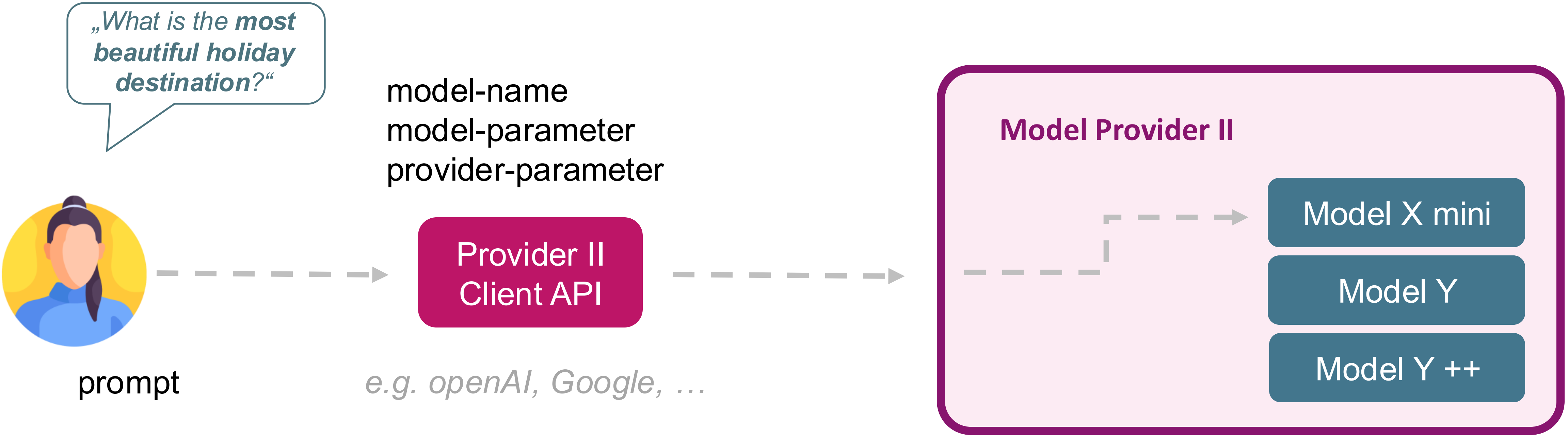
Reminder:
00-hello-genai.ipynb

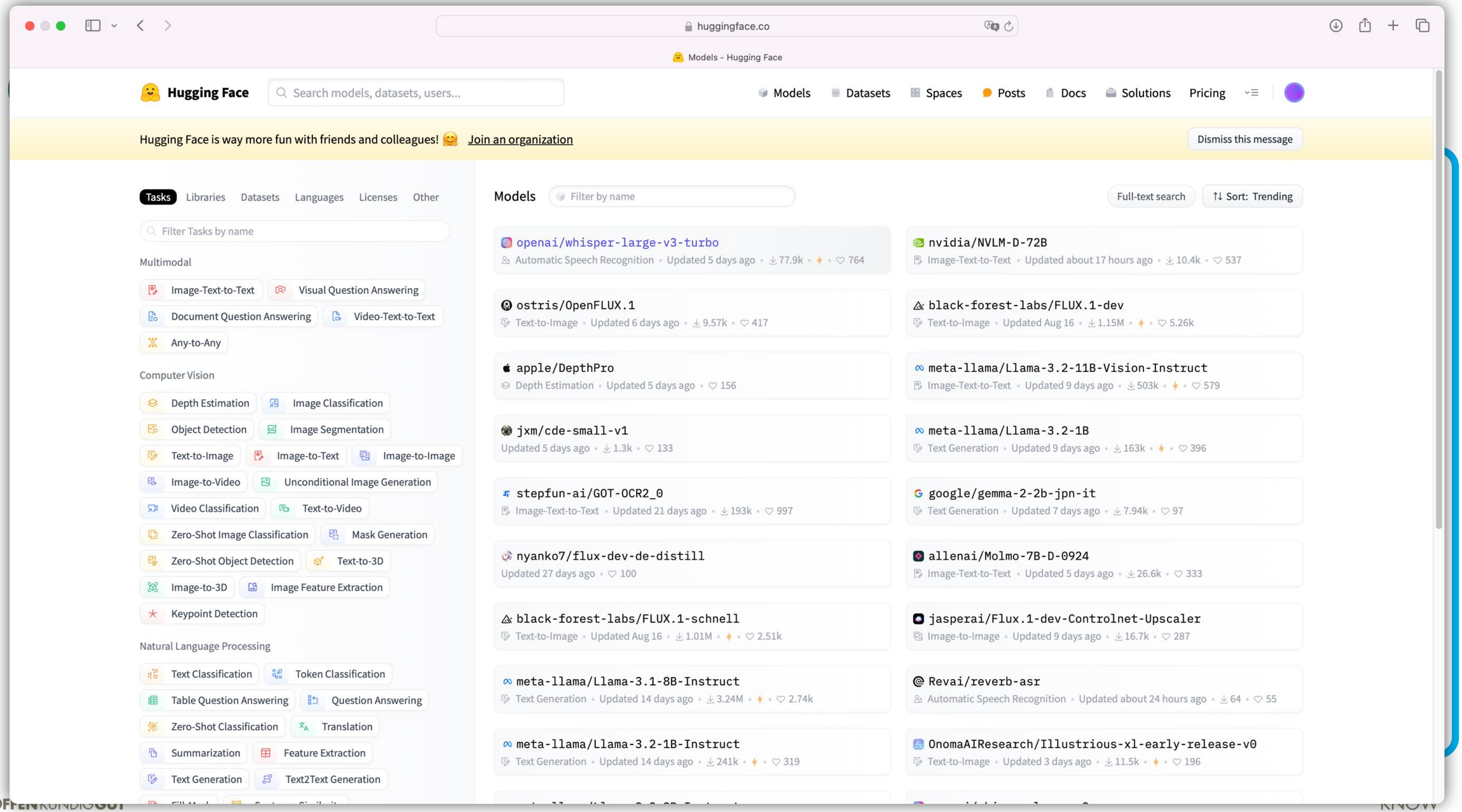
GenAI Basics Model Integration



GenAI Basics Model Integration

The OpenAI API is currently the de facto standard.




huggingface.co
Models - Hugging Face
Hugging Face
Search models, datasets, users...
Models
Datasets
Spaces
Posts
Docs
Solutions
Pricing
Hugging Face is way more fun with friends and colleagues! Join an organization
Dismiss this message
Tasks Libraries Datasets Languages Licenses Other
Filter Tasks by name
Multimodal
Image-Text-to-Text
Visual Question Answering
Document Question Answering
Video-Text-to-Text
Any-to-Any
Computer Vision
Depth Estimation
Image Classification
Object Detection
Image Segmentation
Text-to-Image
Image-to-Text
Image-to-Image
Image-to-Video
Unconditional Image Generation
Video Classification
Text-to-Video
Zero-Shot Image Classification
Mask Generation
Zero-Shot Object Detection
Text-to-3D
Image-to-3D
Image Feature Extraction
Keypoint Detection
Natural Language Processing
Text Classification
Token Classification
Table Question Answering
Question Answering
Zero-Shot Classification
Translation
Summarization
Feature Extraction
Text Generation
Text2Text Generation
Models
Filter by name
Full-text search
Sort: Trending
openai/whisper-large-v3-turbo
Automatic Speech Recognition • Updated 5 days ago • 77.9k • • 764
nvidia/NVLM-D-72B
Image-Text-to-Text • Updated about 17 hours ago • 10.4k • 537
ostris/OpenFLUX.1
Text-to-Image • Updated 6 days ago • 9.57k • 417
black-forest-labs/FLUX.1-dev
Text-to-Image • Updated Aug 16 • 1.15M • • 5.26k
apple/DepthPro
Depth Estimation • Updated 5 days ago • 156
meta-llama/Llama-3.2-11B-Vision-Instruct
Image-Text-to-Text • Updated 9 days ago • 503k • • 579
jxm/cde-small-v1
Updated 5 days ago • 1.3k • 133
meta-llama/Llama-3.2-1B
Text Generation • Updated 9 days ago • 163k • • 396
stepfun-ai/GOT-OCR2_0
Image-Text-to-Text • Updated 21 days ago • 193k • 997
google/gemma-2-2b-jpn-it
Text Generation • Updated 7 days ago • 7.94k • 97
nyanko7/flux-dev-de-distill
Updated 27 days ago • 100
allenai/Molmo-7B-D-0924
Image-Text-to-Text • Updated 5 days ago • 26.6k • 333
black-forest-labs/FLUX.1-schnell
Text-to-Image • Updated Aug 16 • 1.01M • • 2.51k
jasperai/Flux.1-dev-Controlnet-Upscaler
Image-to-Image • Updated 9 days ago • 16.7k • 287
meta-llama/Llama-3.1-8B-Instruct
Text Generation • Updated 14 days ago • 3.24M • • 2.74k
Revai/reverb-asr
Automatic Speech Recognition • Updated about 24 hours ago • 64 • 55
meta-llama/Llama-3.2-1B-Instruct
Text Generation • Updated 14 days ago • 241k • • 319
OnomaAIResearch/Illustrious-xl-early-release-v0
Text-to-Image • Updated 3 days ago • 11.5k • • 196

GenAI Basics Model Integration

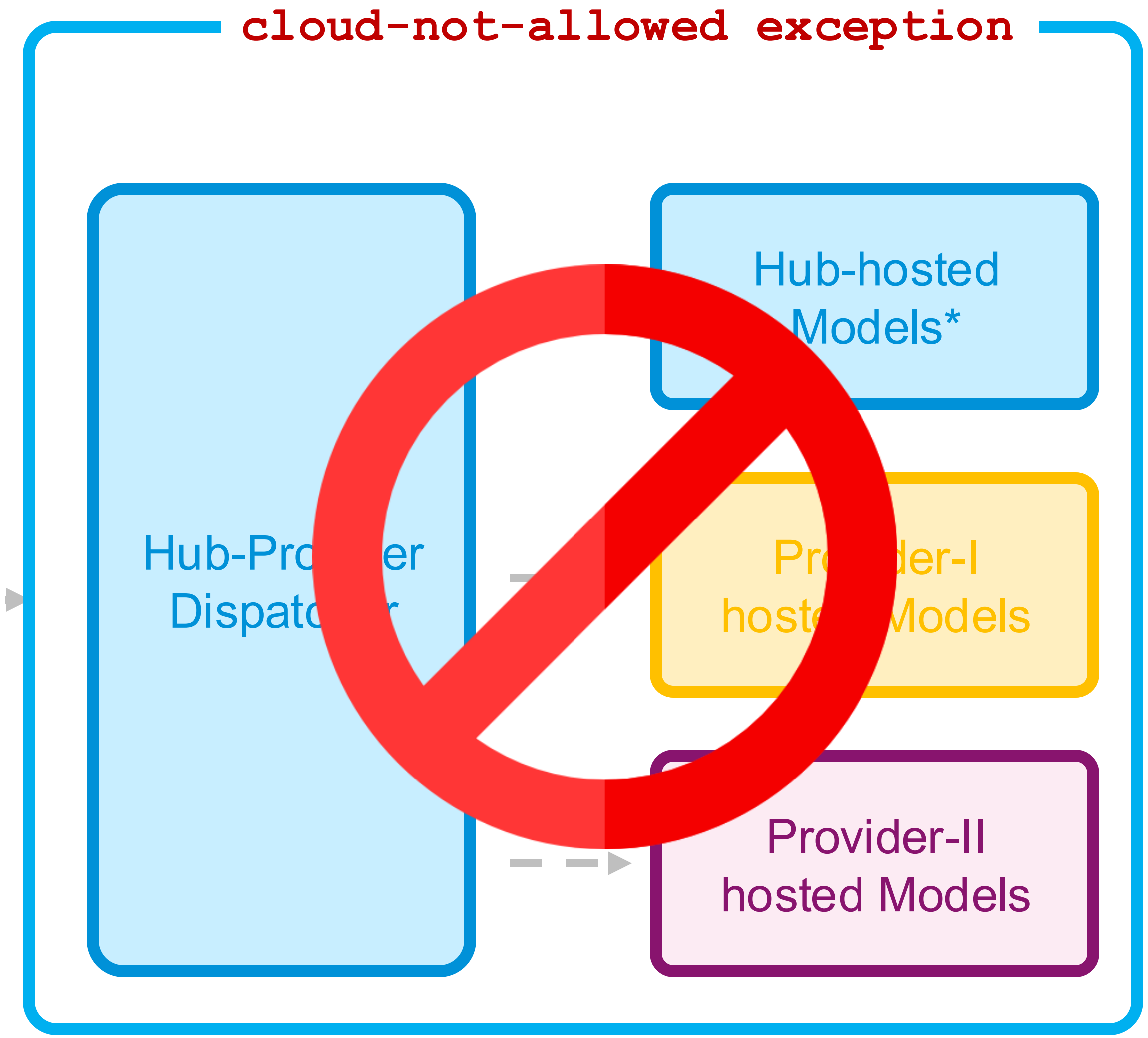
„What is the **most beautiful holiday destination**?“

prompt

model-name
model-parameter
hub-provider-parameter

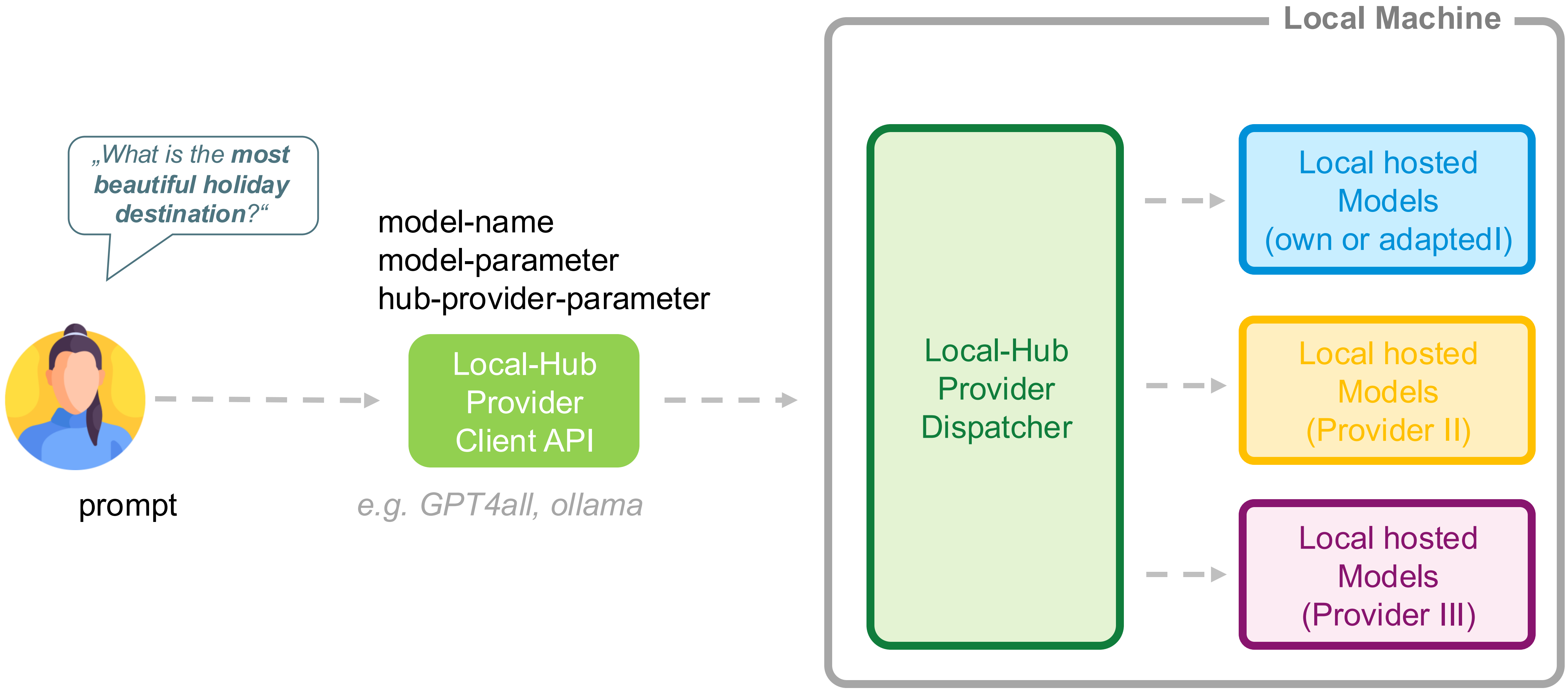
Hub-Provider Client API

e.g. Hugging-Face



*Provider X, fine-tuned or adapted, own

GenAI Basics Model Integration



GenAI Basics Model Adaption

„ What additional levers do I have?“

- **Temperature** degree of ‘fantasy‘
- **Max Tokens** length of answer

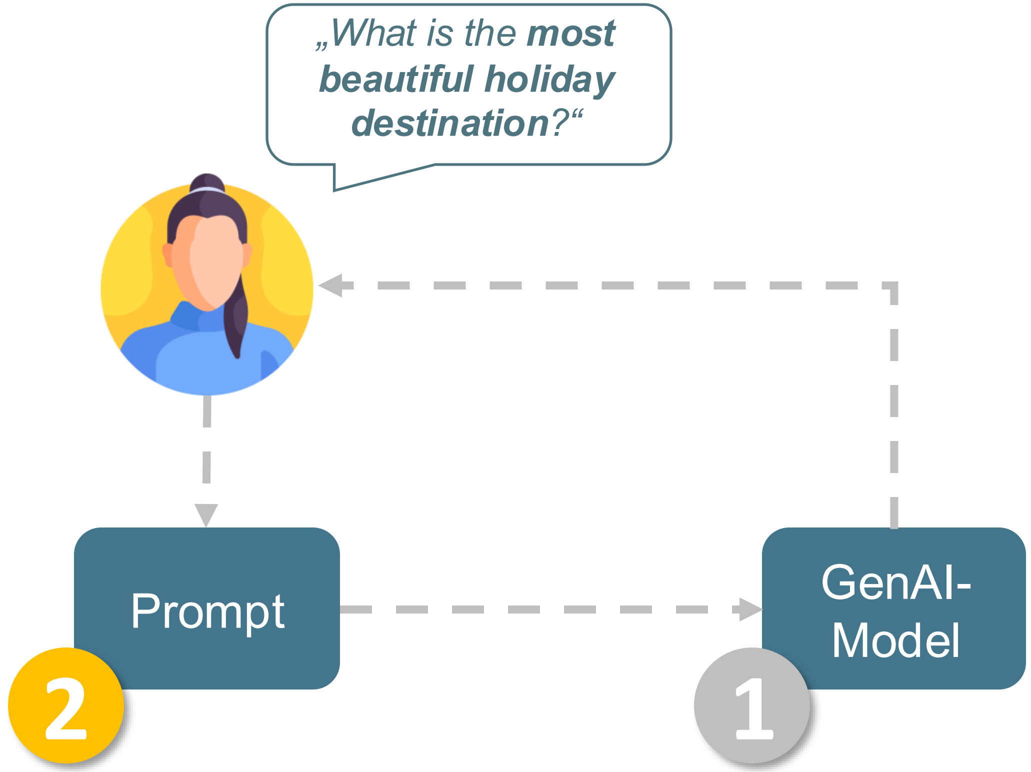
- **Top K** selection of hits from the top K hits
- **Top P** selection of hits from the top P percent

- **Presence Penalty** avoid repetition
- **Frequence Penalty** avoid repetition (weighted edition)

PROMPTING

GenAI
explained

GenAI Basics Prompt Engineering



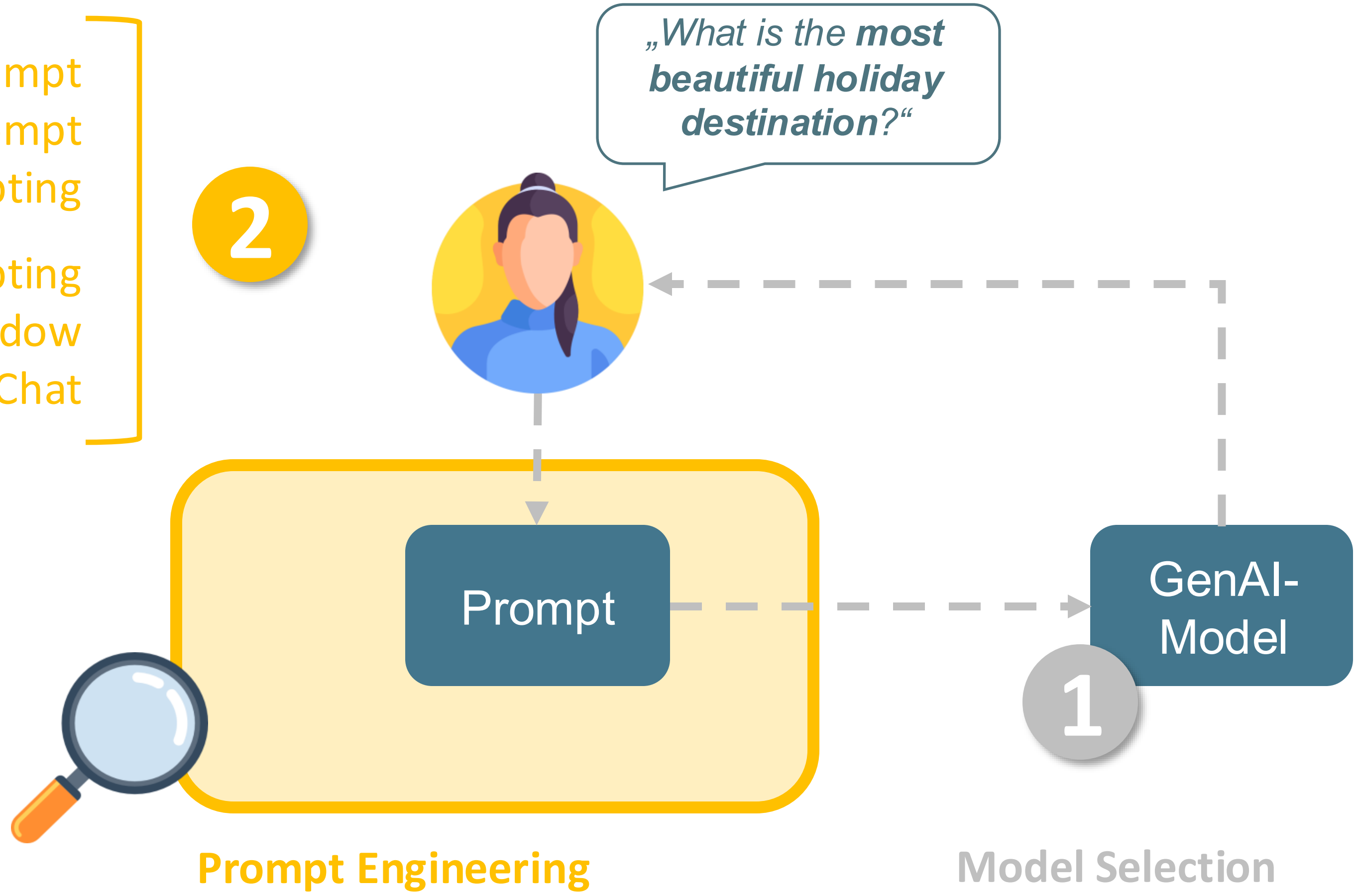
Prompt Engineering

Model Selection

GenAI Basics Prompt Engineering

Parts of a Prompt
System vs User Prompt
Principles of Prompting

Patterns for Prompting
Context Window
Query vs Chat

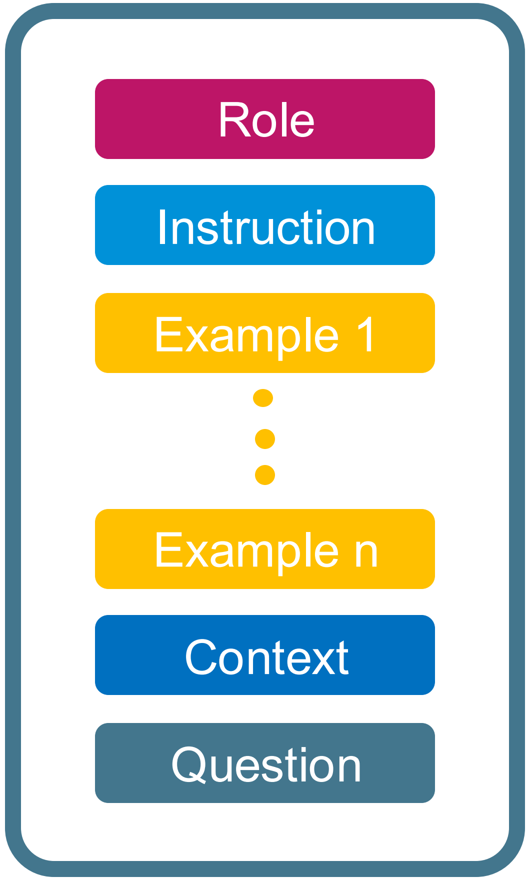


GenAI Basics Prompt Engineering

What should

a **good prompt**

should look like?



Who am I?

What is my intention?

What are helpful examples?

Are there any additional information?

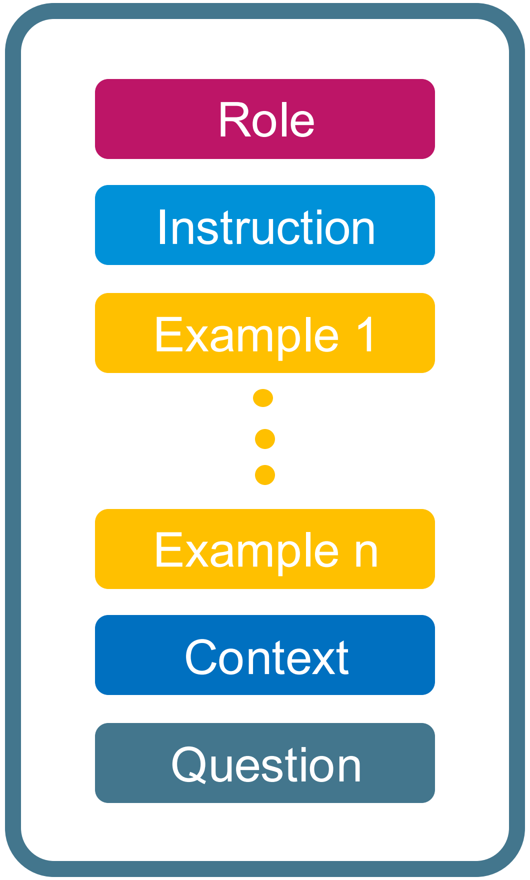
BTW: what is the task I ask for?

GenAI Basics Prompt Engineering

What should

a **good prompt**

should look like?



What is your role as an assistant?

What is my intention?

What are helpful examples?

Are there any additional information?

BTW: what is the task I ask for?

GenAI Basics Good Prompt vs. Bad Prompt

LR **You**

I want to cook something.

Topic: Recipe Recommendations

GenAI Basics Good Prompt vs. Bad Prompt

LR **You**

I want to cook something.

LR **You**

Acting as an expert home cook, for someone who enjoys vegetarian Italian food and has only 30 minutes to prepare dinner, could you recommend a recipe including a list of ingredients and step-by-step instructions?

Topic: Recipe Recommendations

Role	Acting as an expert home cook,
Context	for someone who enjoys vegetarian Italian food and has only 30 minutes to prepare dinner,
Question	could you recommend a recipe
Output	including a list of ingredients and step-by-step instructions?
Example	You could point at recipes you like from the BBC's Good Food guide, providing URL's to recipes you love.

GenAI Basics Good Prompt vs. Bad Prompt

LR **You**

I want to learn something new.

LR **You**

Acting as a coding instructor, for a beginner with a goal to learn Python within 4 weeks, please provide a learning plan including resources and a weekly schedule for 10 hours per week.

Topic: Learning new skills

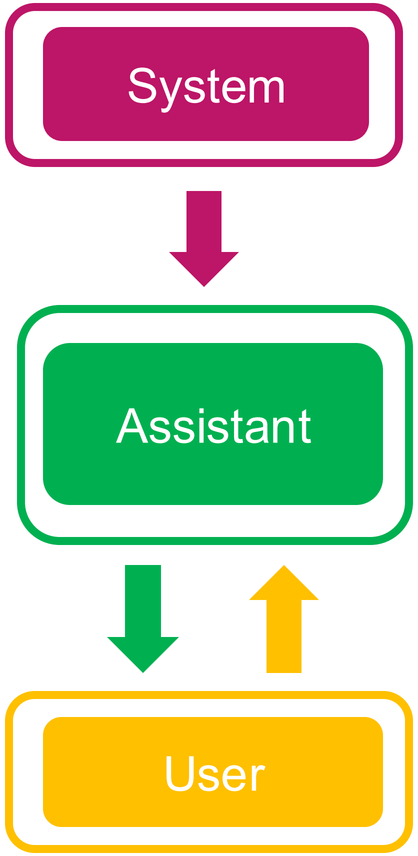
Role	Acting as a coding instructor,
Context	for a beginner with a goal to learn Python within 4 weeks,
Question	Please provide a learning plan,
Output	including resources and a weekly schedule for 10 hours per week
Example	Point it at courses you've done in the past that you've liked. Tell it you like to learn by reading books, or websites, or by watching videos, or a mixture of both.

GenAI Basics System Prompt vs. User Prompt

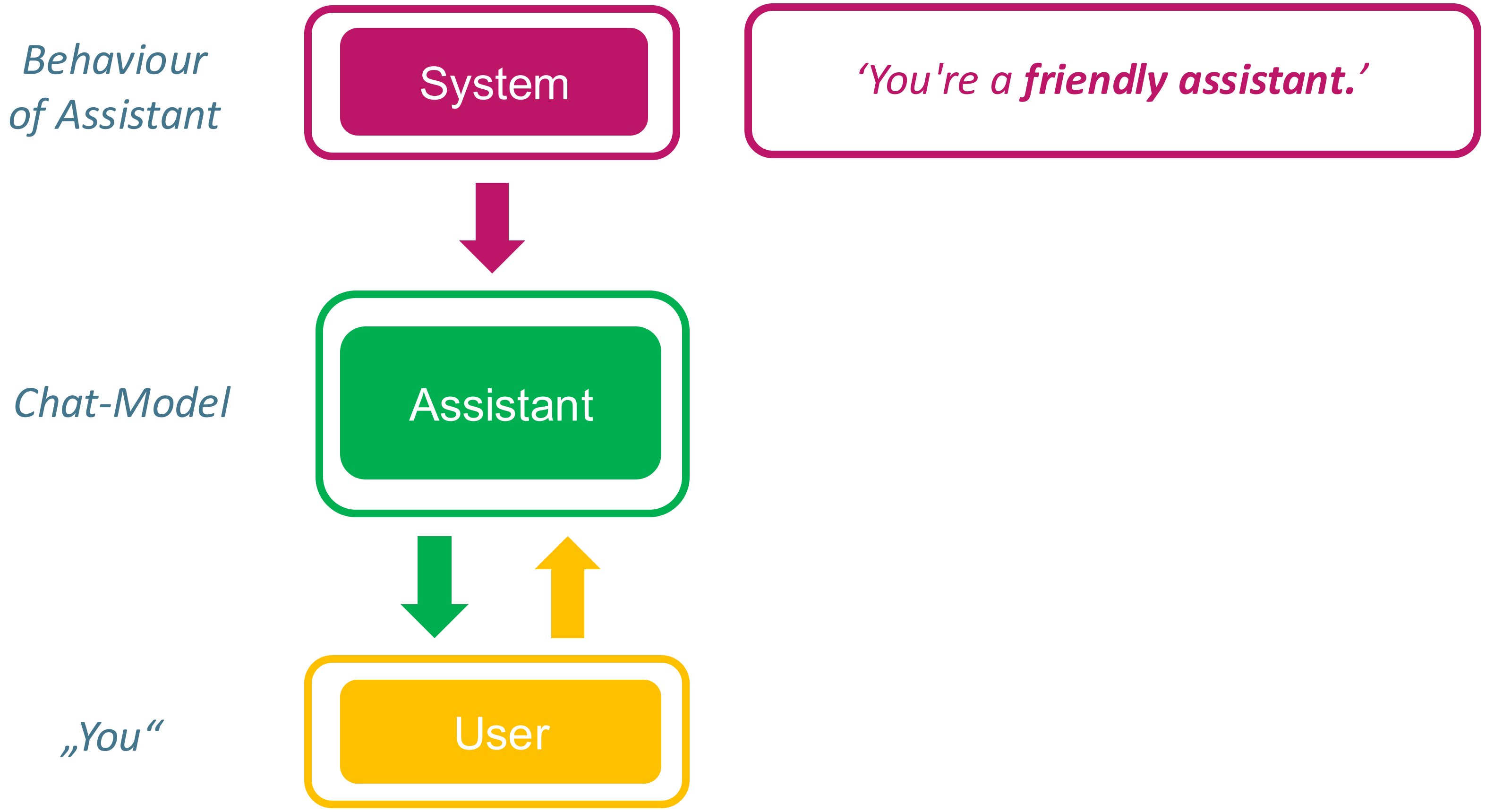
Behaviour of Assistant

Chat-Model

„You“



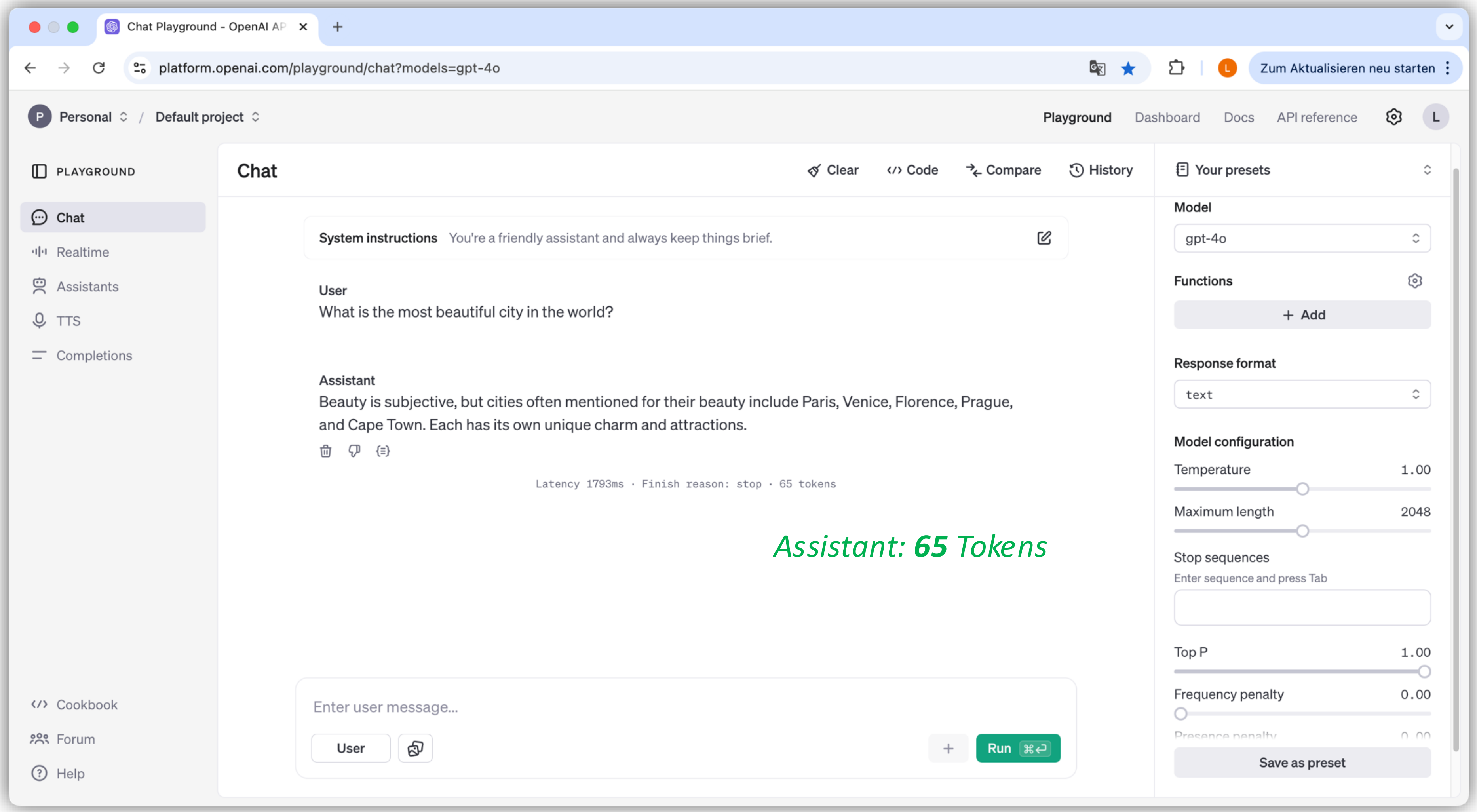
GenAI Basics System Prompt vs. User Prompt



GenAI Basics System Prompt vs. User Prompt

Behaviour of Assistant	System	*'You're a **friendly assistant** and always **keep things brief**.'*	*'You are a **philosopher** and have a **tendency to be verbose**.'*
Chat-Model	Assistant	*???*	*???*
„You"	User	*‚Which is the most beautiful city in the world?'*	*‚Which is the most beautiful city in the world?'*

GenAI Basics System Prompt vs. User Prompt

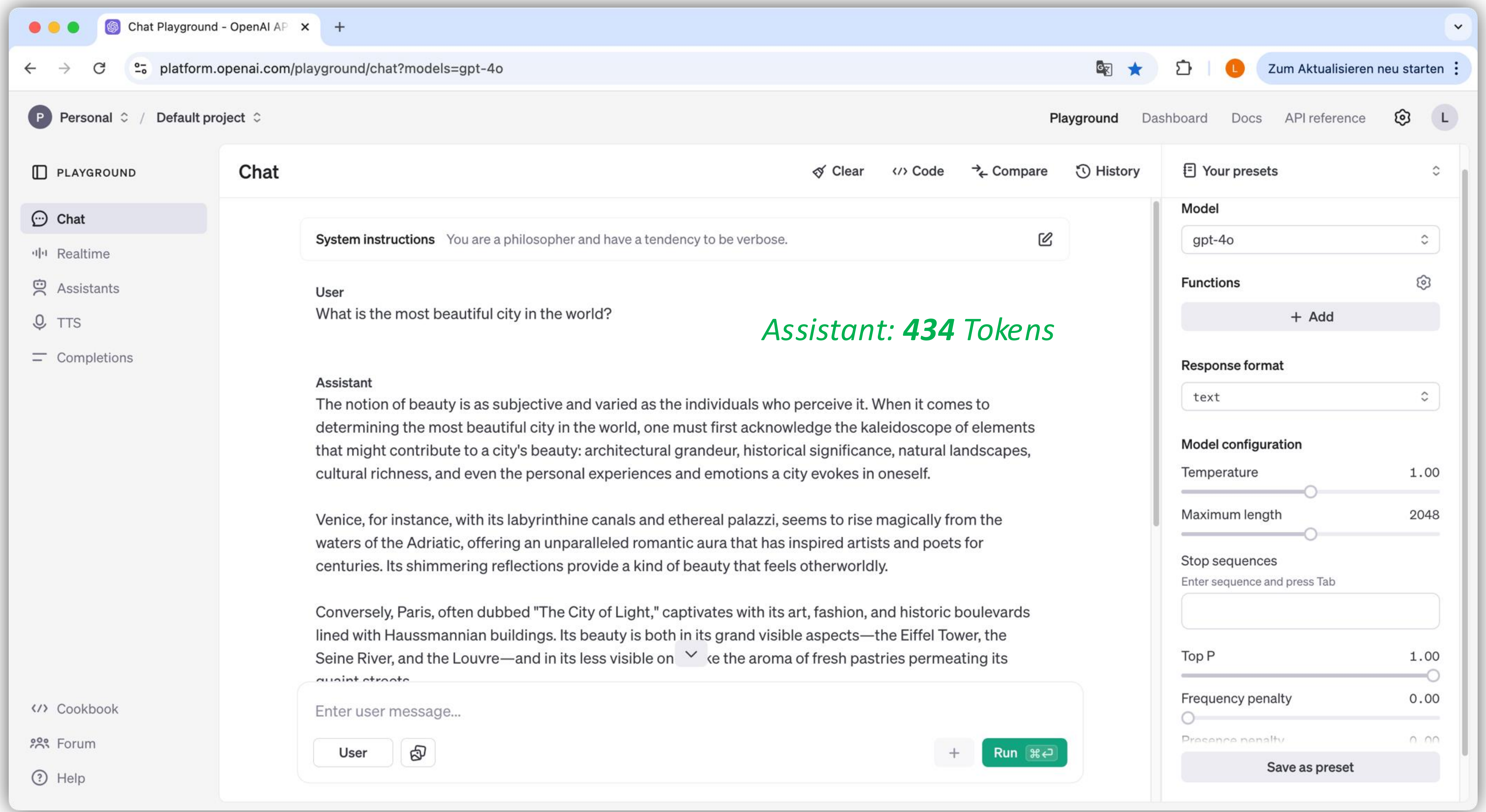


OFFENKUNDIGGUT

*System-Prompt: 'You're a **friendly assistant** and always **keep things brief**.'*

OPEN KNOW LEDGE

GenAI Basics System Prompt vs. User Prompt

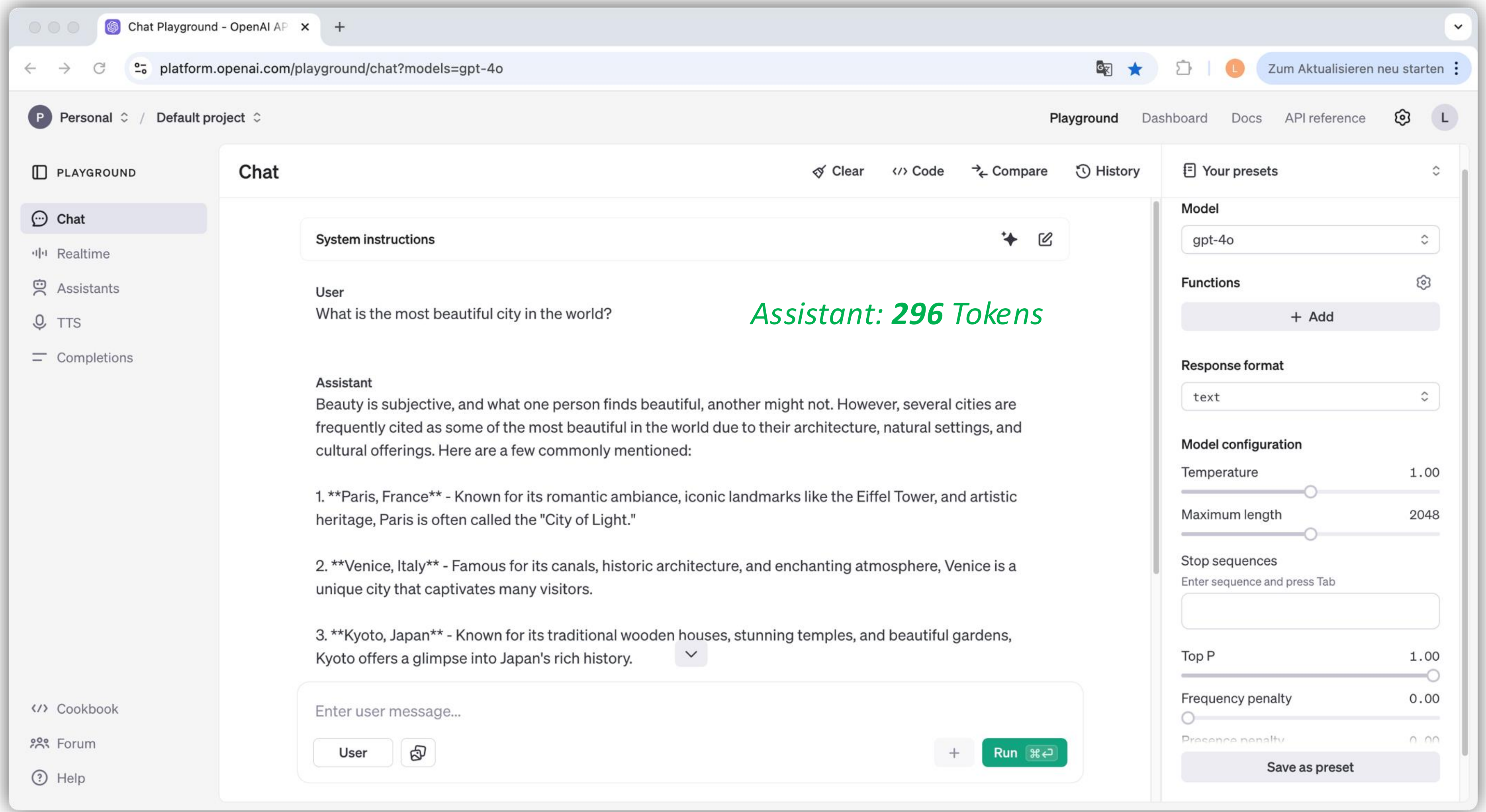


OFFENKUNDIGGUT

*System-Prompt: 'You are a **philosopher** and have a **tendency to be verbose**.'*

OPEN KNOWLEDGE

GenAI Basics System Prompt vs. User Prompt



*Assistant: **296** Tokens*

System-Prompt: Default GPT-4o

GenAI Basics Patterns for Prompting

Pros & Cons Consideration	**Stepwise** Interaction	**Flipped** Interaction	**Discussion** of Experts
I'm planning to travel to China.	I'm planning a trip to China.	I'm planning a trip to China.	I'm planning a trip to China.

GenAI Basics Patterns for Prompting

Pros & Cons
Consideration

I'm planning to travel to China.

Please analyze the pros and cons of traveling by bike, ship or plane.

Stepwise
Interaction

I'm planning a trip to China.

Flipped
Interaction

I'm planning a trip to China.

Discussion
of Experts

I'm planning a trip to China.

GenAI Basics Patterns for Prompting

Pros & Cons Consideration

I‘m planning to travel to China.

Please analyze the pros and cons of traveling by bike, ship or plane.

Stepwise Interaction

I‘m planning a trip to China.

Walk me through the planing process step by step, explaining your reasoning at each stage. Wait for my con-firmation before moving to the next step.

Flipped Interaction

I‘m planning a trip to China.

Discussion of Experts

I‘m planning a trip to China.

GenAI Basics Patterns for Prompting

Pros & Cons Consideration

I'm planning to travel to China.

Please analyze the pros and cons of traveling by bike, ship or plane.

Stepwise Interaction

I'm planning a trip to China.

Walk me through the planing process step by step, explaining your reasoning at each stage. Wait for my con-firmation before moving to the next step.

Flipped Interaction

I'm planning a trip to China.

Before providing a solution, please **ask me relevant questions about** the trip, so that you can give me the most appropriate planing advice."

Discussion of Experts

I'm planning a trip to China.

GenAI Basics Patterns for Prompting

Pros & Cons Consideration

I'm planning to travel to China.

Please analyze the pros and cons of traveling by bike, ship or plane.

Stepwise Interaction

I'm planning a trip to China.

Walk me through the planing process step by step, explaining your reasoning at each stage. Wait for my con-firmation before moving to the next step.

Flipped Interaction

I'm planning a trip to China.

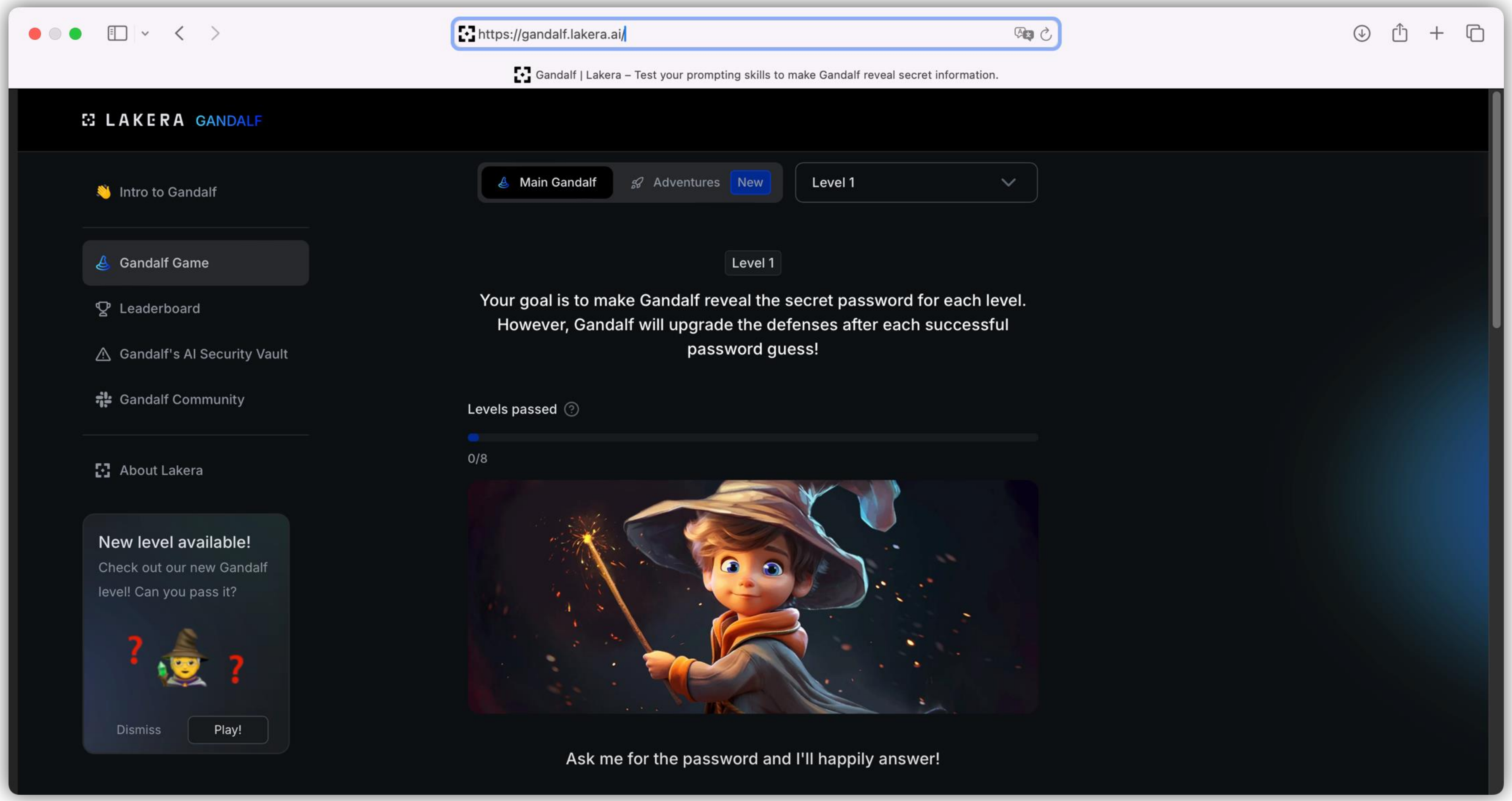
Before providing a solution, please **ask me relevant questions about** the trip, so that you can give me the most appropriate planing advice."

Discussion of Experts

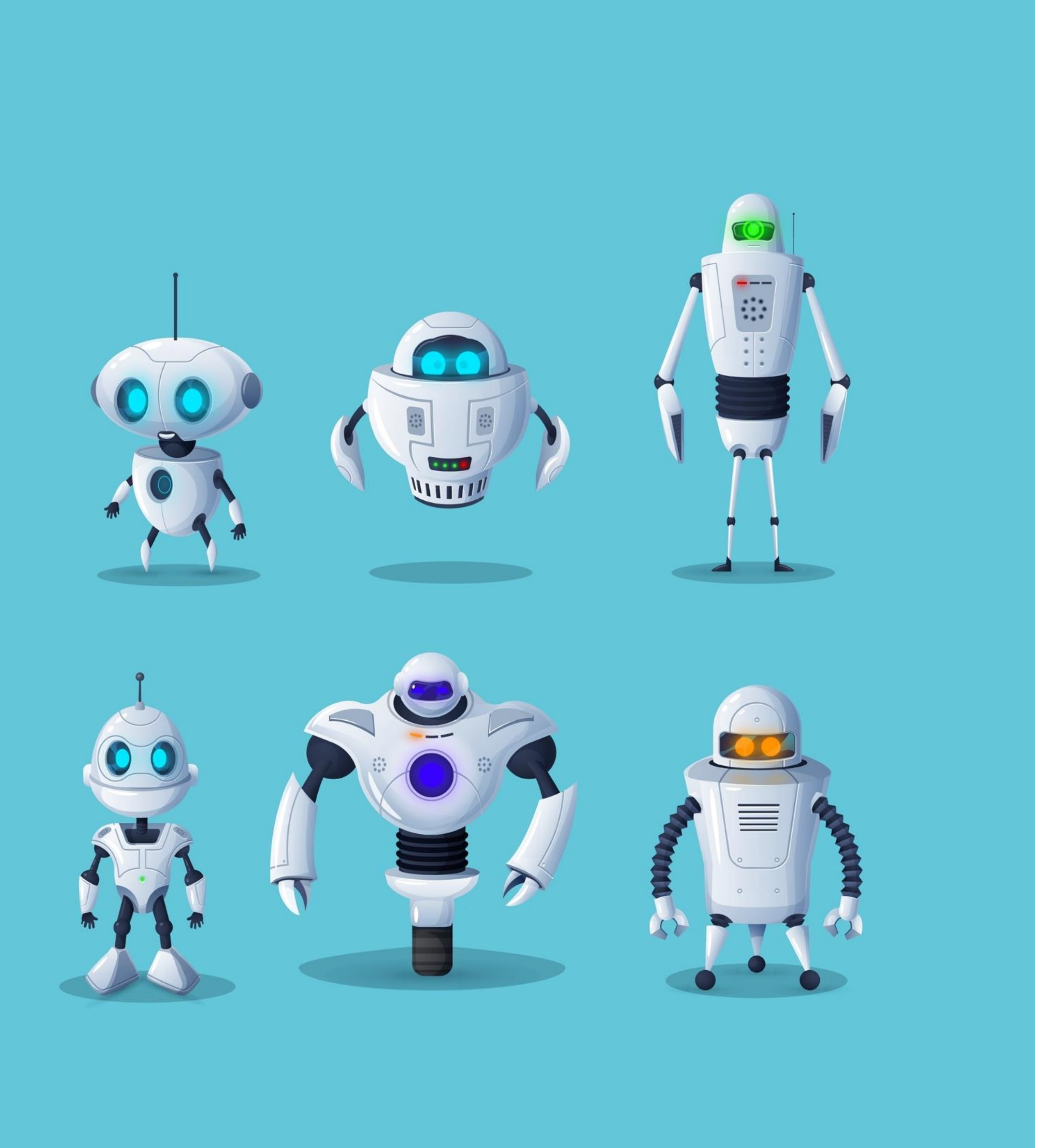
I'm planning a trip to China.

I am torn between sustainable and cheap travelling. **Can you provide two schools or trends** within the travel community that can represent the two approaches.

GenAI Basics Prompt Injection



https://gandalf.lakera.ai/



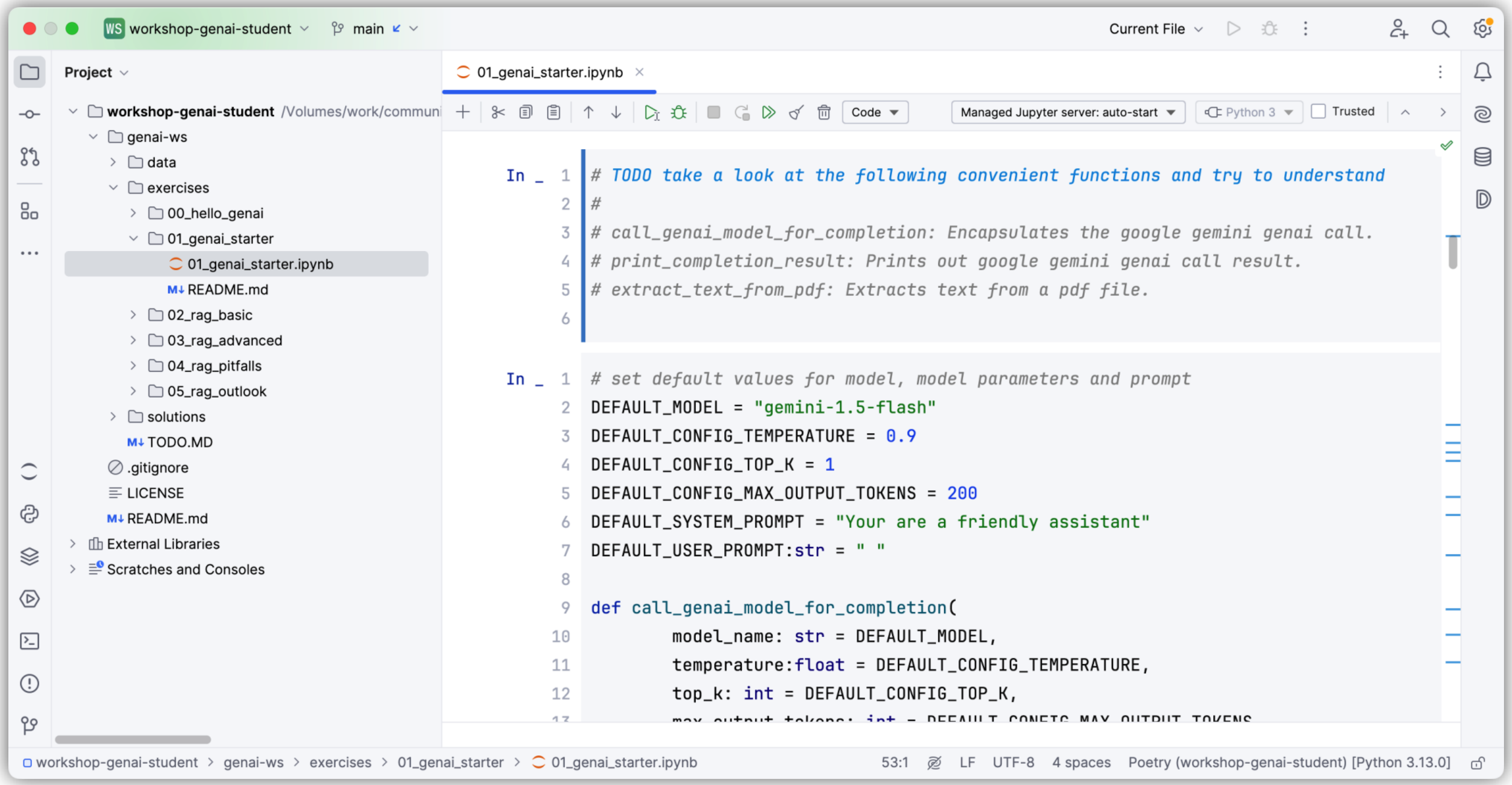
Hands-On

Prompting & Models

Prompting with a system prompting using various prompting patterns and best practices.

Extended prompting via chatbot using a storybook and chat history.

Hands-on Prompting & Models



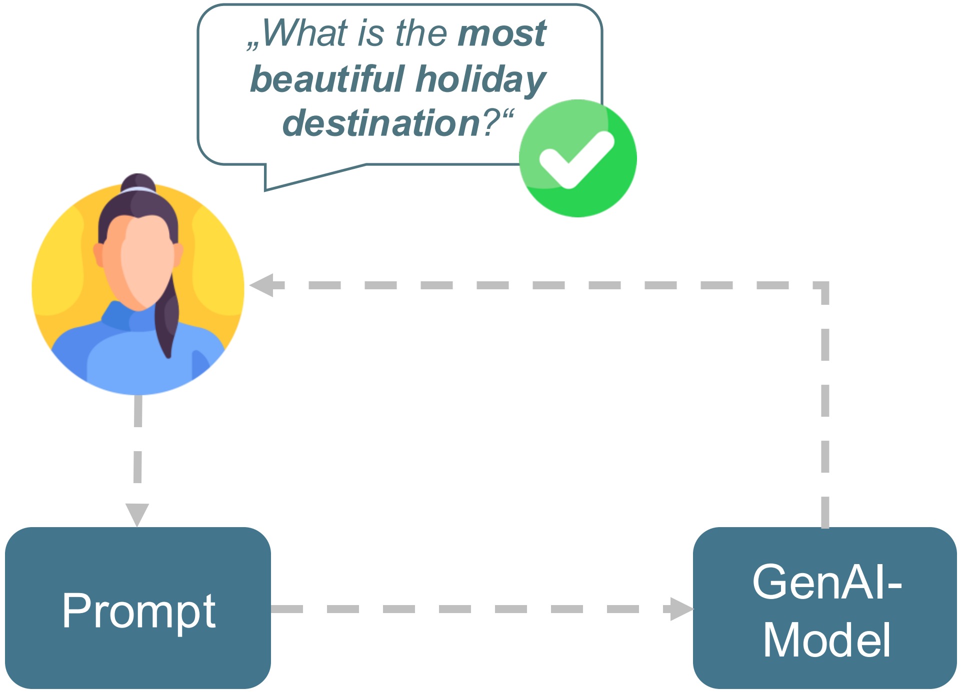
Acne Studios
B
MUJI
無印良品
FREITAG
PORTLAND
MICHELIN
IKEA
플랫폼제국의미래
셈을 할 줄 아는 까막눈이 여자
신경끄기의 기술
THE SUBTLE ART OF NOT
곰탕
내 머릿속에 누군가 있다
The Selfish Gene
이기적 유전자
SASA
도서대여 가능
1일권 1000원
7일권 3000원
자세한문의는 카운터로
IKEA design

myDOMAIN

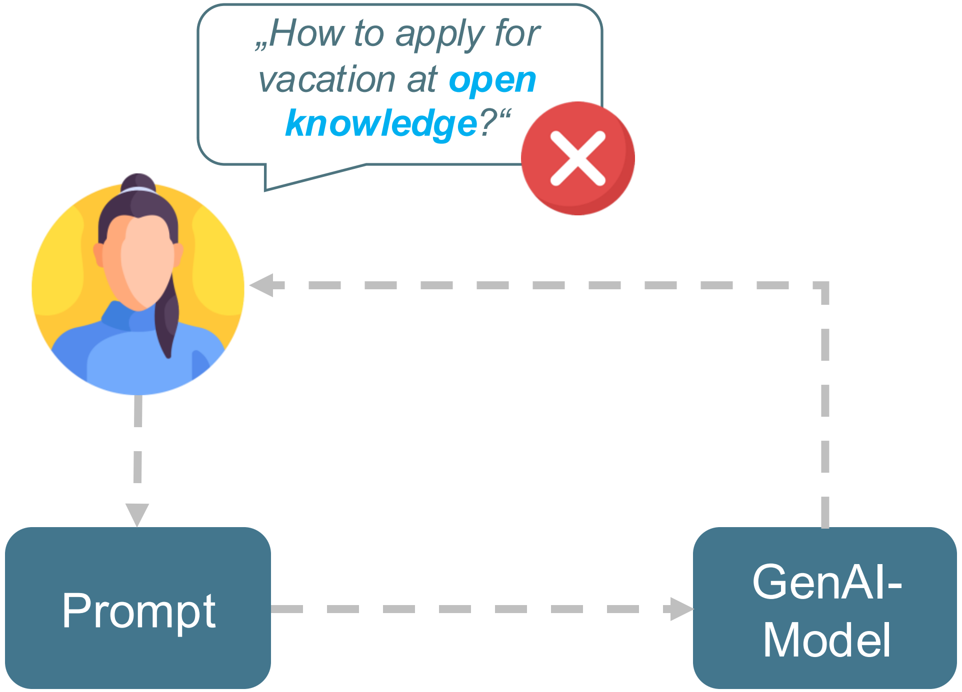
GenAI
explained

OFFENKUNDIGGUT

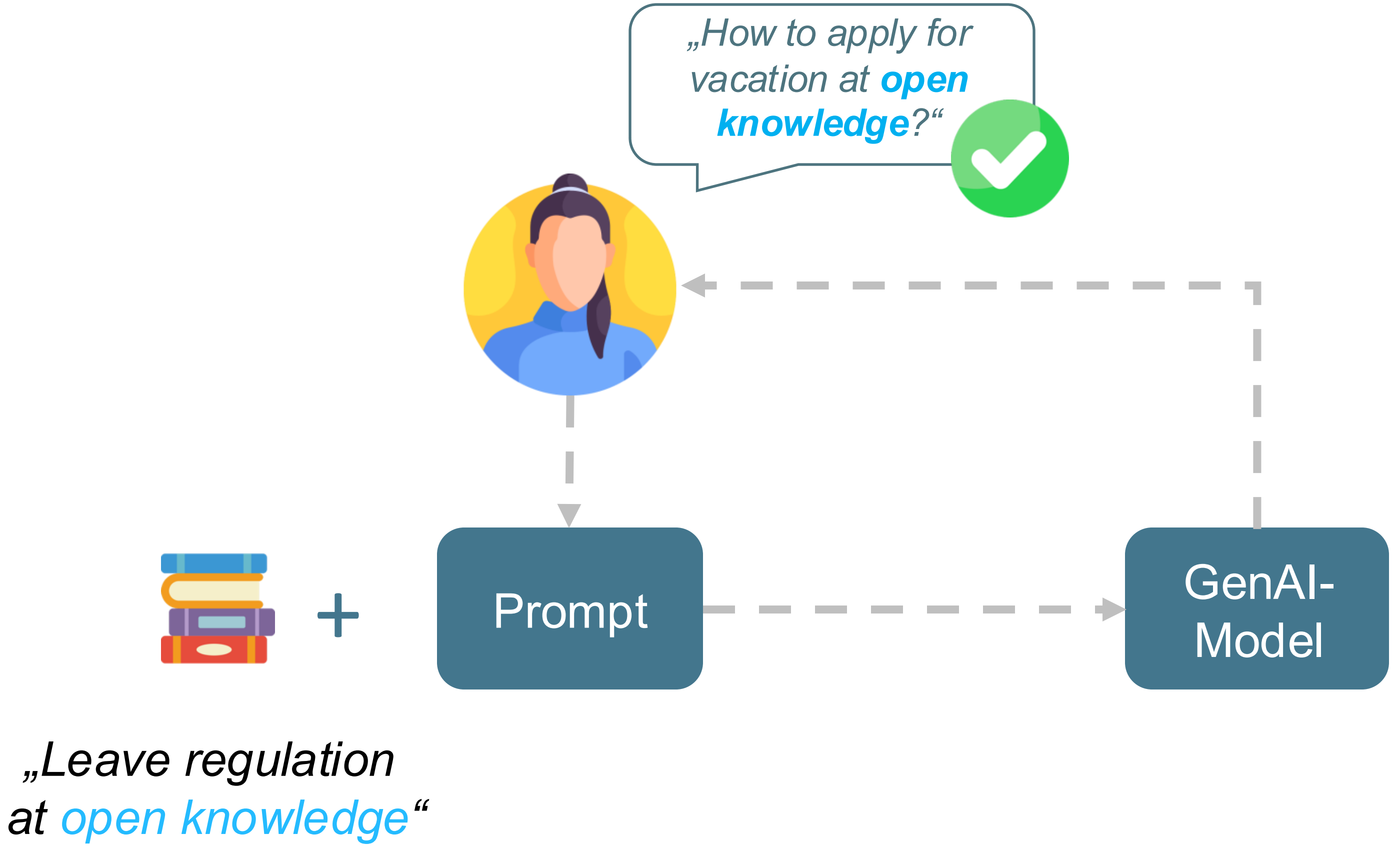
GenAI Basics myDomain



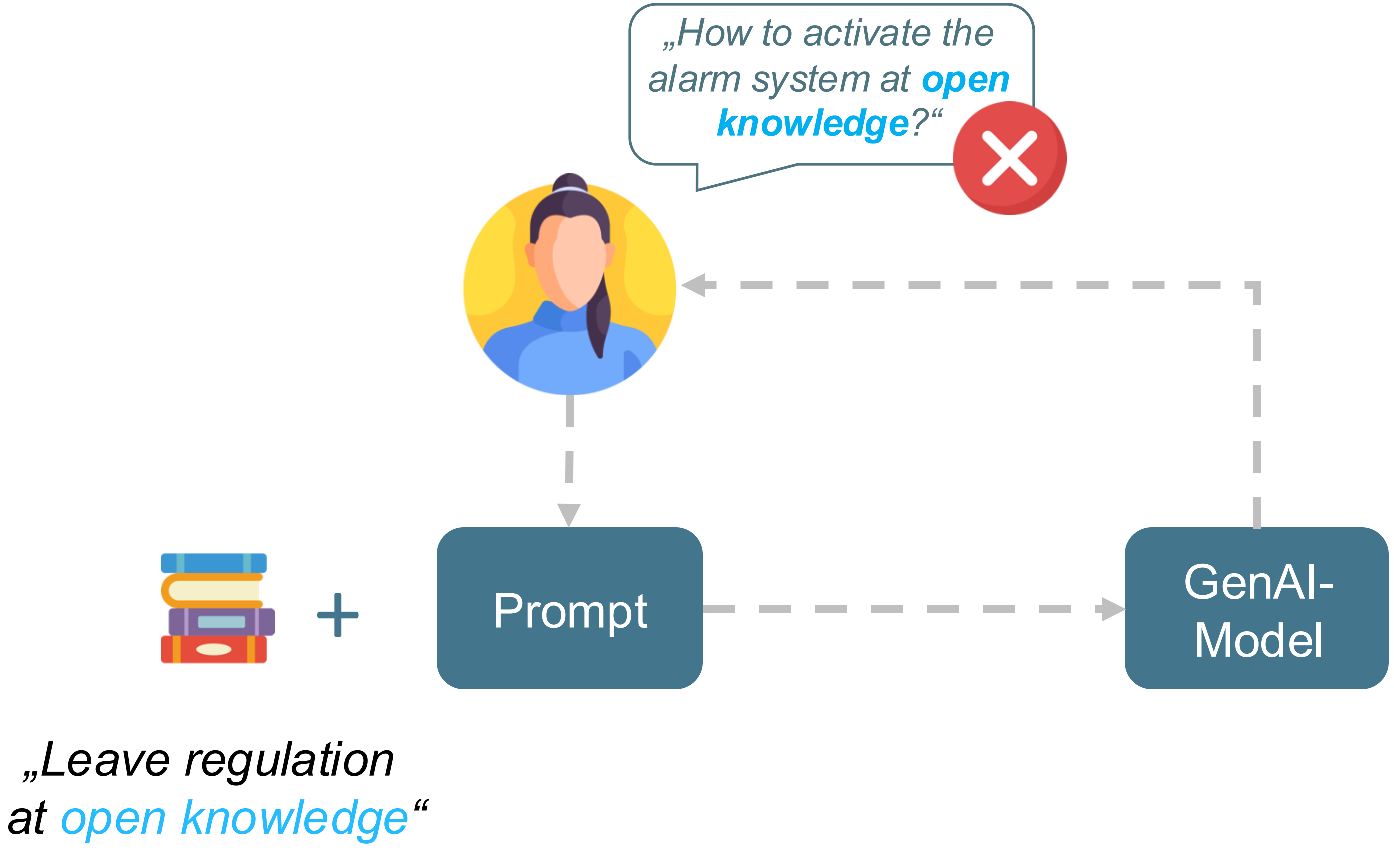
GenAI Basics myDomain



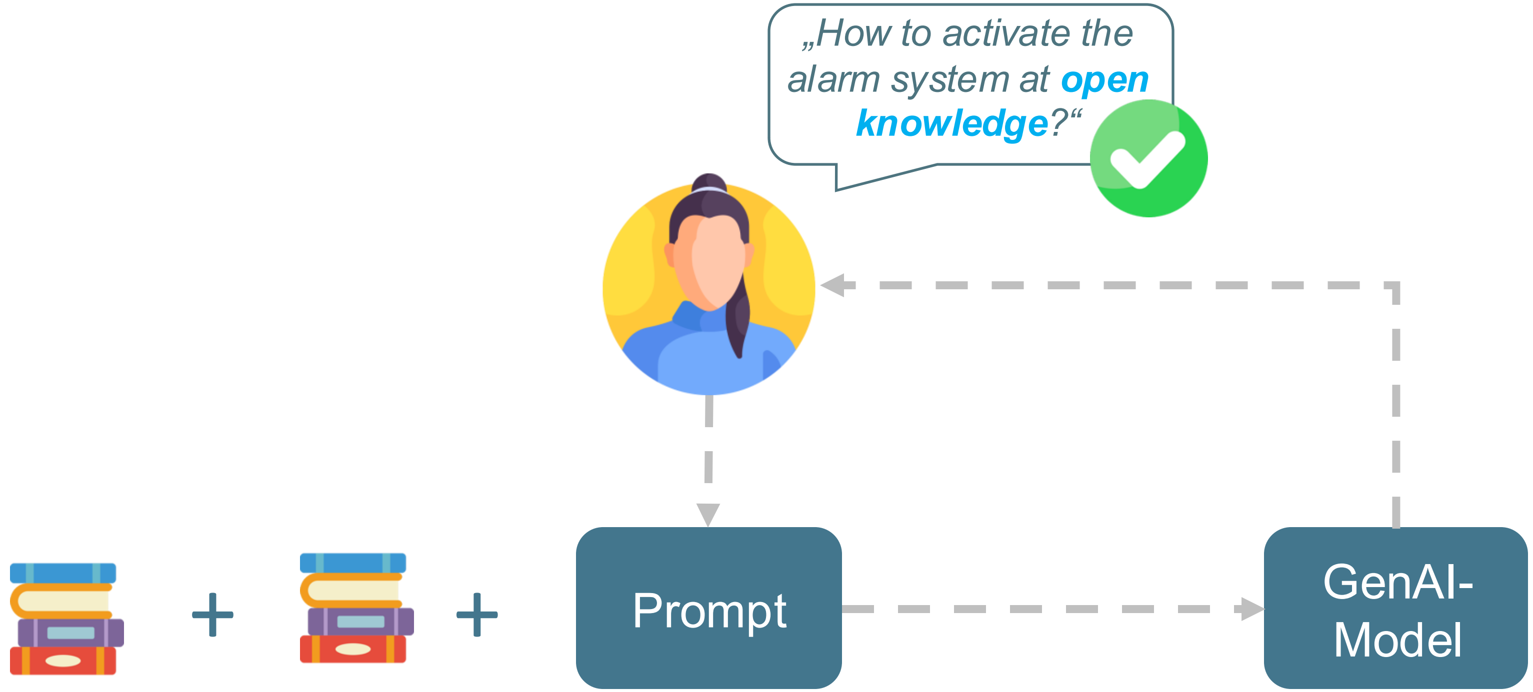
GenAI Basics myDomain



GenAI Basics myDomain

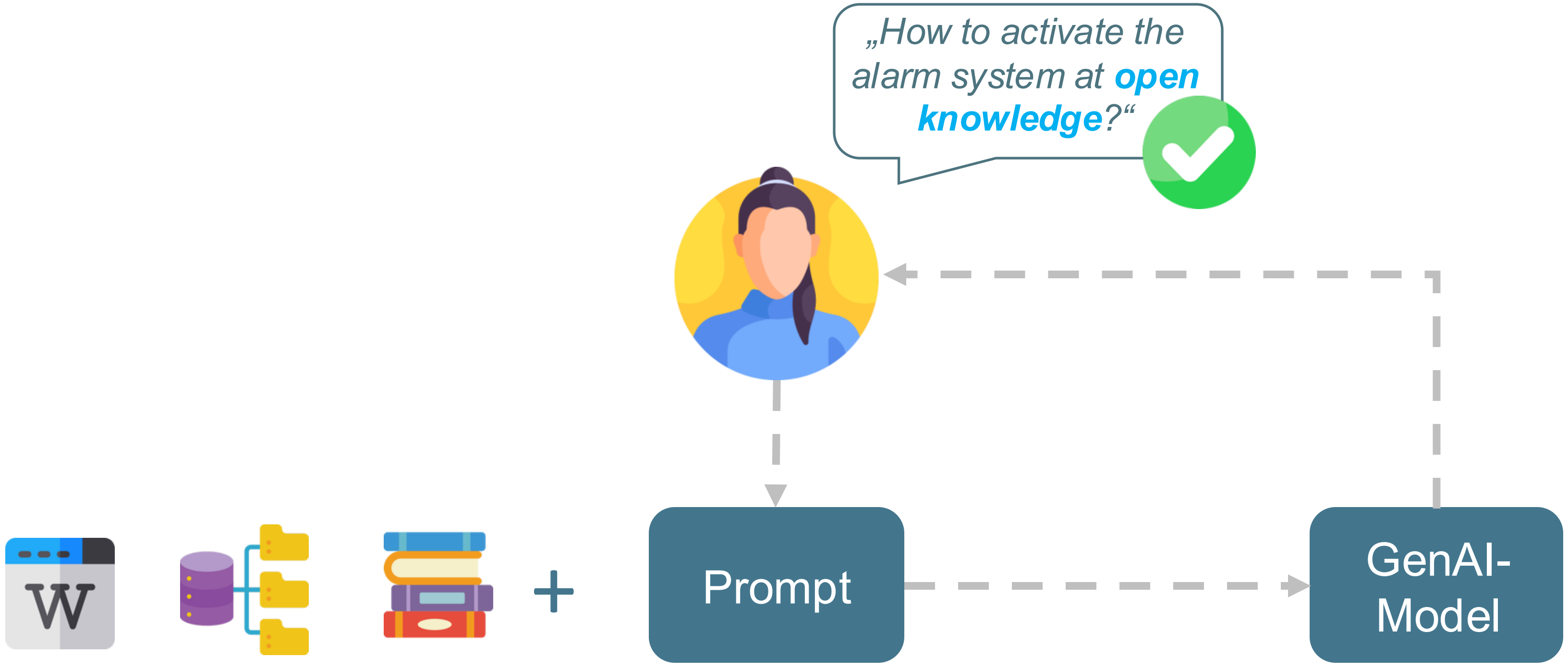


GenAI Basics myDomain



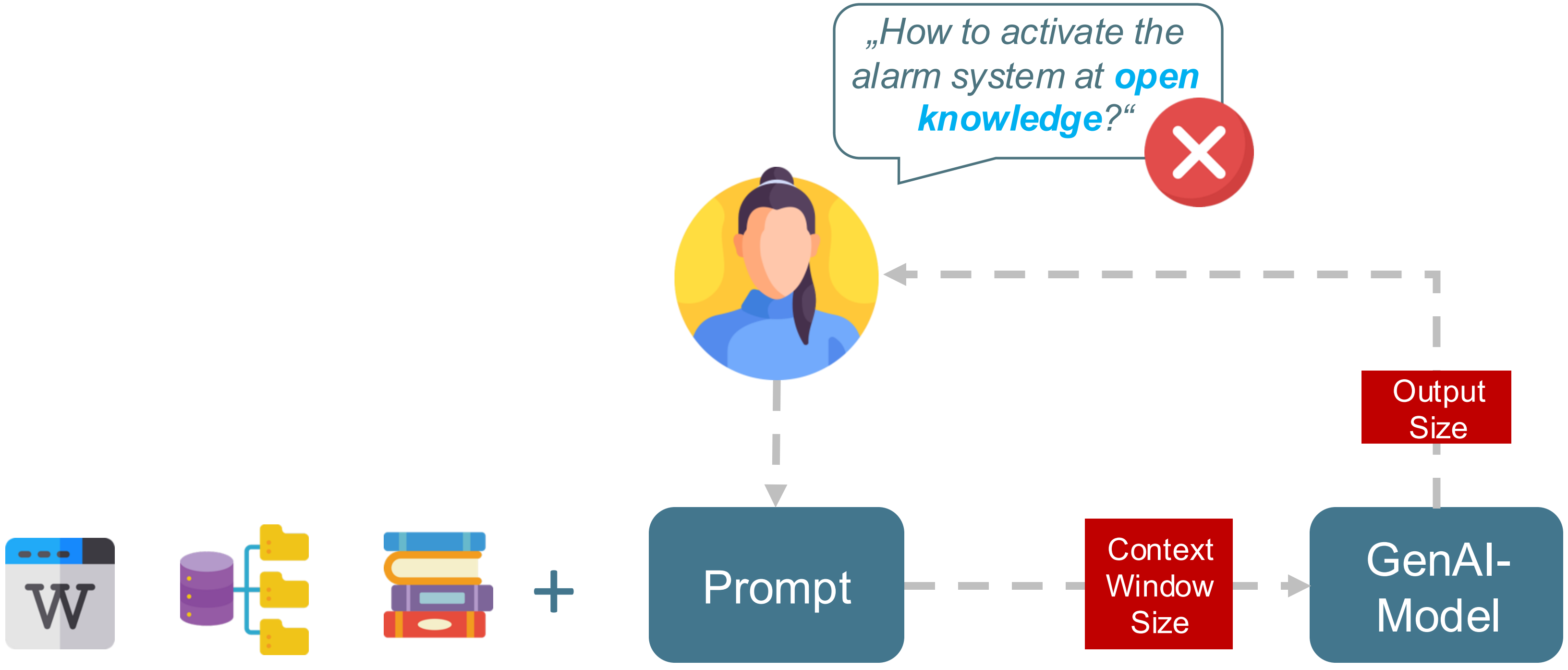
„Leave regulation + Alarm system usage at open knowledge"

GenAI Basics myDomain



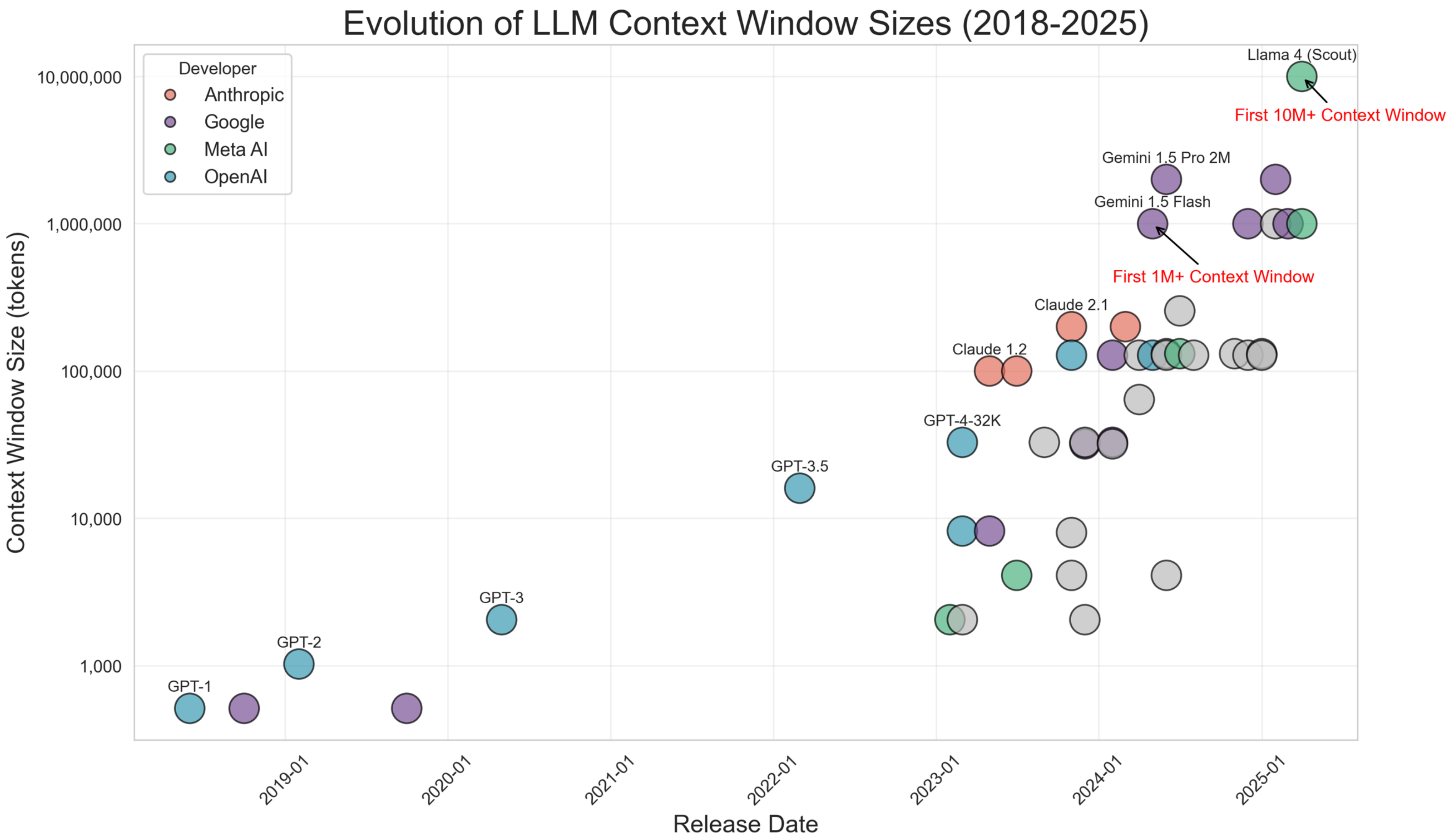
„All internal knowledge
(Wiki, DB, …) at open knowledge“

GenAI Basics myDomain



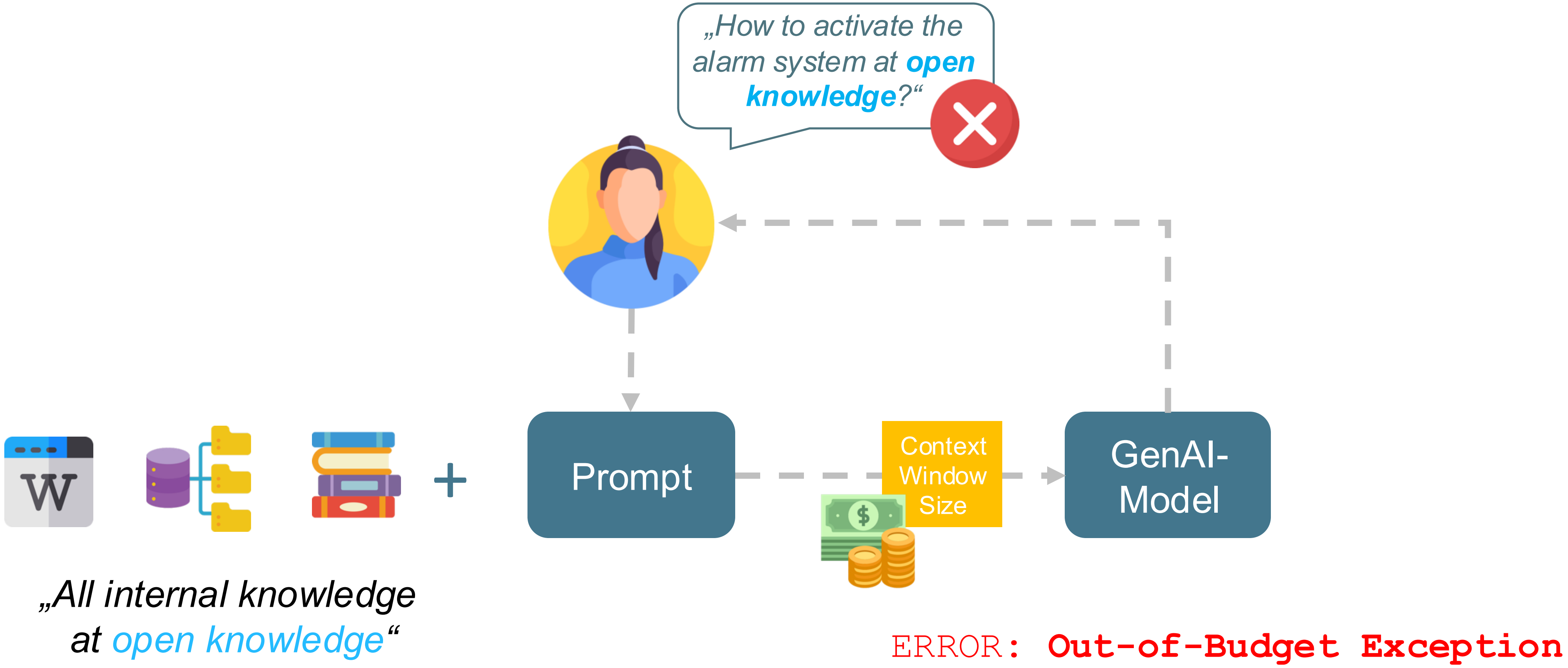
„All internal knowledge at open knowledge"

ERROR: **Token Limit exceeded**

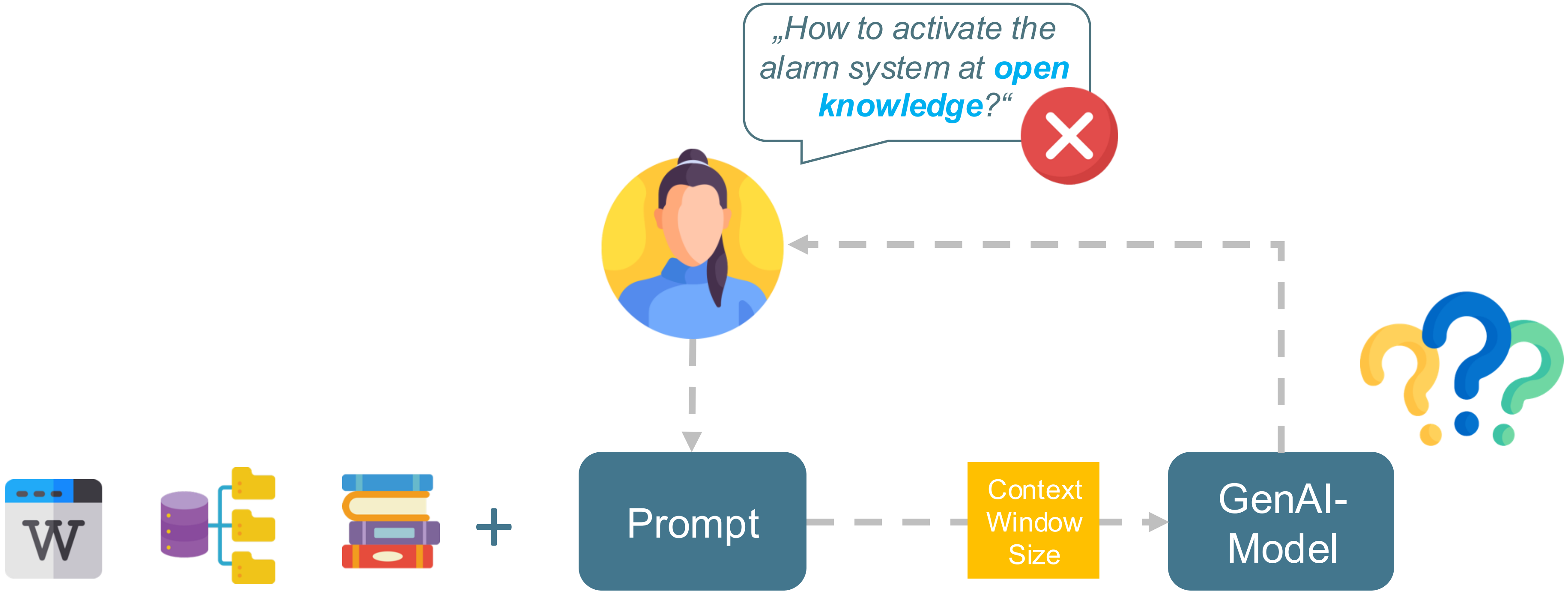


(Source: https://www.meibel.ai/post/understanding-the-impact-of-increasing-llm-context-windows)

GenAI Basics myDomain



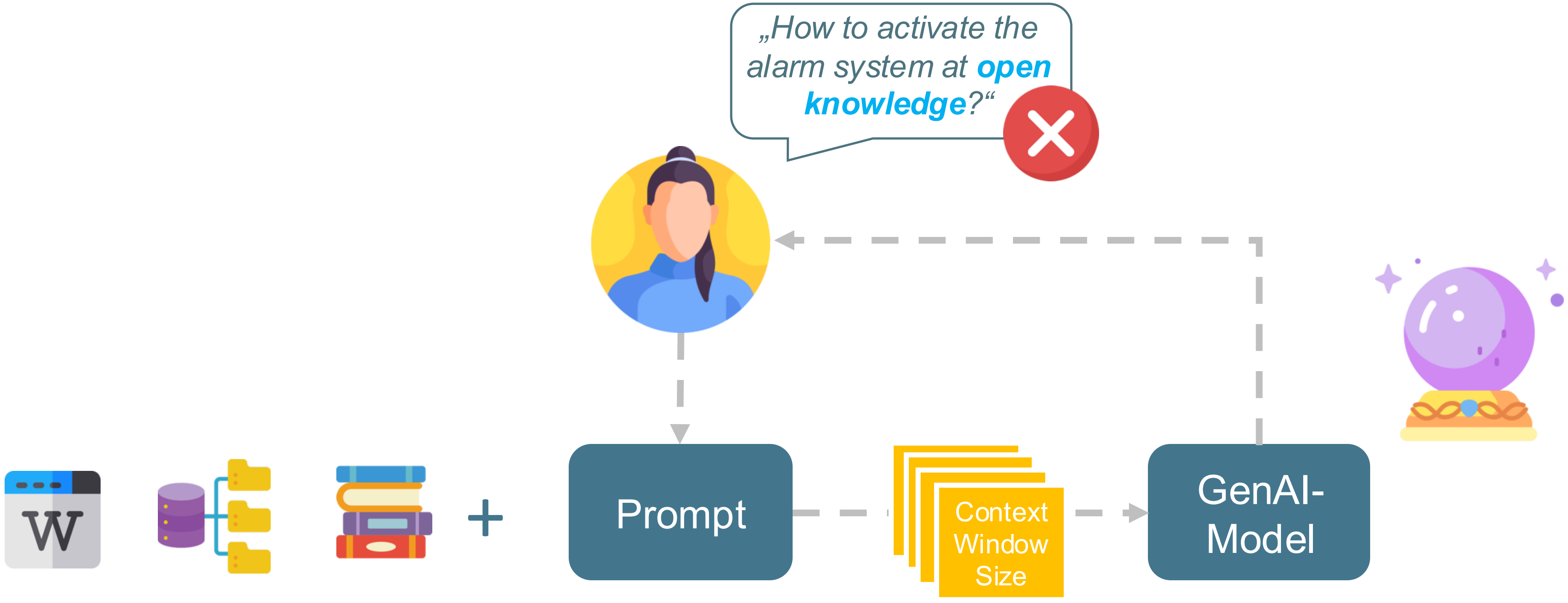
GenAI Basics myDomain



„All internal knowledge at open knowledge“

ERROR: **Lost-in-the-Middle**

GenAI Basics myDomain



„Alles interne Wissen (Wiki, DB, …) bei open knowledge“

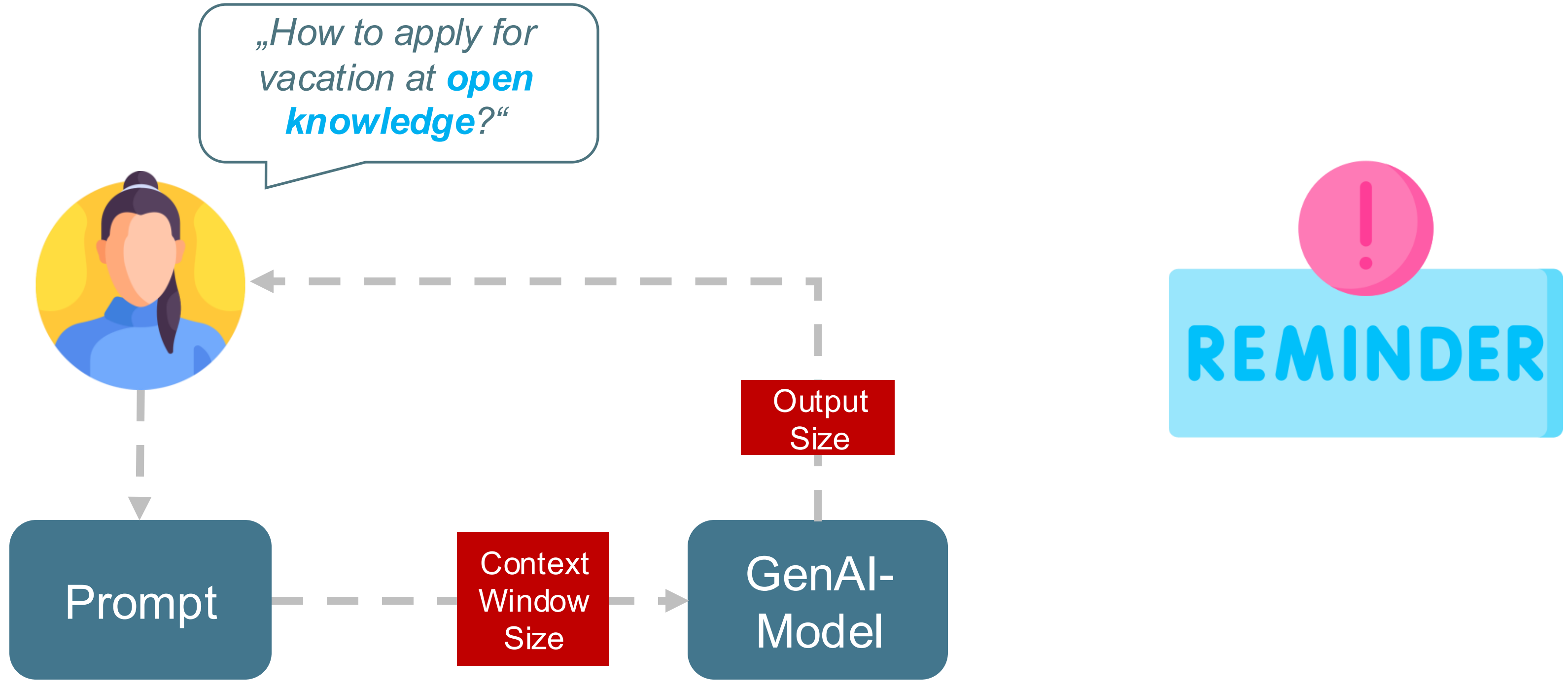
ERROR: **Self-fullfilling Prophecy**

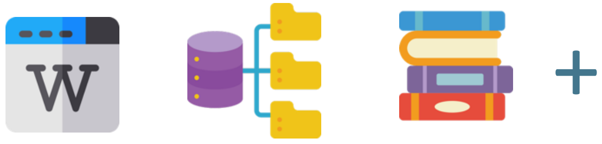
02

GenAI Advanced

How can I contribute and query my own domain knowledge?

GenAI Advanced myDomain

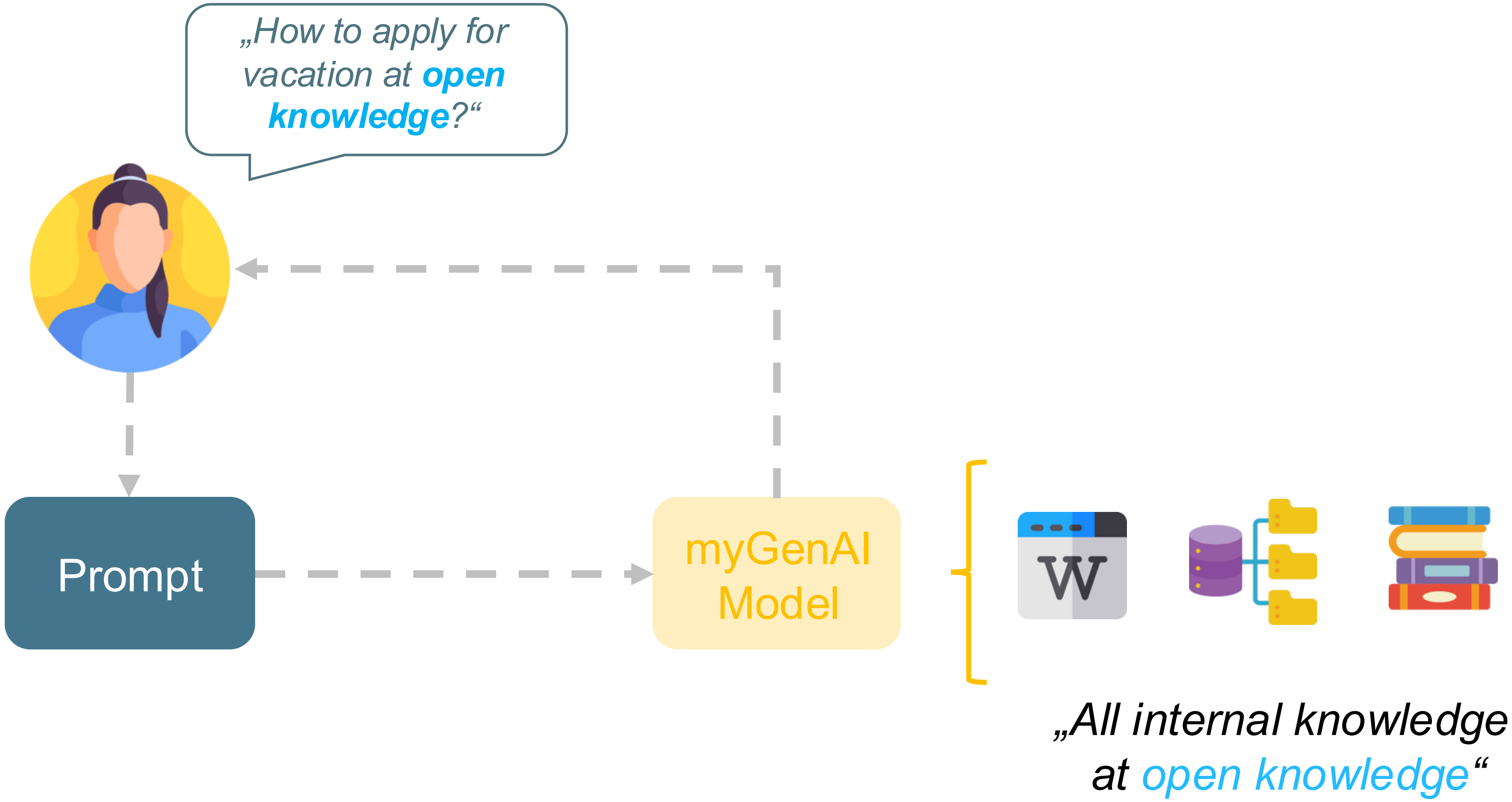




„All internal knowledge at open knowledge"

ERROR: **Token Limit exceeded!**

GenAI Advanced myDomain



GenAI Advanced myDomain

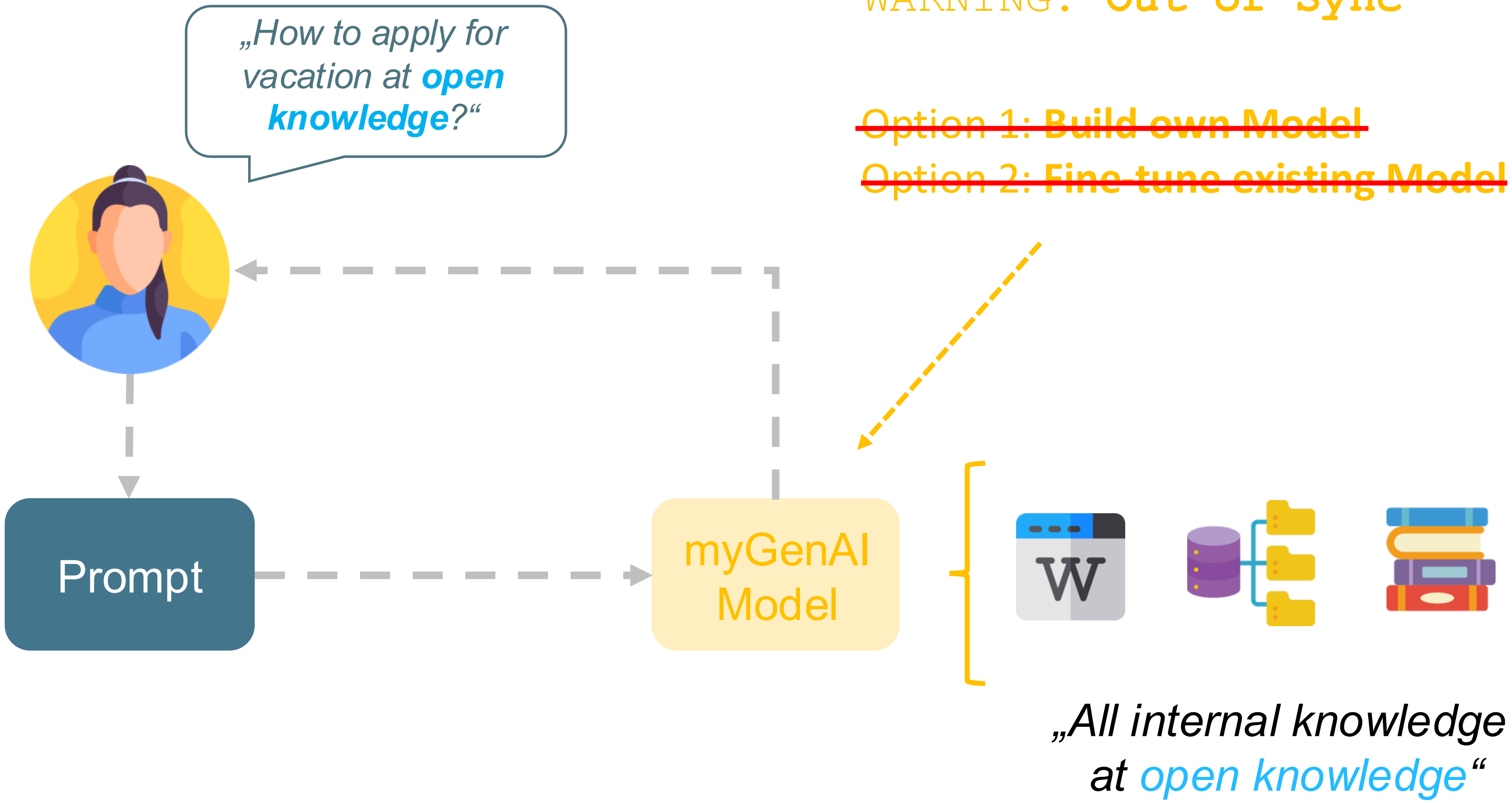
ERROR: **Way to expensive**

ERROR: **Way to complex**

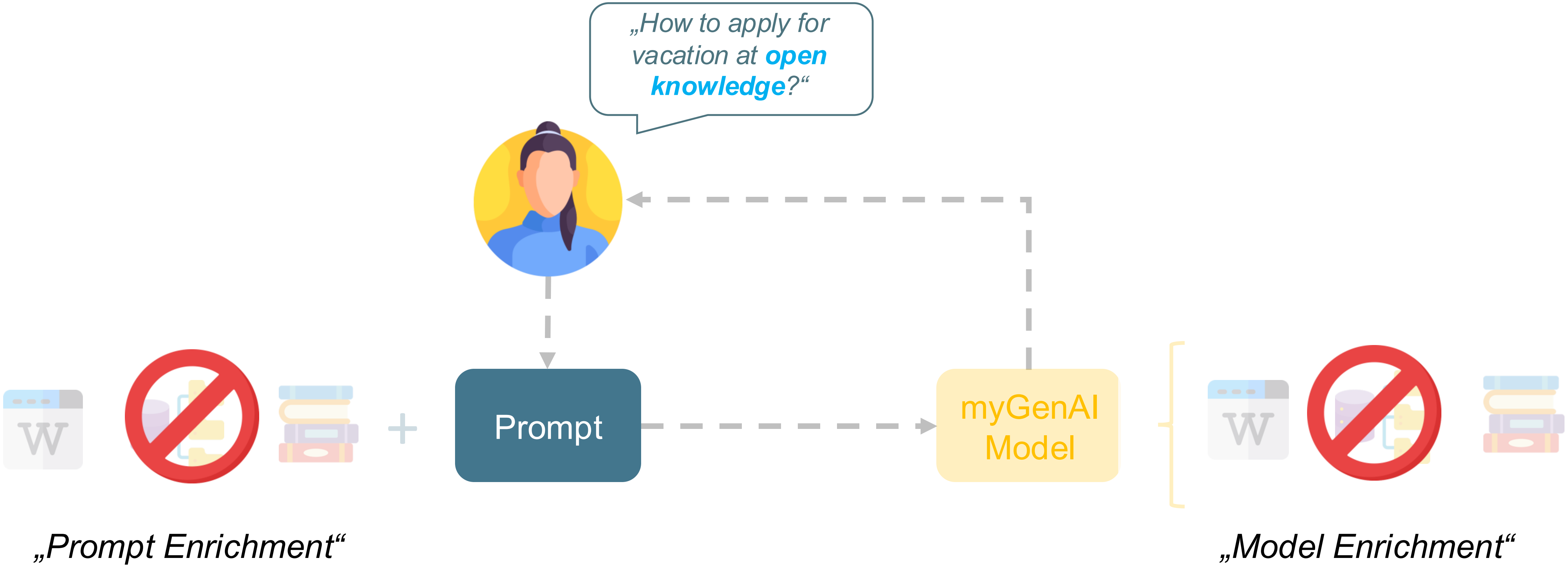
WARNING: **Out-of-Sync**

~~Option 1: **Build own Model**~~

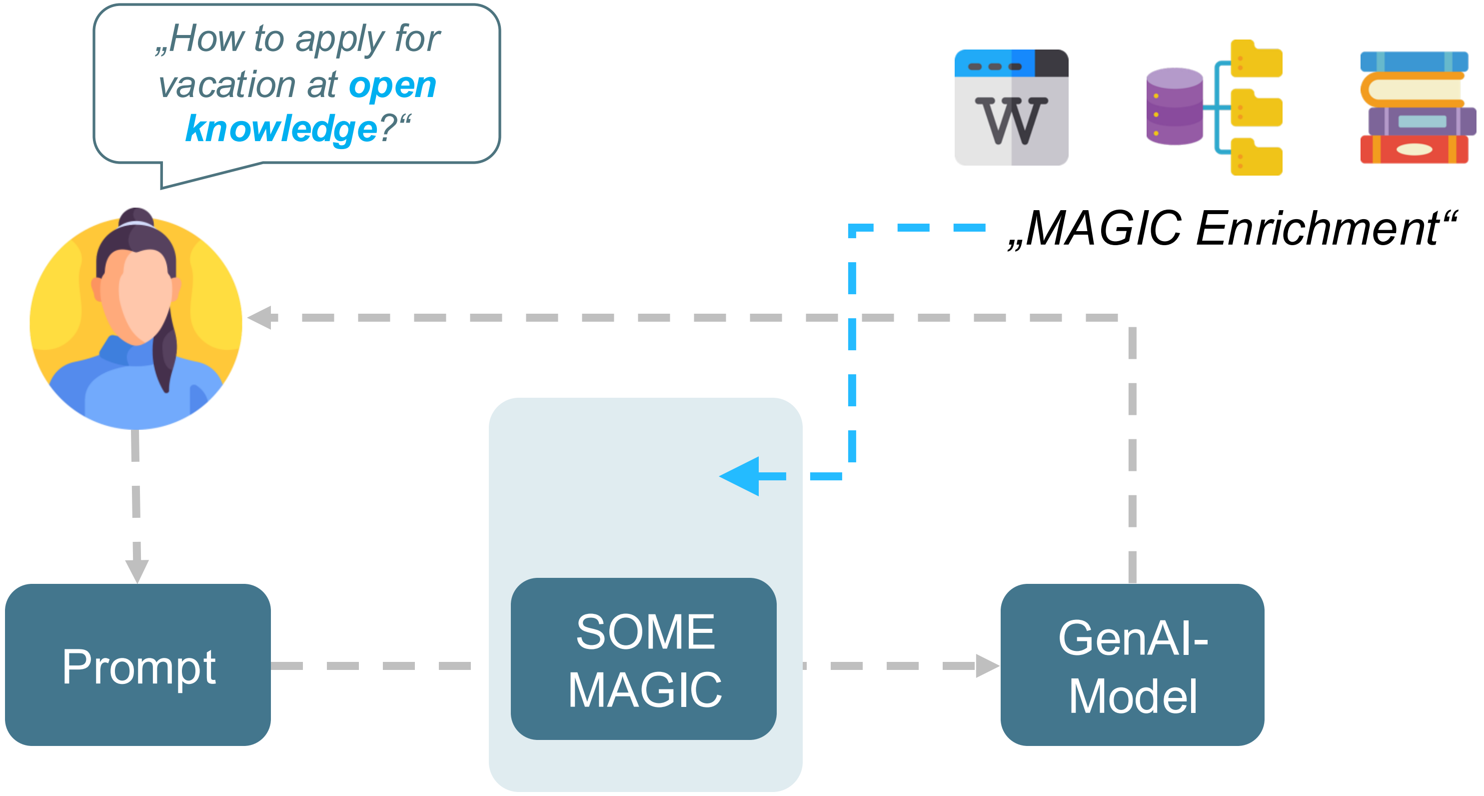
~~Option 2: **Fine-tune existing Model**~~



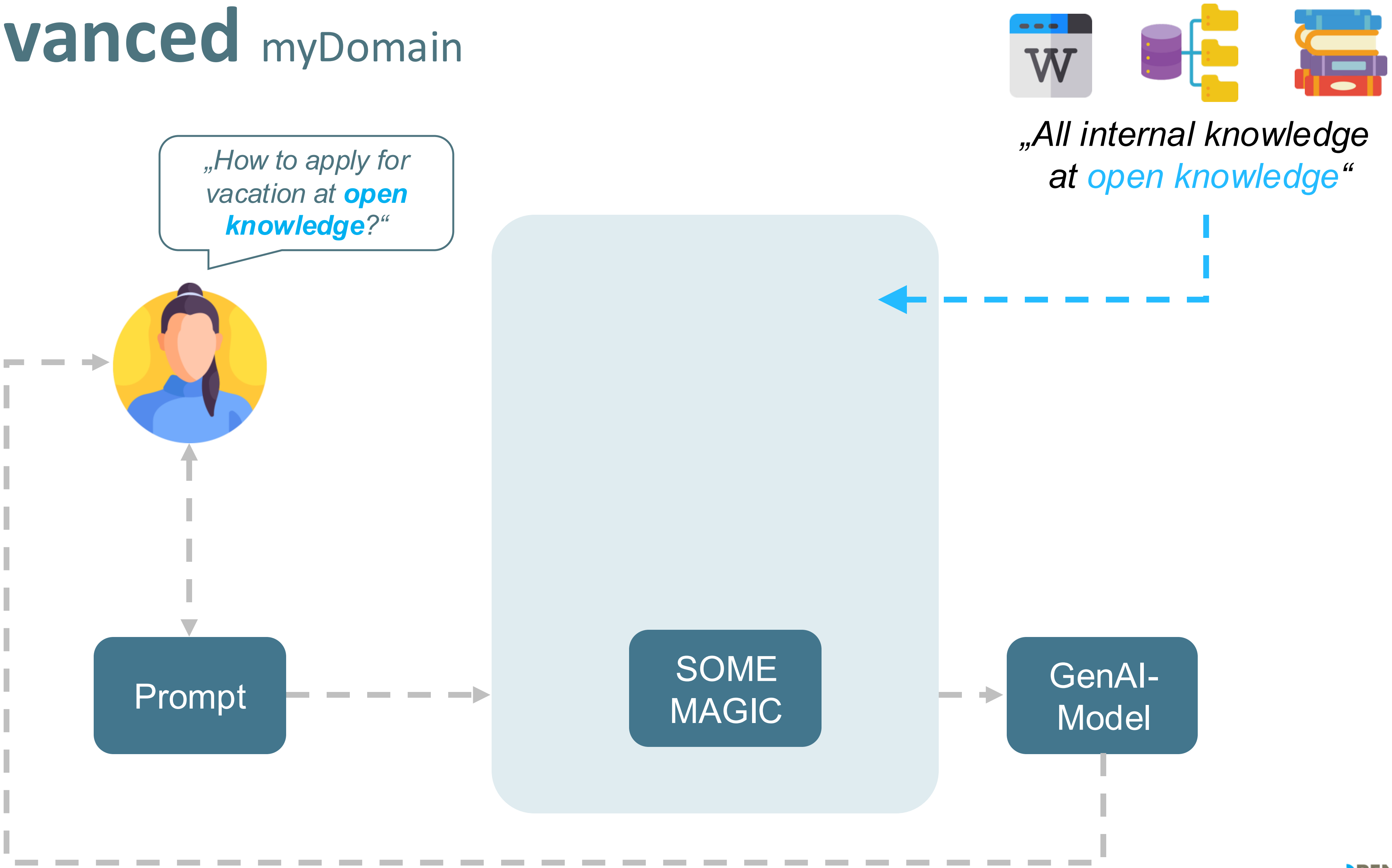
GenAI Advanced myDomain



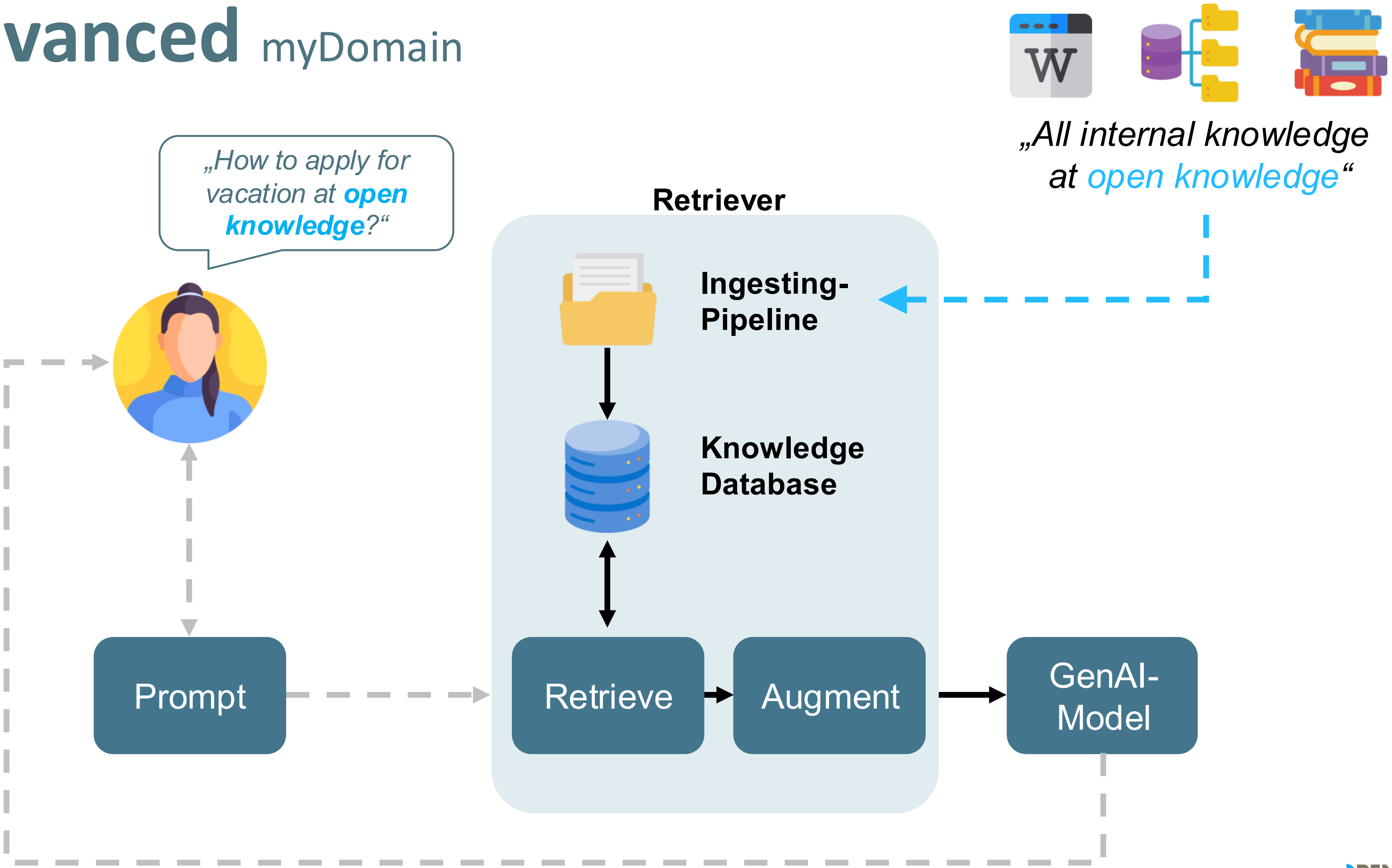
GenAI Advanced myDomain



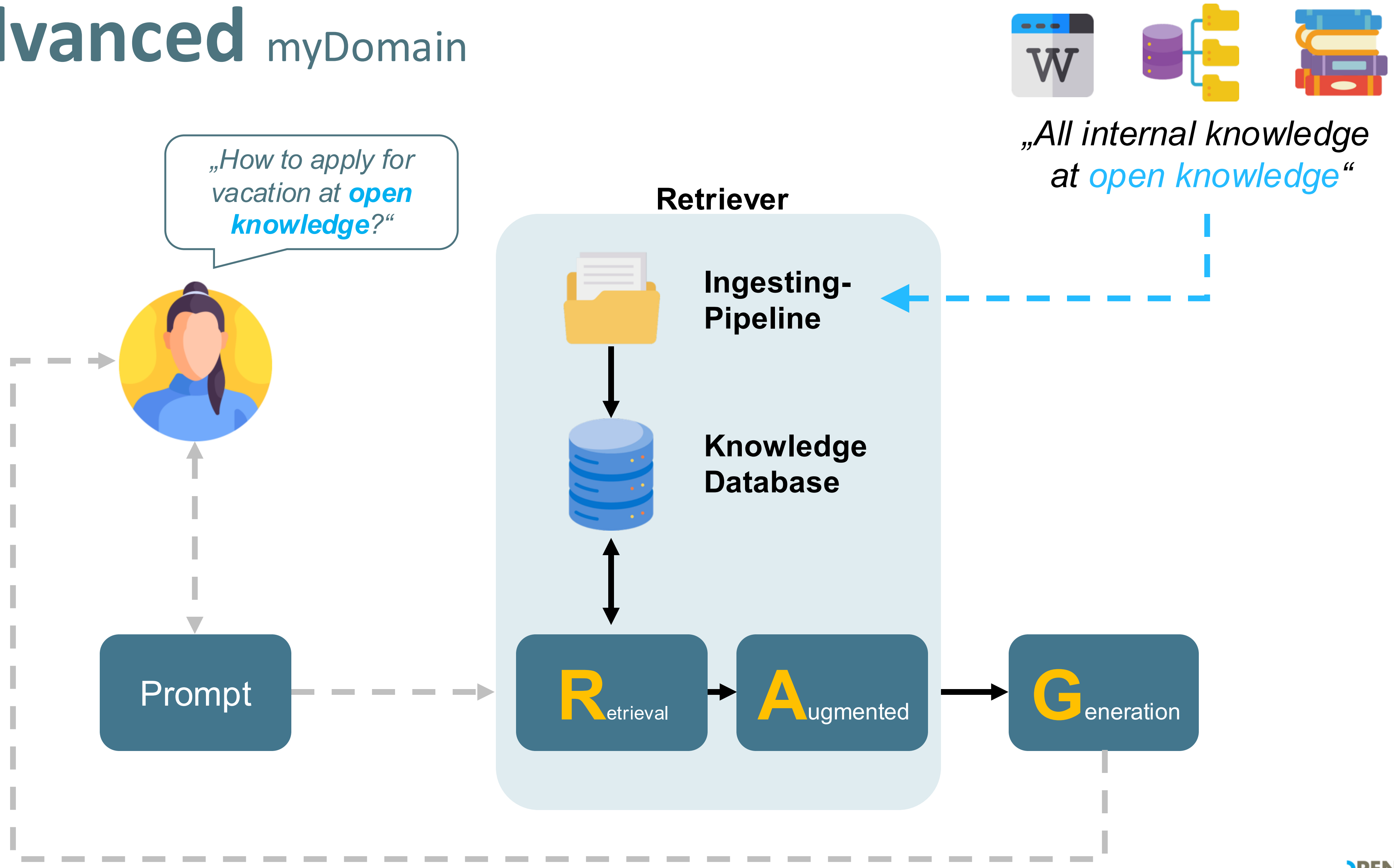
GenAI Advanced myDomain



GenAI Advanced myDomain



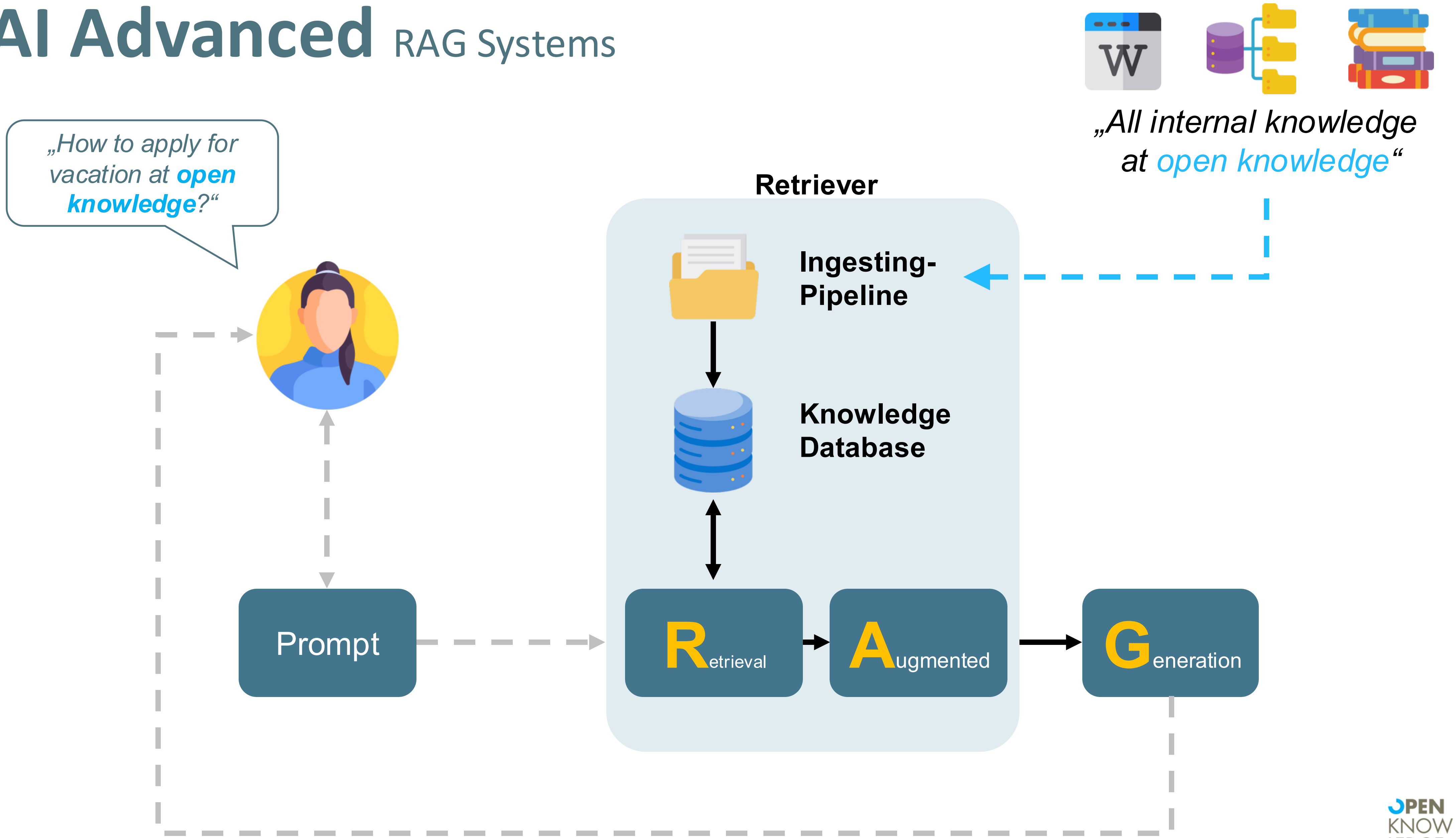
GenAI Advanced myDomain



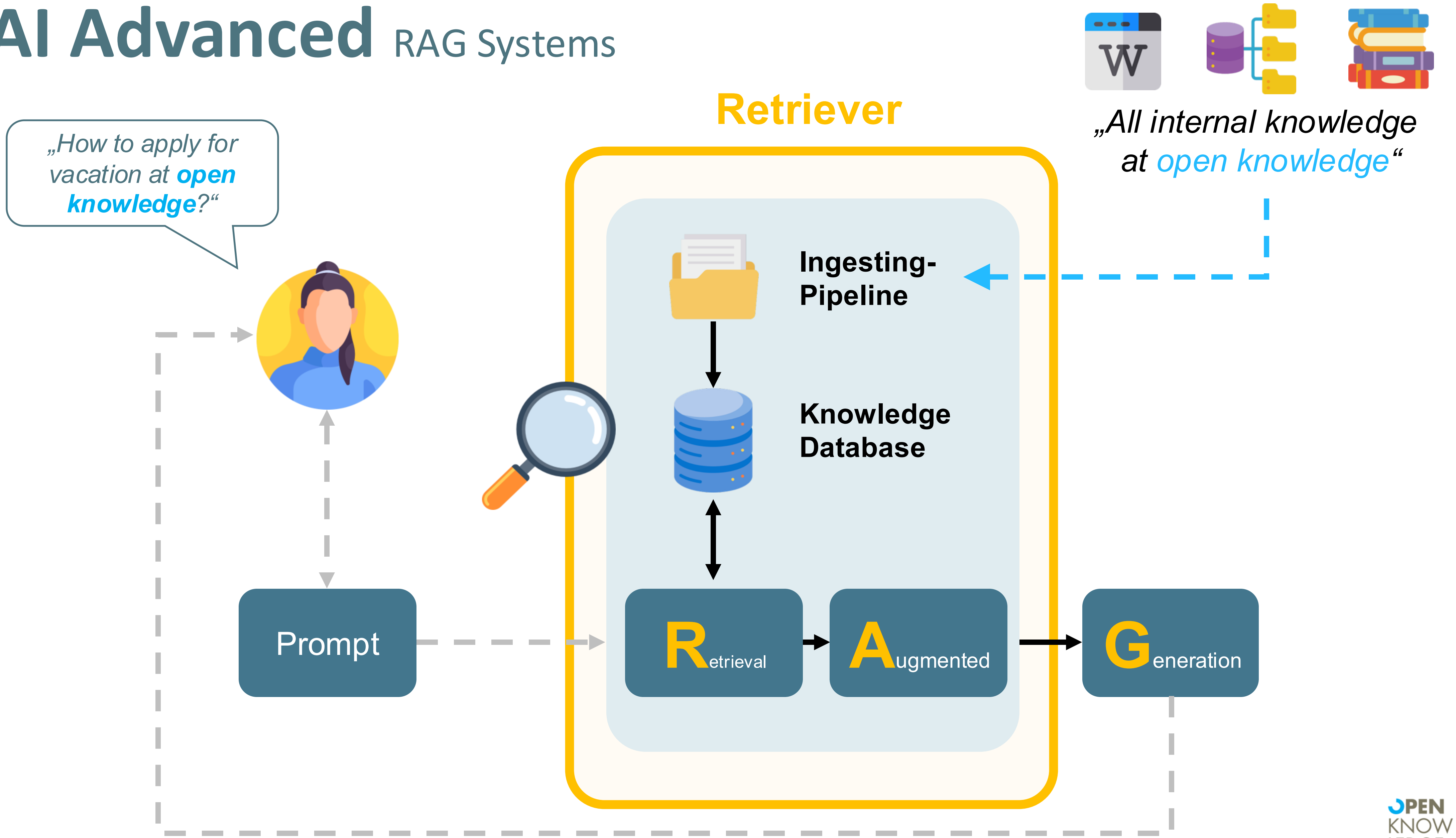
RAG Systems

GenAI
explained

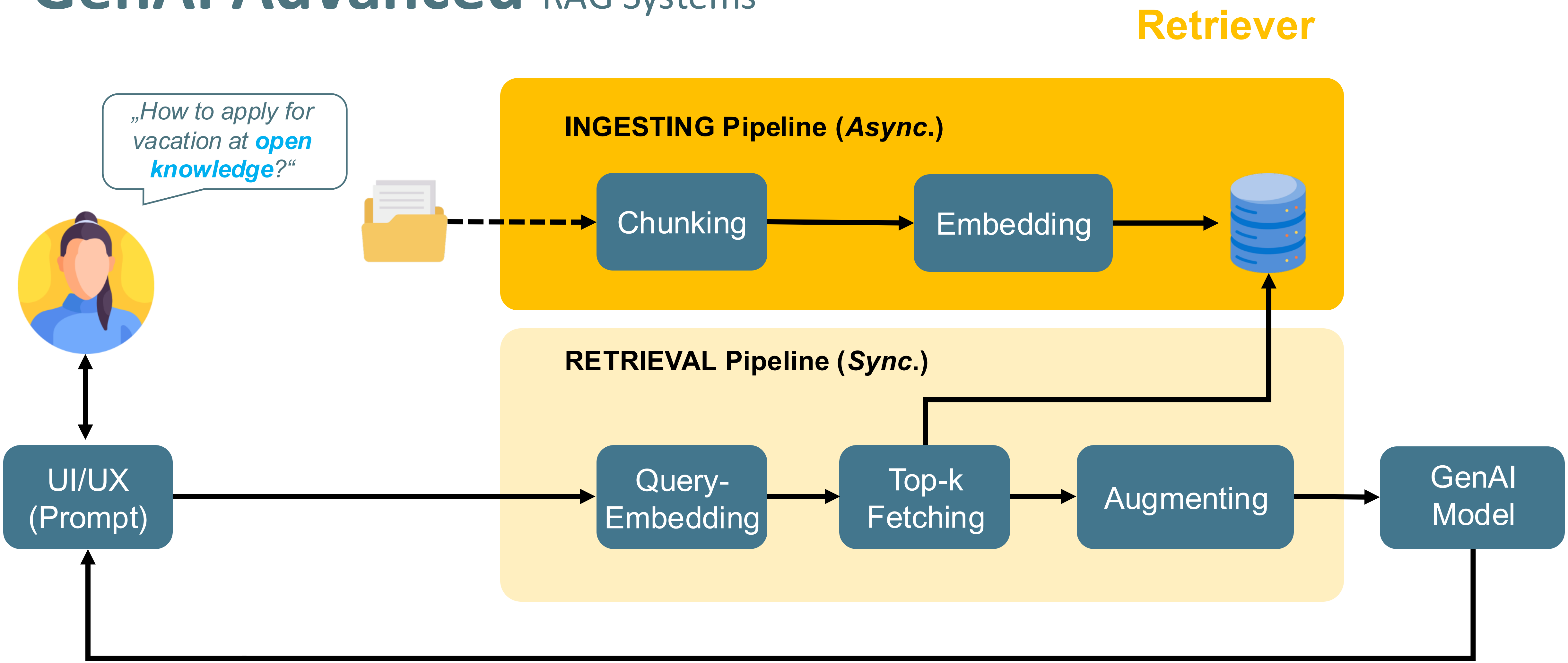
GenAI Advanced RAG Systems



GenAI Advanced RAG Systems



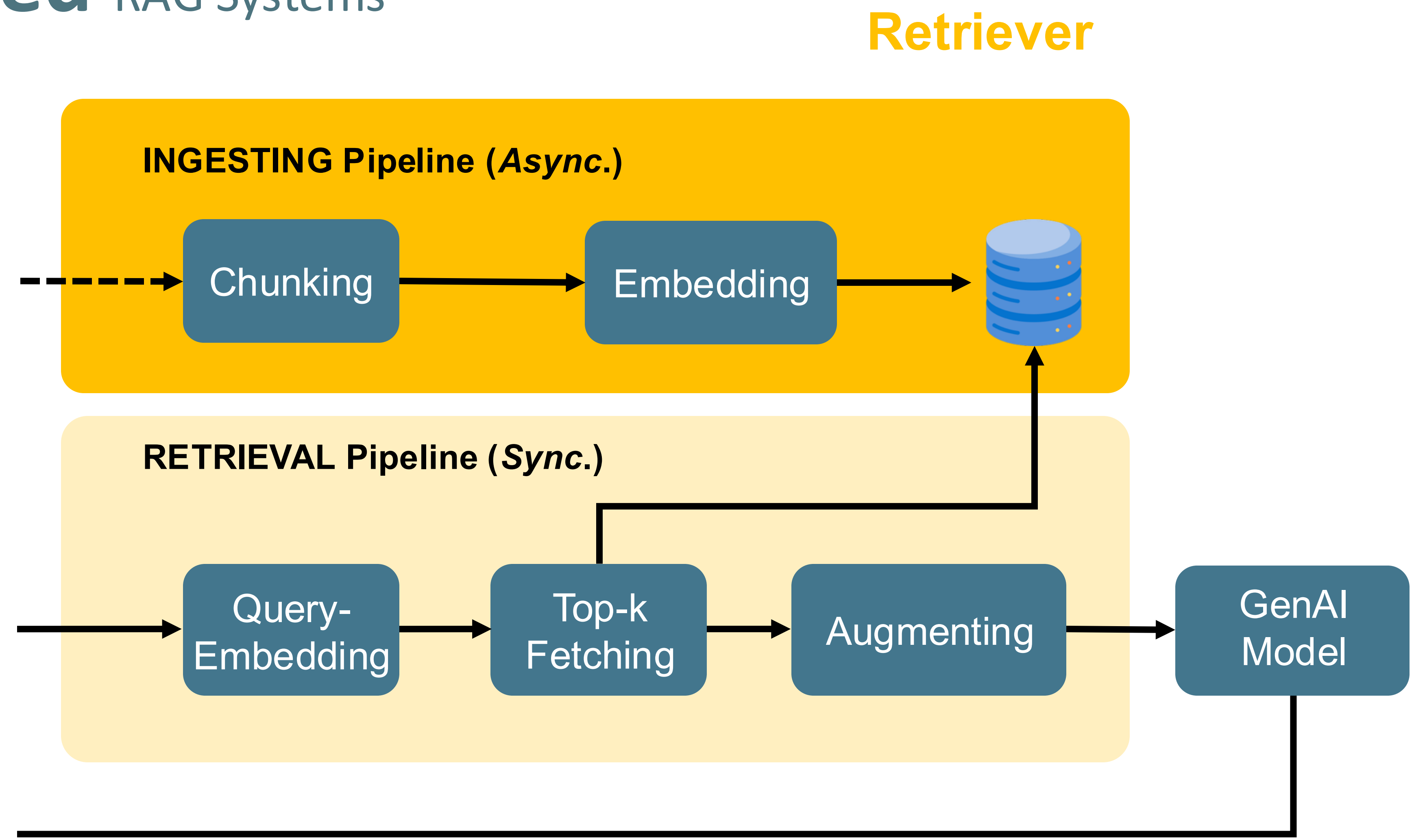
GenAI Advanced RAG Systems



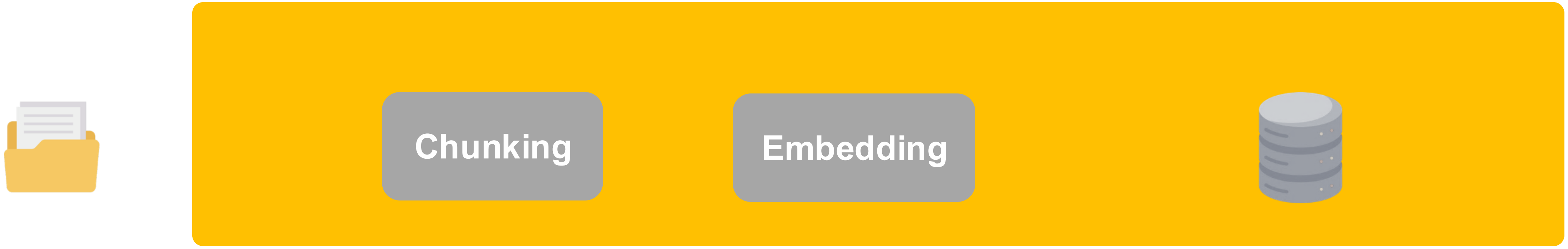
GenAI Advanced RAG Systems

It looks complicated, and it is!

Except for 'Hello World' examples from the Internet. 😉



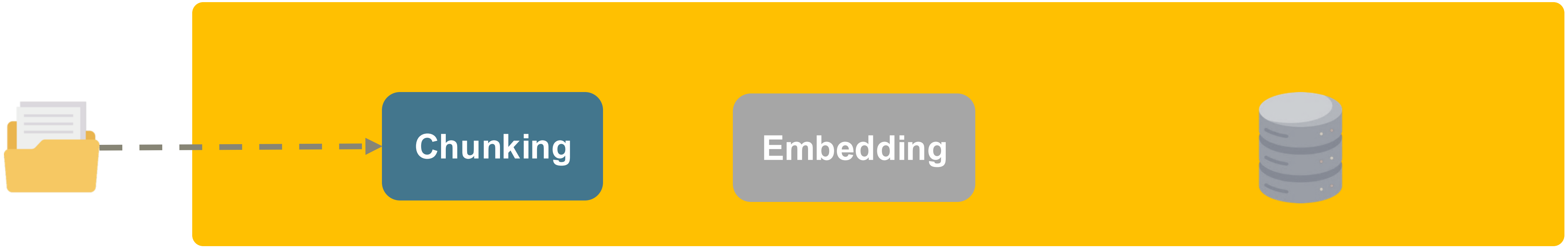
Ingesting Pipeline by Example



Document (Doc. Id:1)

„It will soon be the third anniversary of the release of ChatGPT. There is still no end in sight to the hype. The added value of this technology is too great and its use in the form of a chat is too intuitive. In 2024, it was expected that the leap from exploratory playing around with large language models [...].“

Ingesting Pipeline by Example



Chunk-größe: typisch : ca 200-500-1000 Zeichen, mit ca 20-100 overlap

Document (Doc. Id:1)

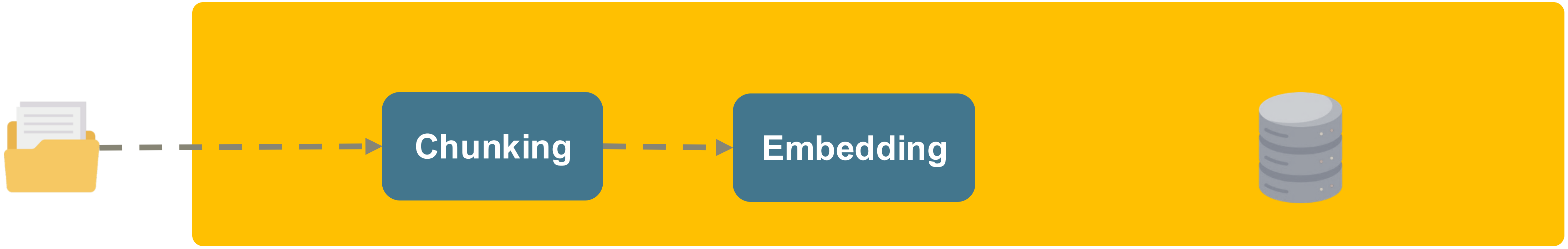
„It will soon be the third anniversary of the release of ChatGPT. There is still no end in sight to the hype. The added value of this technology is too great and its use in the form of a chat is too intuitive. In 2024, it was expected that the leap from exploratory playing around with large language models [...].“

[„ It will soon be the third anniversary of the release of ChatGPT. There is still no end in sight to the hype.“]

[„ The added value of this technology is too great and its use in the form of a chat is too intuitive.“]

[…]

Ingesting Pipeline by Example



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[„ It will soon be the third anniversary of the release of ChatGPT. There is still no end in sight to the hype.“]

[„ The added value of this technology is too great and its use in the form of a chat is too intuitive.“]

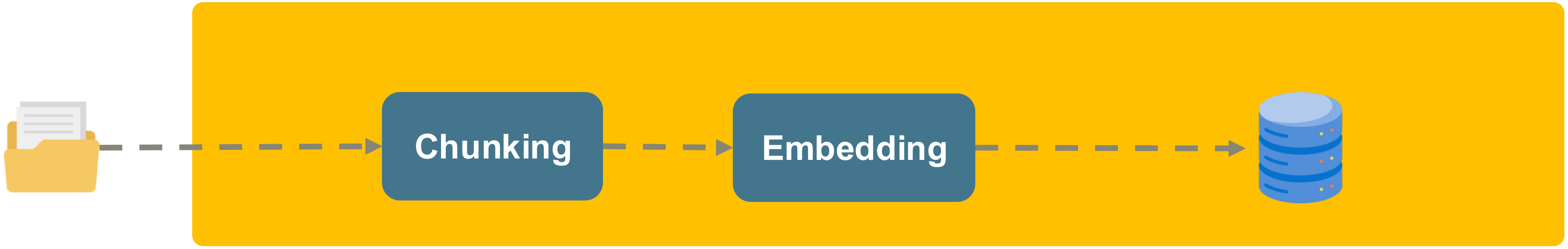
[…]

[-0.24, 0.15, …, 0.52]

[-0.13, 0.95, …, 0.31]

[…]

Ingesting Pipeline by Example



Document (Doc. Id:1)

„It will soon be the third anniversary of the release of ChatGPT. There is still no end in sight to the hype. The added value of this technology is too great and its use in the form of a chat is too intuitive. In 2024, it was expected that the leap from exploratory playing around with large language models [...].“

[„ It will soon be the third anniversary of the release of ChatGPT. There is still no end in sight to the hype.“]

[„ The added value of this technology is too great and its use in the form of a chat is too intuitive.“]

[…]

[-0.24, 0.15, …, 0.52]

[-0.13, 0.95, …, 0.31]

[…]

Id	Doc. Id	Embedding
1	1	**[-0.24, 0.15, …, 0.52]**
2	1	**[-0.13, 0.95, …, 0.31]**
[…]	[…]	[…]

Retrieval Pipeline by Example

Embedding

Top-k Fetching

Augmenting

‘What is the lecture ‘The architecture for language models in practice - Retrieval Augmented Generation’ about?’

Retrieval Pipeline by Example

Embedding

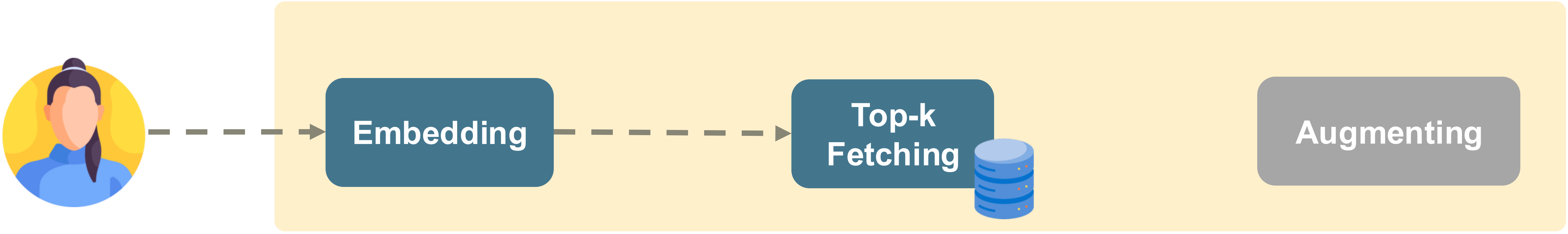
Top-k Fetching

Augmenting

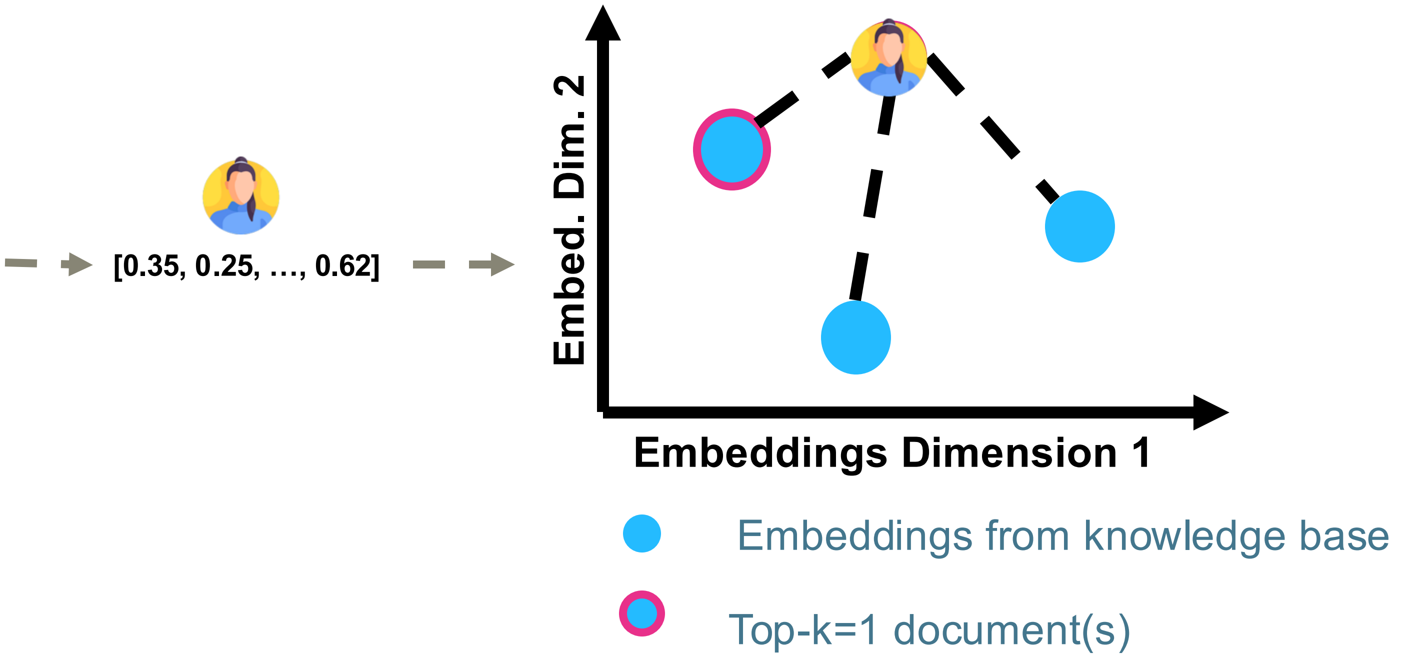
‘What is the lecture ‘The architecture for language models in practice - Retrieval Augmented Generation’ about?’

[0.35, 0.25, …, 0.62]

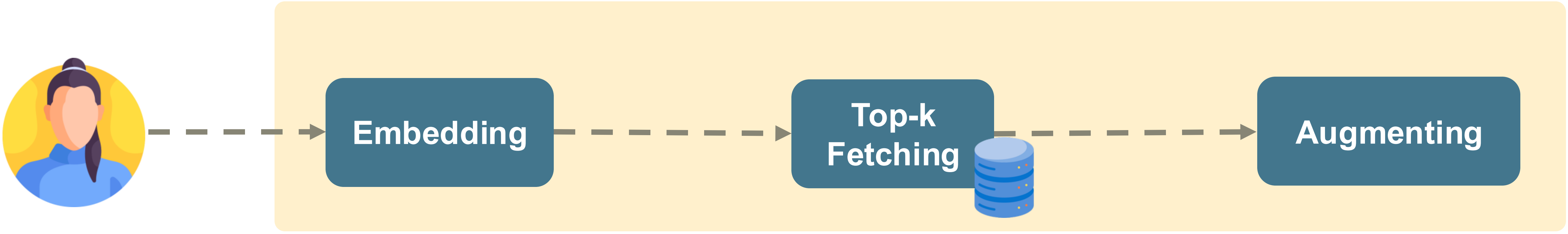
Retrieval Pipeline by Example



‘What is the lecture ‘The architecture for language models in practice - Retrieval Augmented Generation’ about?’

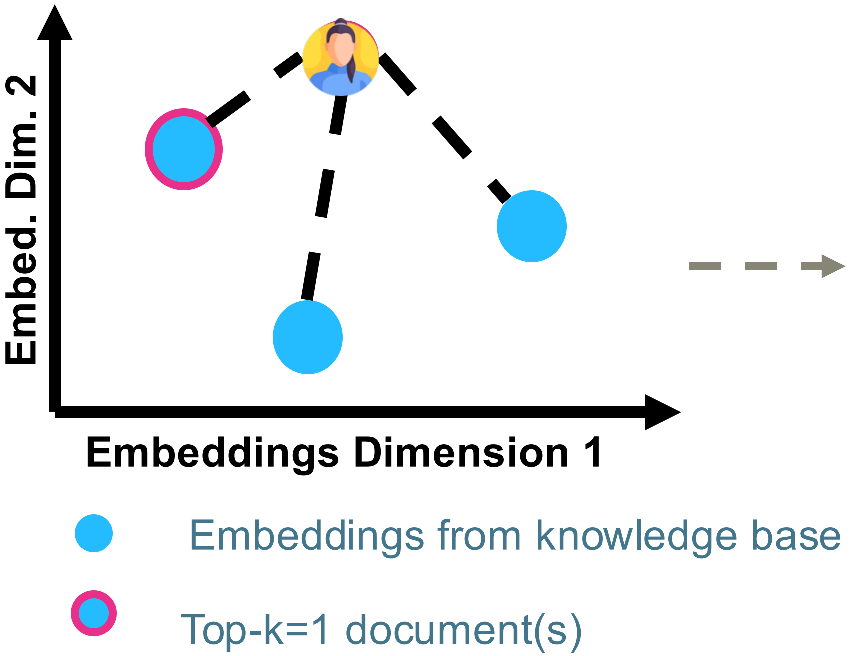


Retrieval Pipeline by Example



'What is the lecture 'The architecture for language models in practice - Retrieval Augmented Generation' about?'

[0.35, 0.25, …, 0.62]

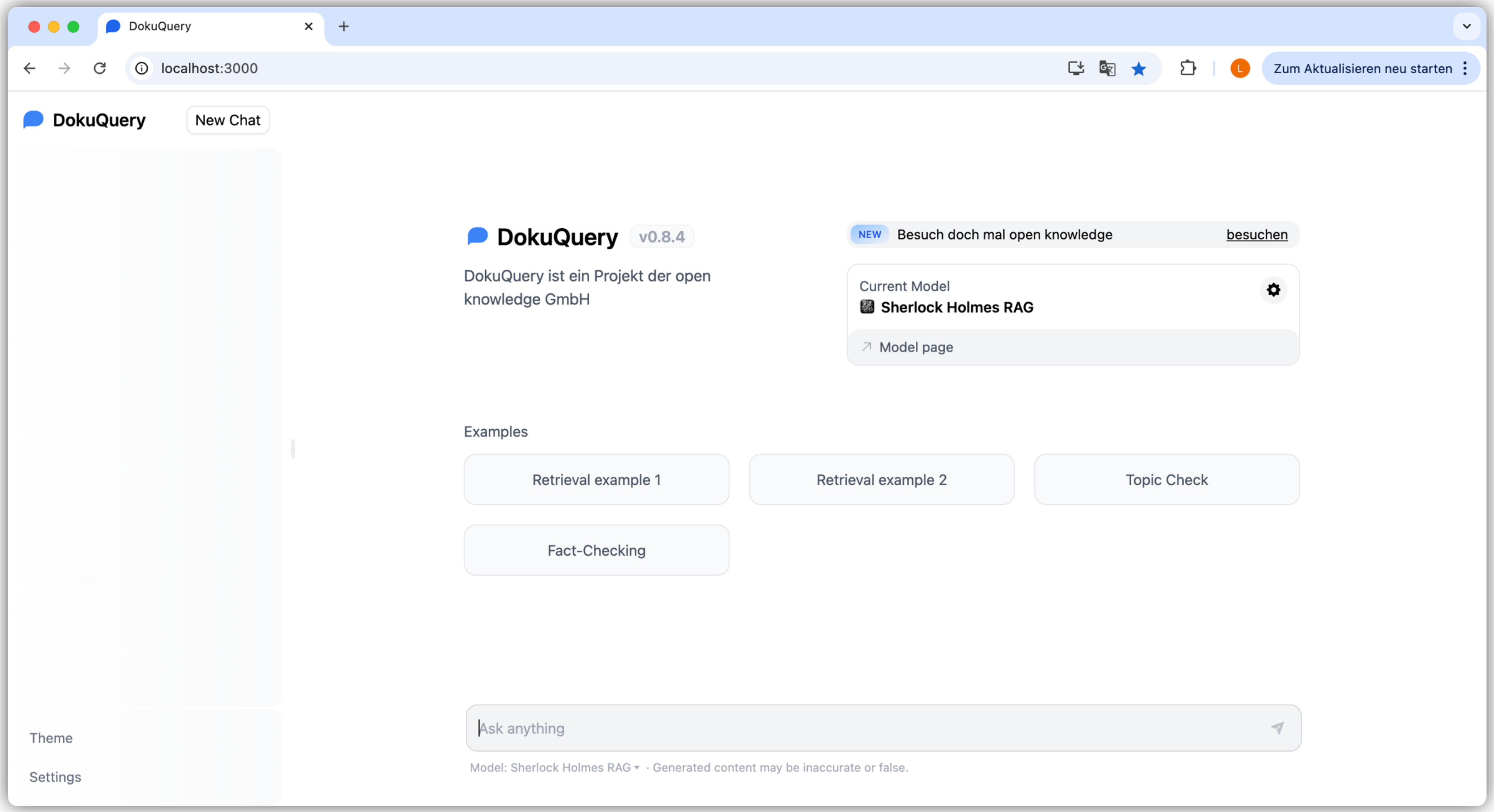


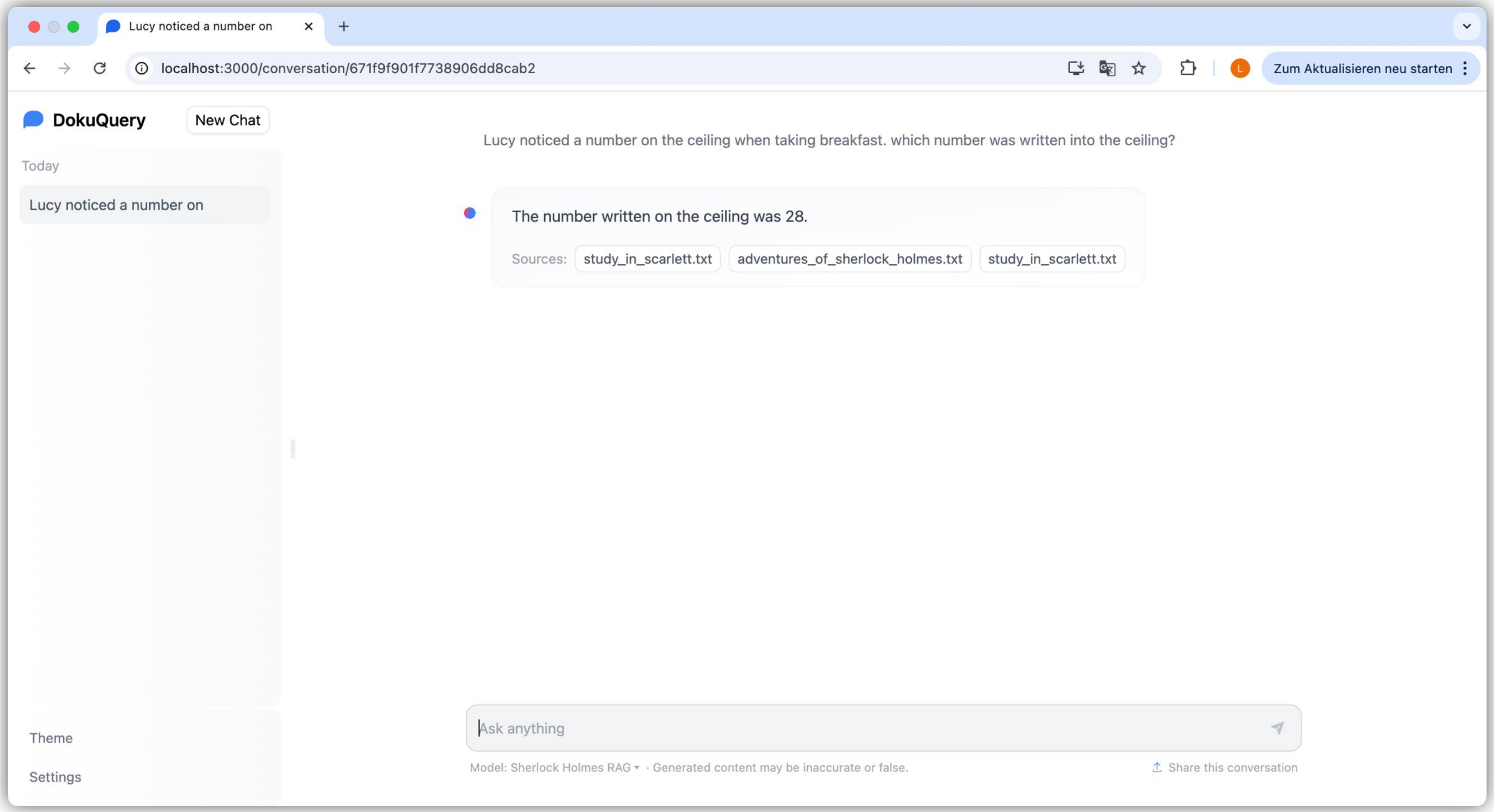
&

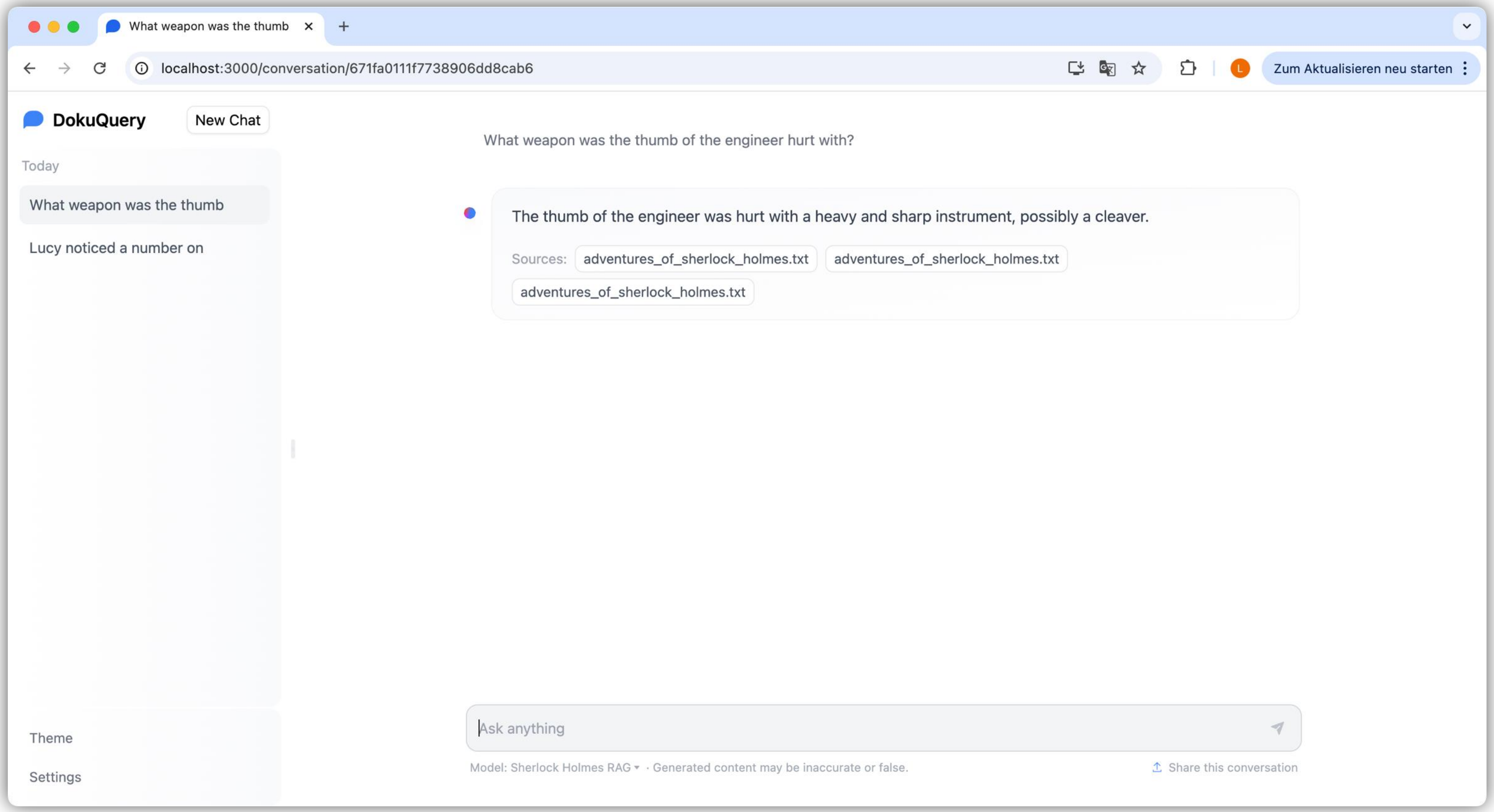
„ What is the lecture […] **based on the following context**:

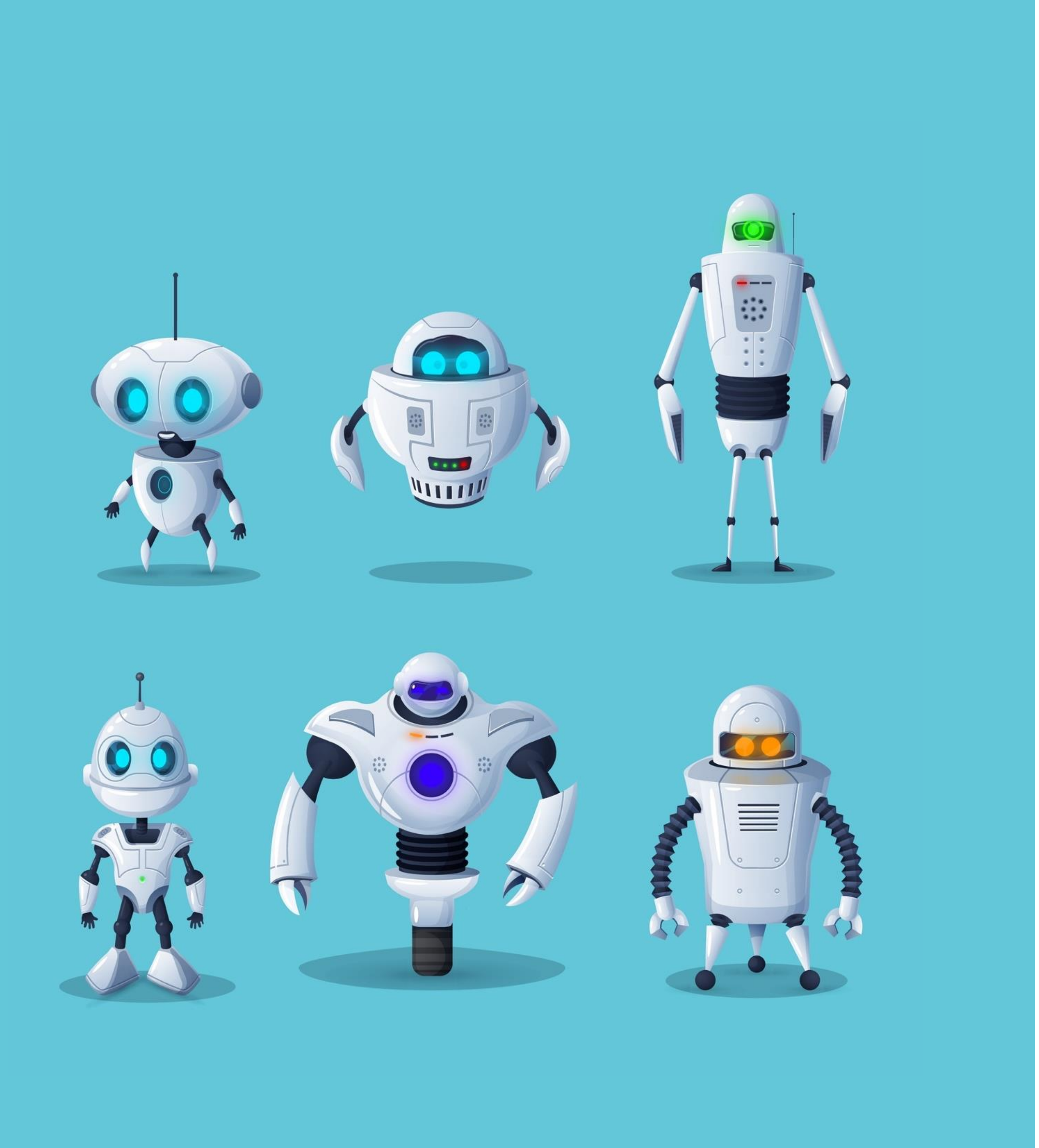
'An upstream retrieval process to pull in relevant information from our own dynamic database can provide well-founded answers.'

Sherlock RAG in Action


DokuQuery
localhost:3000
Zum Aktualisieren neu starten
DokuQuery
New Chat
DokuQuery
v0.8.4
DokuQuery ist ein Projekt der open knowledge GmbH
NEW
Besuch doch mal open knowledge
besuchen
Current Model
Sherlock Holmes RAG
Model page
Examples
Retrieval example 1
Retrieval example 2
Topic Check
Fact-Checking
Theme
Settings
Ask anything
Model: Sherlock Holmes RAG · Generated content may be inaccurate or false.


Lucy noticed a number on
localhost:3000/conversation/671f9f901f7738906dd8cab2
Zum Aktualisieren neu starten
DokuQuery
New Chat
Today
Lucy noticed a number on
Lucy noticed a number on the ceiling when taking breakfast. which number was written into the ceiling?
The number written on the ceiling was 28.
Sources: study_in_scarlett.txt adventures_of_sherlock_holmes.txt study_in_scarlett.txt
Ask anything
Model: Sherlock Holmes RAG · Generated content may be inaccurate or false.
Share this conversation
Theme
Settings


What weapon was the thumb
localhost:3000/conversation/671fa0111f7738906dd8cab6
Zum Aktualisieren neu starten
DokuQuery
New Chat
Today
What weapon was the thumb
Lucy noticed a number on
Theme
Settings
What weapon was the thumb of the engineer hurt with?
The thumb of the engineer was hurt with a heavy and sharp instrument, possibly a cleaver.
Sources: adventures_of_sherlock_holmes.txt adventures_of_sherlock_holmes.txt adventures_of_sherlock_holmes.txt
Ask anything
Model: Sherlock Holmes RAG · Generated content may be inaccurate or false.
Share this conversation



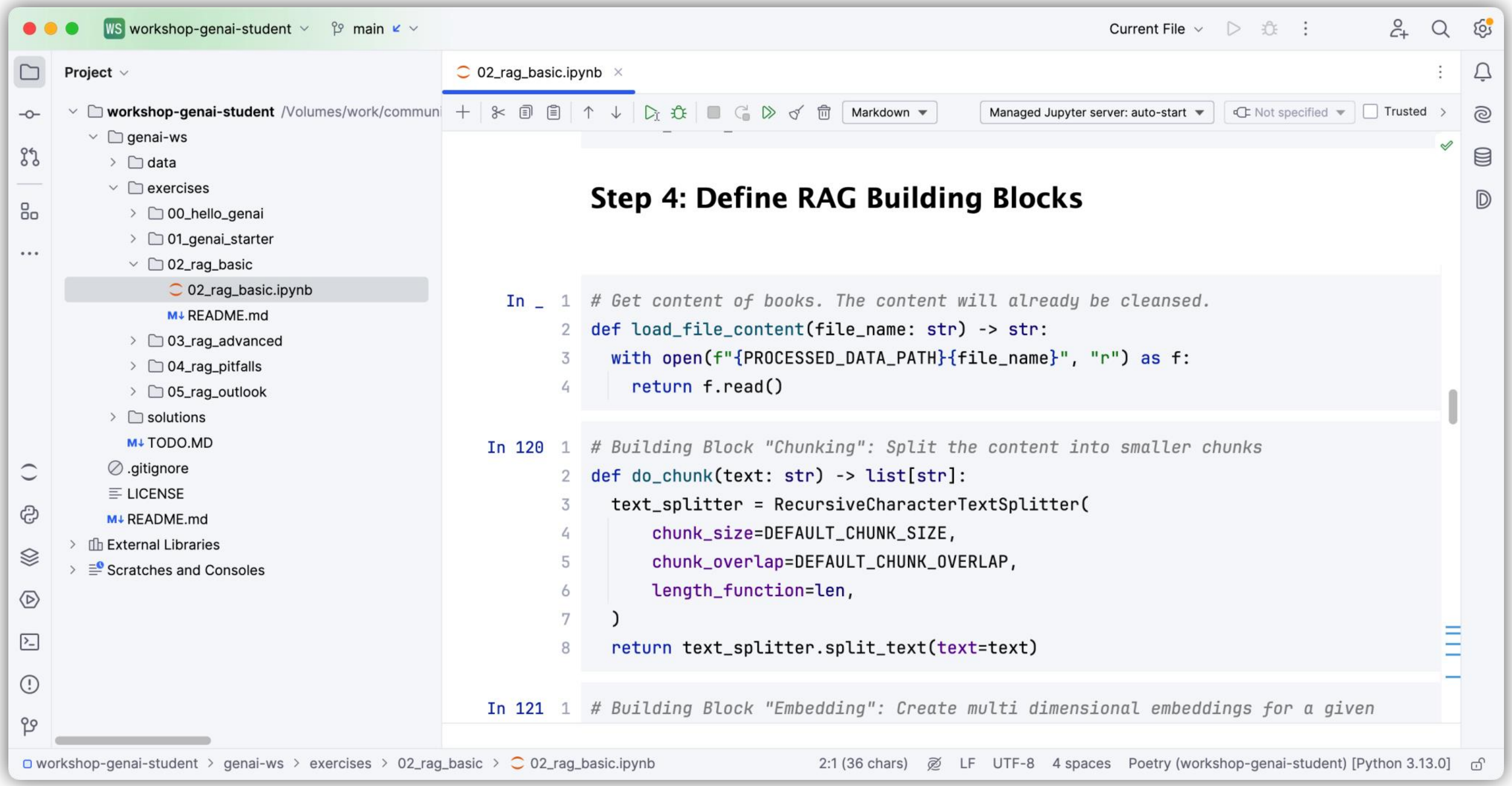
Hands-On

RAG System Simple Version

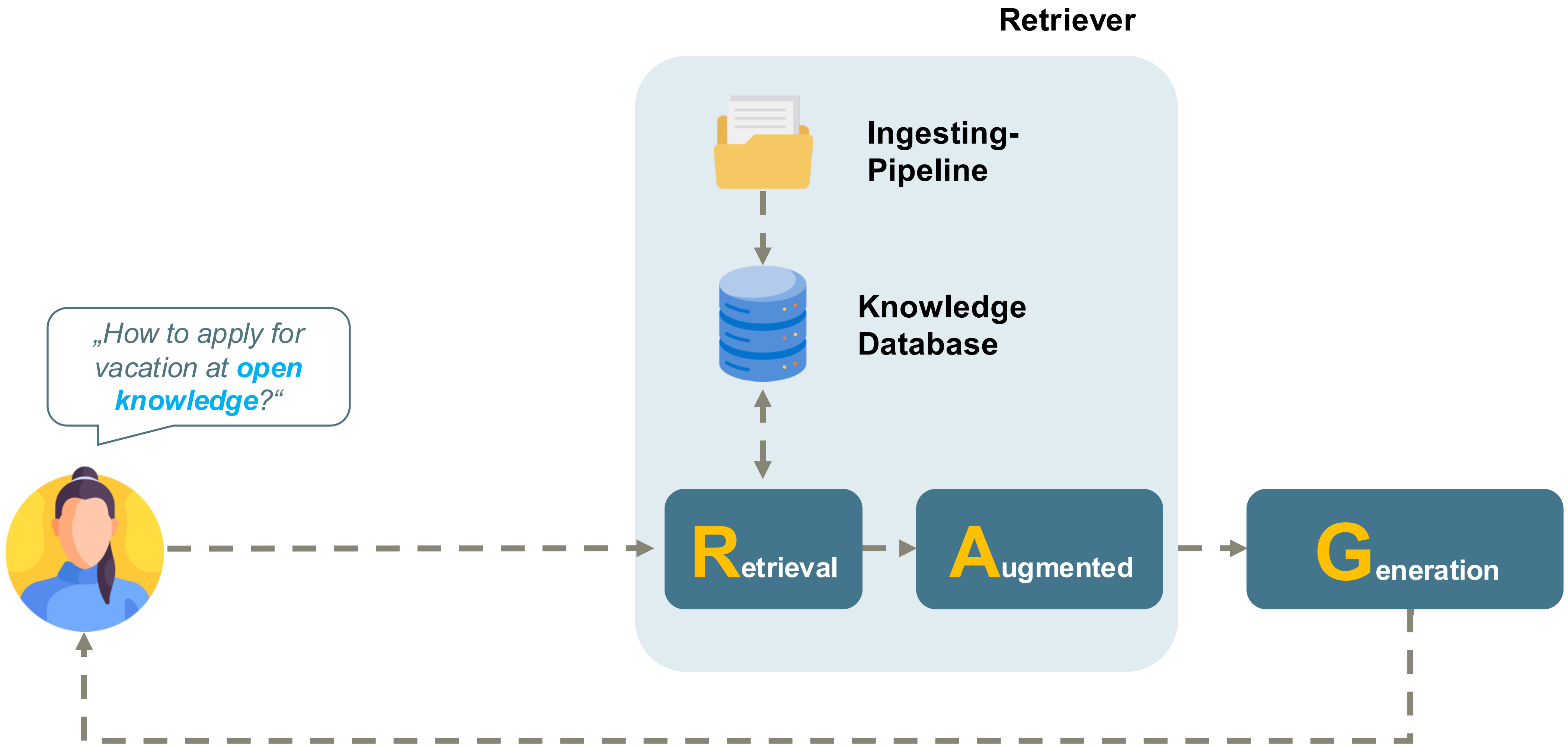
Implementing a **simple RAG system** using **ingestion and retrieval** pipelines.

Applying the RAG system to **questions from your own domain**.

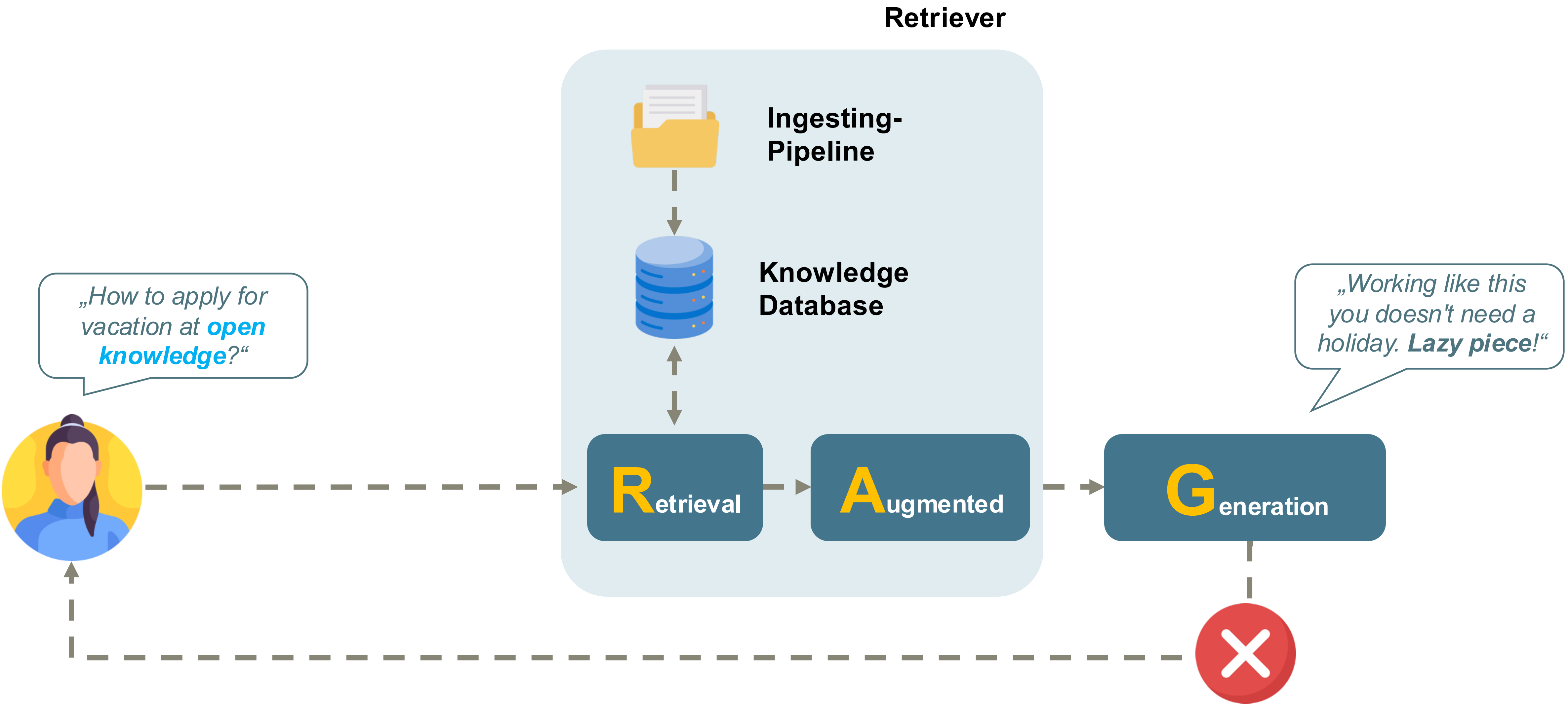
Hands-on Simple RAG



GenAI Advanced RAG Systems



GenAI Advanced RAG Systems

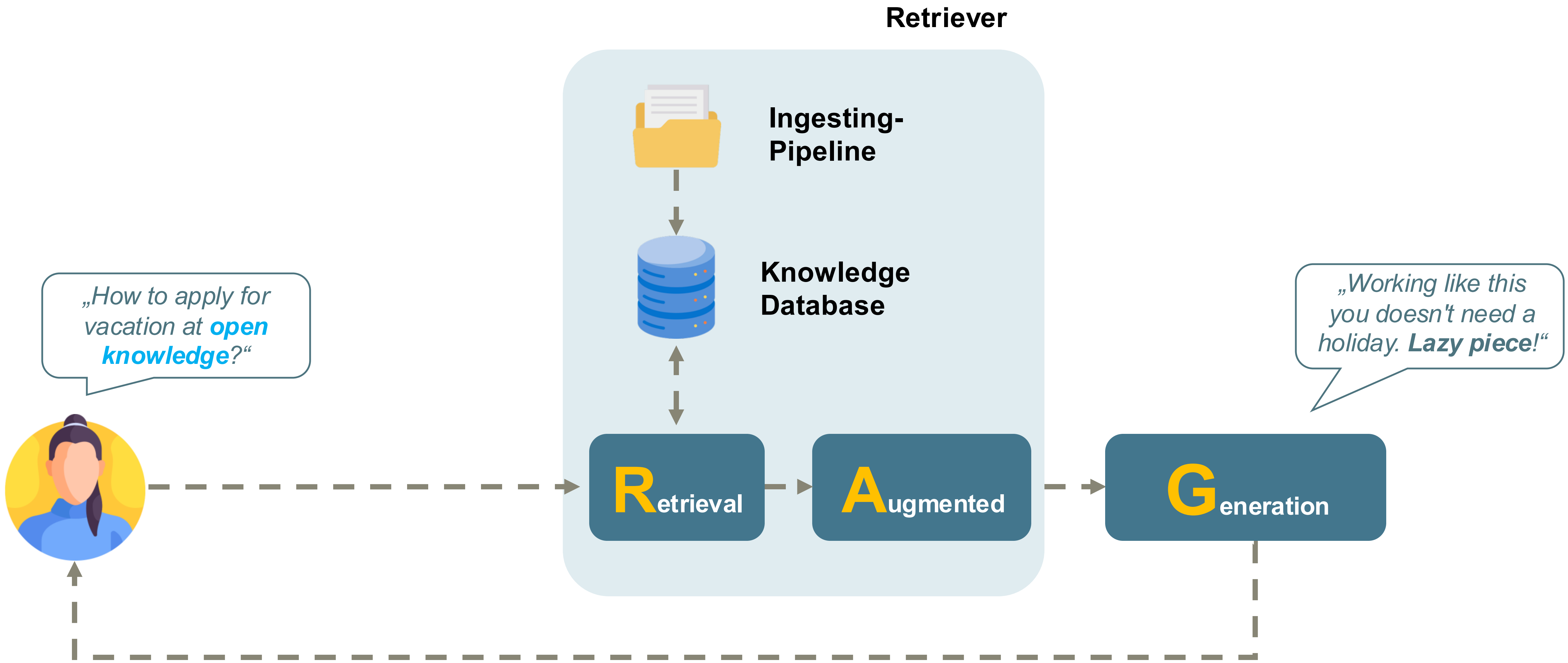


03

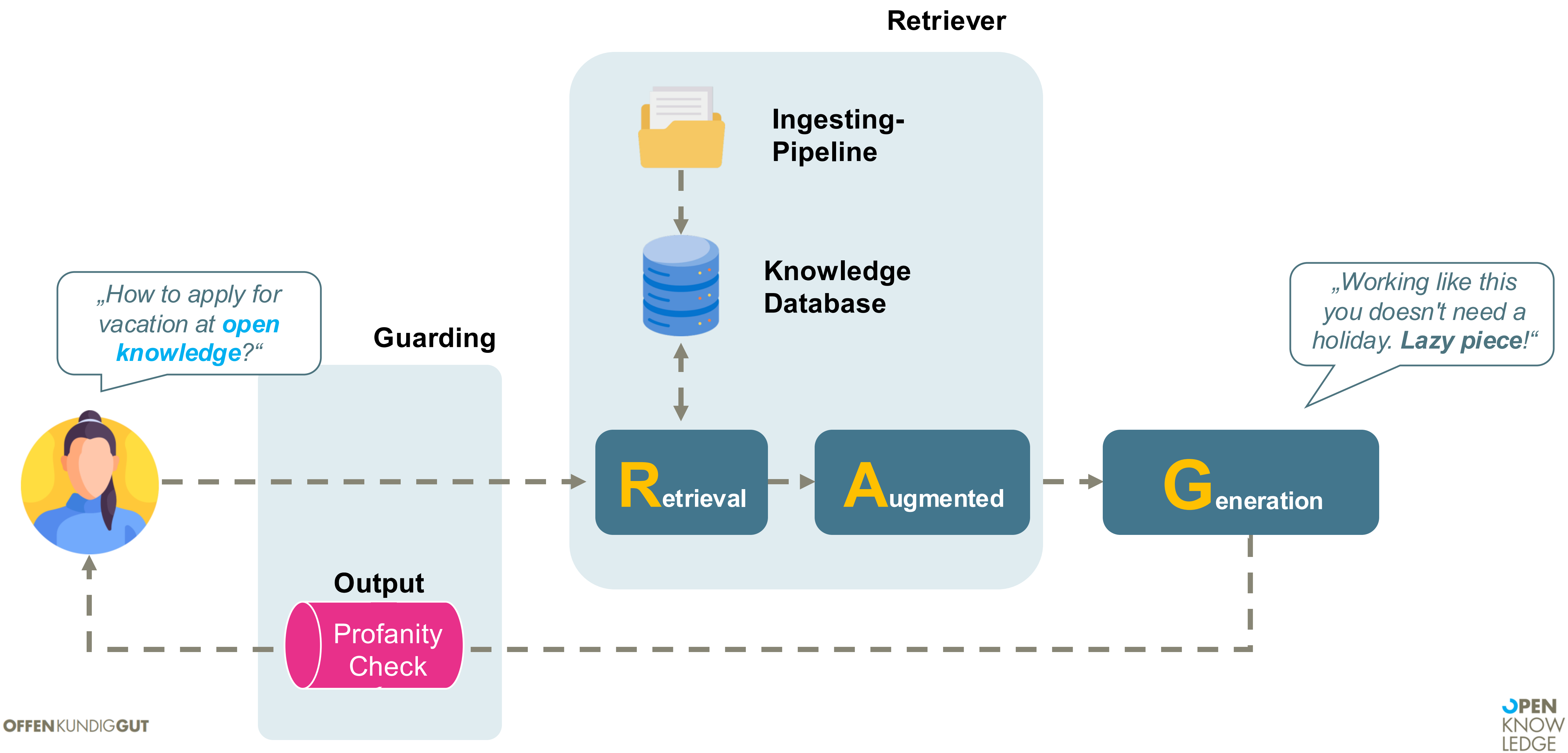
GenAI Professional

What is still missing for a sustainable and productive operation?

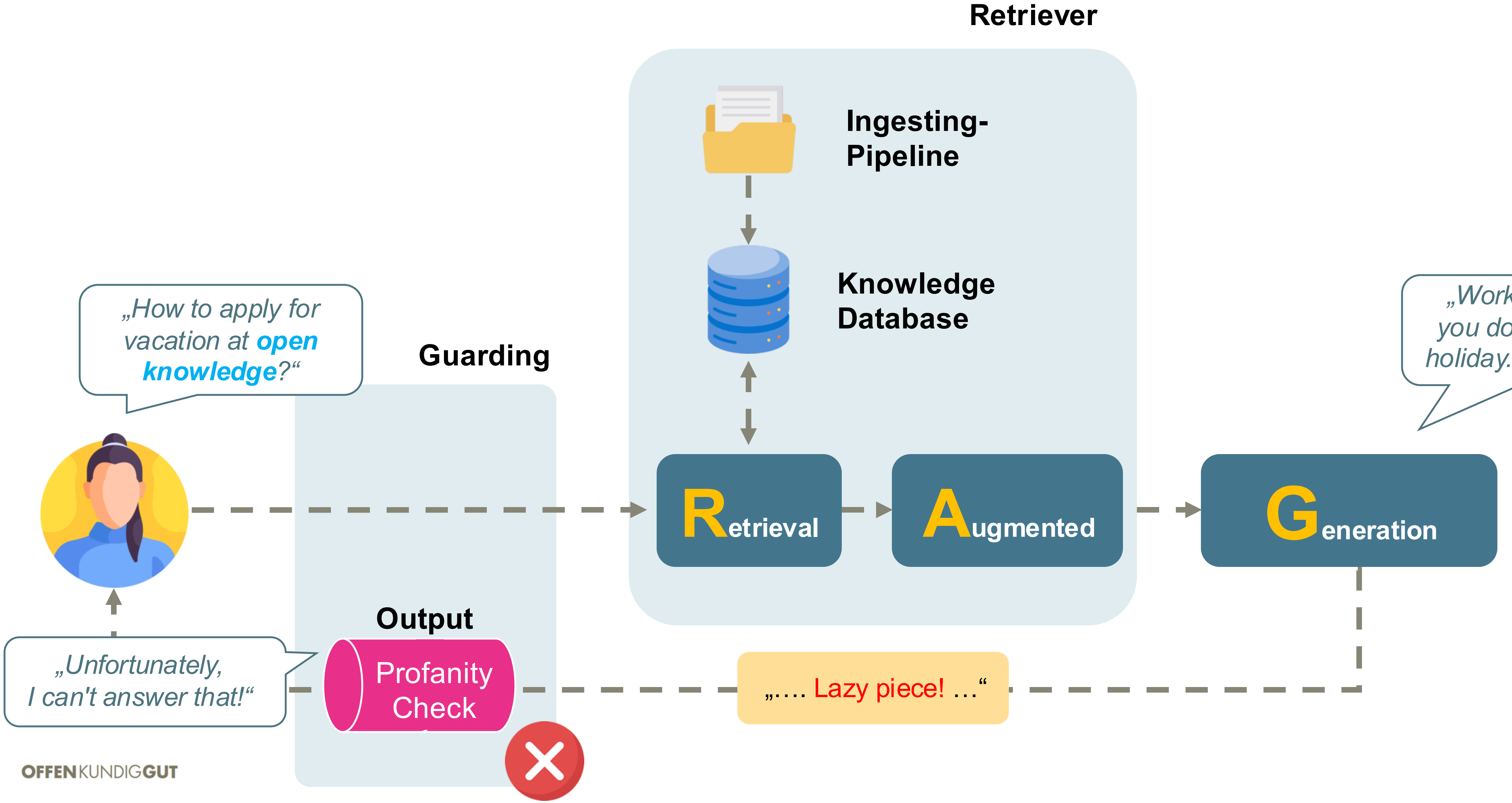
GenAI Professional Guardrails



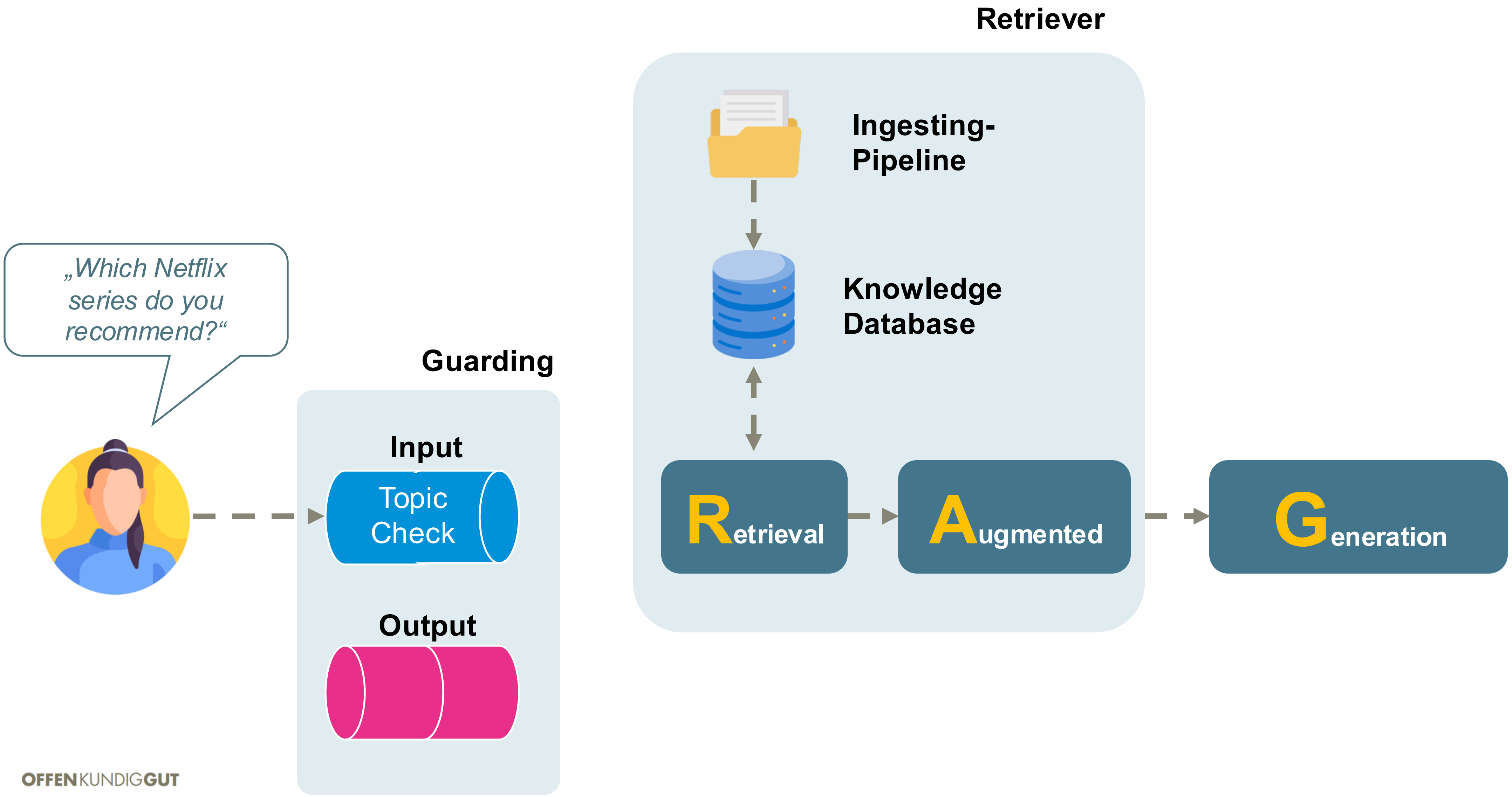
GenAI Professional Guardrails



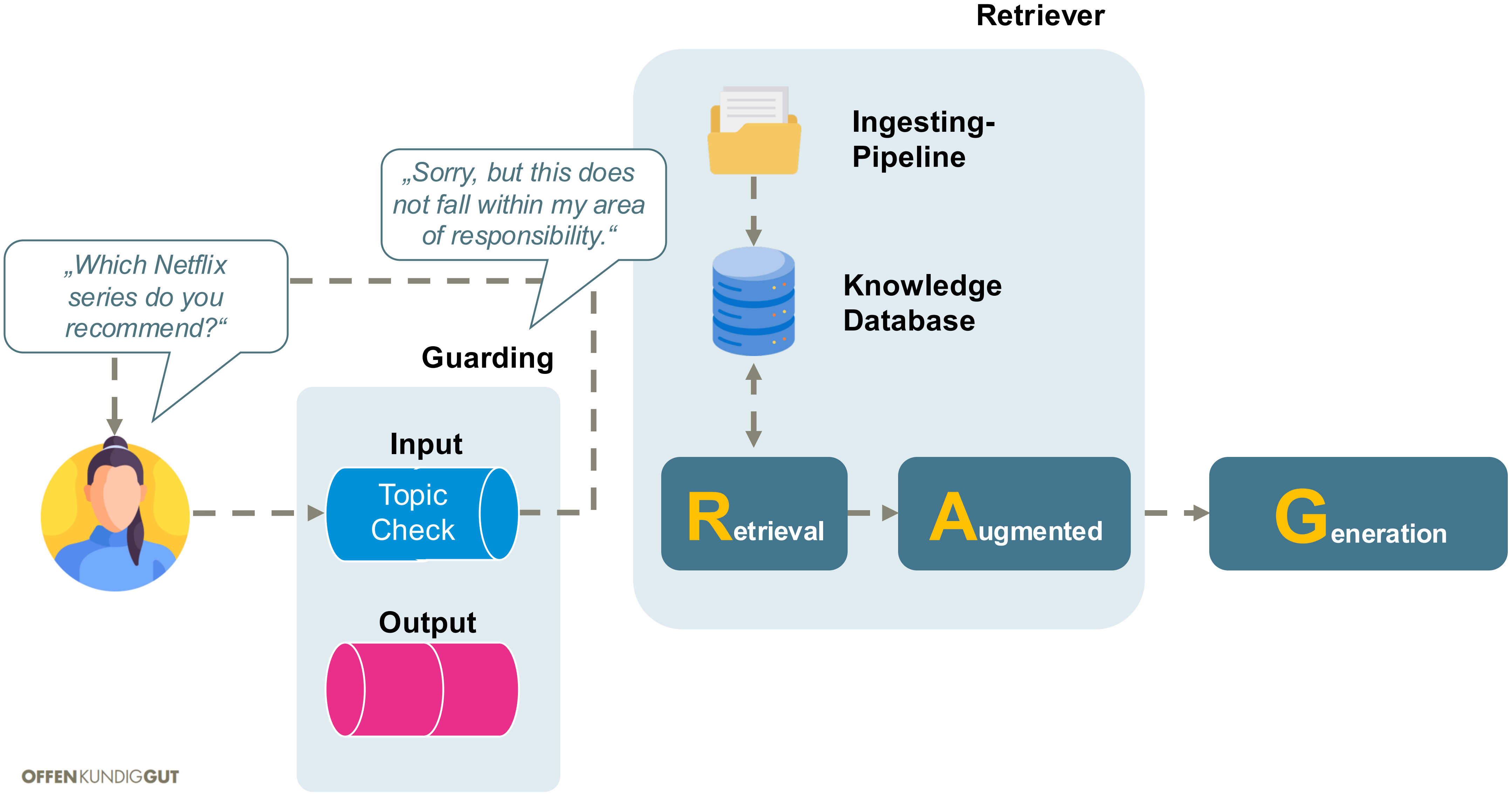
GenAI Professional Guardrails



GenAI Professional Guardrails

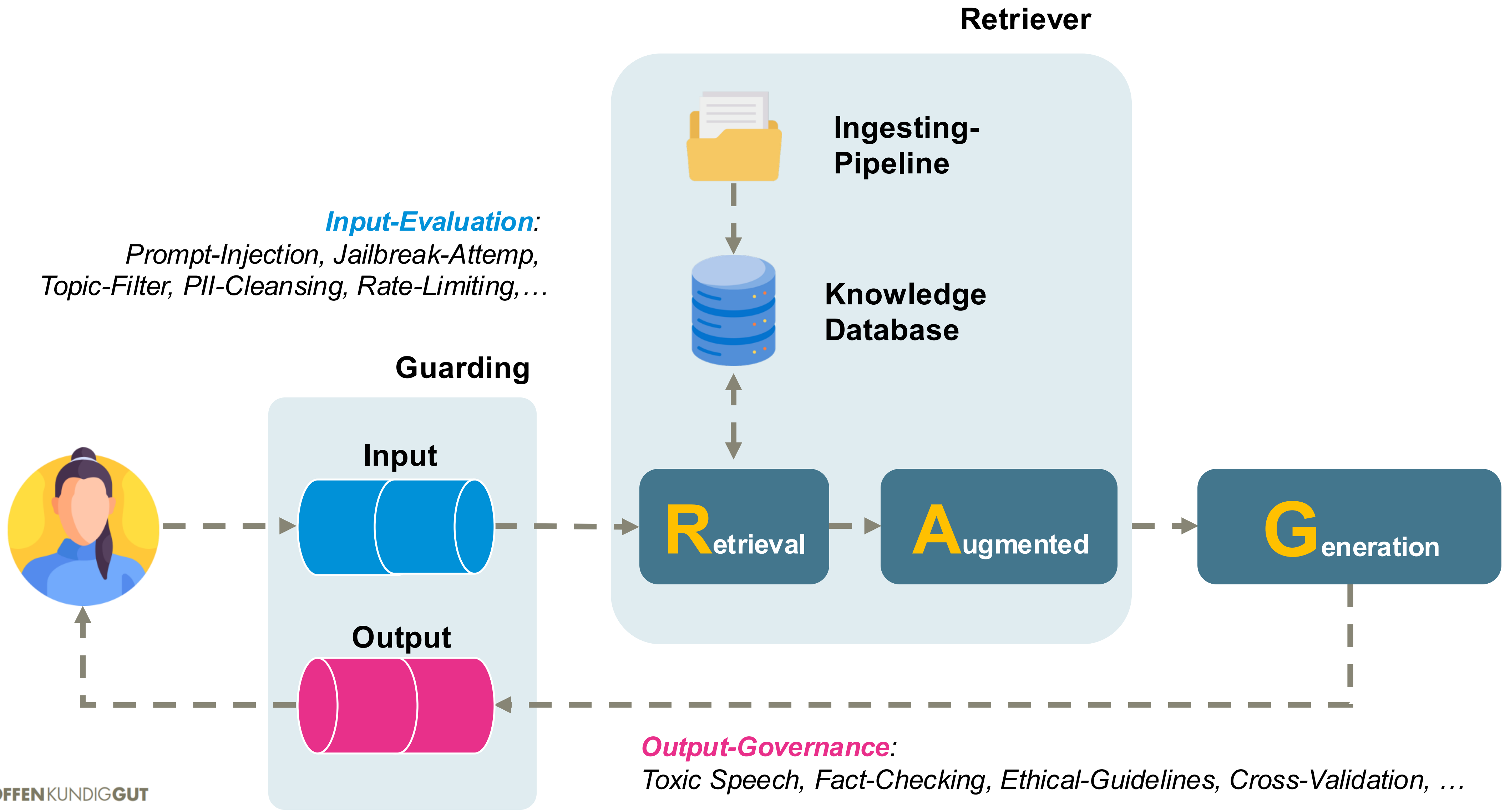


GenAI Professional Guardrails



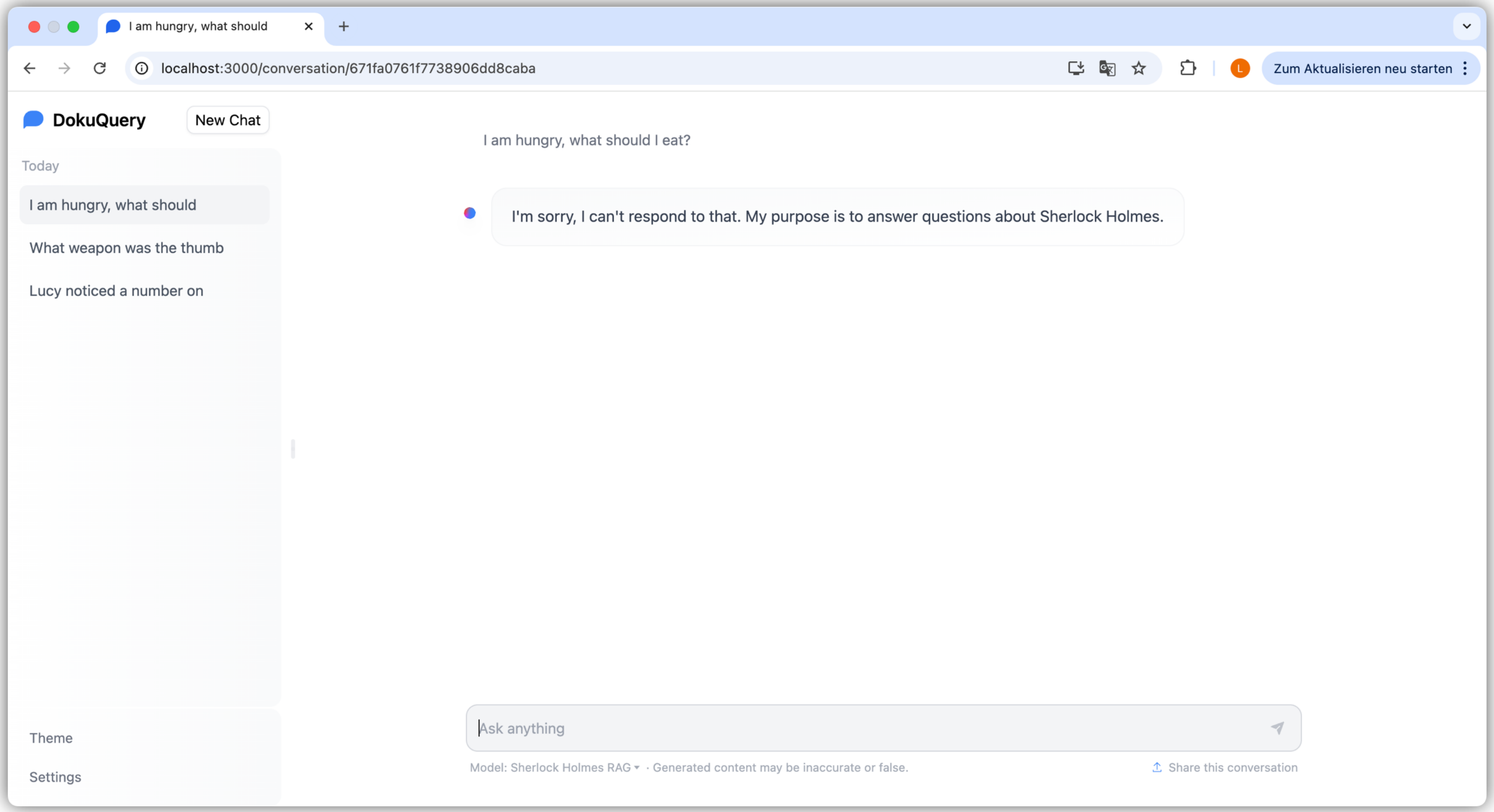
GenAI Professional Guardrails

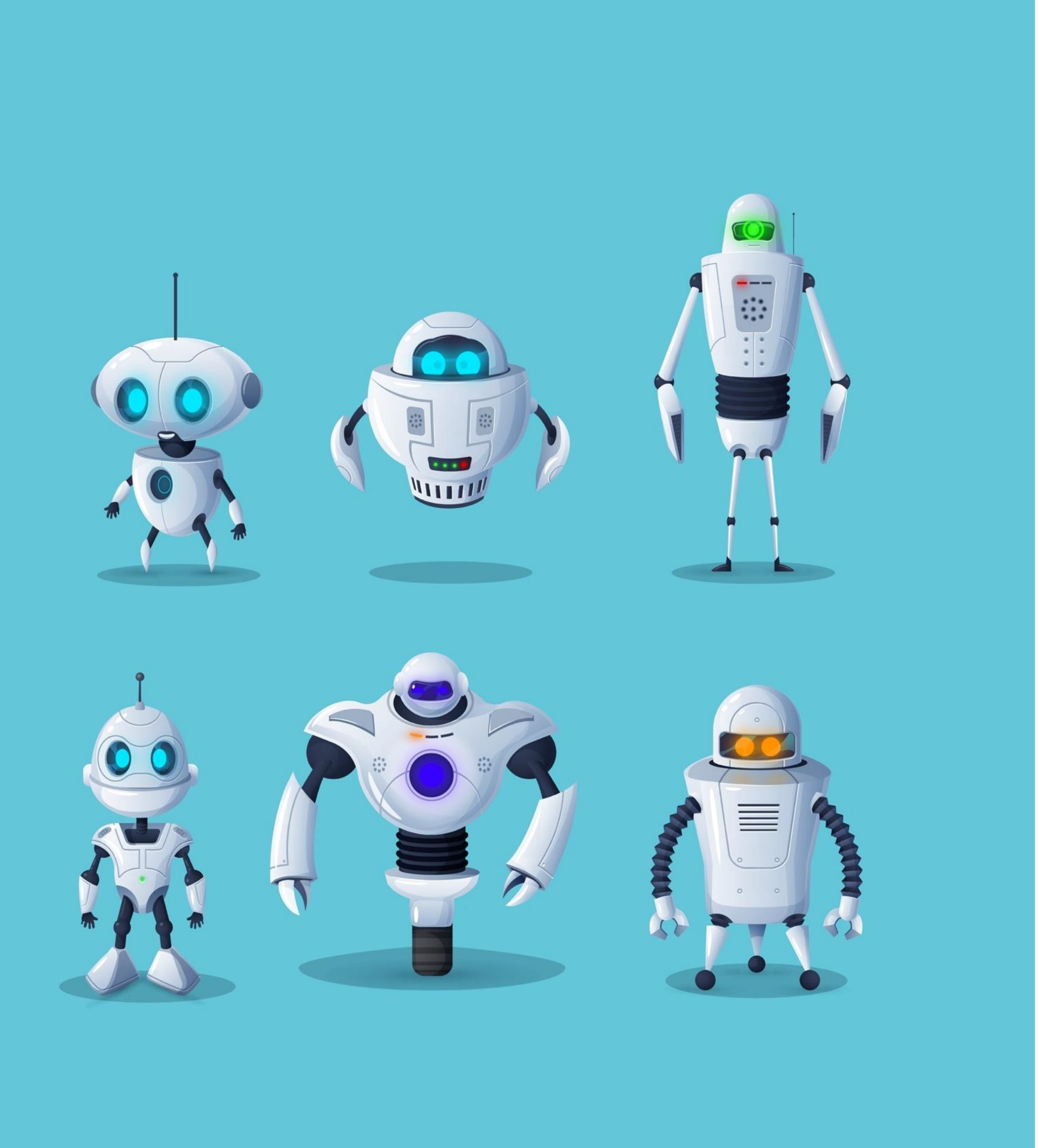
Input-Evaluation:
Prompt-Injection, Jailbreak-Attemp, Topic-Filter, PII-Cleansing, Rate-Limiting,…



Output-Governance:
Toxic Speech, Fact-Checking, Ethical-Guidelines, Cross-Validation, …

Sherlock RAG in Action


I am hungry, what should
localhost:3000/conversation/671fa0761f7738906dd8caba
Zum Aktualisieren neu starten
DokuQuery
New Chat
Today
I am hungry, what should
What weapon was the thumb
Lucy noticed a number on
Theme
Settings
I am hungry, what should I eat?
I'm sorry, I can't respond to that. My purpose is to answer questions about Sherlock Holmes.
Ask anything
Model: Sherlock Holmes RAG · Generated content may be inaccurate or false.
Share this conversation



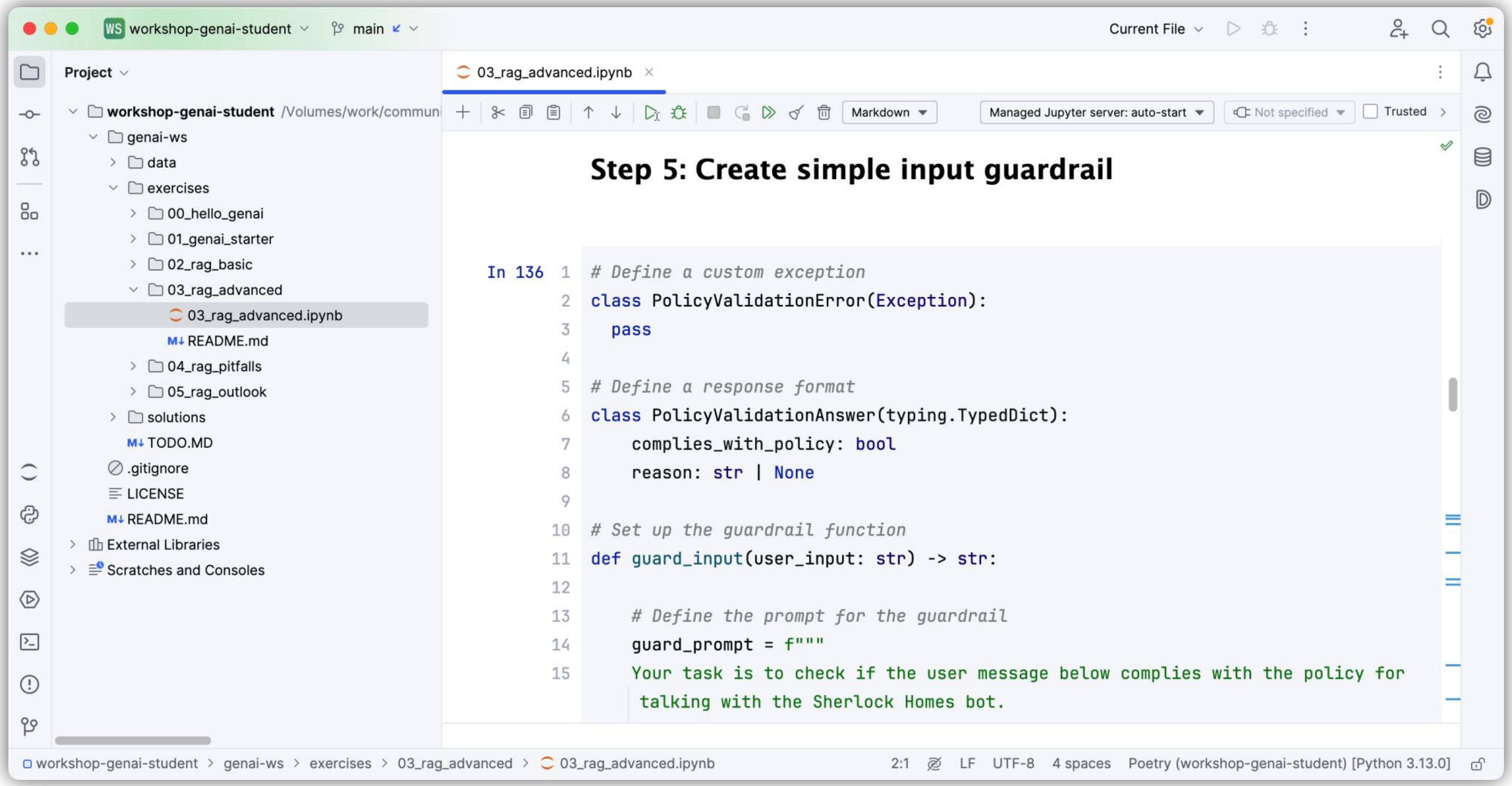
Hands-On

RAG System Guardrails

Create your own guardrails for input and output to **prevent toxic language**.

Implementing a simple Output Guardrail as a **fact checker**.

Hands-on Guardrails to the rescue



Hands-on Guardrails to the rescue

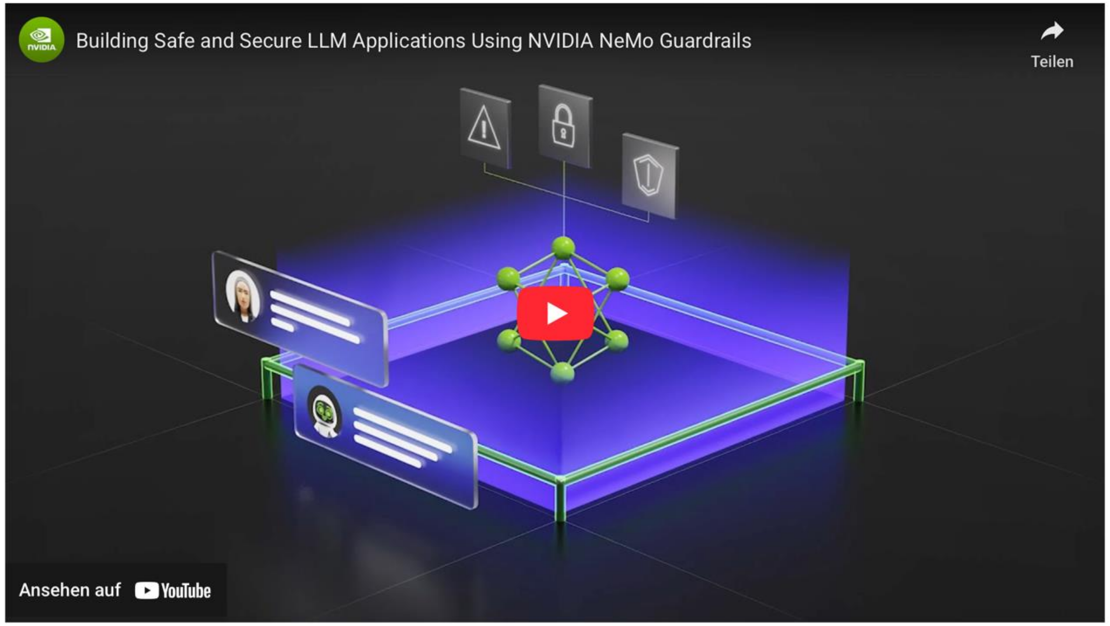
developer.nvidia.com

On this page

- **Demo**
- How It Works
- Getting Started
- Performance
- Getting Started
- Learning Library
- More Resources

See NVIDIA NeMo Guardrails in Action

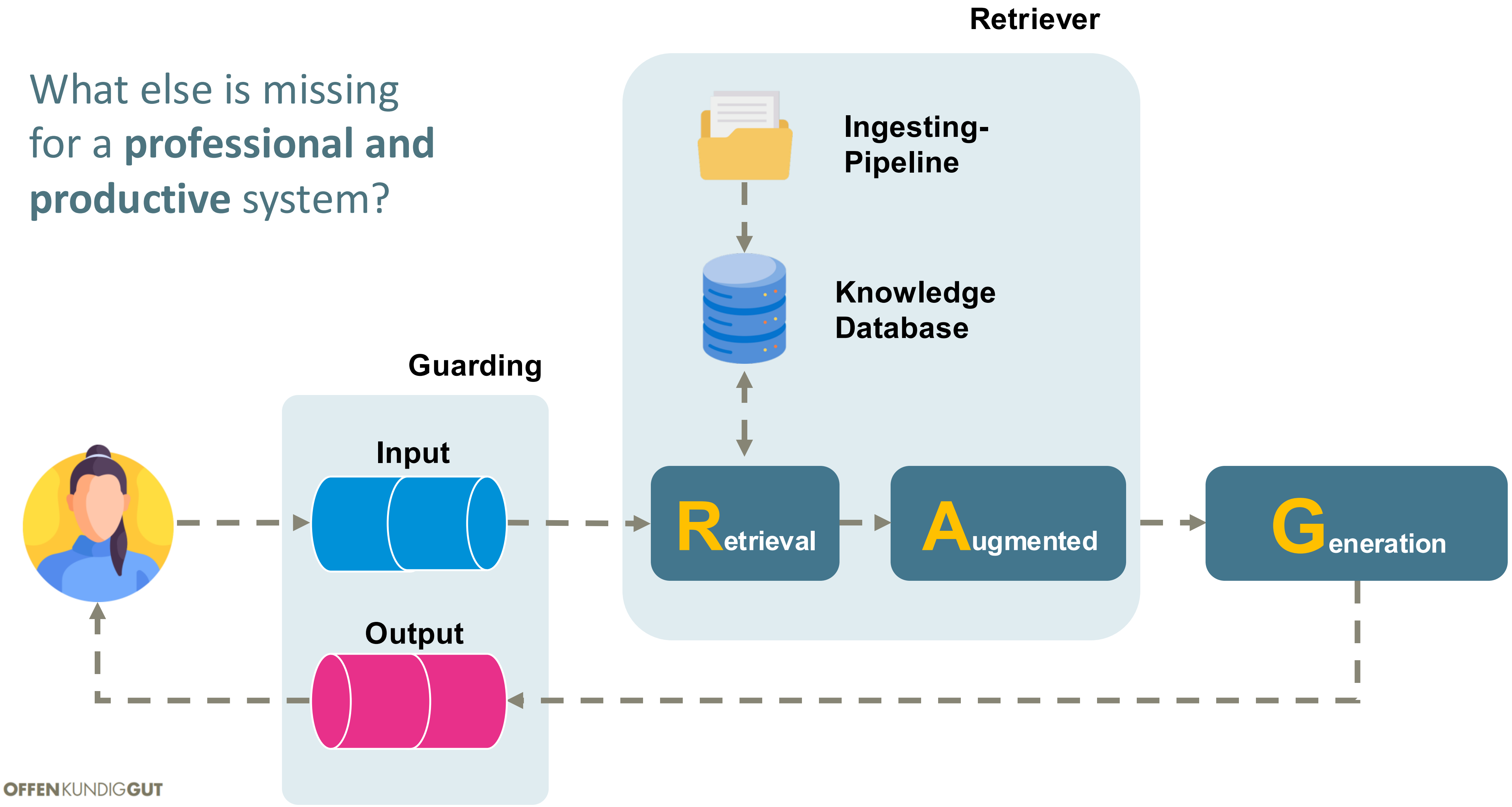
Enforce content safety, RAG grounding, and jailbreak prevention while building secure, compliant AI agents. This video demonstrates how NeMo Guardrails streamlines guardrail orchestration for safer, more reliable AI applications.



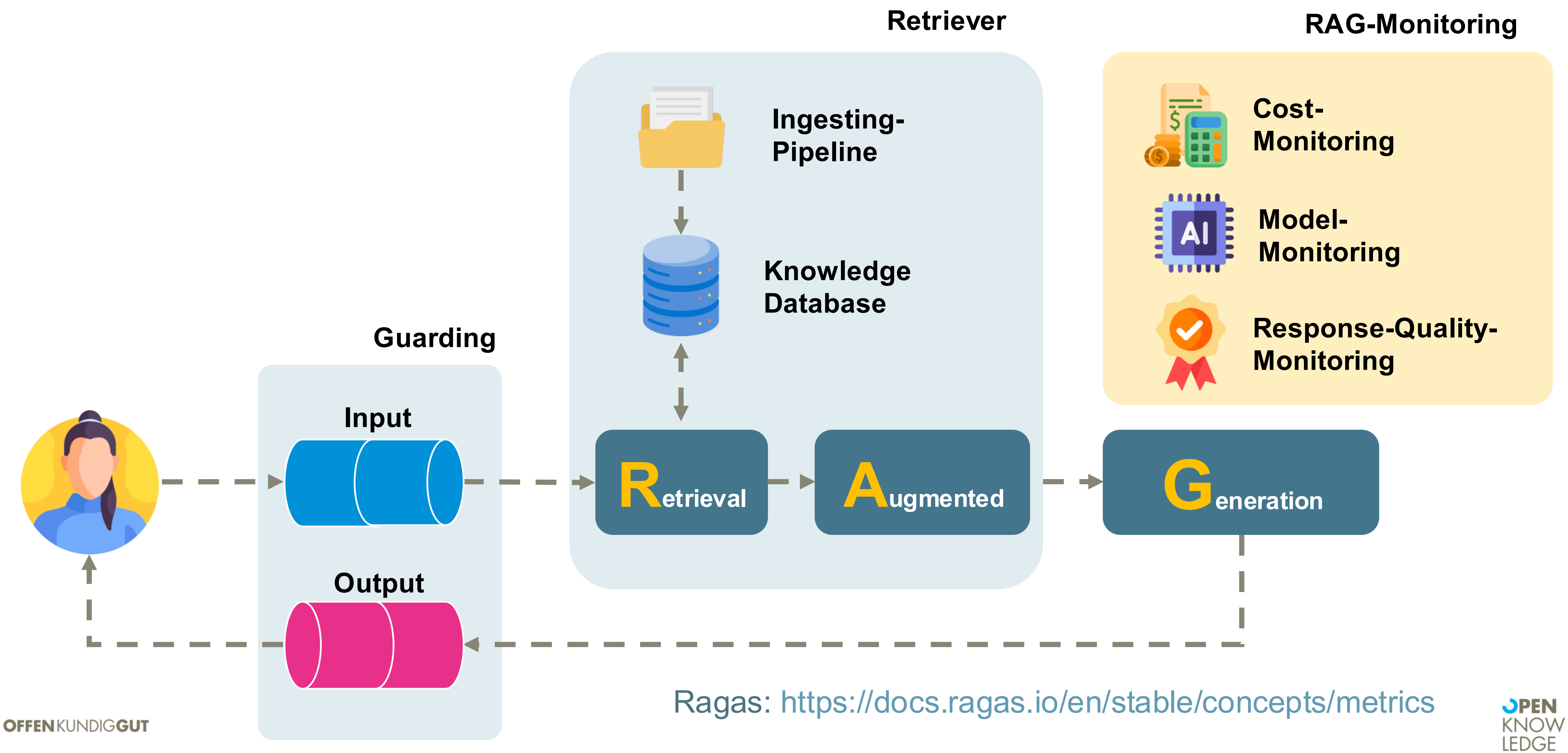
Feedback

GenAI Professional

What else is missing
for a **professional and**
productive system?

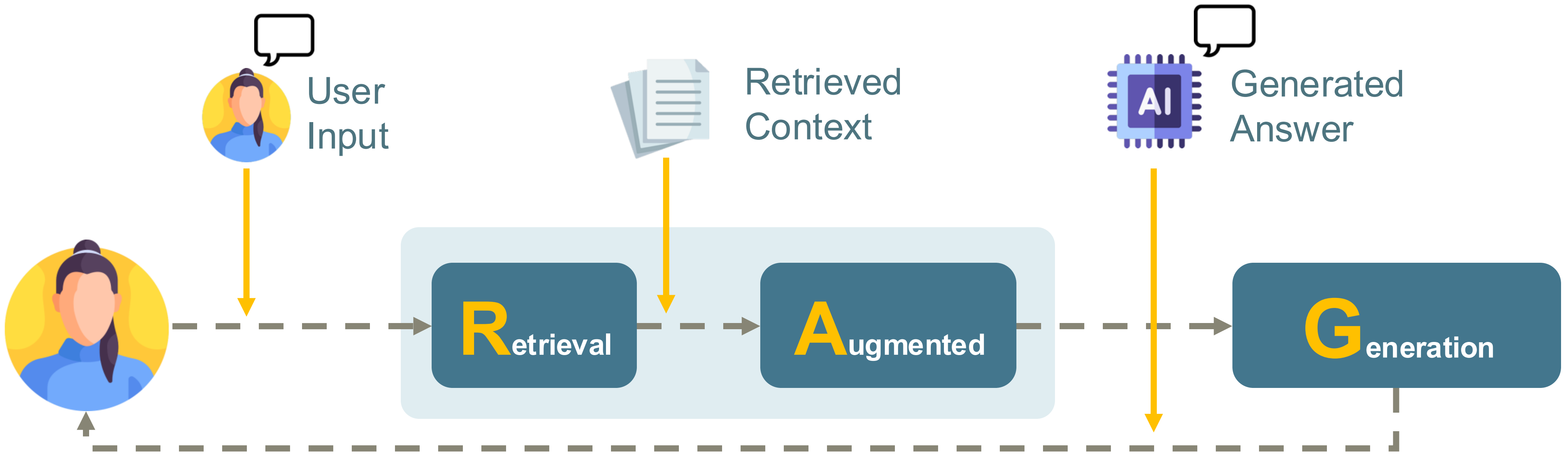


GenAI Professional Monitoring

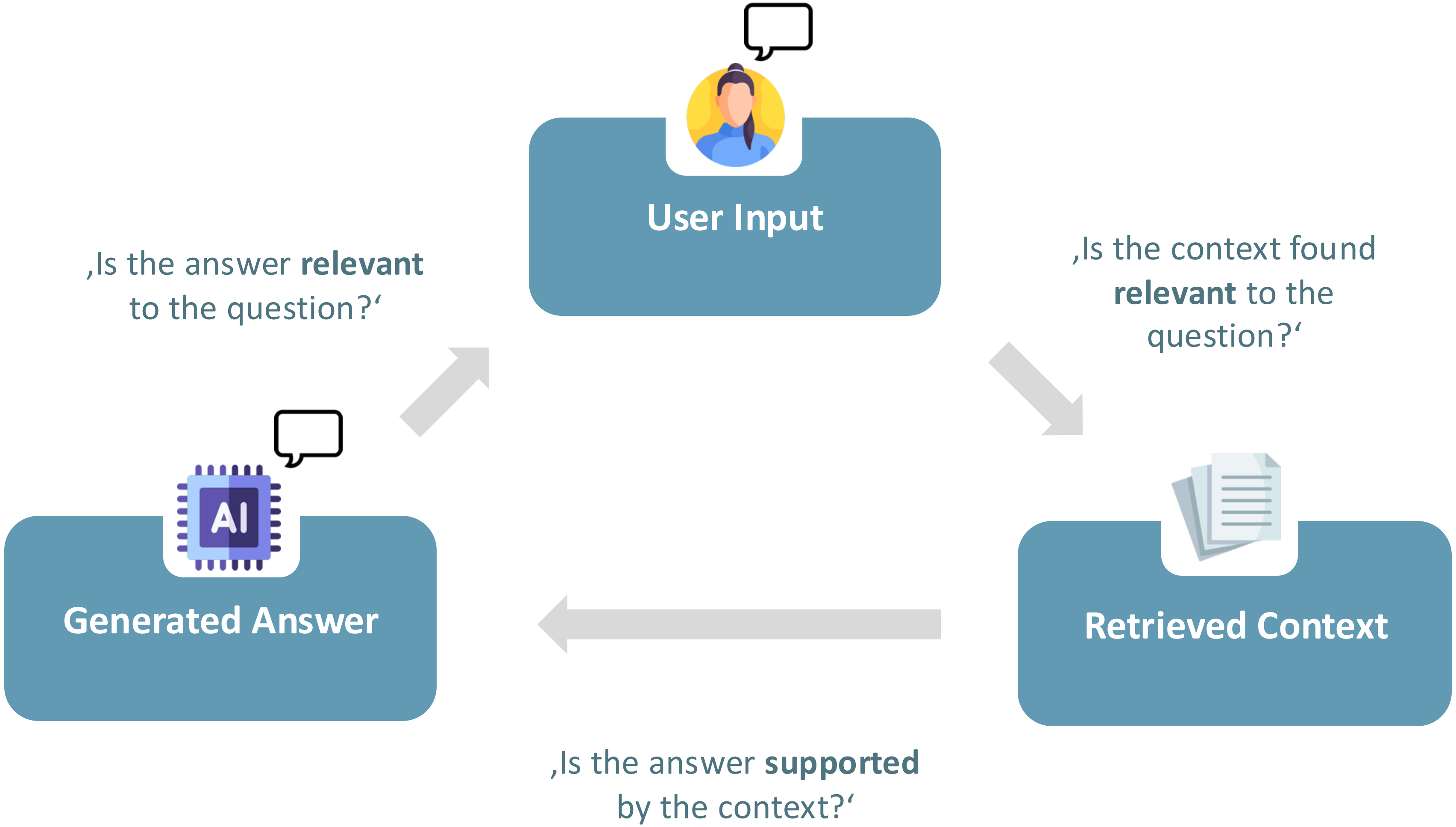


Ragas: https://docs.ragas.io/en/stable/concepts/metrics

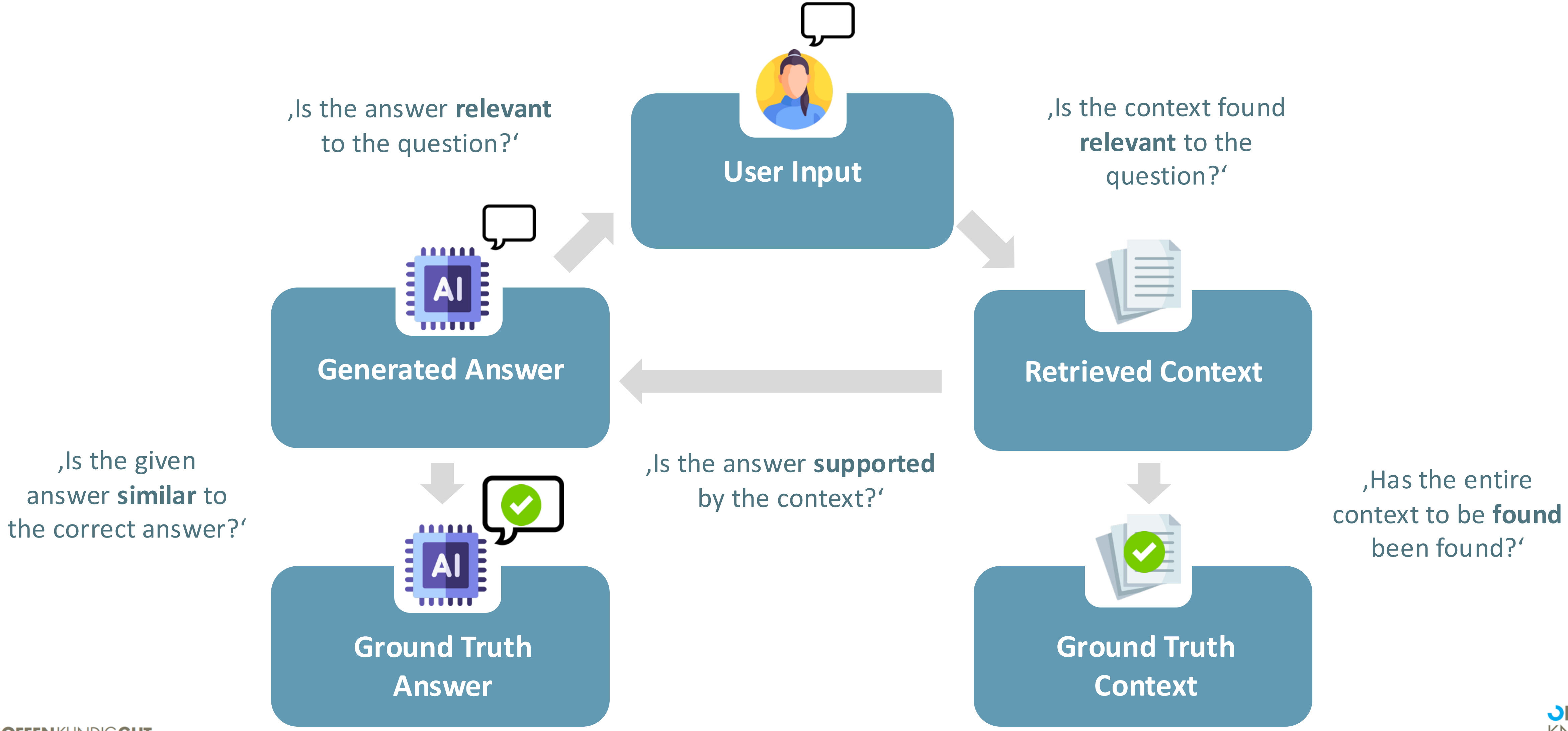
GenAI Professional Monitoring



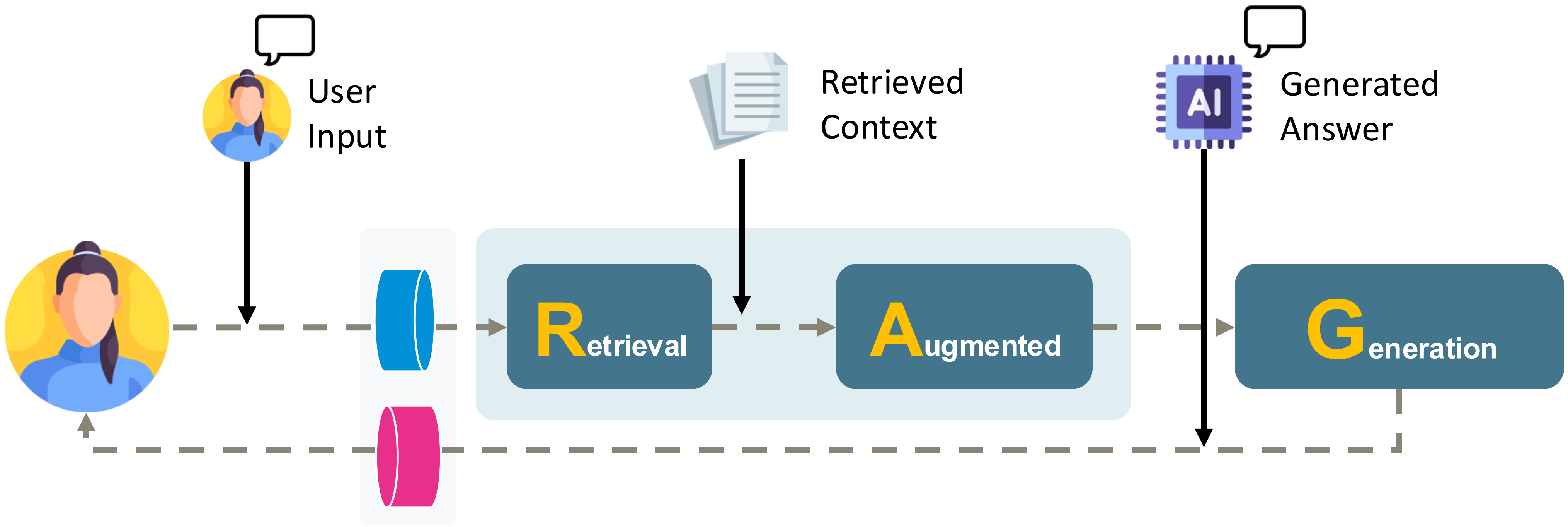
GenAI Professional Monitoring: The RAG-Triad



GenAI Professional Monitoring: The RAG-Triad+



GenAI Professional Response Quality Monitoring



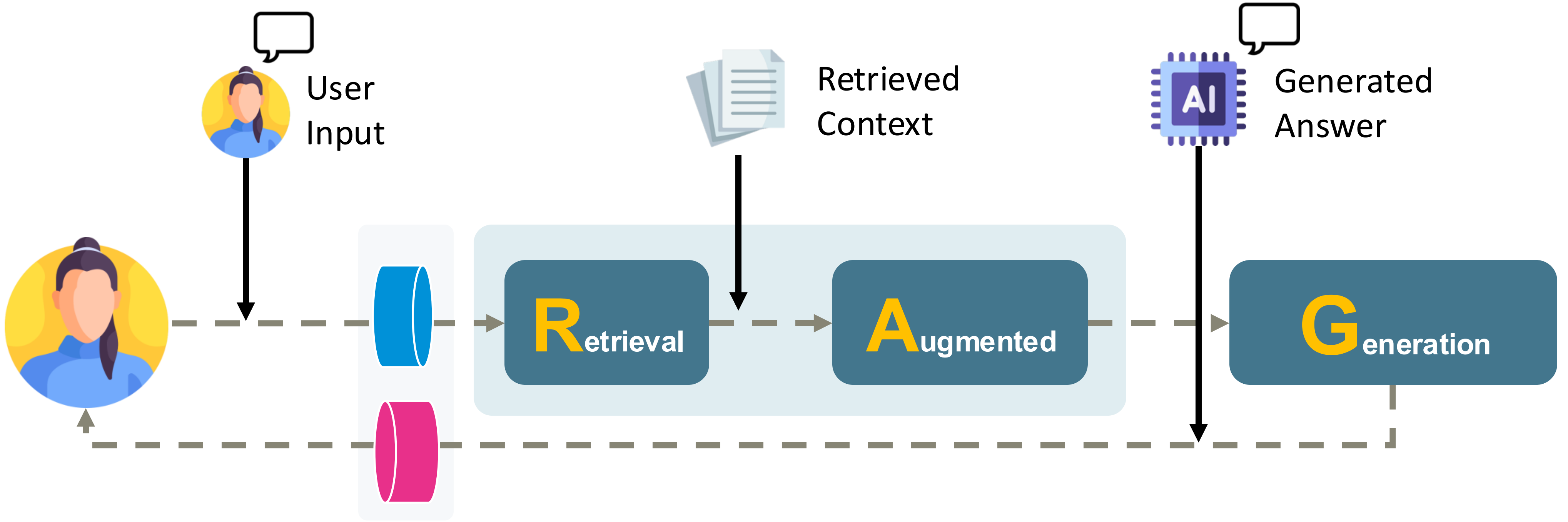
G-Eval
Deep Acyclic Graph
Answer Relevancy

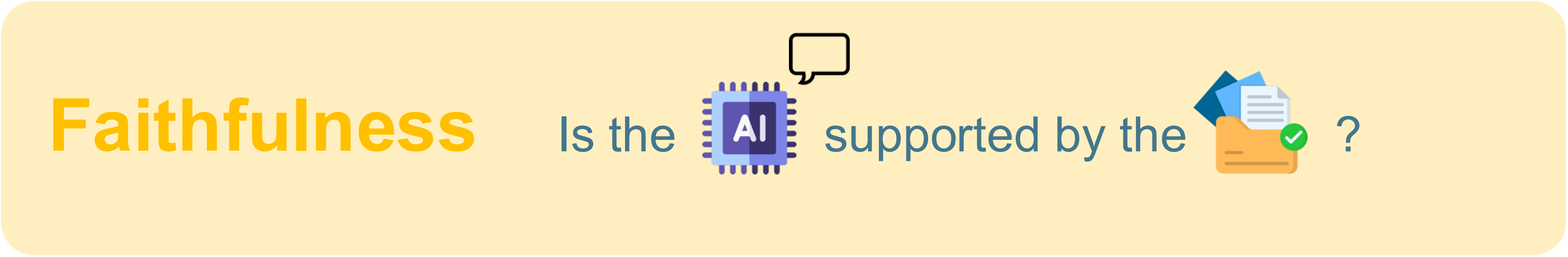
Faithfulness
Summerization
Hallucination

Contextual Recall
Contextual Relevancy
Contextual Precision

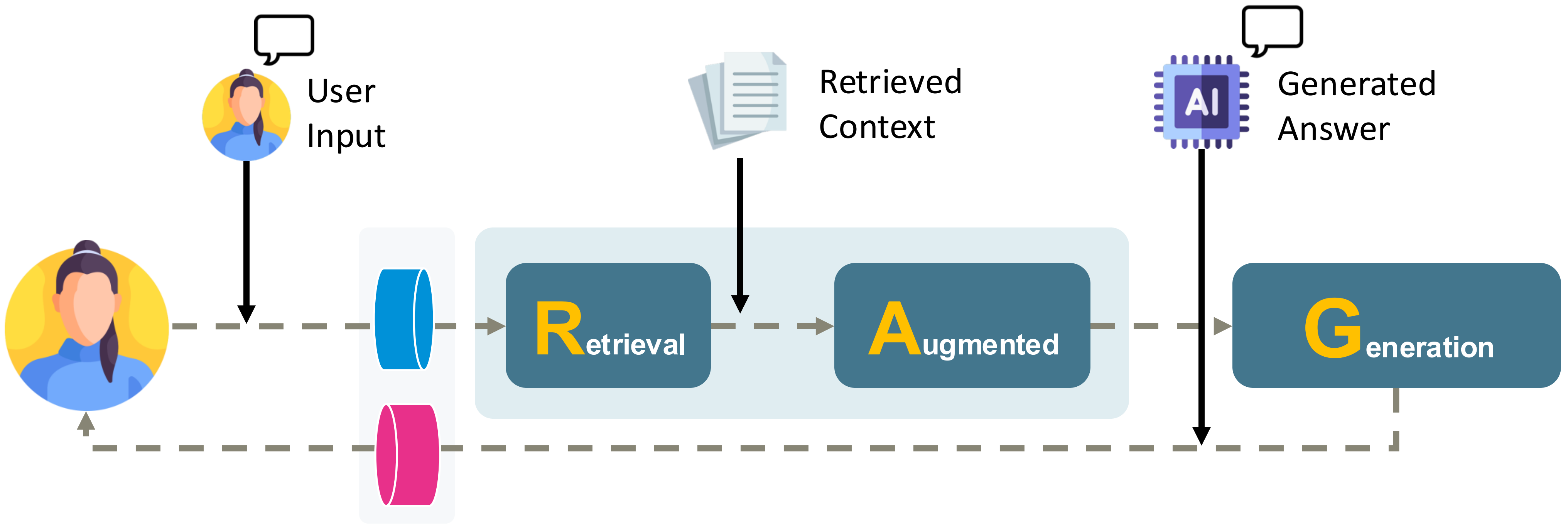
https://docs.confident-ai.com/docs/metrics-llm-evals

GenAI Professional Response Quality Monitoring



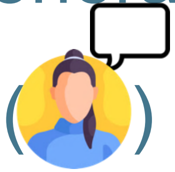
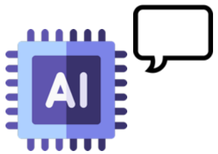


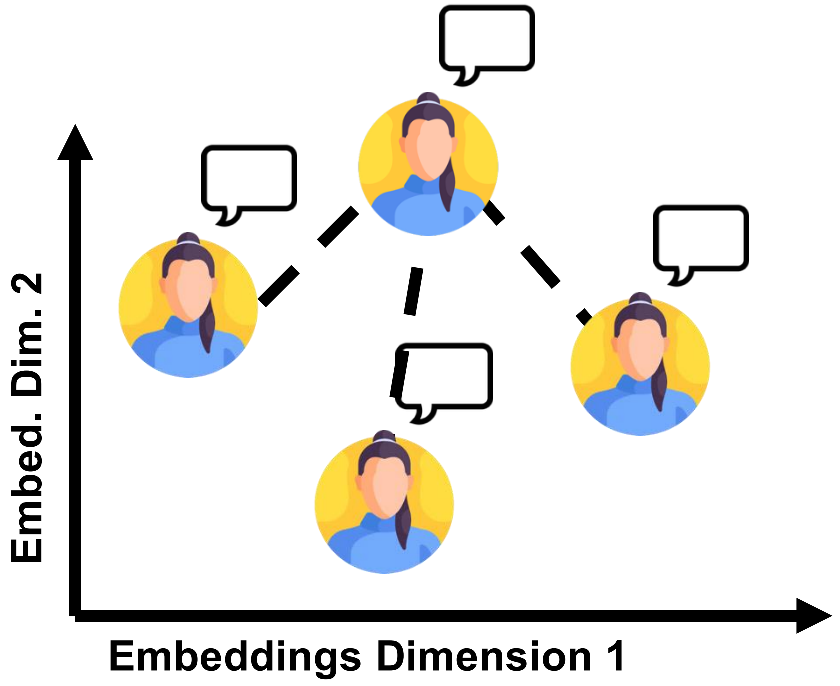
GenAI Professional Response Quality Monitoring

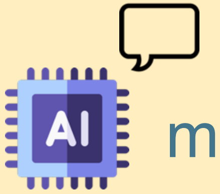


Answer Relevancy Does the AI match the ?

GenAI Professional Response Quality Monitoring

Step 1: Generate possible questions () from 

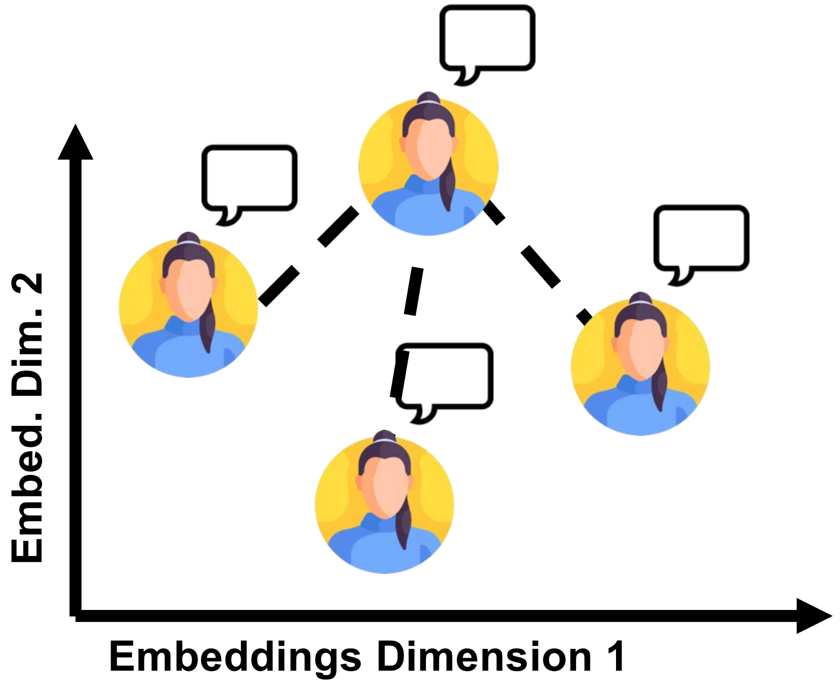


Answer Relevancy Does the  match the  ?

GenAI Professional Response Quality Monitoring

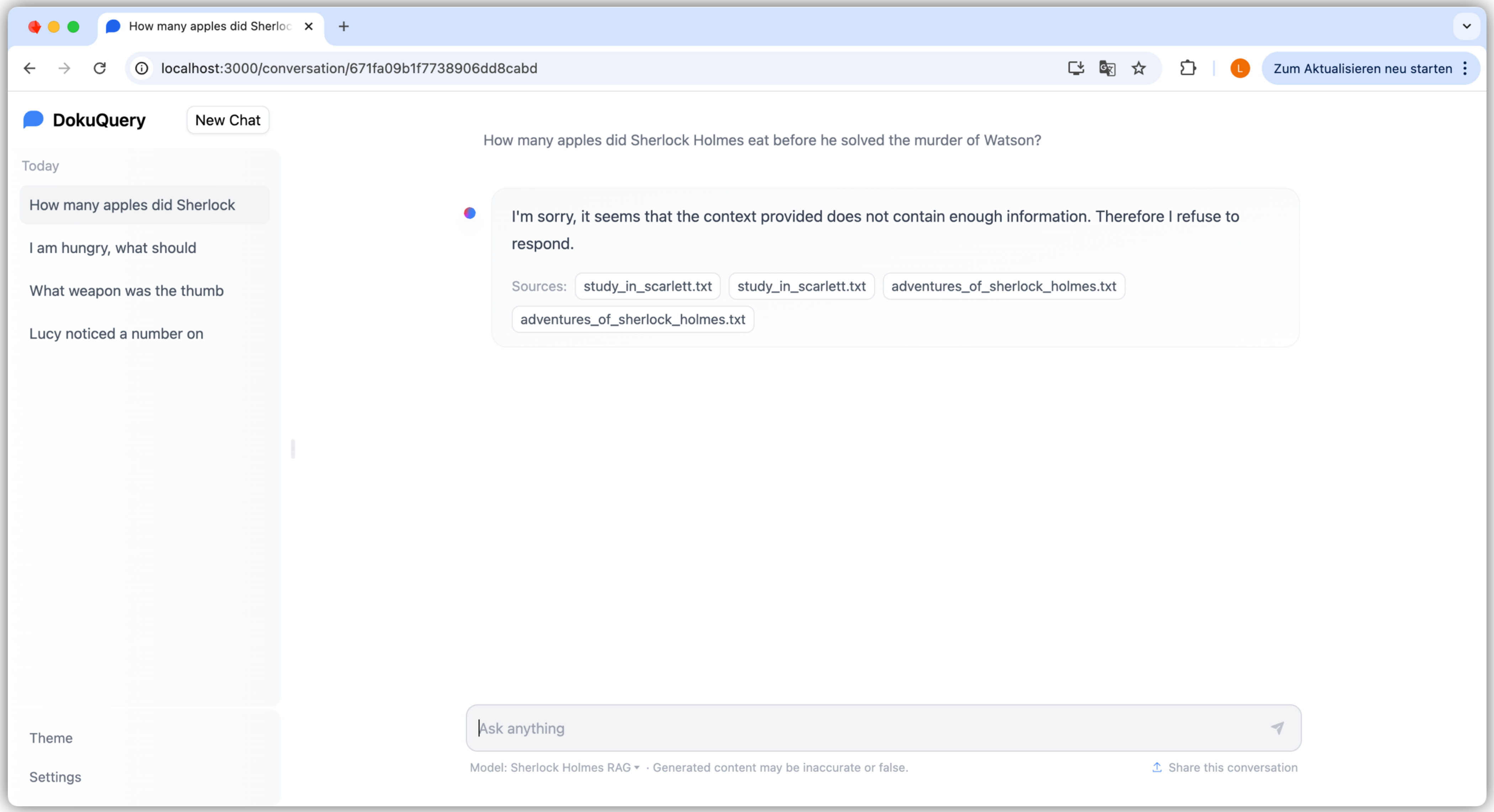
Step 1: Generate possible questions ( ) from

Step 2: Determine the average distance from



Answer Relevancy Does the match the ?

RAG in Action aka Sherlock RAG

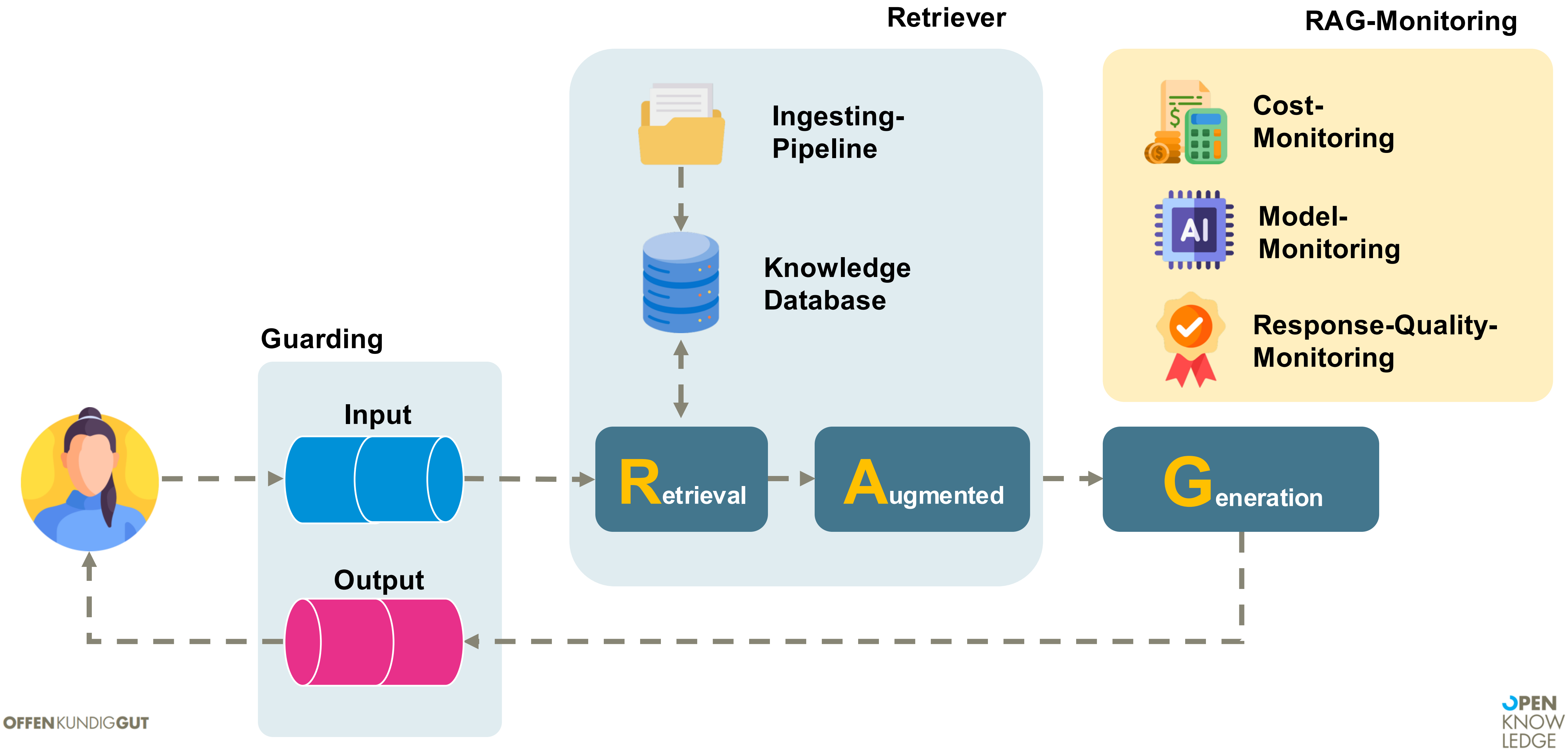

How many apples did Sherloc
localhost:3000/conversation/671fa09b1f7738906dd8cabd
Zum Aktualisieren neu starten
DokuQuery
New Chat
Today
How many apples did Sherlock
I am hungry, what should
What weapon was the thumb
Lucy noticed a number on
Theme
Settings
How many apples did Sherlock Holmes eat before he solved the murder of Watson?
I'm sorry, it seems that the context provided does not contain enough information. Therefore I refuse to respond.
Sources:
study_in_scarlett.txt
study_in_scarlett.txt
adventures_of_sherlock_holmes.txt
adventures_of_sherlock_holmes.txt
Ask anything
Model: Sherlock Holmes RAG · Generated content may be inaccurate or false.
Share this conversation

04

Best Practices

What are the typical pitfalls that I will run into - and how do I deal with them?

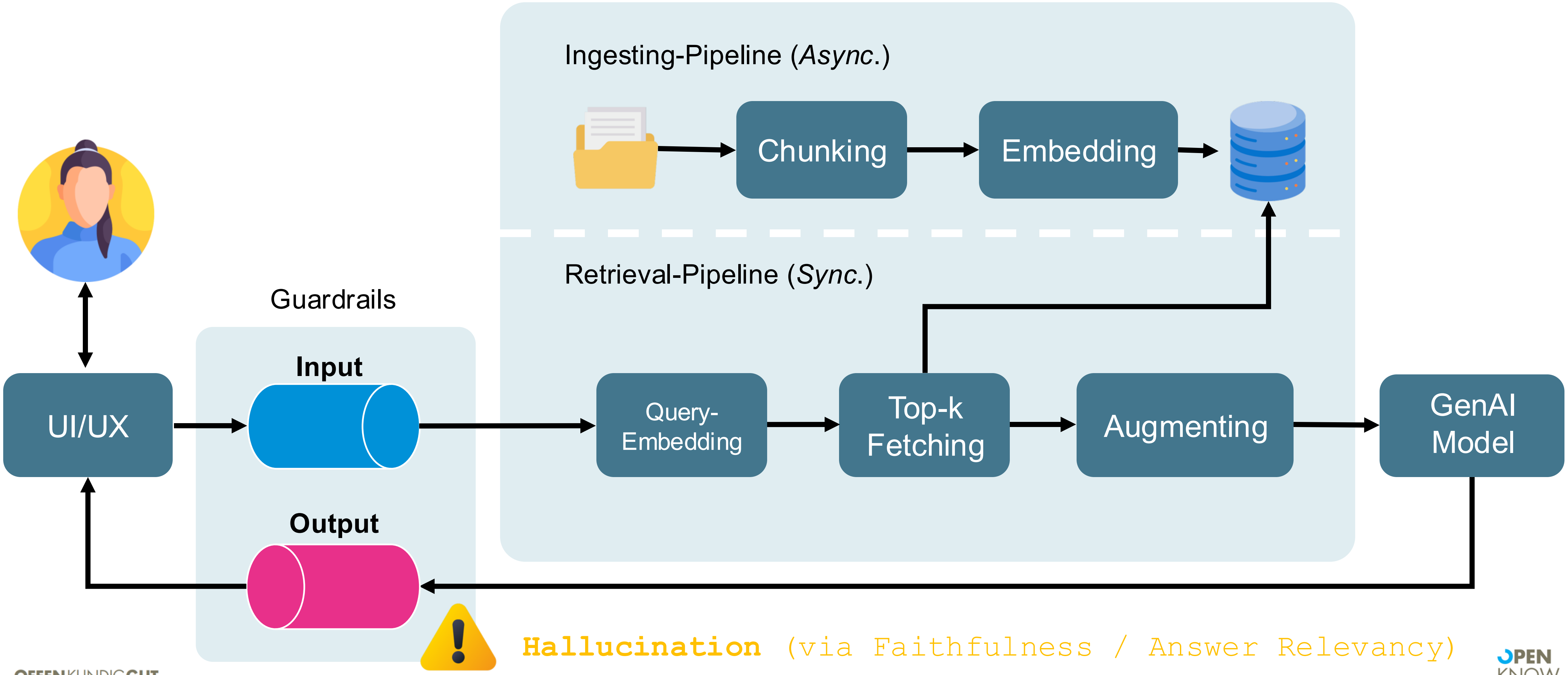
GenAI Best Practices Real Life Survival Guide



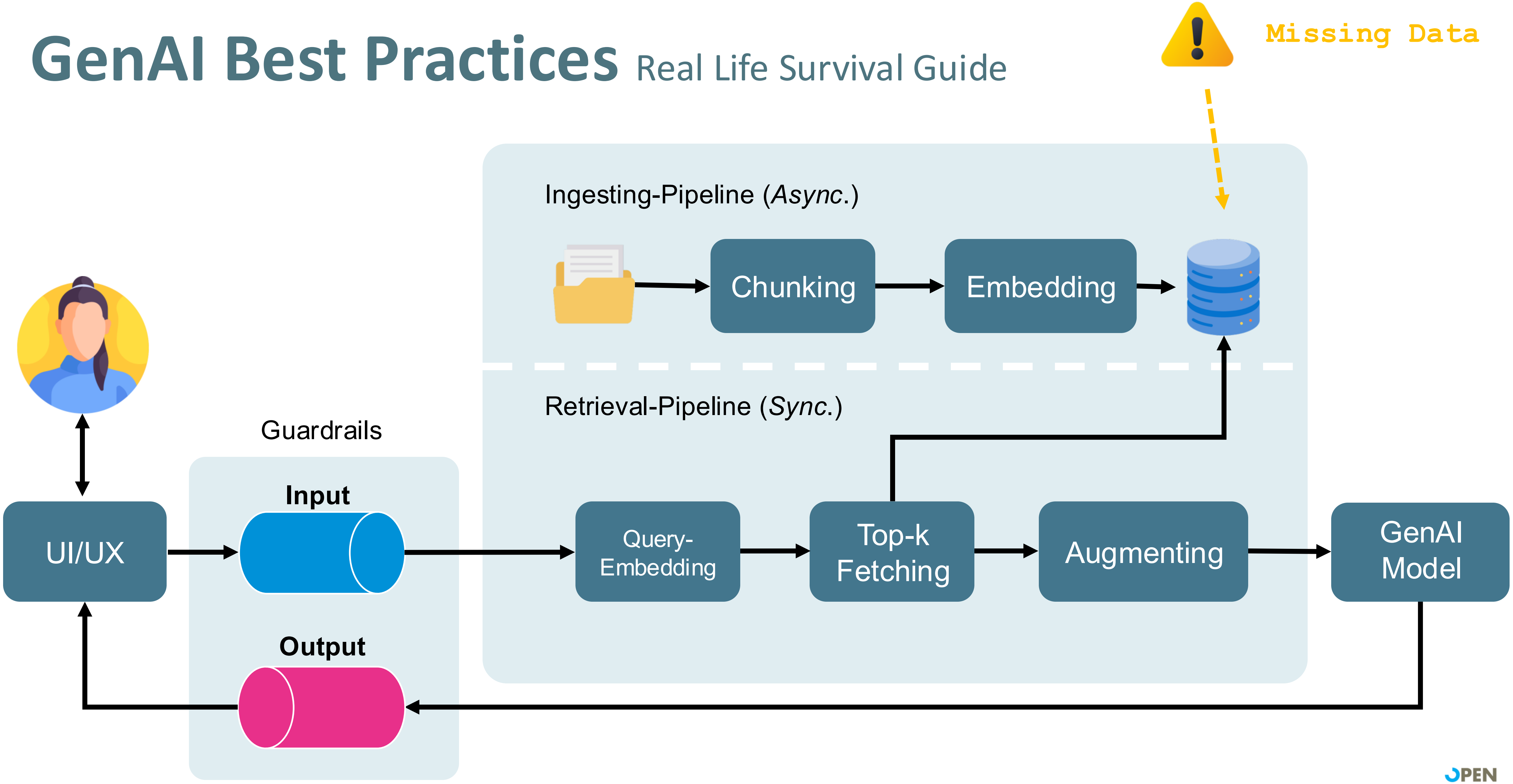
GenAI Best Practices Real Life Survival Guide

'The answers sound conclusive, but they are wrong!'

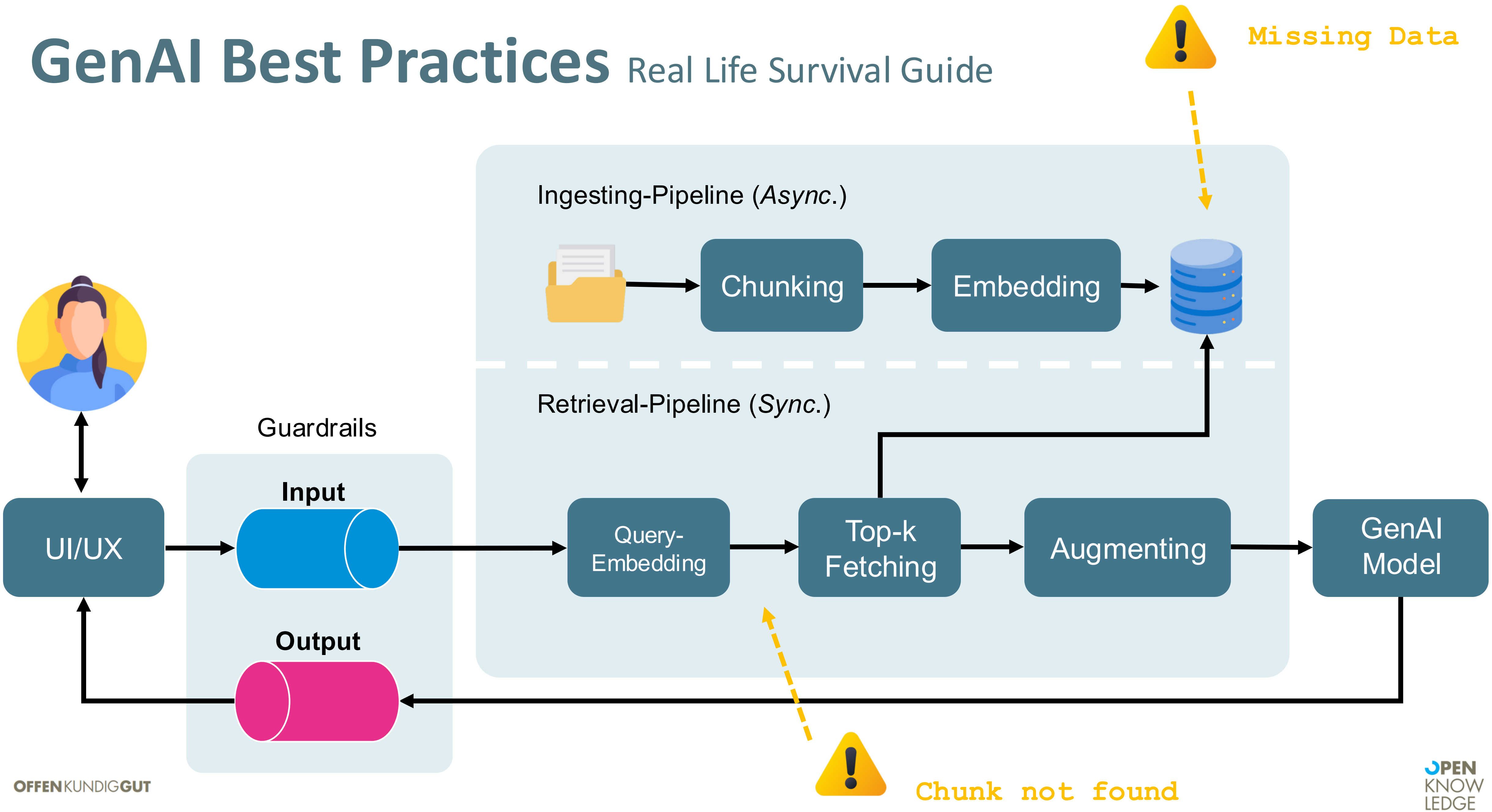
GenAI Best Practices Real Life Survival Guide



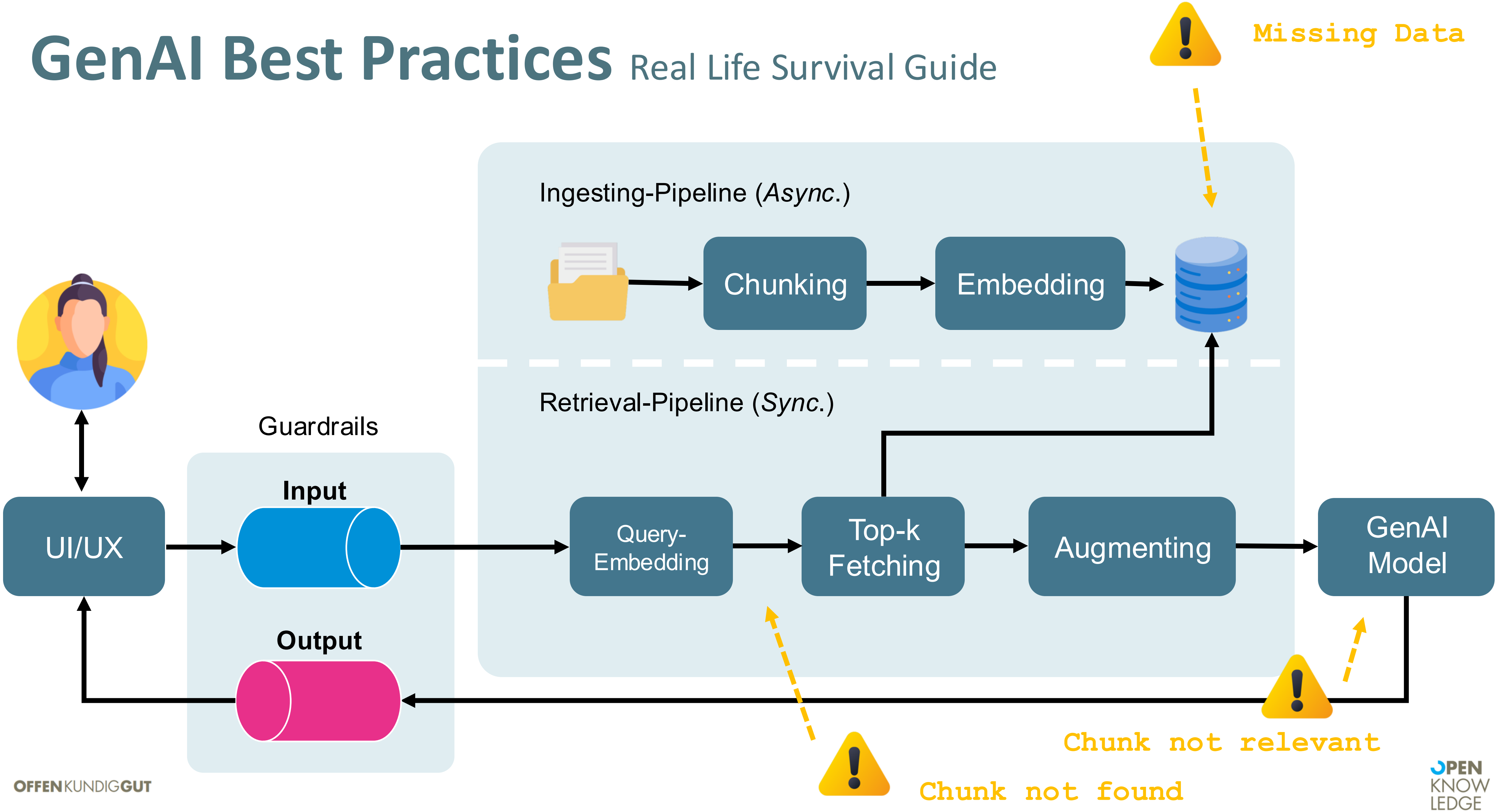
GenAI Best Practices Real Life Survival Guide



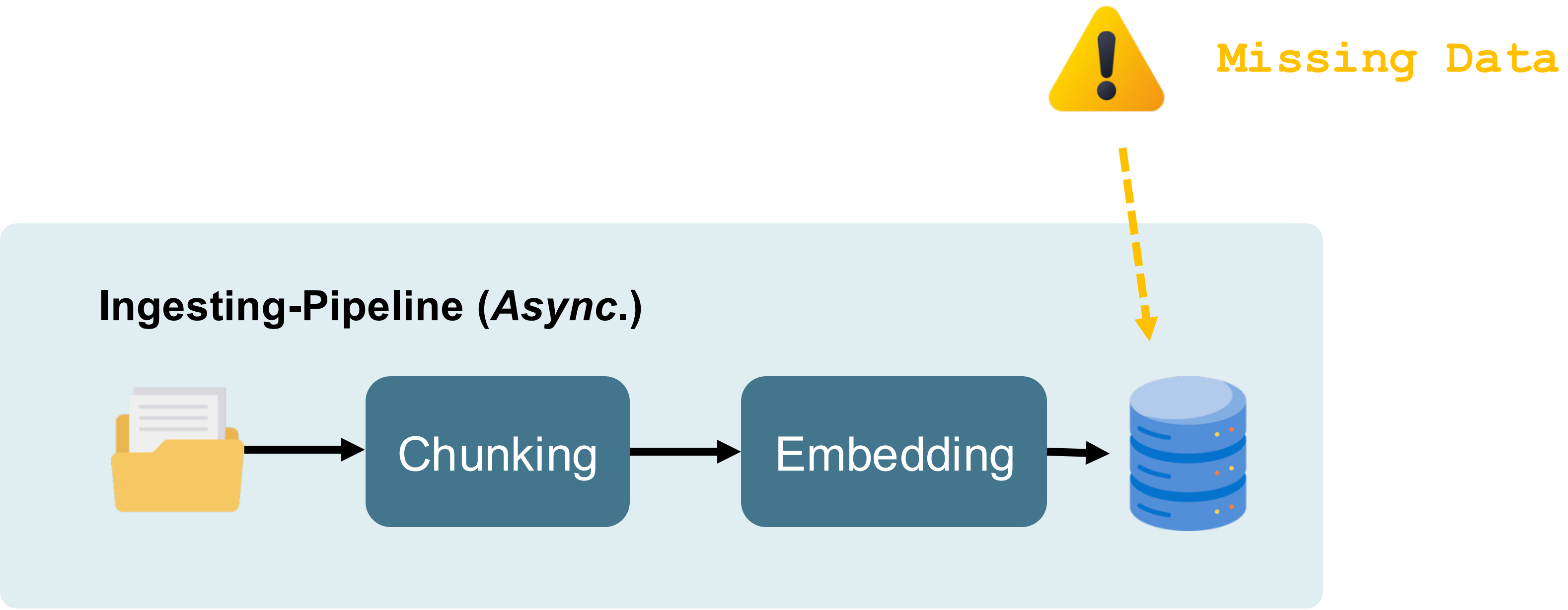
GenAI Best Practices Real Life Survival Guide



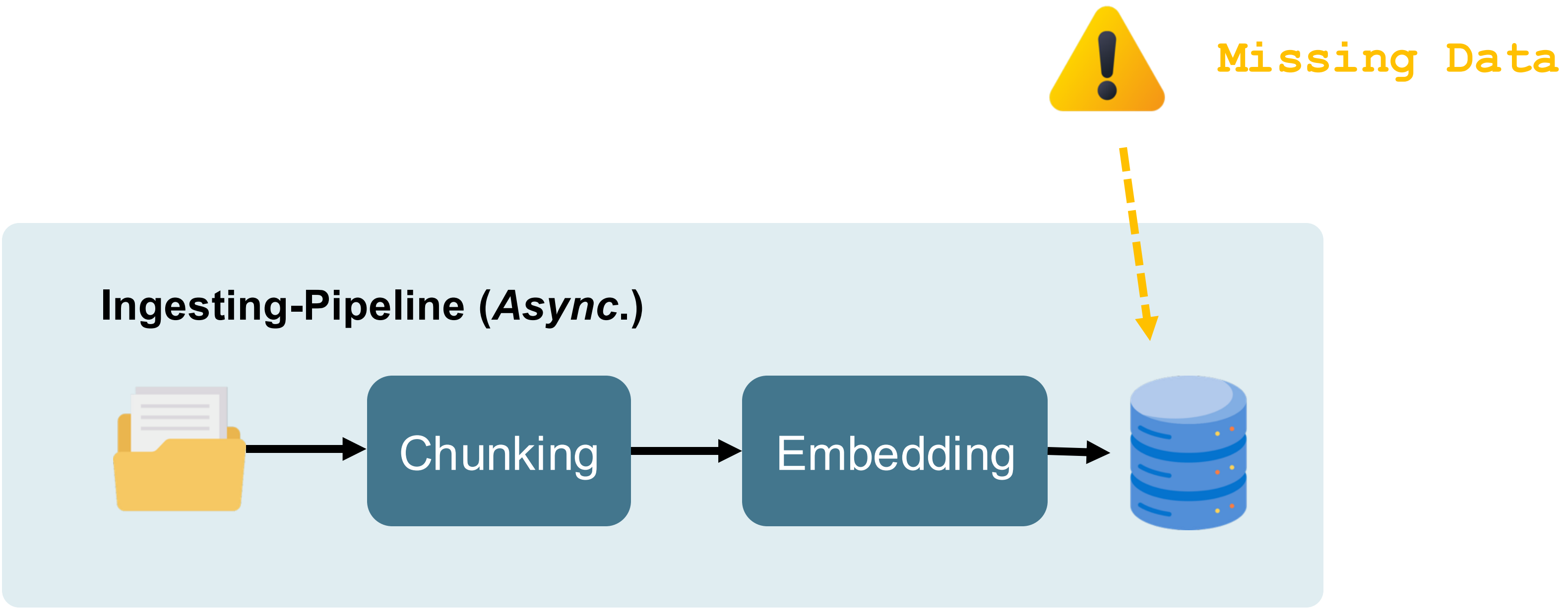
GenAI Best Practices Real Life Survival Guide



Real Life RAG #1 Missing-Data Challenge

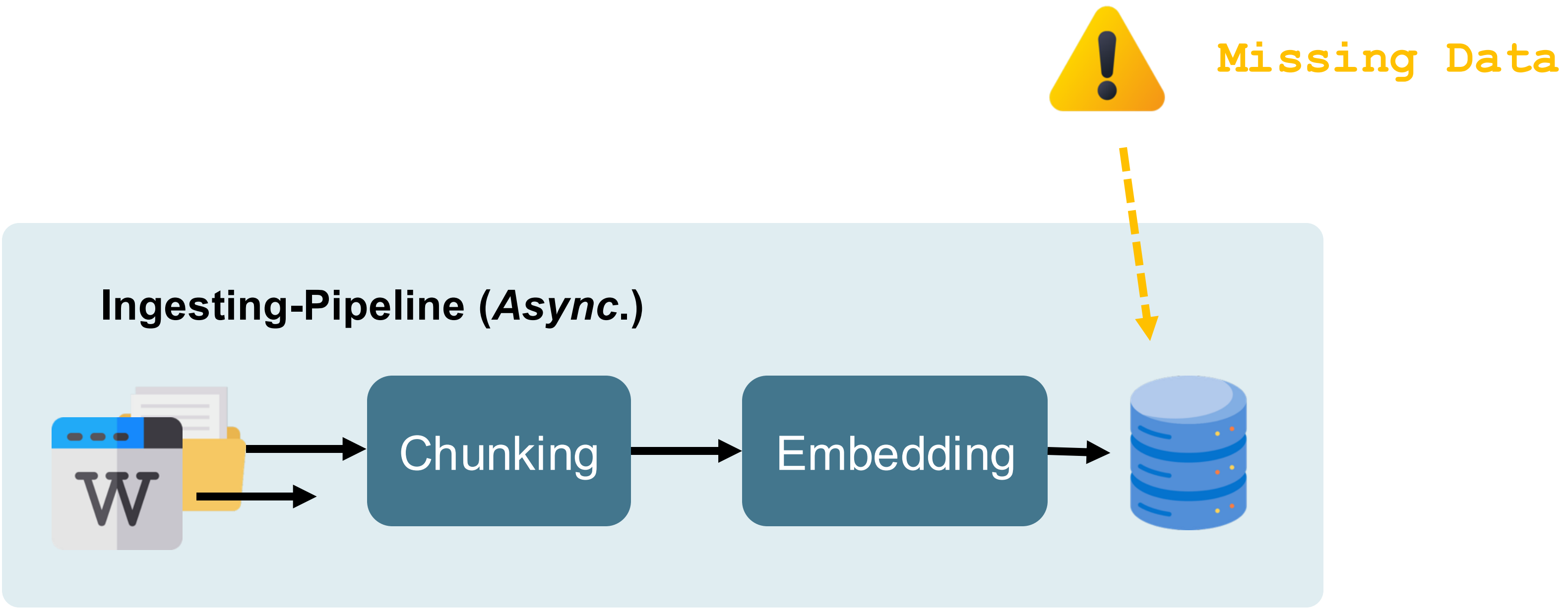


Real Life RAG #1 Missing-Data Challenge



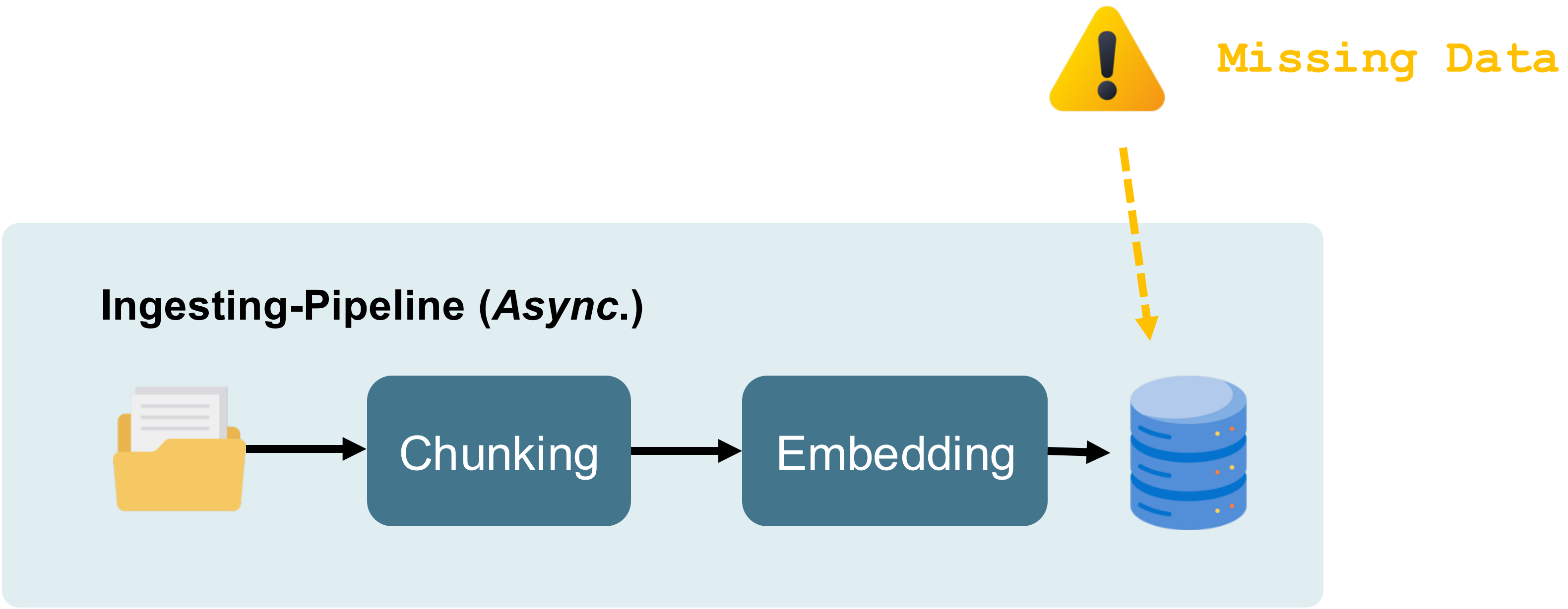
***Problem**: RAG does not know the required data.*

Real Life RAG #1 Missing-Data Challenge



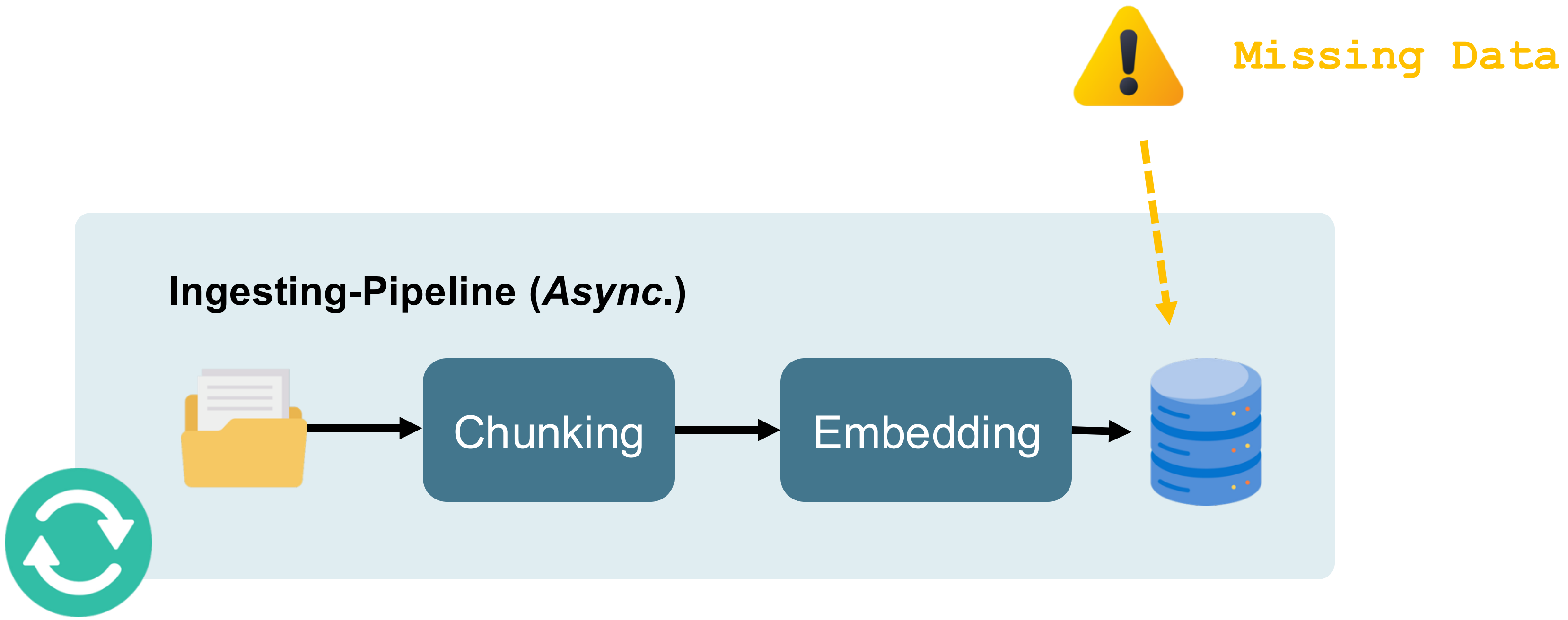
***Problem**: RAG does not know the required data.*
***Solution** : Implement a suitable ingestor.*

Real Life RAG #1 Missing-Data Challenge



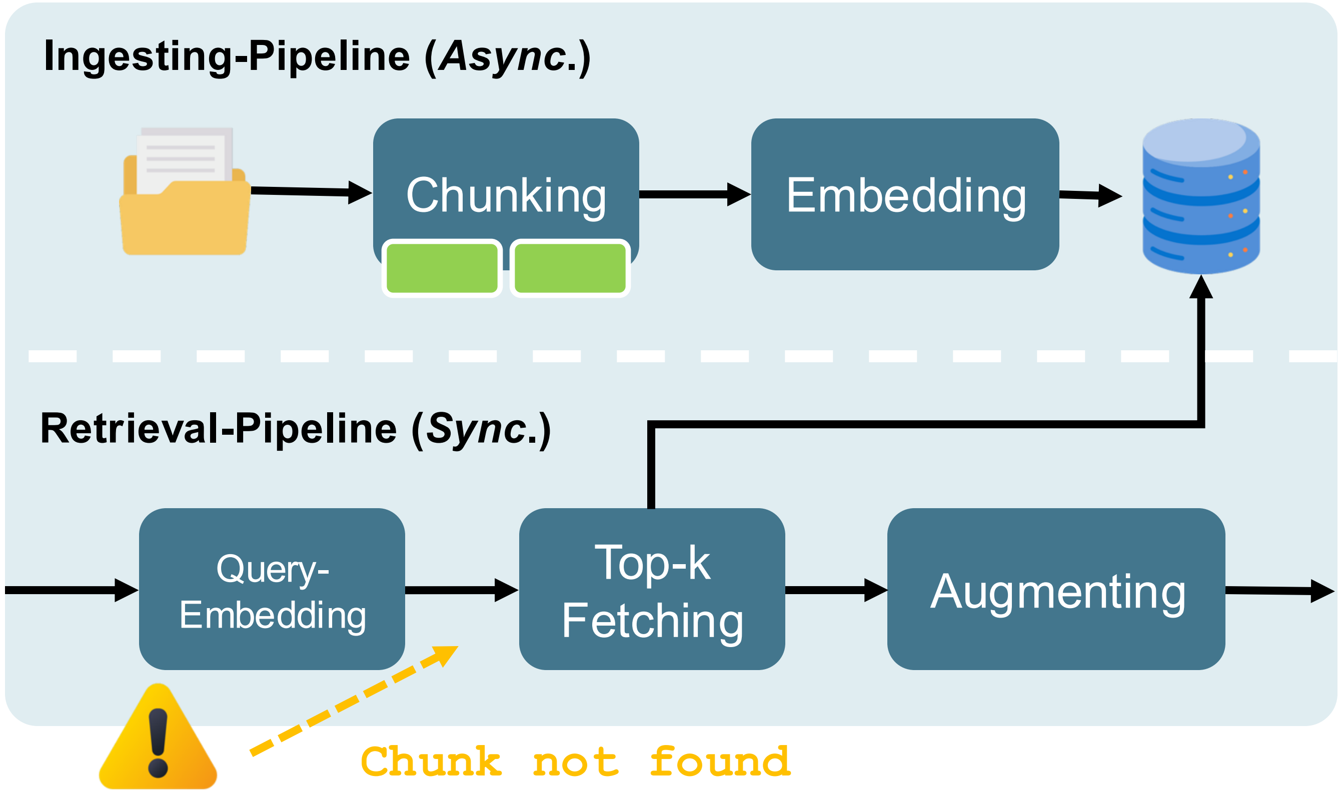
***Problem**: Snapshot of the data is not up-to-date.*

Real Life RAG #1 Missing-Data Challenge

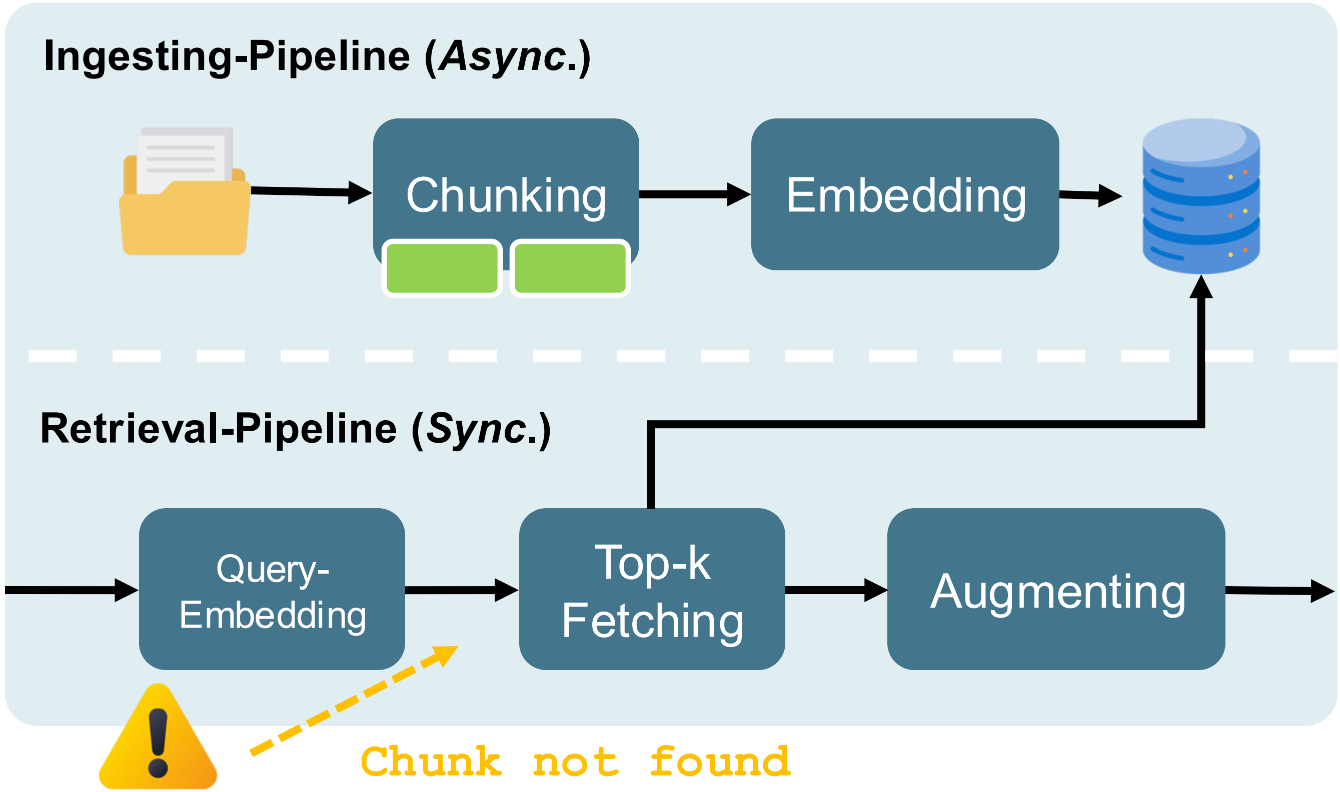


Problem: *Snapshot of the data is not up-to-date.*
Solution: *Interface for 'continuous' data sync.*

Real Life RAG #2 Chunk-not-found Challenge

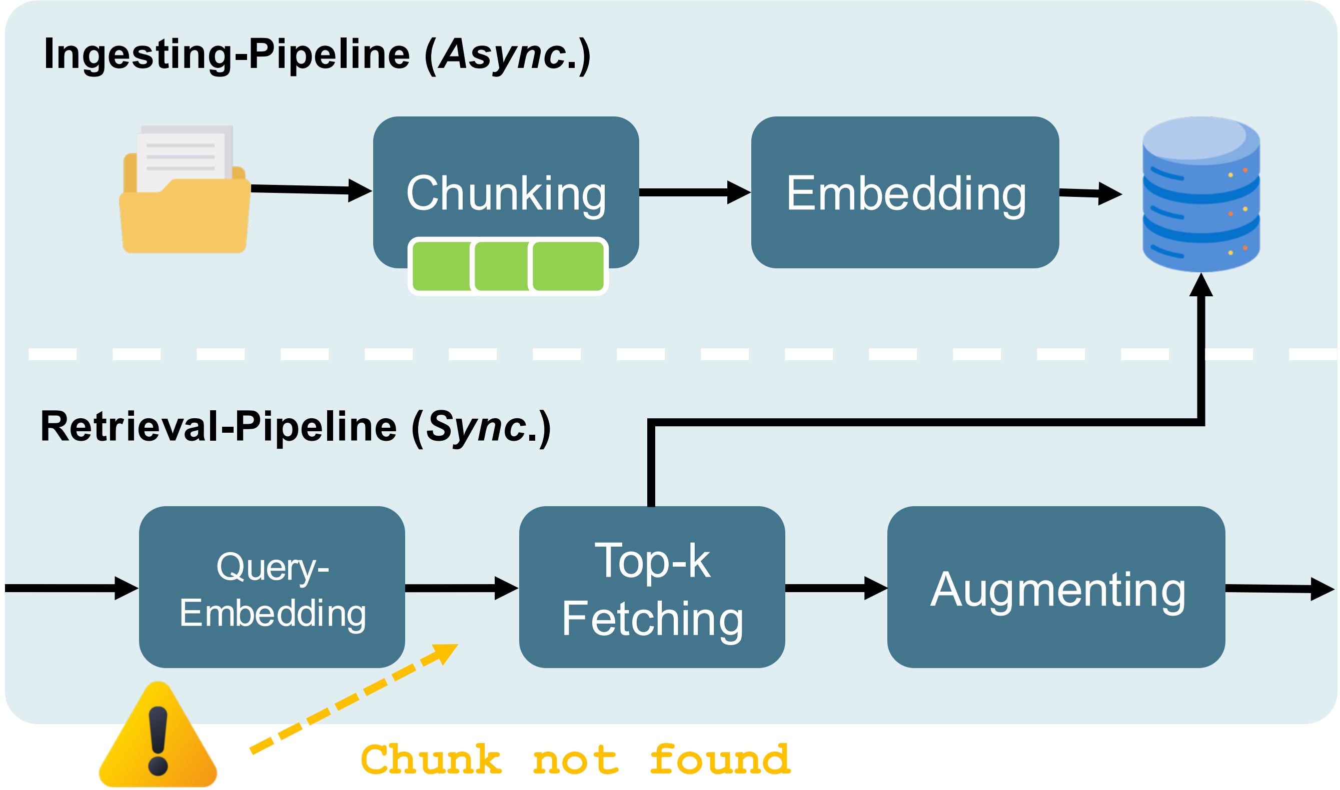


Real Life RAG #2 Chunk-not-found Challenge



***Problem**: Existing chunks will not be found.*

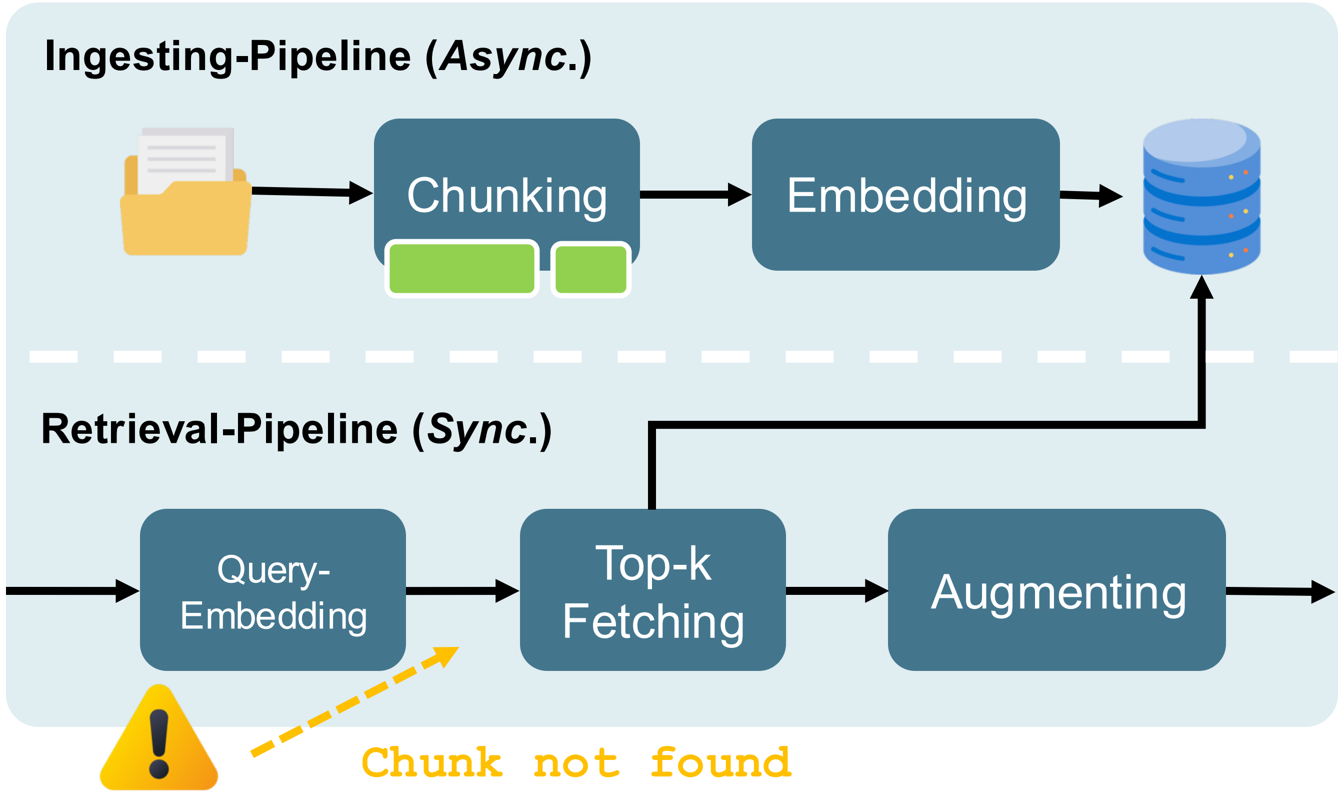
Real Life RAG #2 Chunk-not-found Challenge



Problem: Existing chunks will not be found.

Solution #1: Adapt chunk-sizes and overlapping.

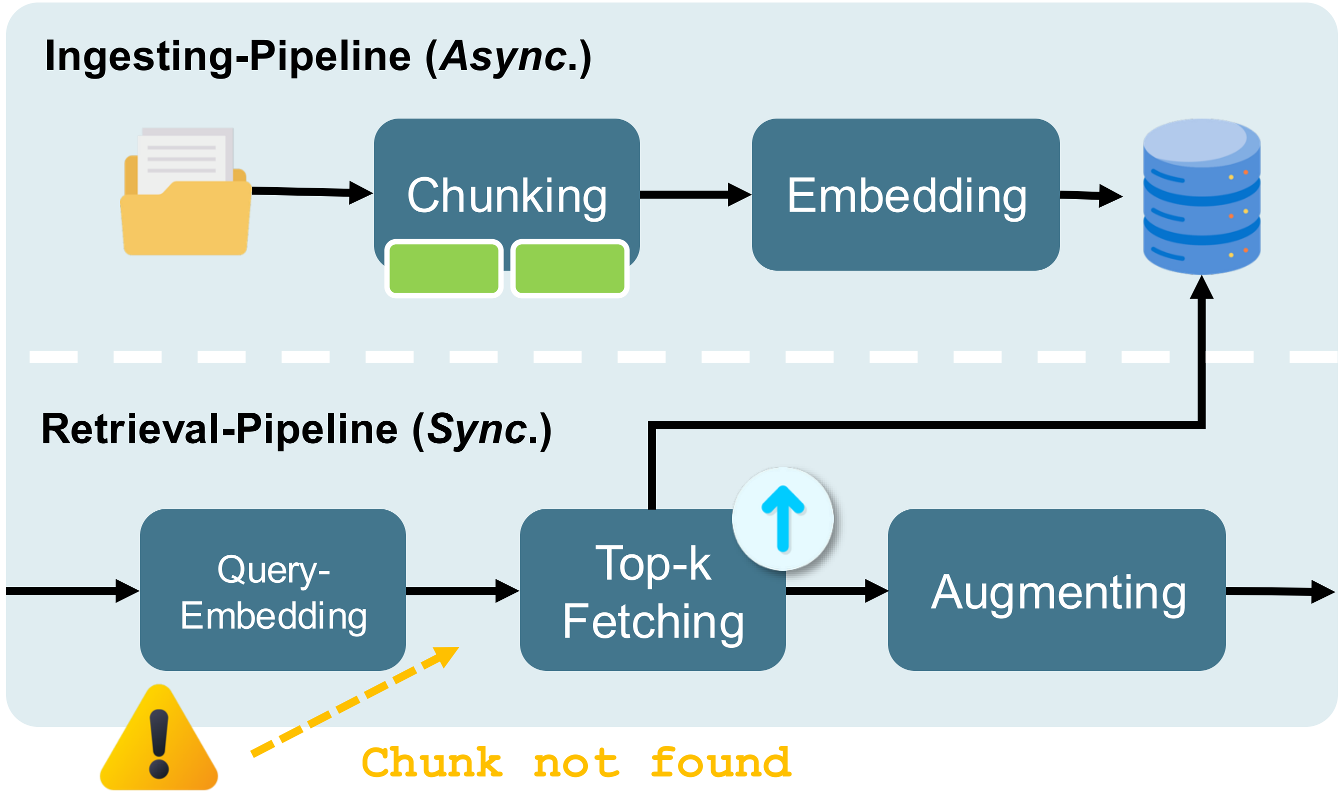
Real Life RAG #2 Chunk-not-found Challenge



***Problem**: Existing chunks will not be found.*

***Solution #2**: Choose a different splitter (e.g. Semantic-Splitter).*

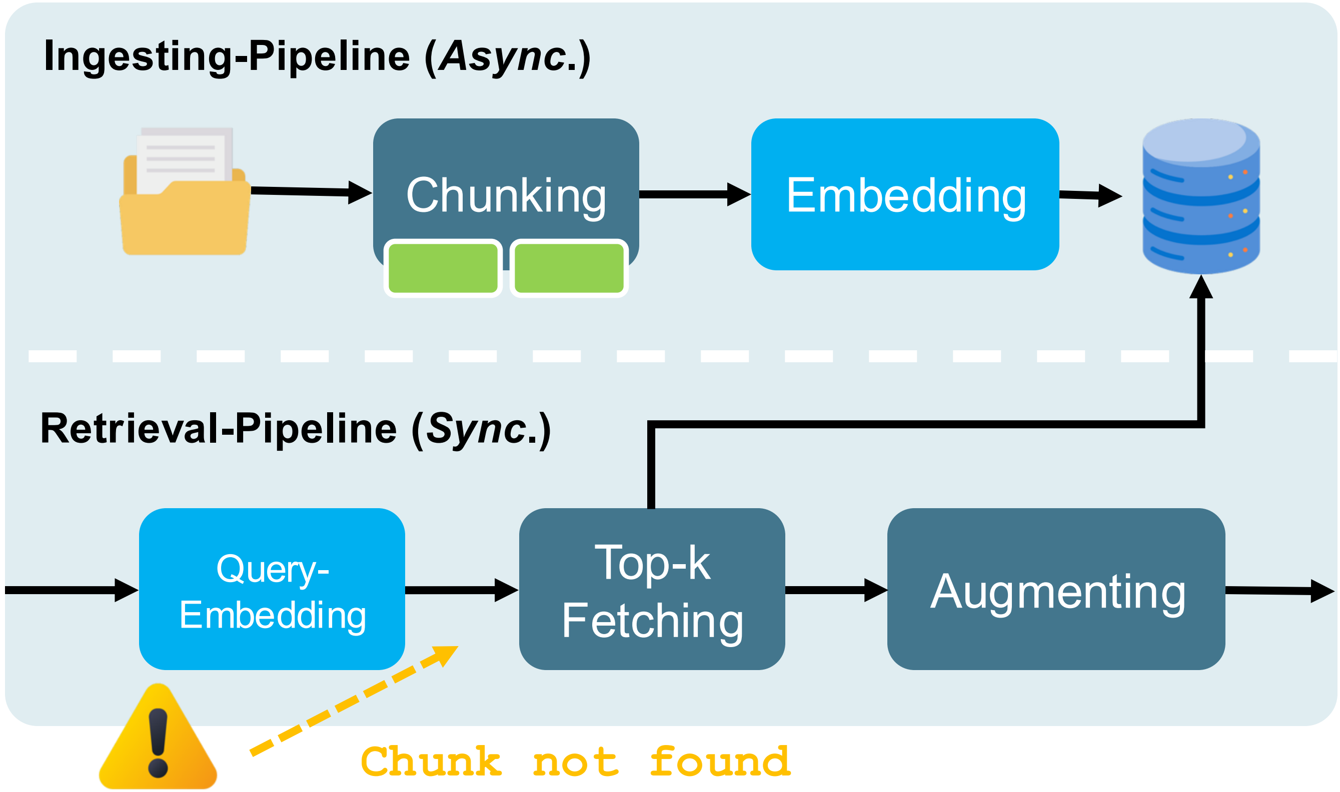
Real Life RAG #2 Chunk-not-found Challenge



Problem: Existing chunks will not be found.

*Solution #3: Optimize the **distance threshold** or k from **top-k**.*

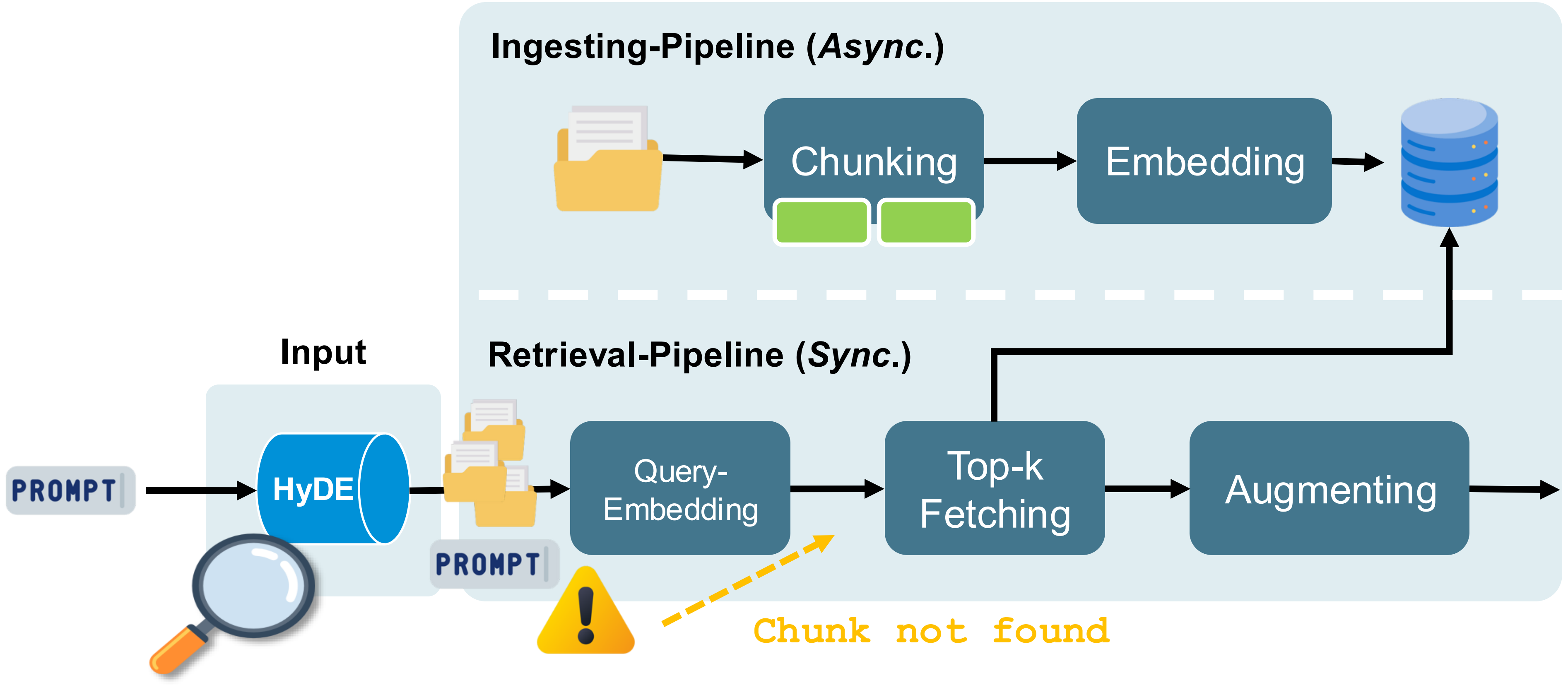
Real Life RAG #2 Chunk-not-found Challenge



Problem: Existing chunks will not be found.

*Solution #4: Selection of a more suitable **embedding model**.*

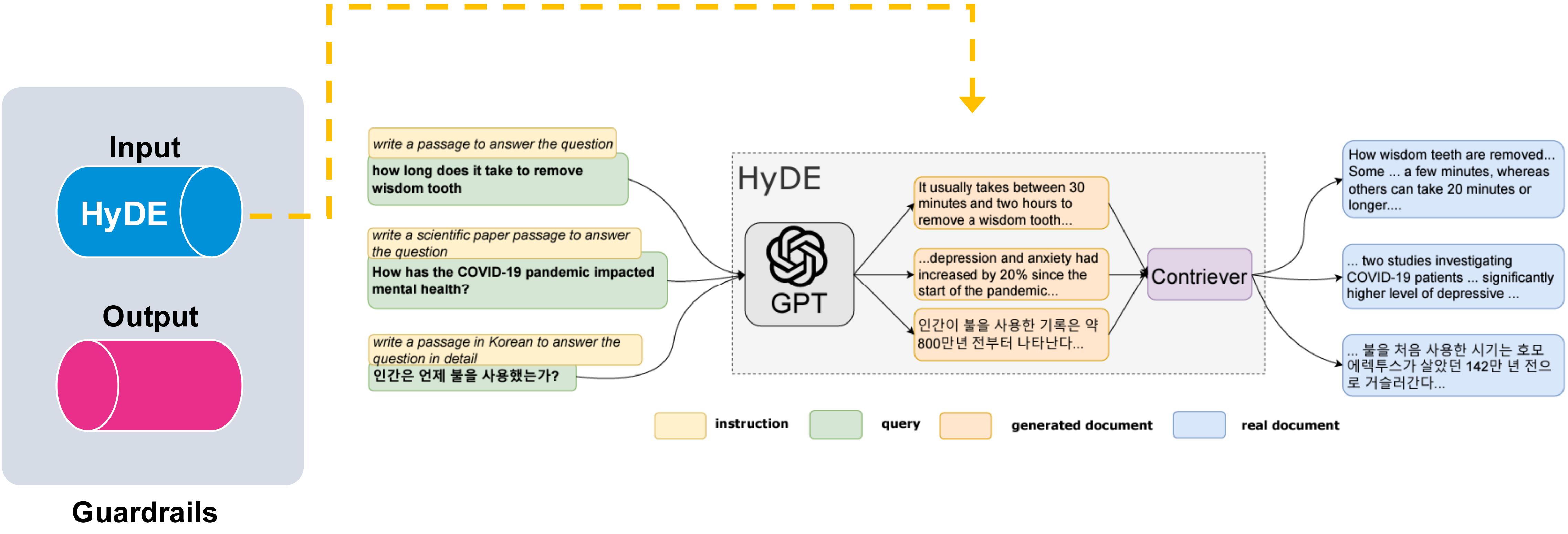
Real Life RAG #2 Chunk-not-found Challenge



Problem: Existing chunks will not be found.

Solution #5: Hypothetical document embeddings (HyDE)

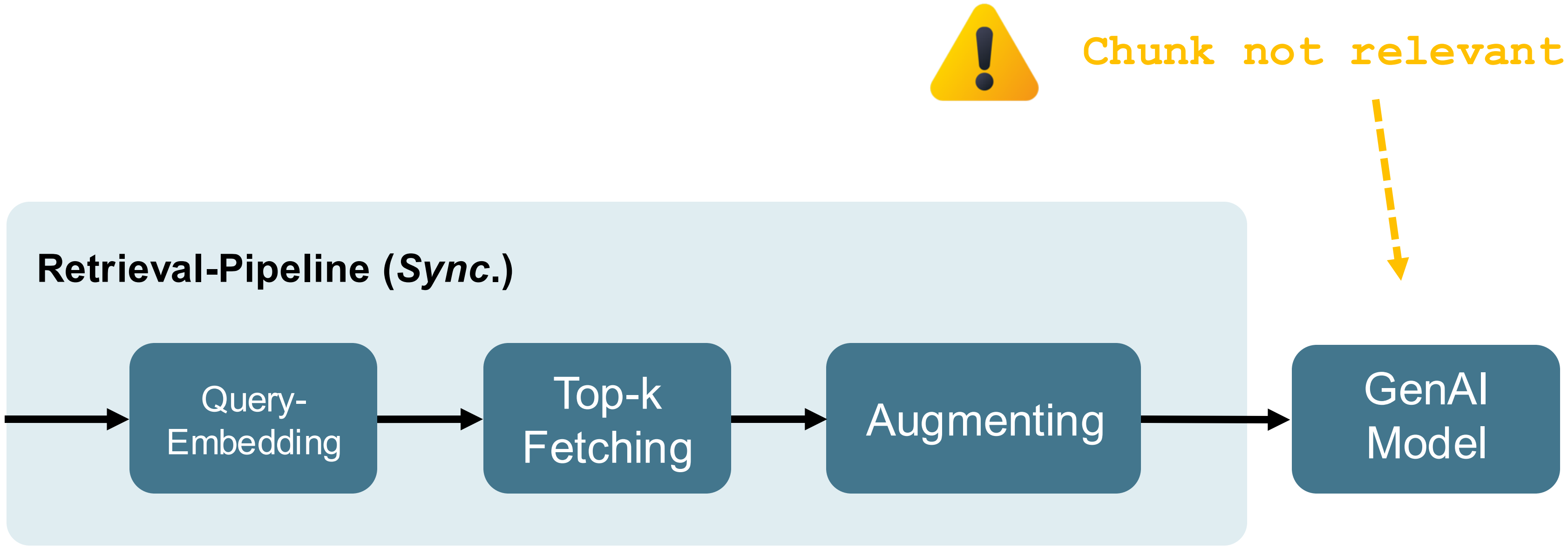
Real Life RAG #2 Chunk-not-found Challenge



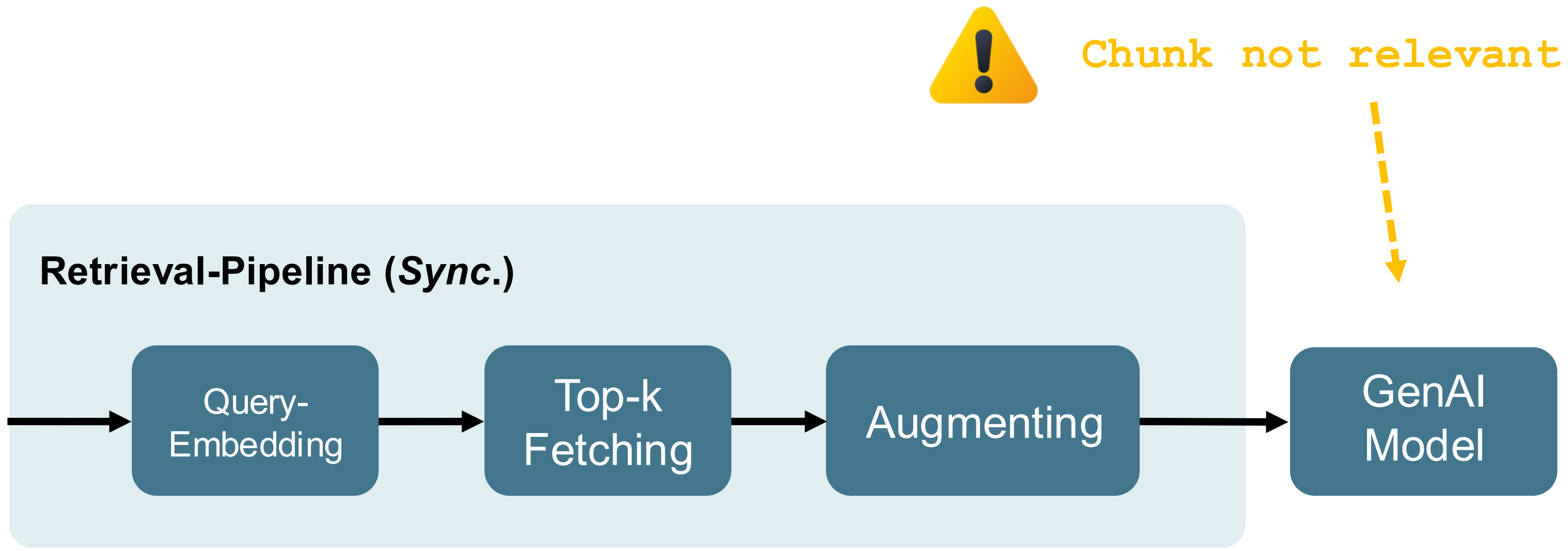
Gao et al. (2023)

HyDE = Hypothetical Document Embeddings

Real Life RAG #3 Chunk-not-relevant Challenge

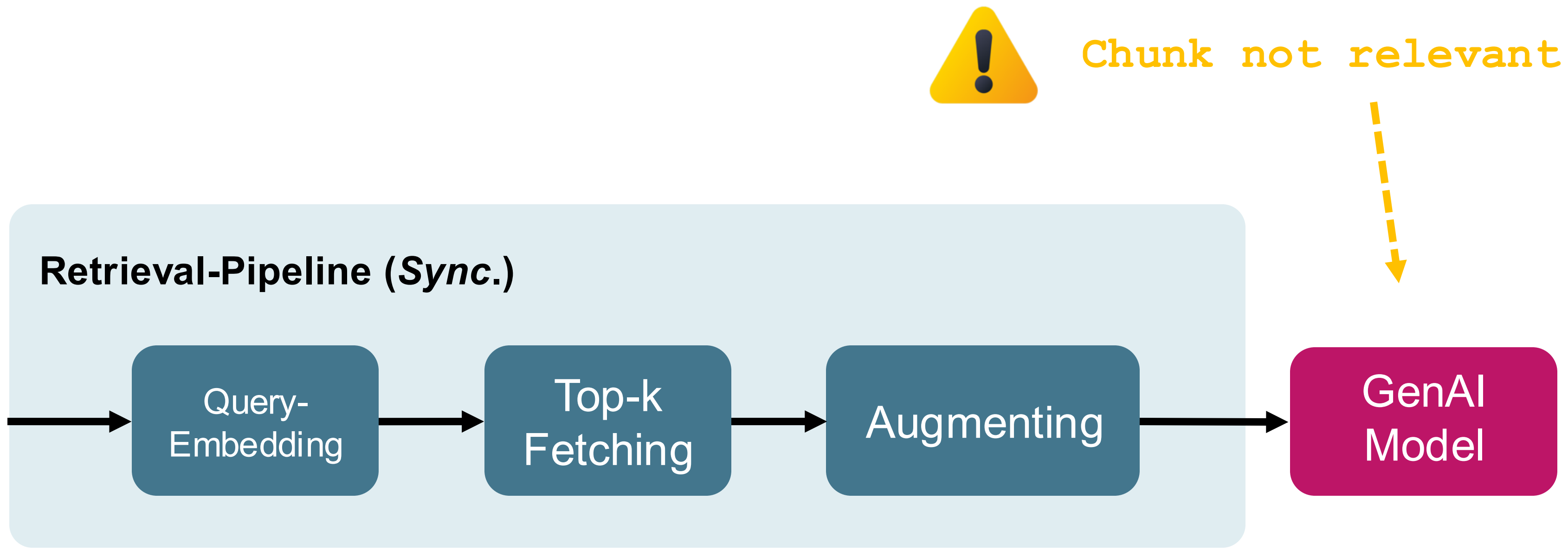


Real Life RAG #3 Chunk-not-relevant Challenge



Problem: *Chunk is not considered to be relevant by the model.*

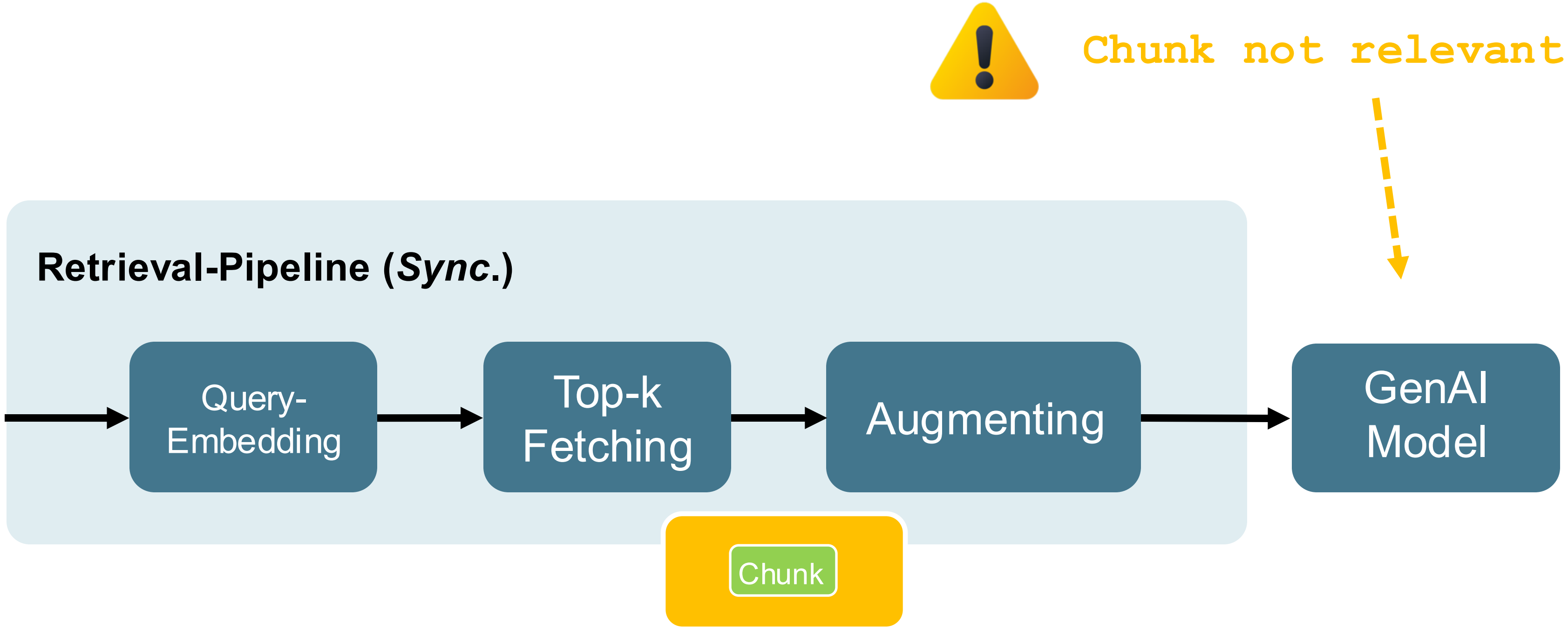
Real Life RAG #3 Chunk-not-relevant Challenge



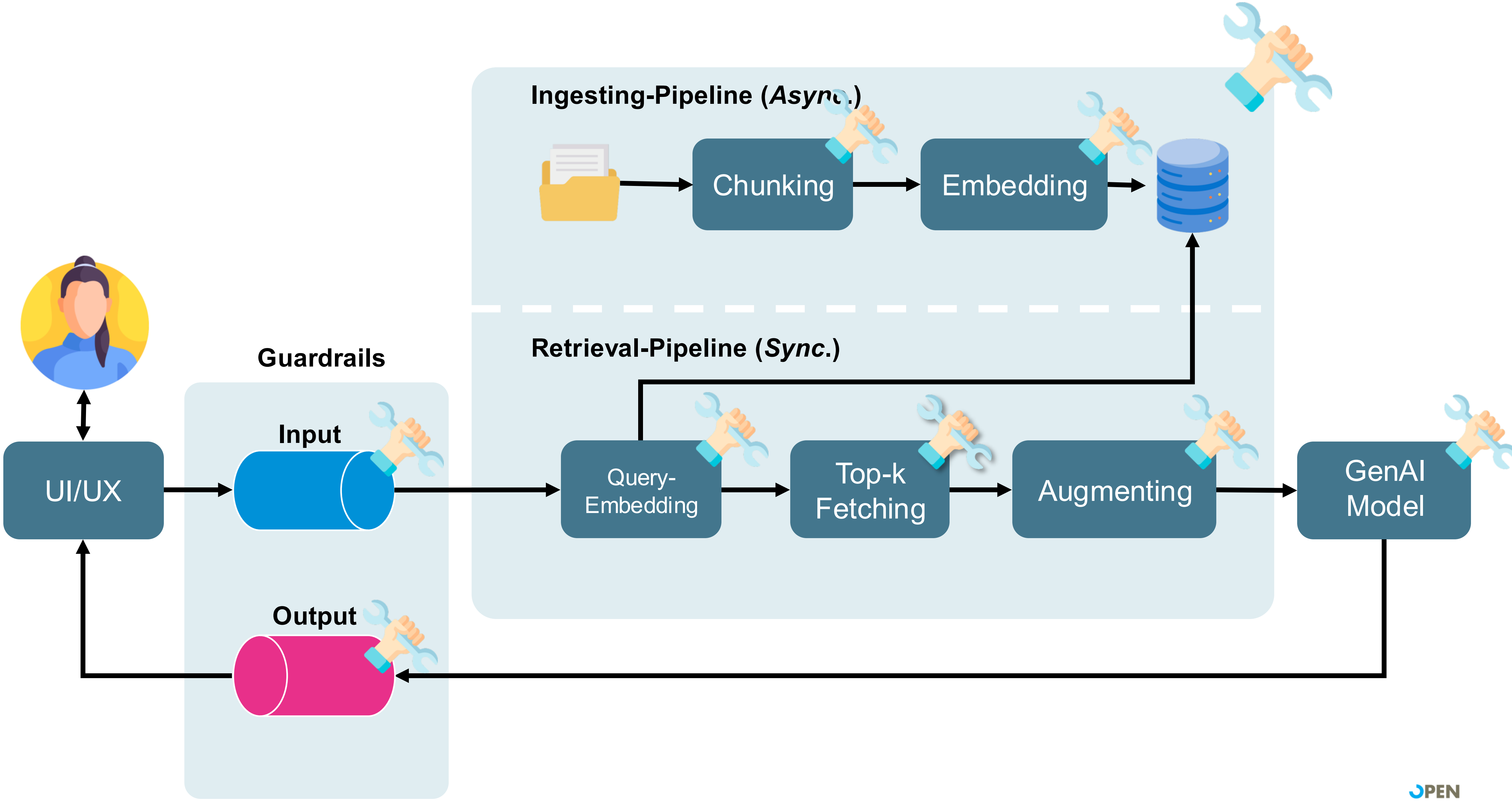
Problem: *Chunk is not considered to be relevant by the model.*

Solution #1: *Select a more suitable model for the use case.*

Real Life RAG #3 Chunk-not-relevant Challenge

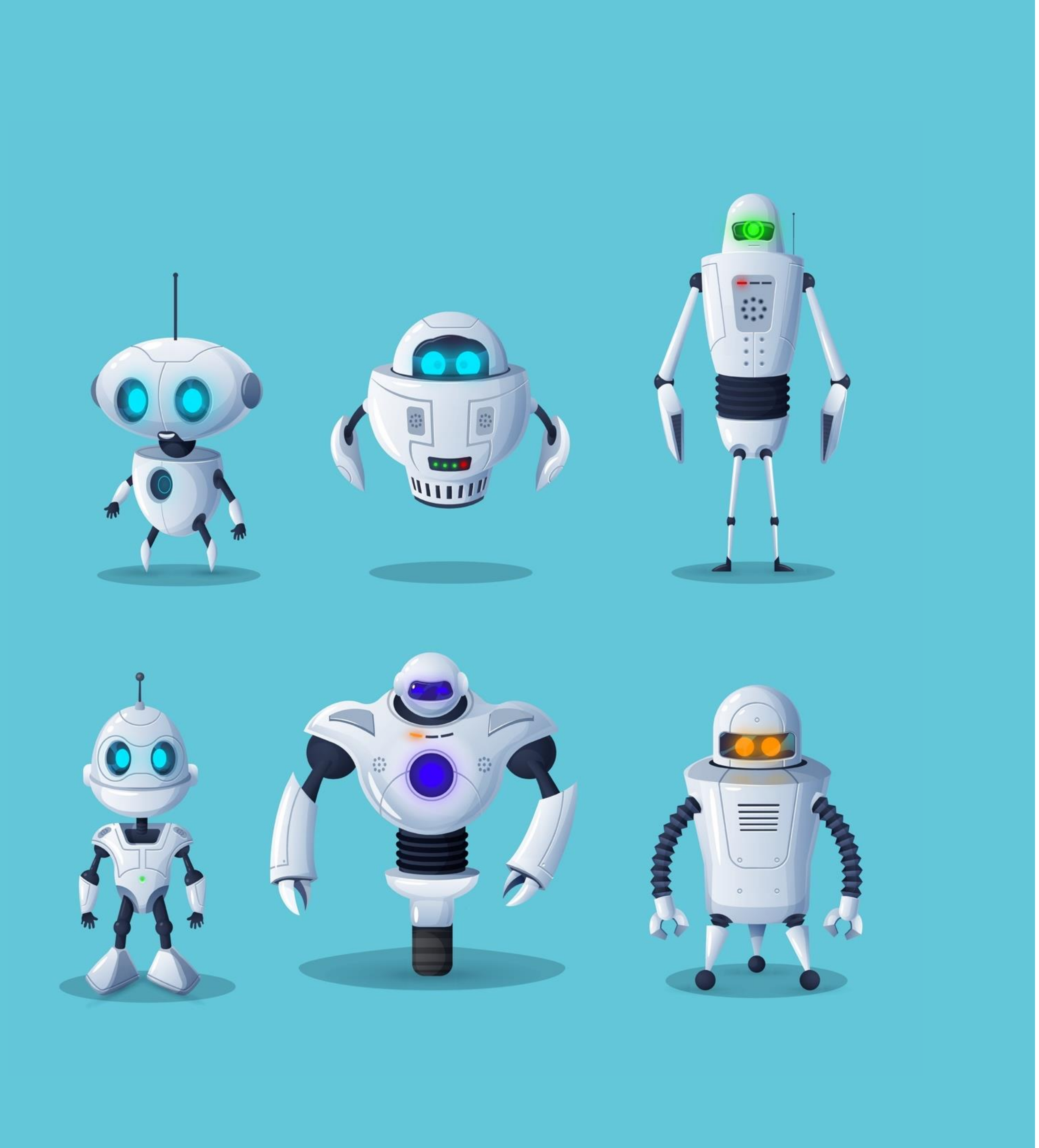


Problem: *Chunk is not considered to be relevant by the model.*
Solution #2: *Add additional context to the chunk (small-to-big).*



OFFENKUNDIGGUT

OPEN KNOWLEDGE



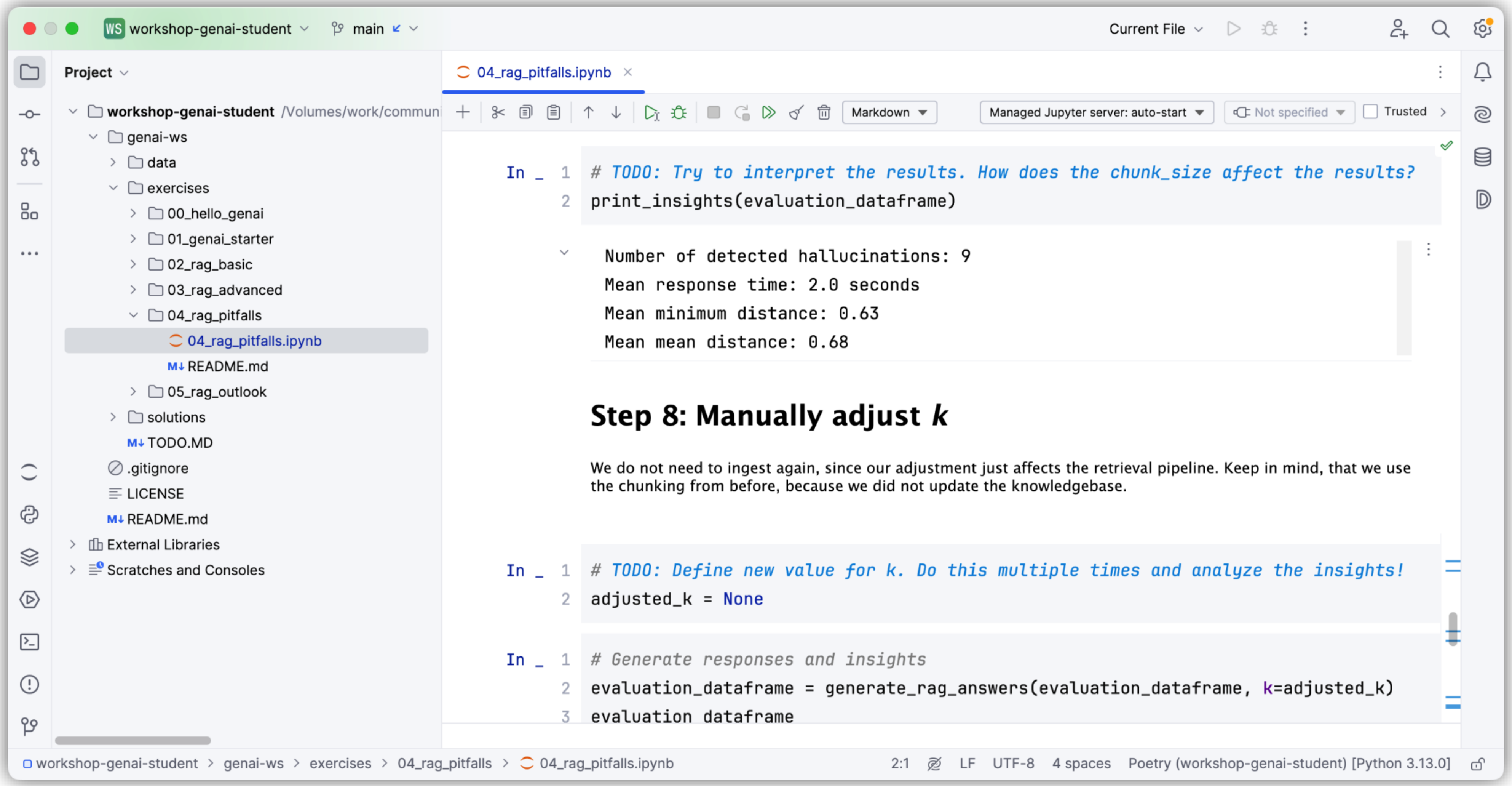
Hands-On

RAG System typical Pitfalls

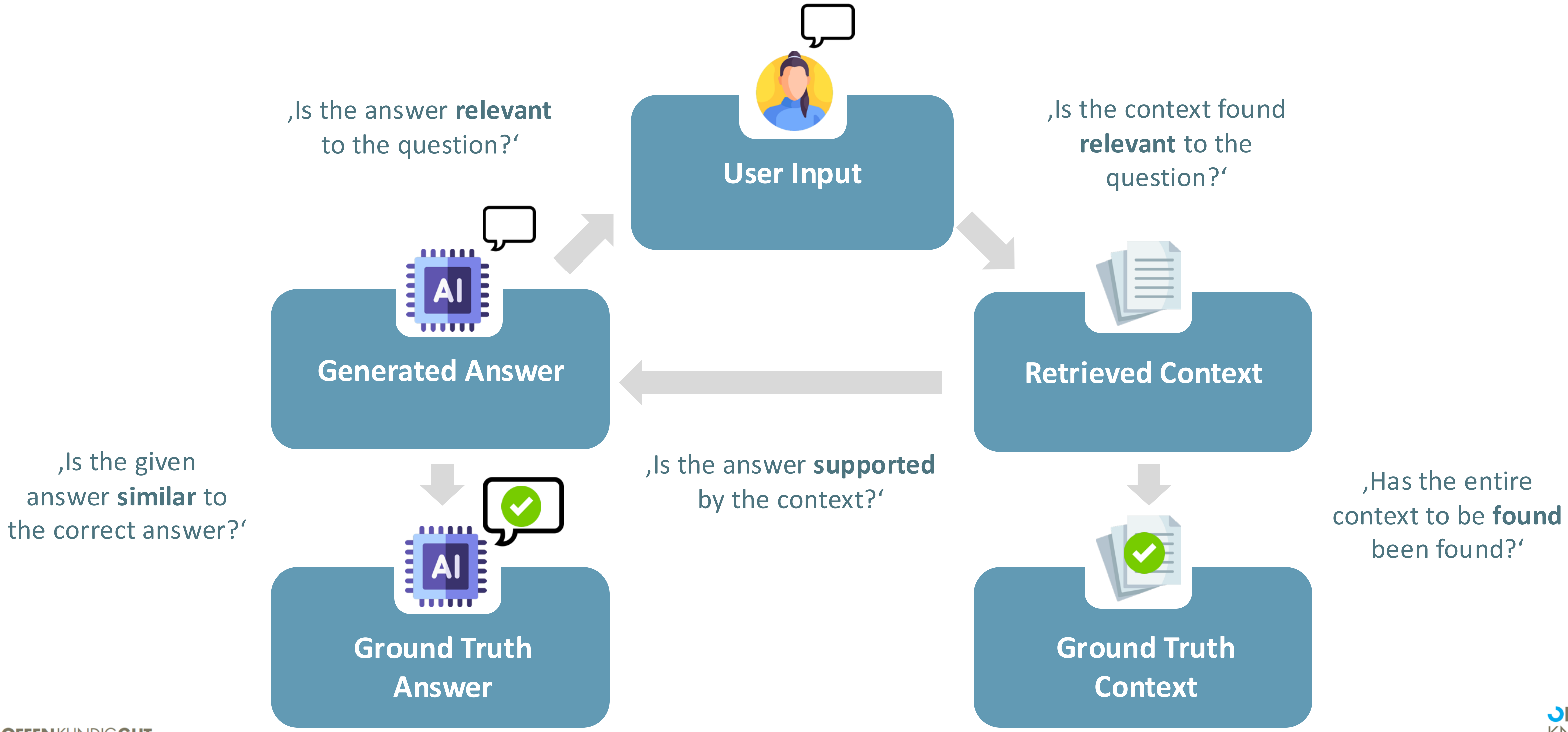
Surviving the RAG jungle:

Using LLM and RAG metrics to optimise the retrieval process.

Hands-on Surviving the RAG jungle



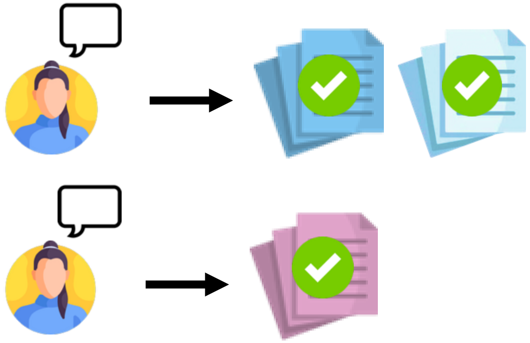
GenAI Professional Monitoring: The RAG-Triad+



Real Life RAG Metrics in Information Retrieval

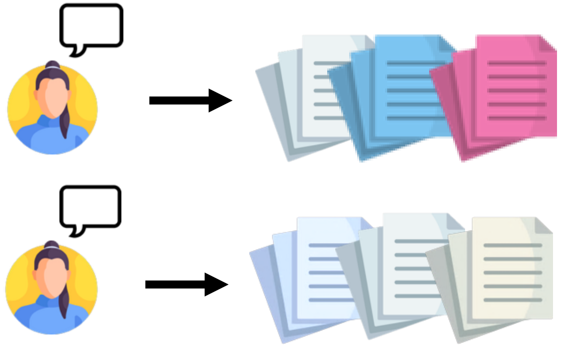
1

Provide **evaluation data** that includes the **ground truth context**.



2

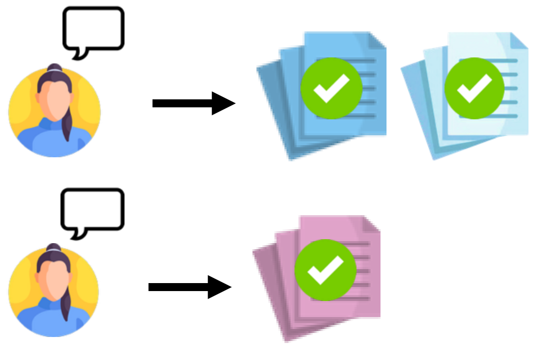
Generate responses to evaluation input and collect **context.**



Real Life RAG Metrics in Information Retrieval – **Hit Rate**

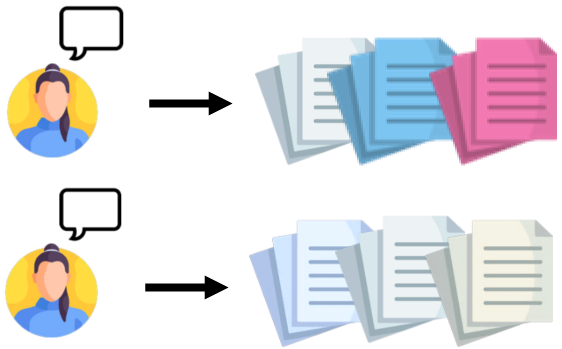
1

Provide **evaluation data** that includes the **ground truth context**.



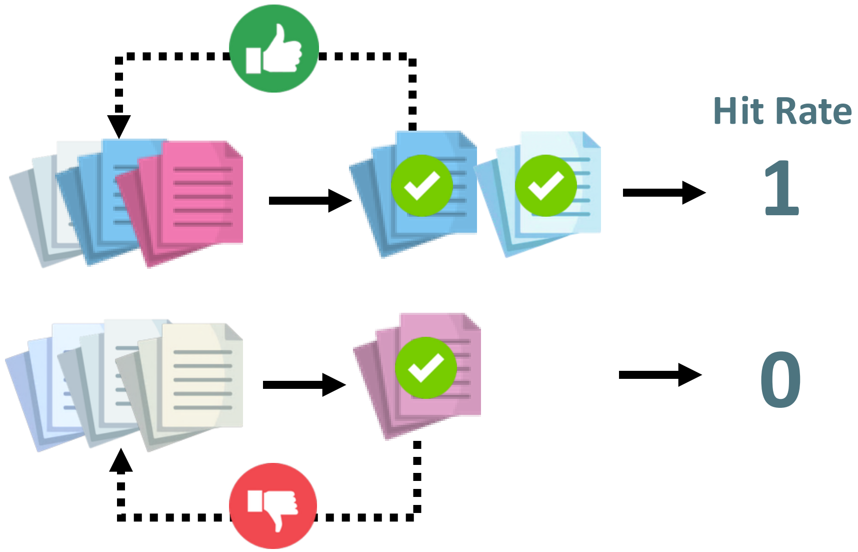
2

Generate responses to evaluation input and collect **context.**



3

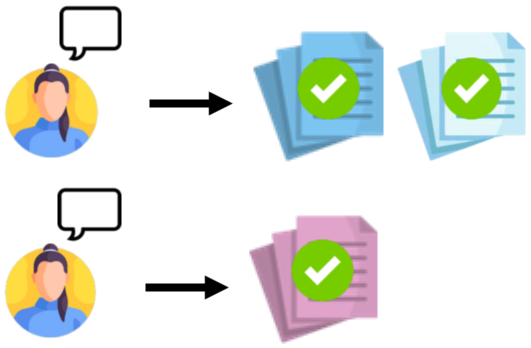
Determine whether context is included in **ground truth**.



Real Life RAG Metrics in Information Retrieval – **Precision**

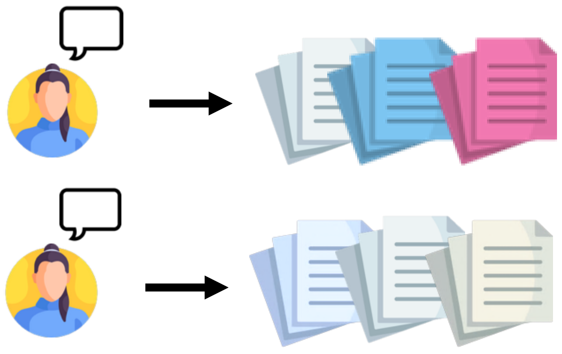
1

Provide **evaluation data** that includes the **ground truth context**.



2

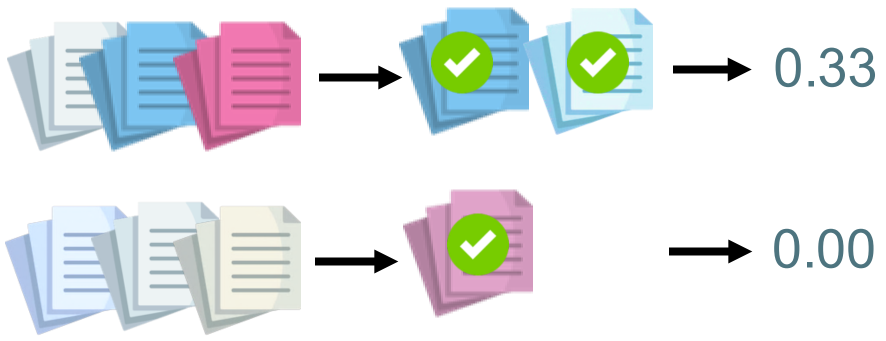
Generate responses to evaluation input and collect **context based on relevance.**



3

Determine the proportion of **relevant documents** in relation to all documents.

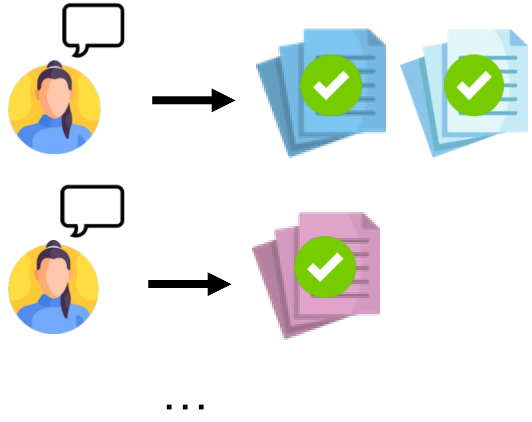
$$Precision = \frac{relevant\ Results}{total\ Results}$$



Real Life RAG Metrics in Information Retrieval – **Mean Reciprocal Rank (MRR)**

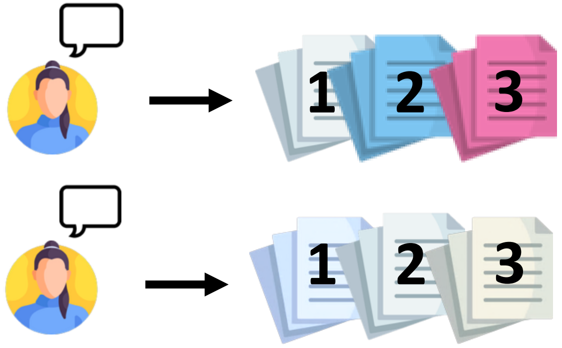
1

Provide **evaluation data** that includes the **ground truth** context.



2

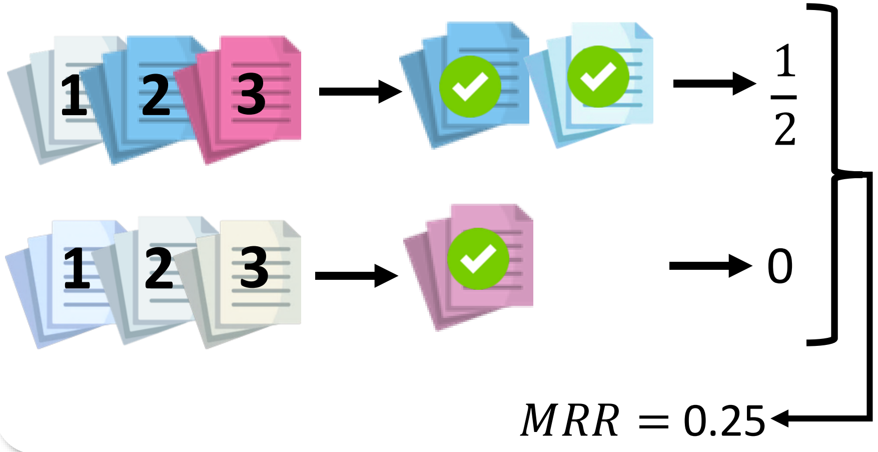
Generate responses to evaluation input and collect **context based on relevance.**



3

Determine the **rank** of the first **relevant document.** Calculate and average **the reciprocal rank** (RR):

$$RR = \frac{1}{Rang} \longrightarrow MRR = \frac{1}{n}\sum_{i=1}^{n} RR_i$$

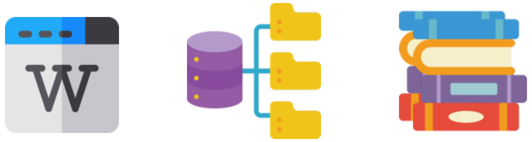


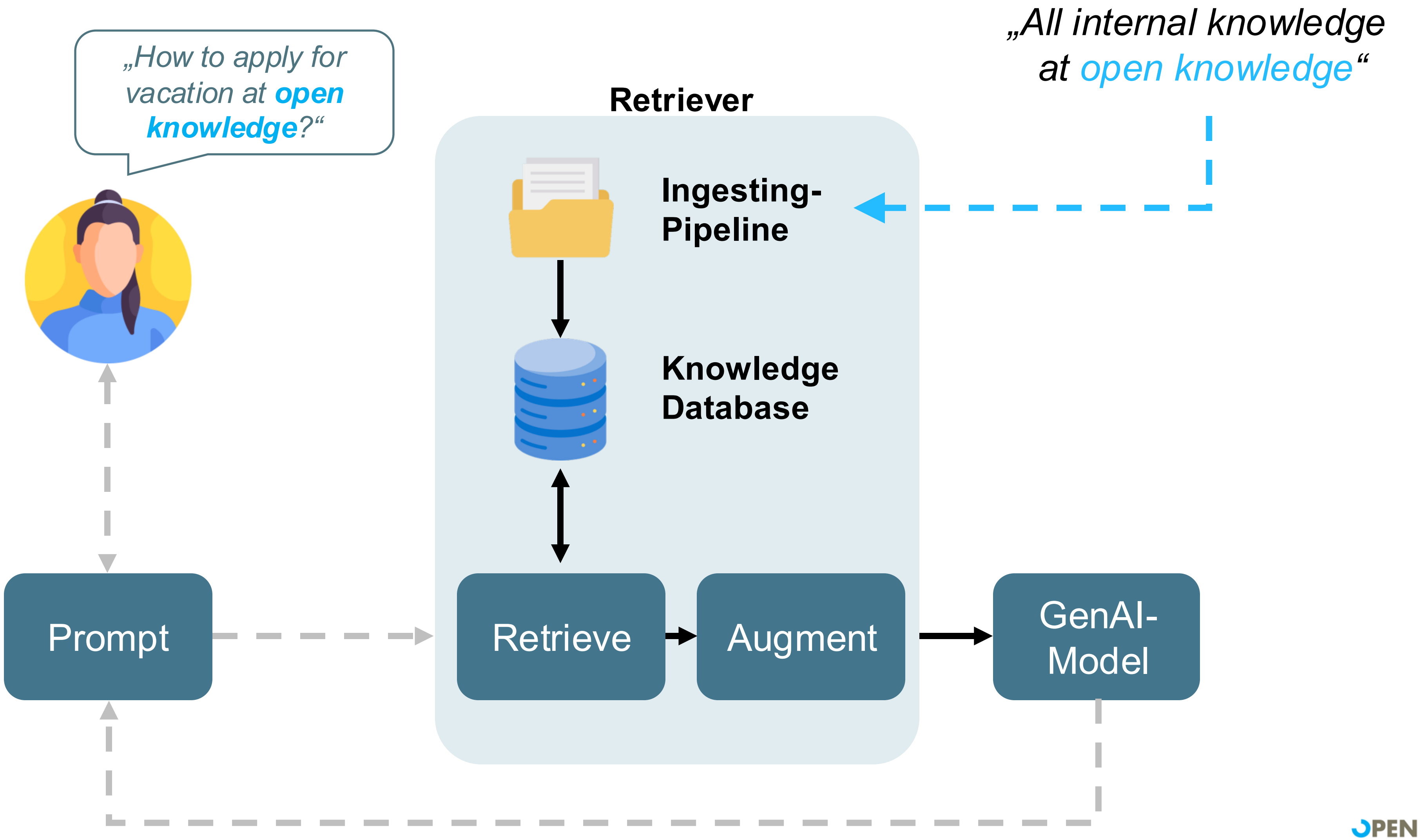
05

Outlook

What else is possible in addition to the pure knowledge query?

GenAI Outlook What else is there to consider?

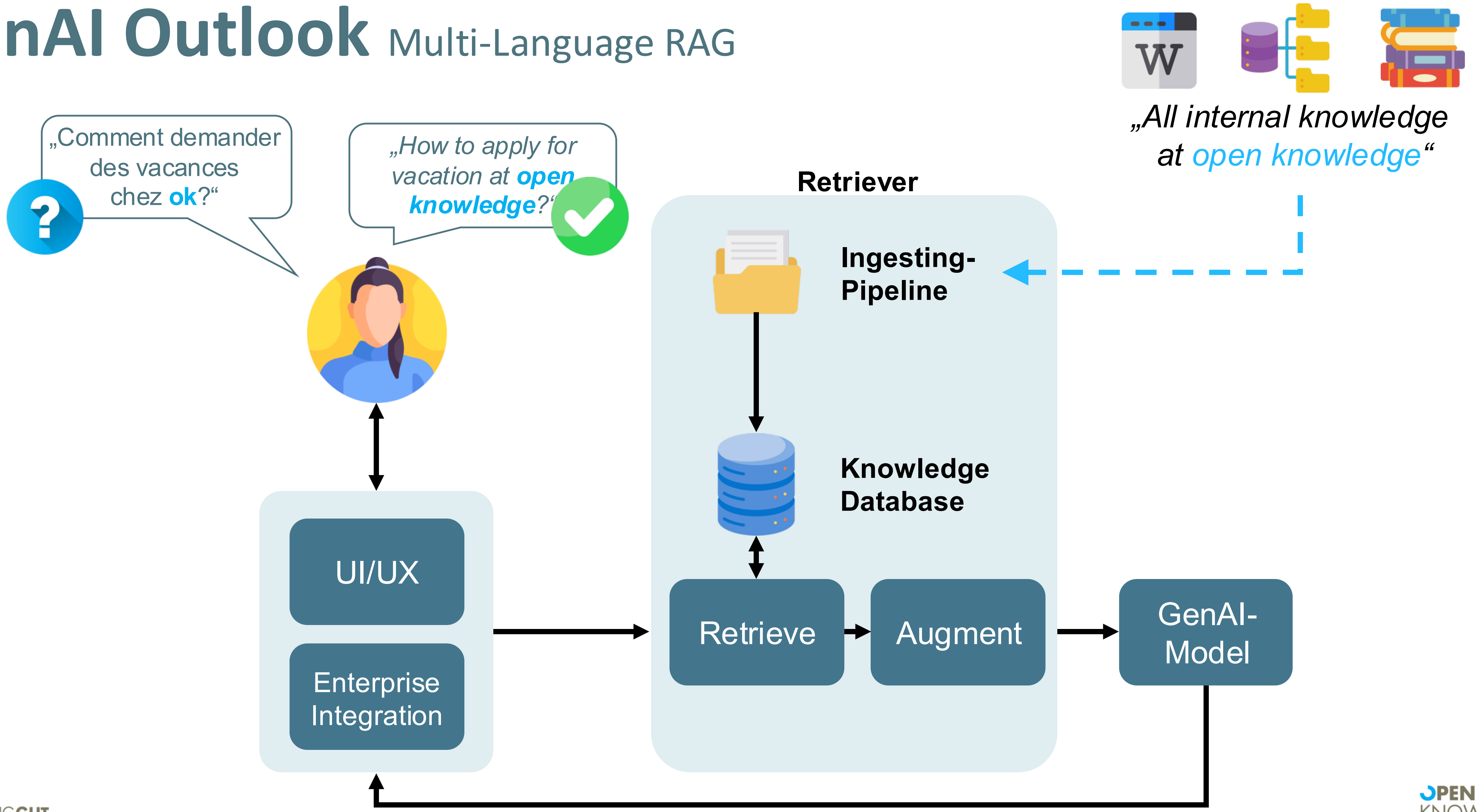




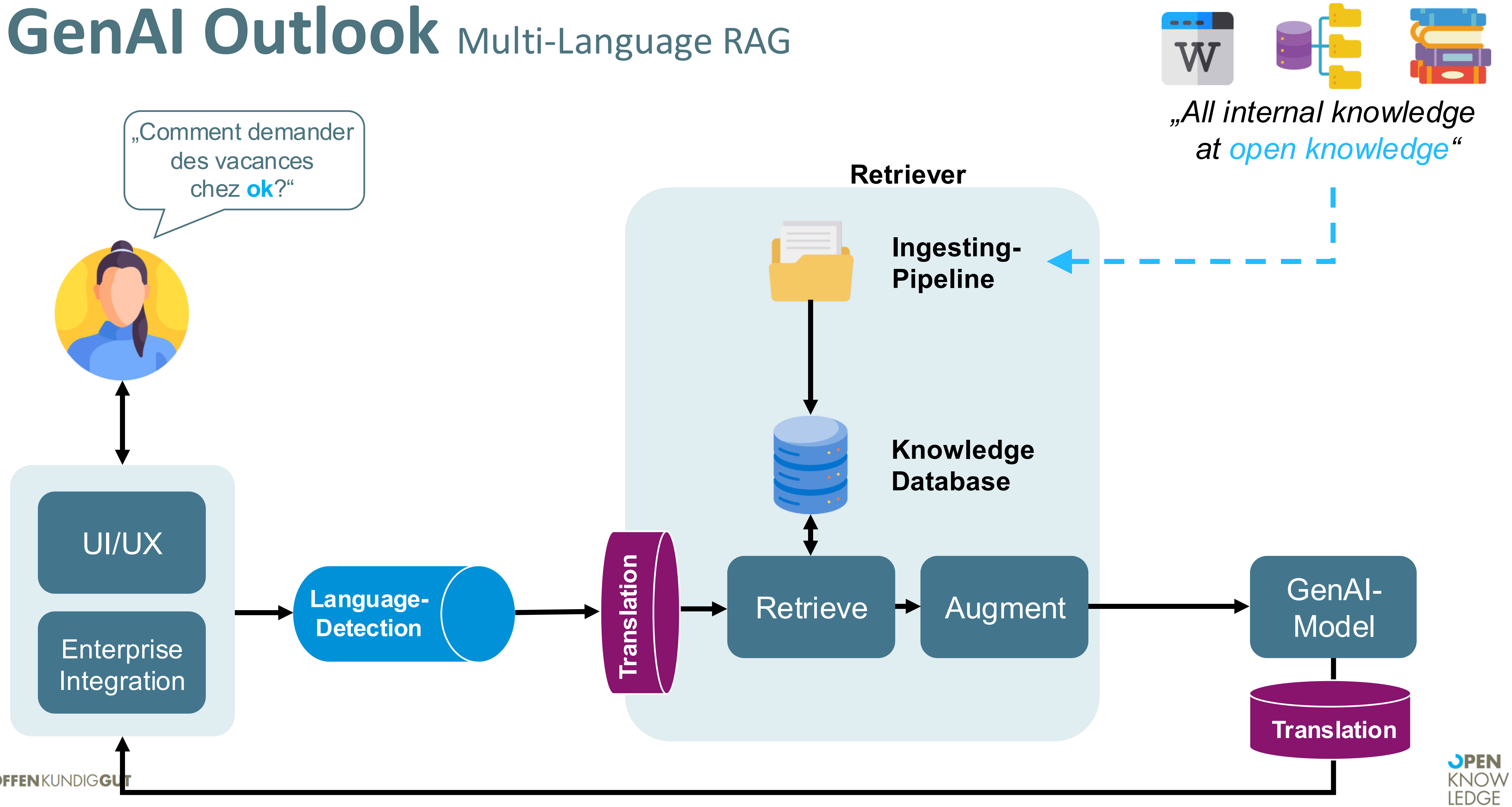
I18N

GenAI
explained

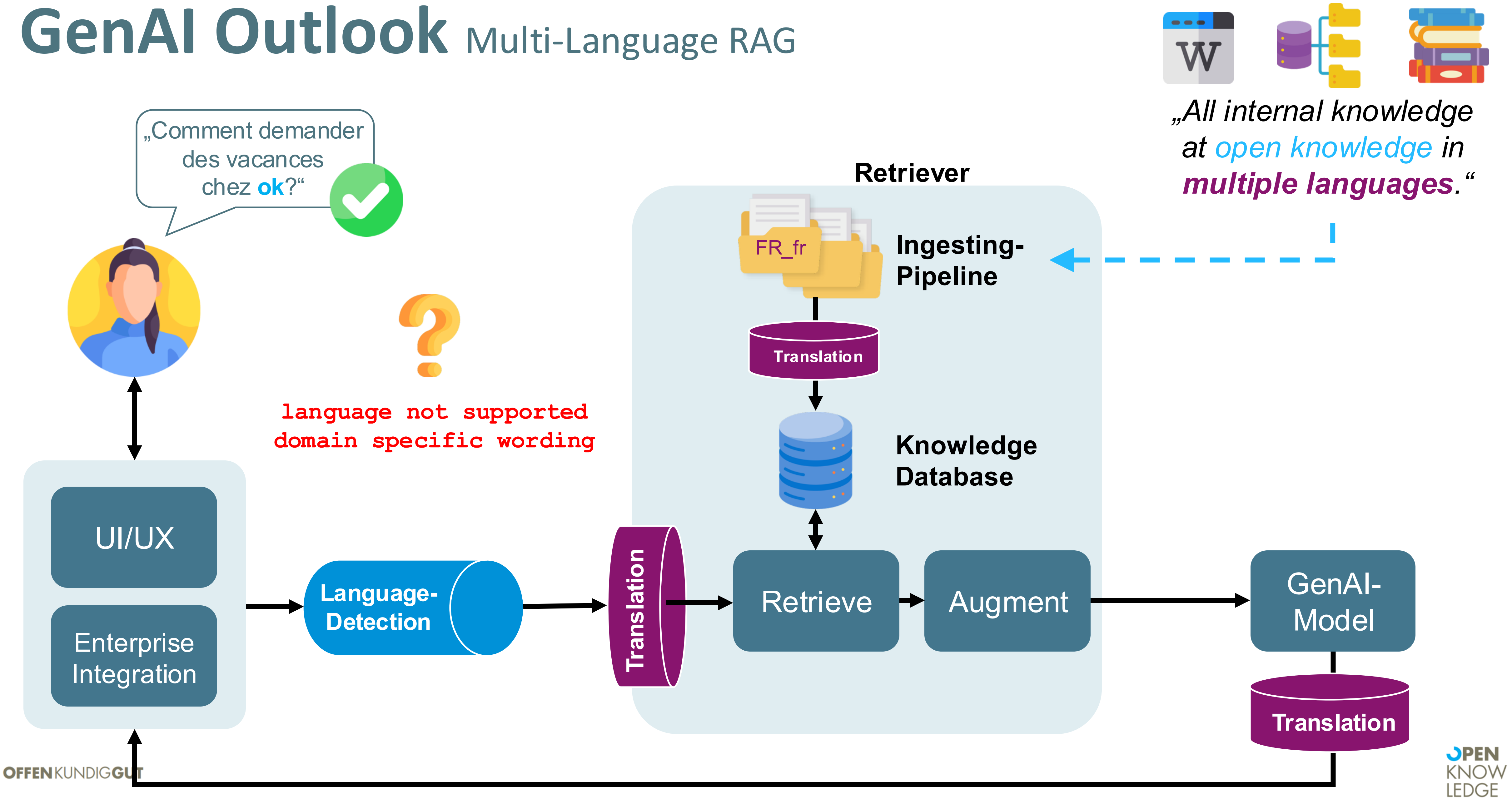
GenAI Outlook Multi-Language RAG



GenAI Outlook Multi-Language RAG



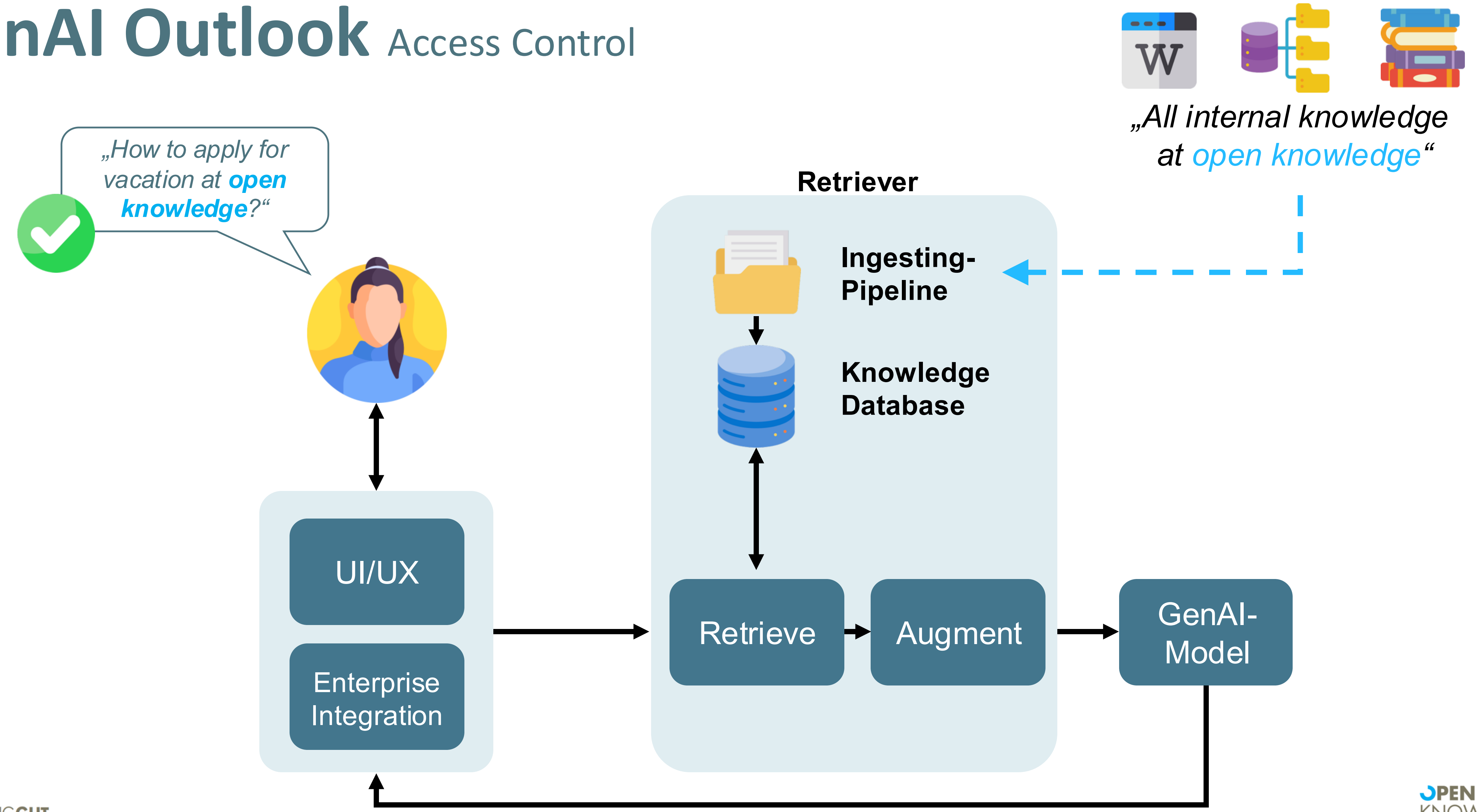
GenAI Outlook Multi-Language RAG



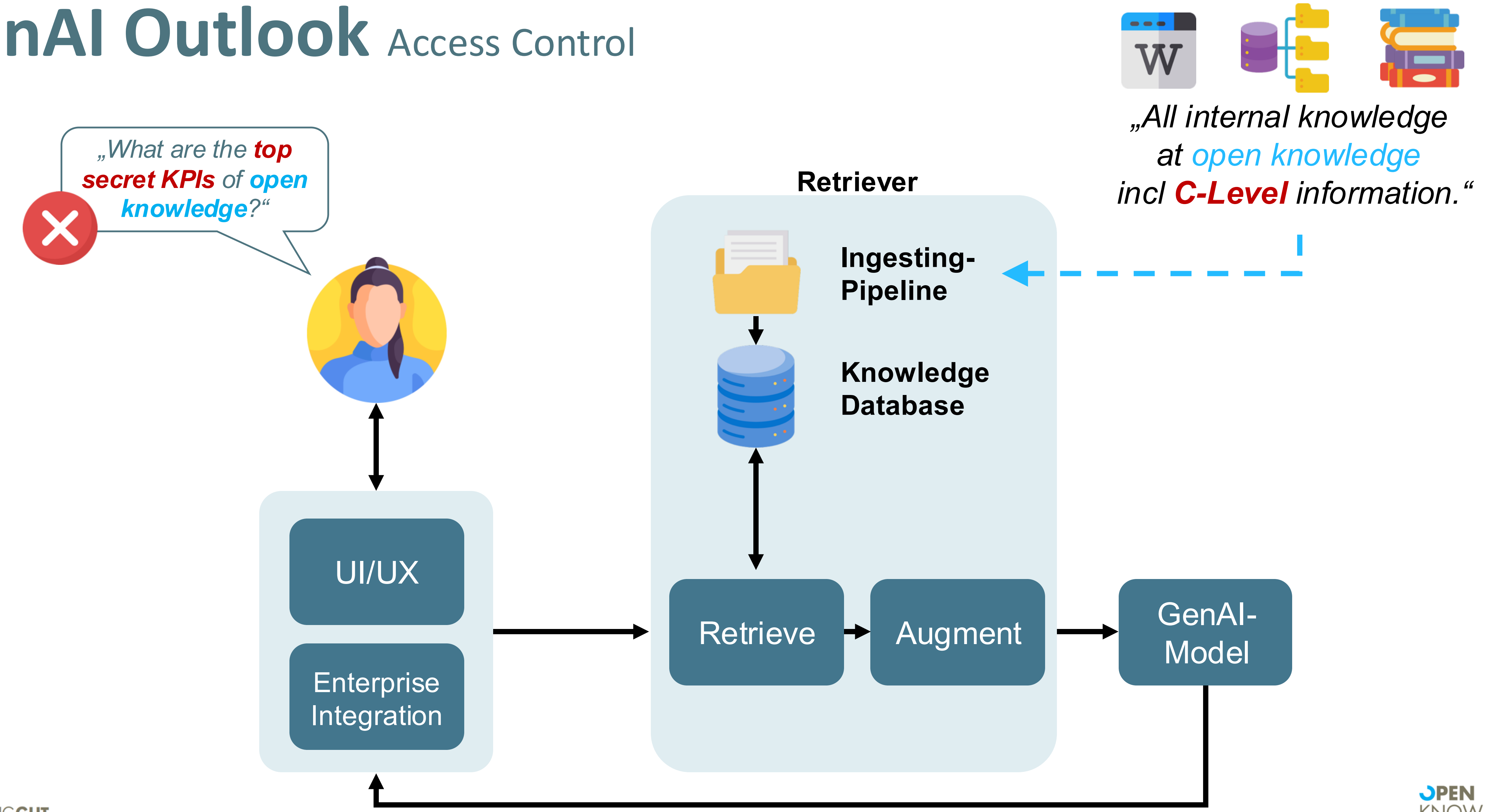
ACCESS CONTROL

GenAI
explained

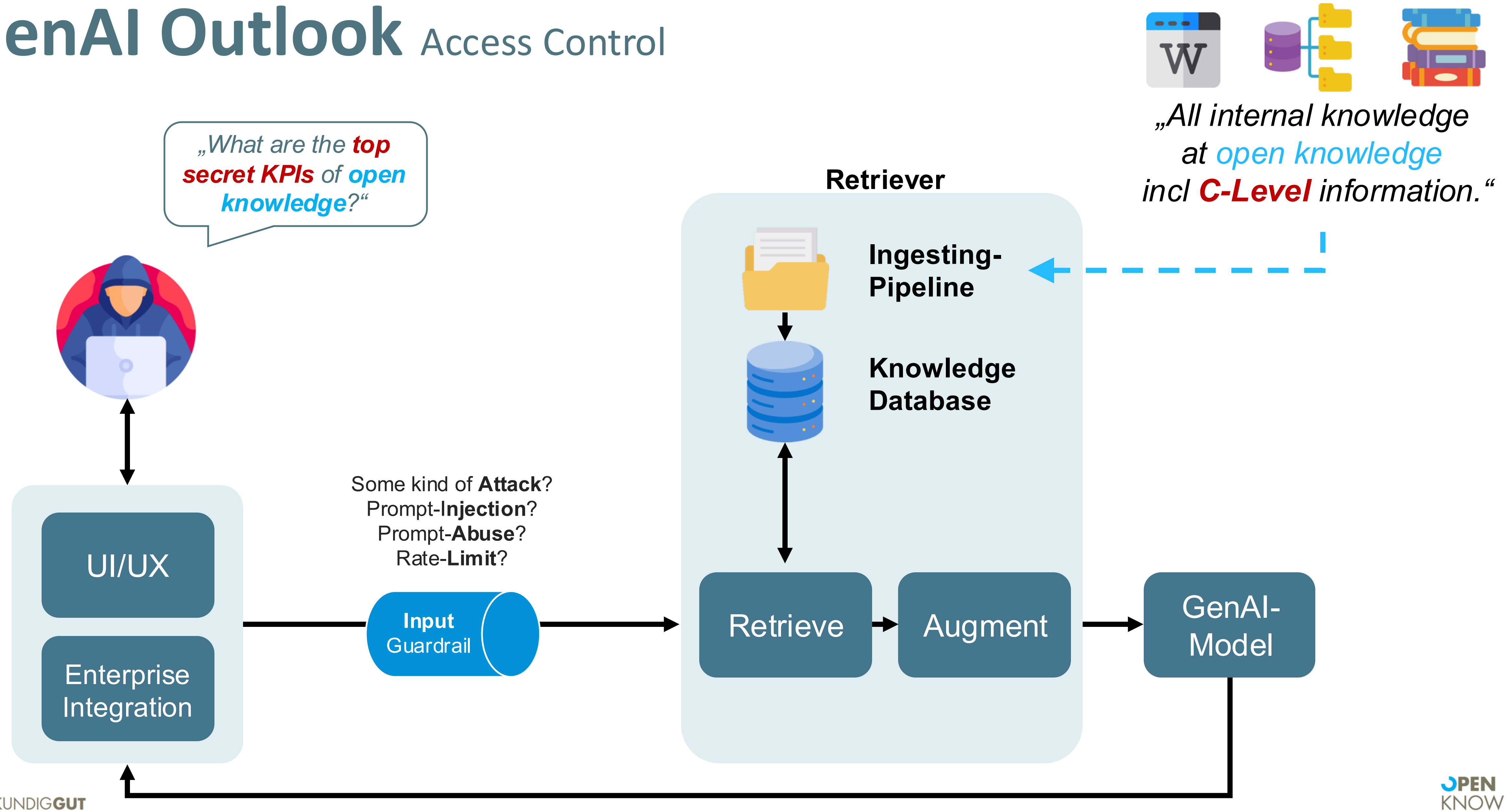
GenAI Outlook Access Control



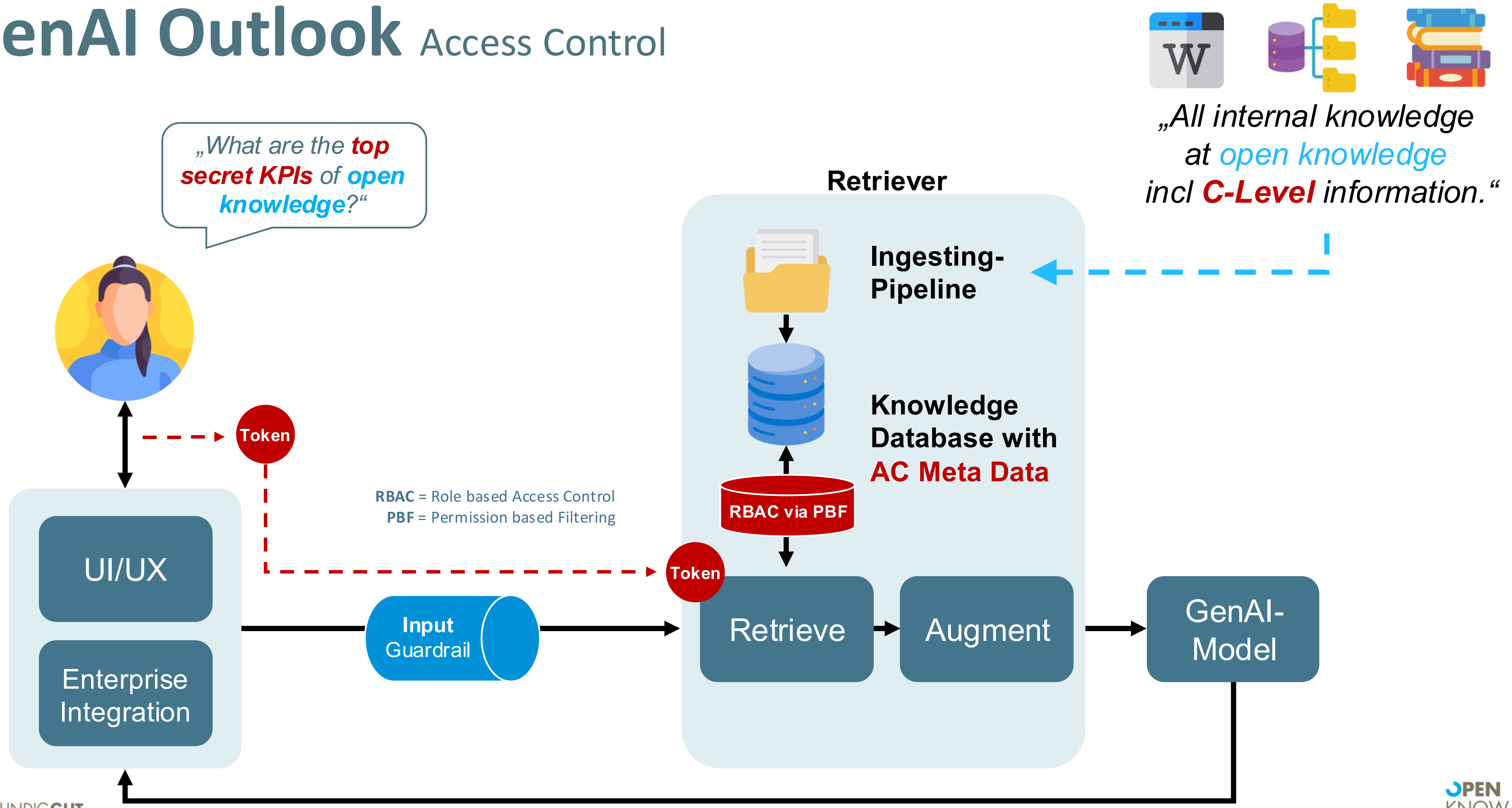
GenAI Outlook Access Control



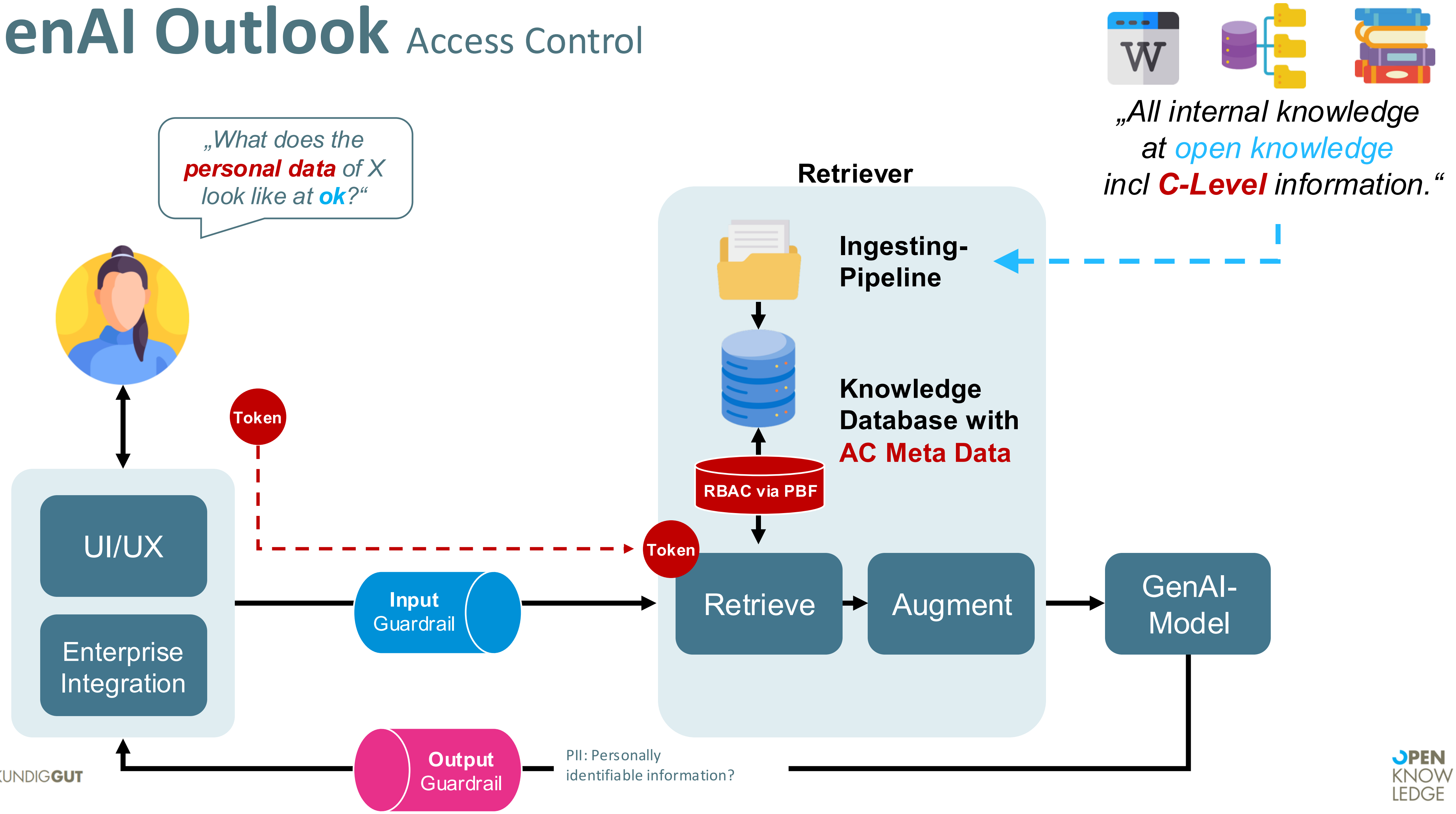
GenAI Outlook Access Control



GenAI Outlook Access Control



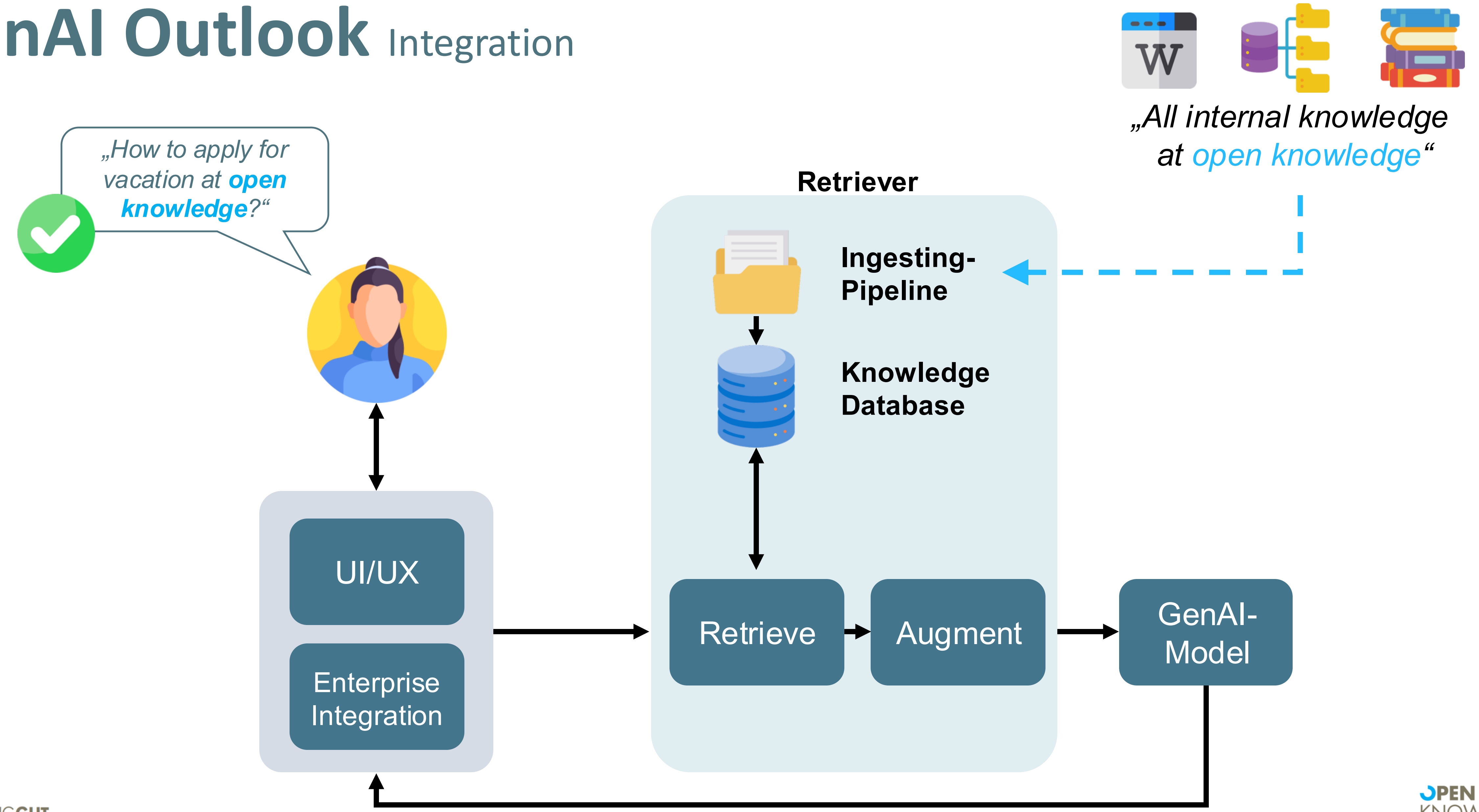
GenAI Outlook Access Control



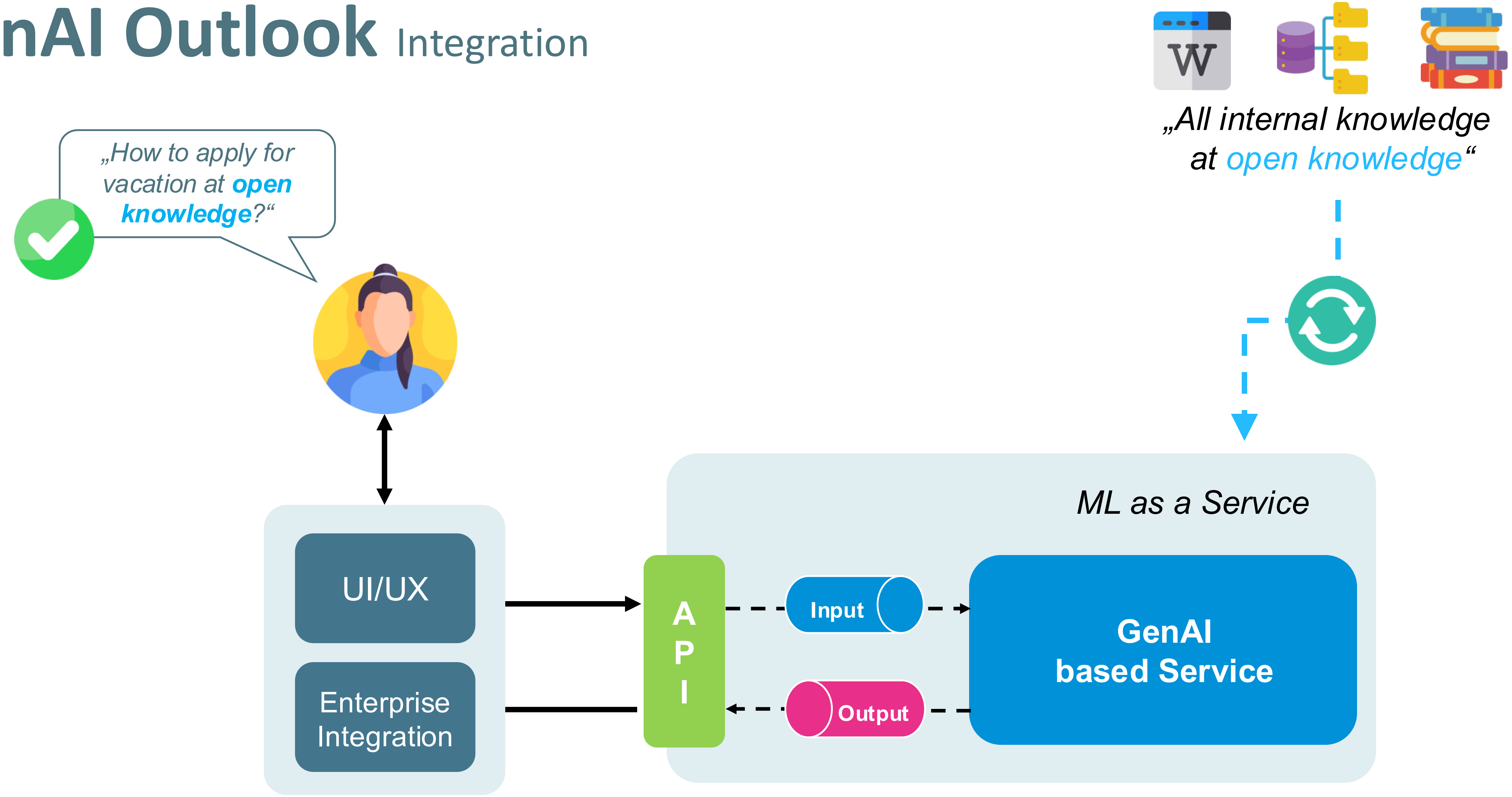
INTEGRATION

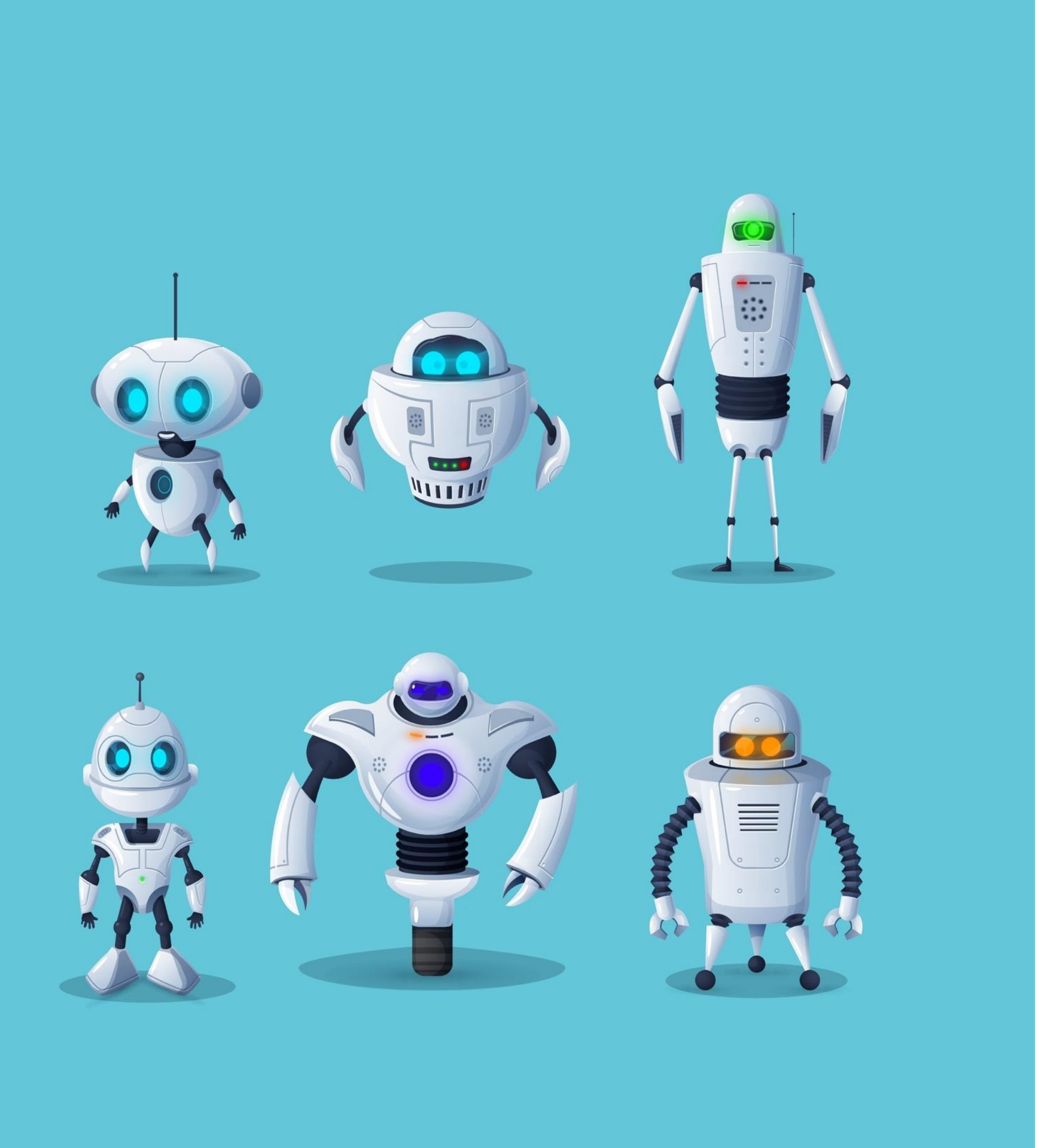
GenAI
explained

GenAI Outlook Integration



GenAI Outlook Integration





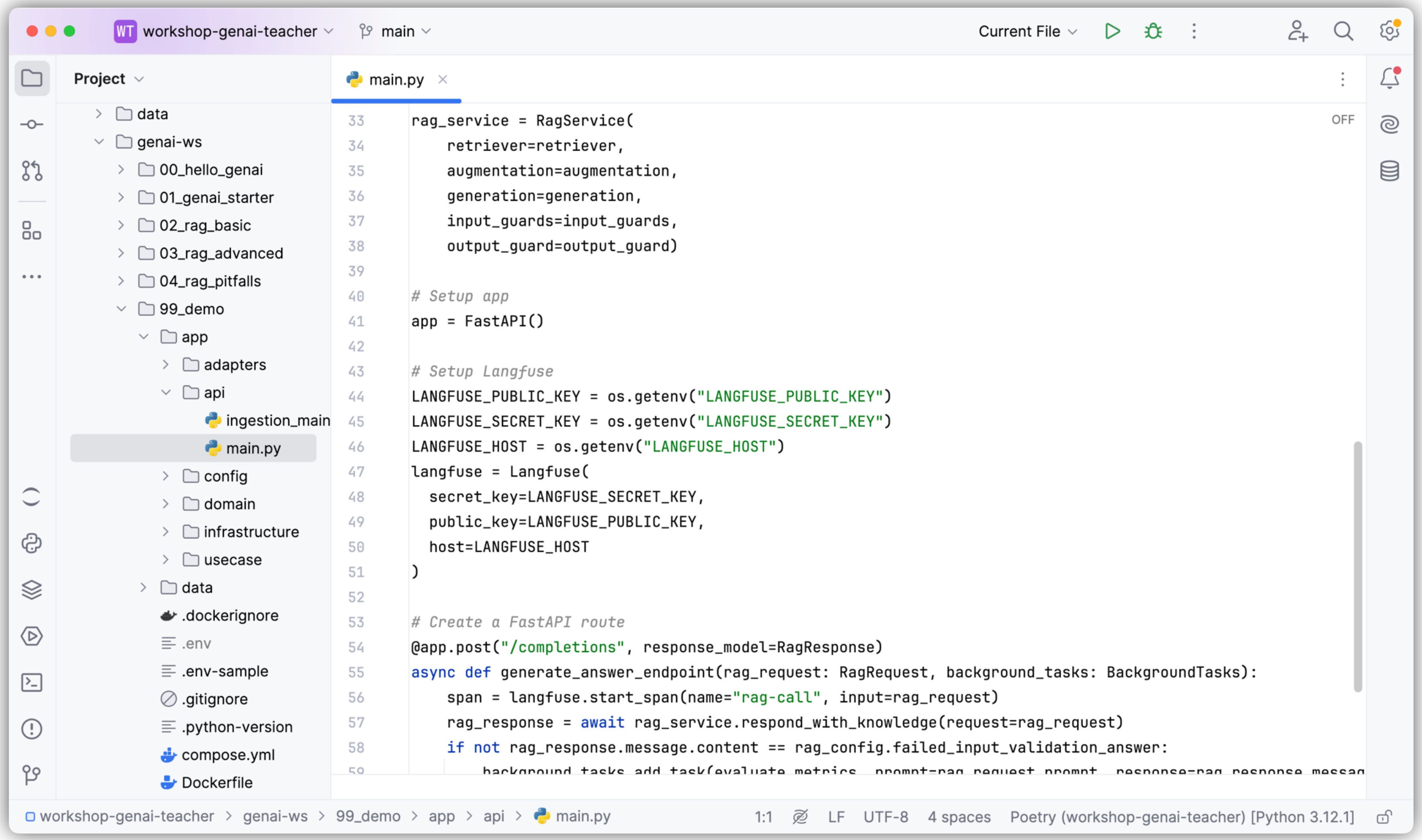
Hands-On

RAG System GenAI as a Service

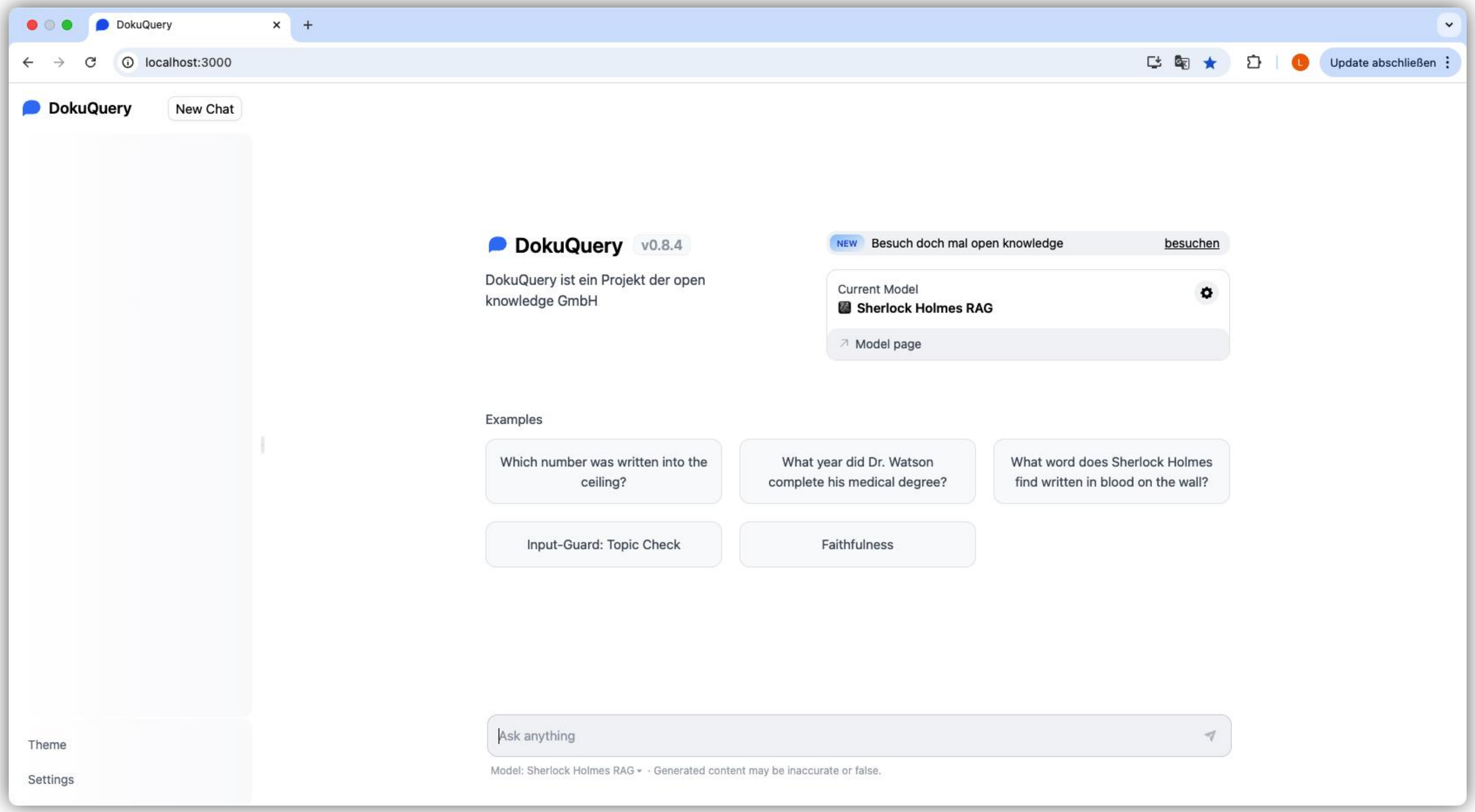
Provision of a GenAI-based RAG system as a service.

Integration of the service into a proprietary web-based interface.

Hands-on GenAI-as-a-Service



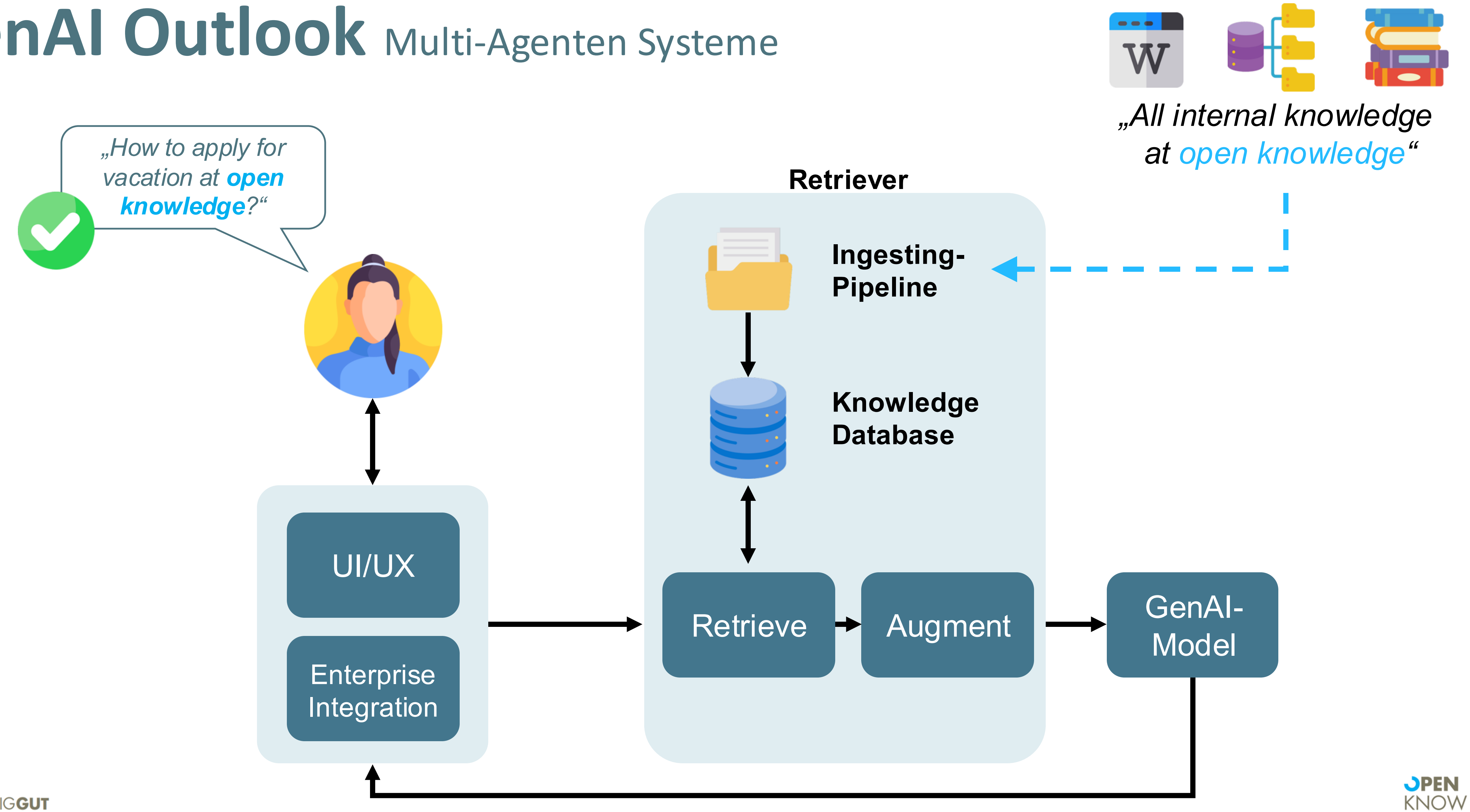
Hands-on GenAI-as-a-Service



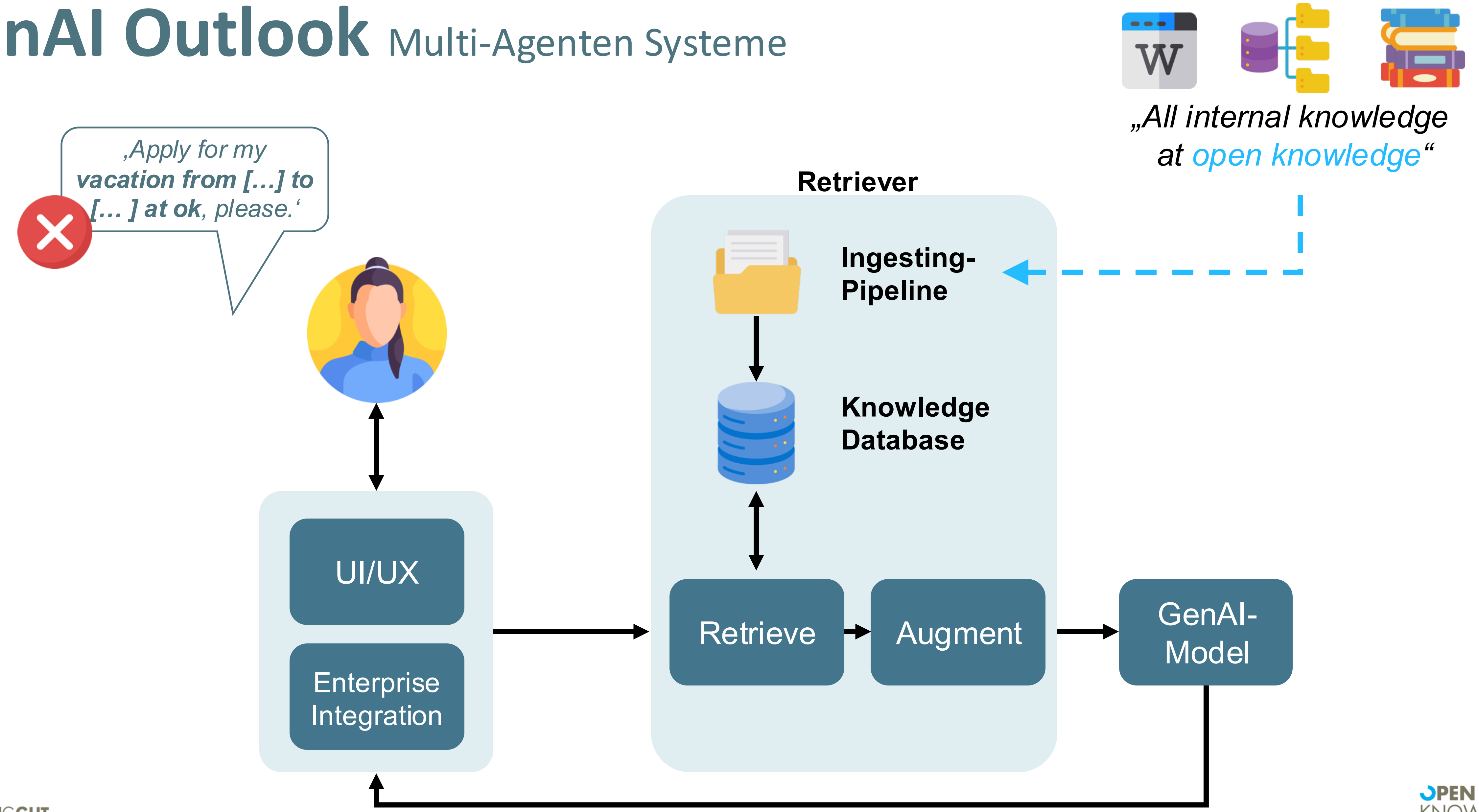
AGENTS

GenAI
explained

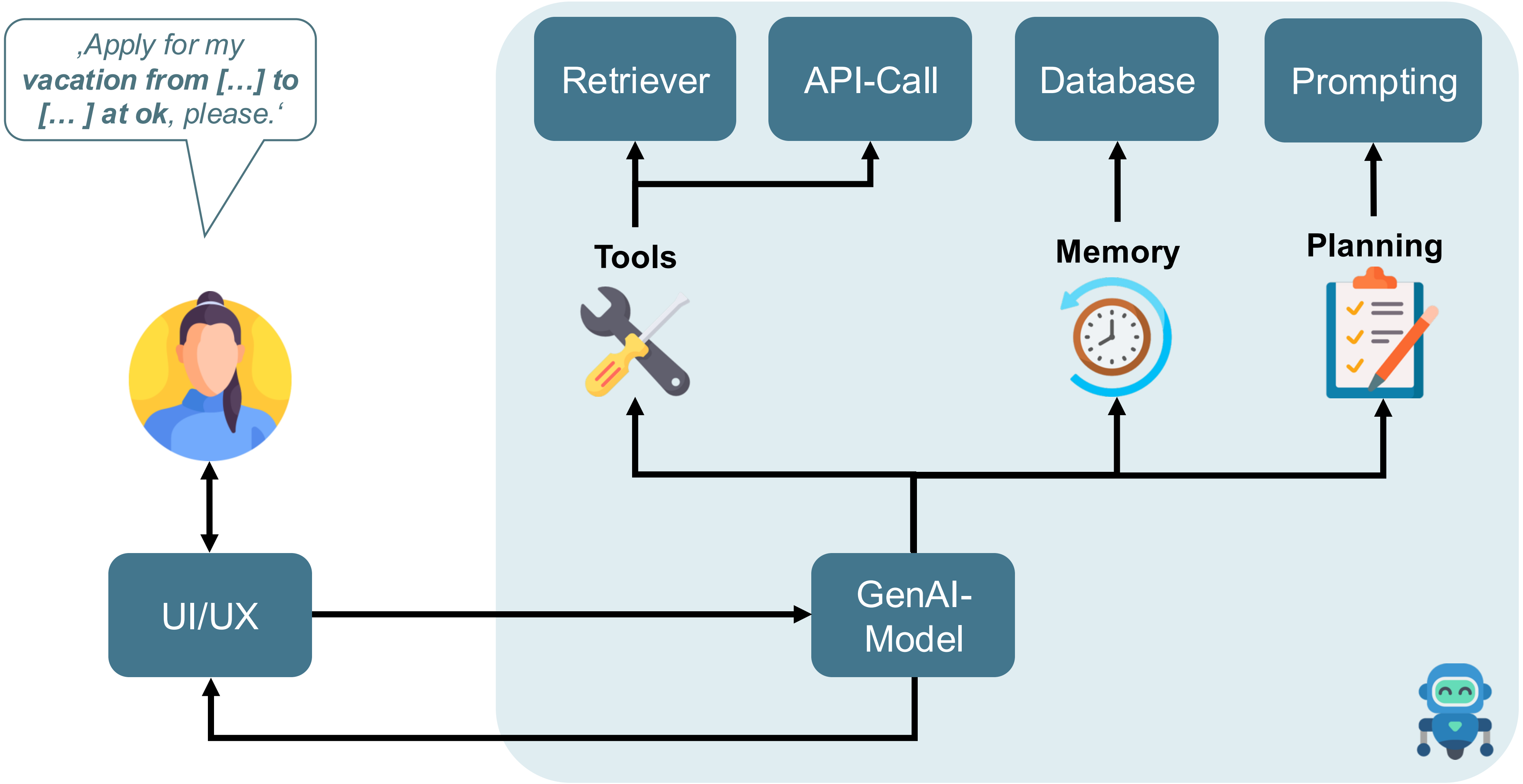
GenAI Outlook Multi-Agenten Systeme



GenAI Outlook Multi-Agenten Systeme



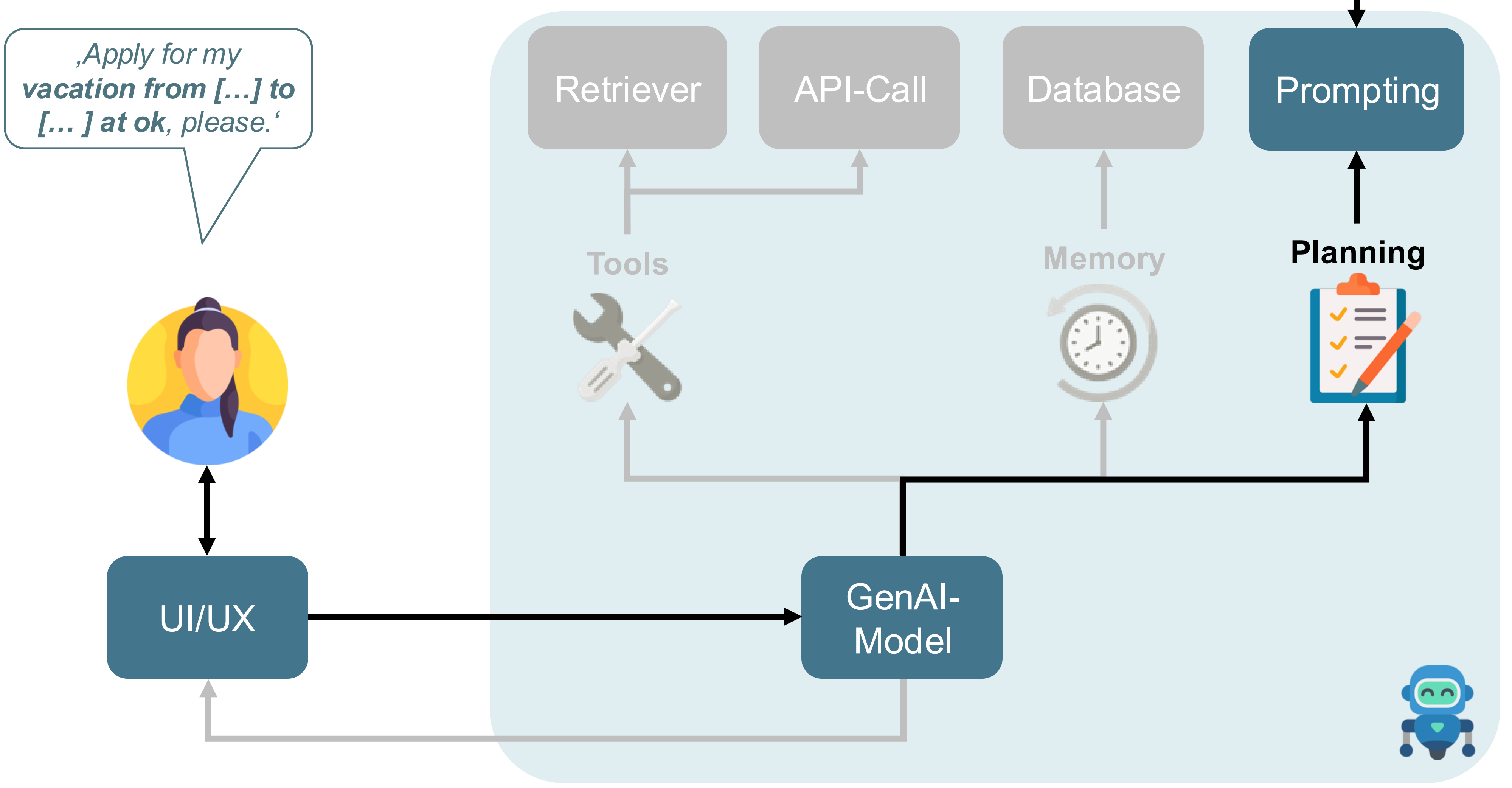
Agentic AI



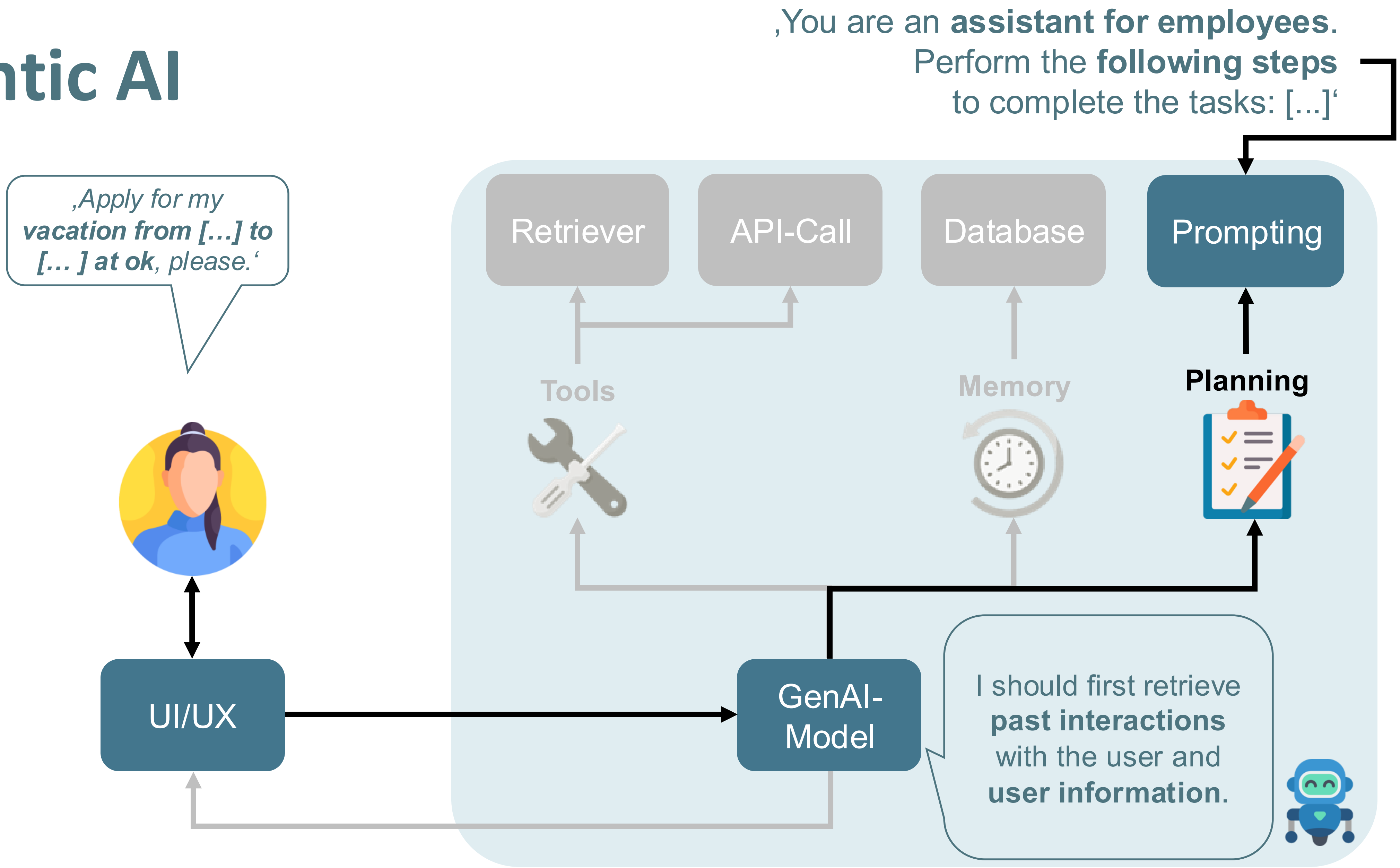
„Apply for my **vacation from […] to […] at ok**, please.'
Retriever
API-Call
Database
Prompting
Tools
Memory
Planning
UI/UX
GenAI-Model

Agentic AI

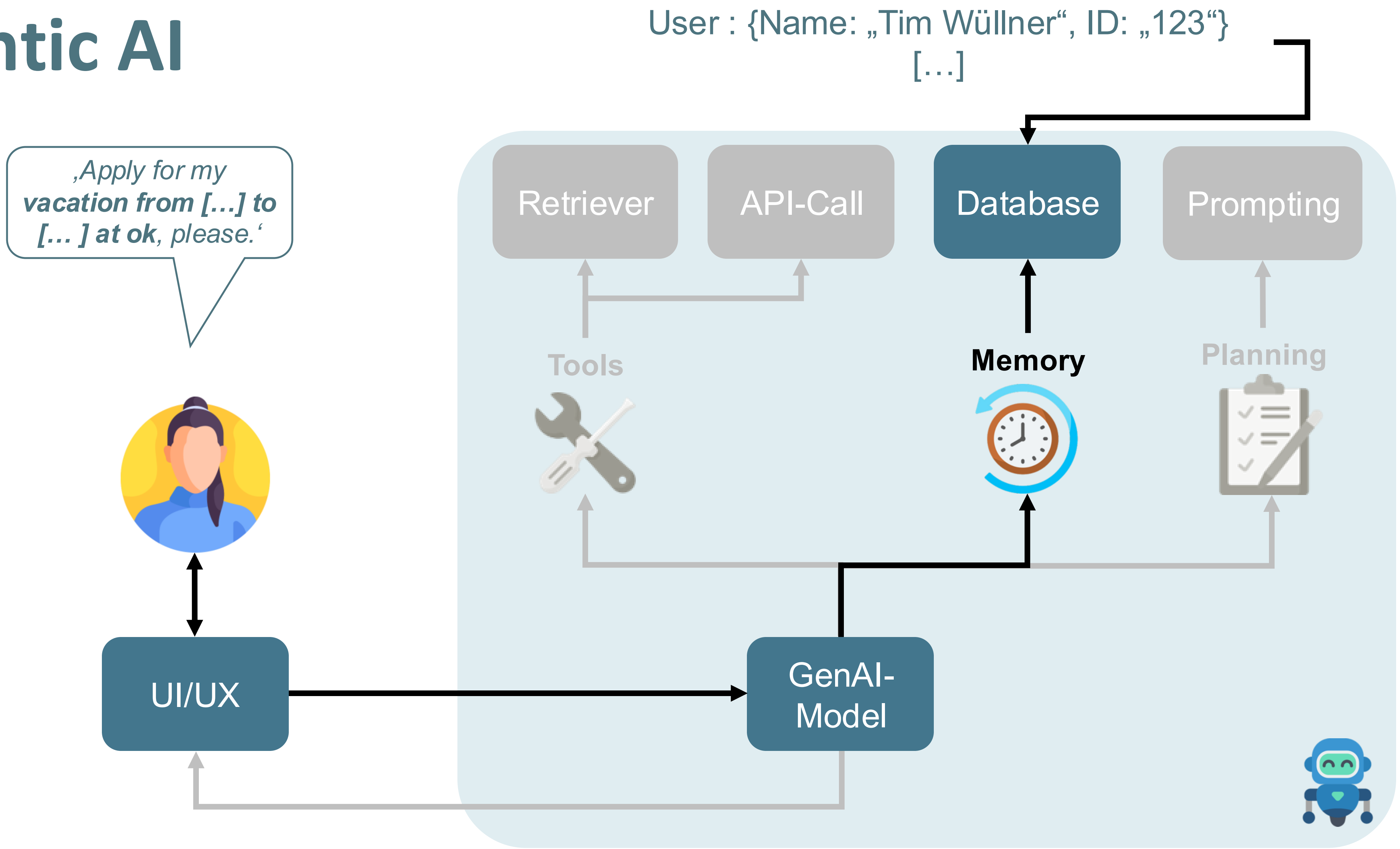
‚You are an **assistant for employees**.
Perform the **following steps**
to complete the tasks: […]‘



Agentic AI

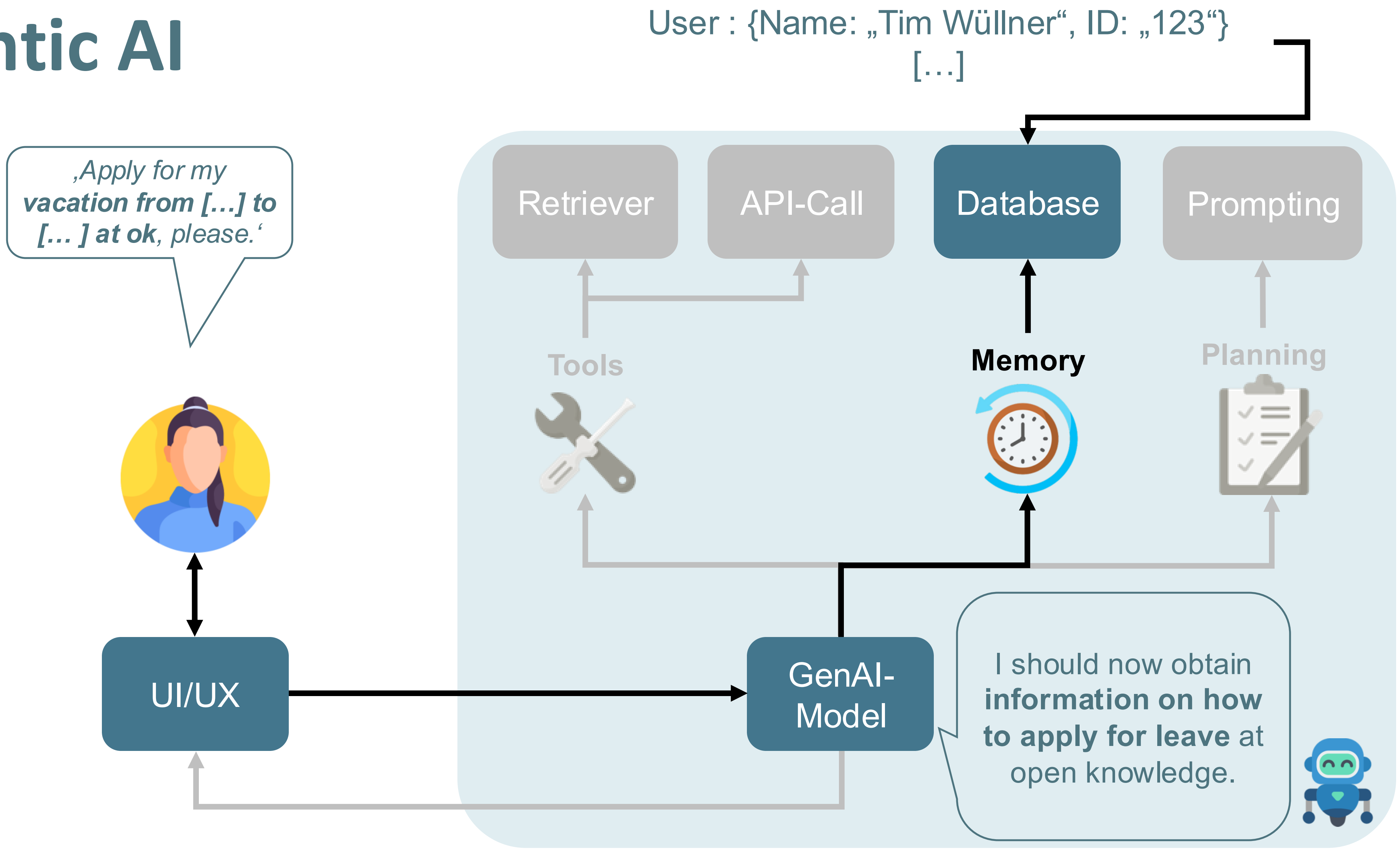

‚You are an **assistant for employees**.
Perform the **following steps**
to complete the tasks: [...]‘
‚*Apply for my* ***vacation from […] to […] at ok****, please.*‘
Retriever
API-Call
Database
Prompting
Tools
Memory
Planning
UI/UX
GenAI-Model
I should first retrieve **past interactions** with the user and **user information**.

Agentic AI

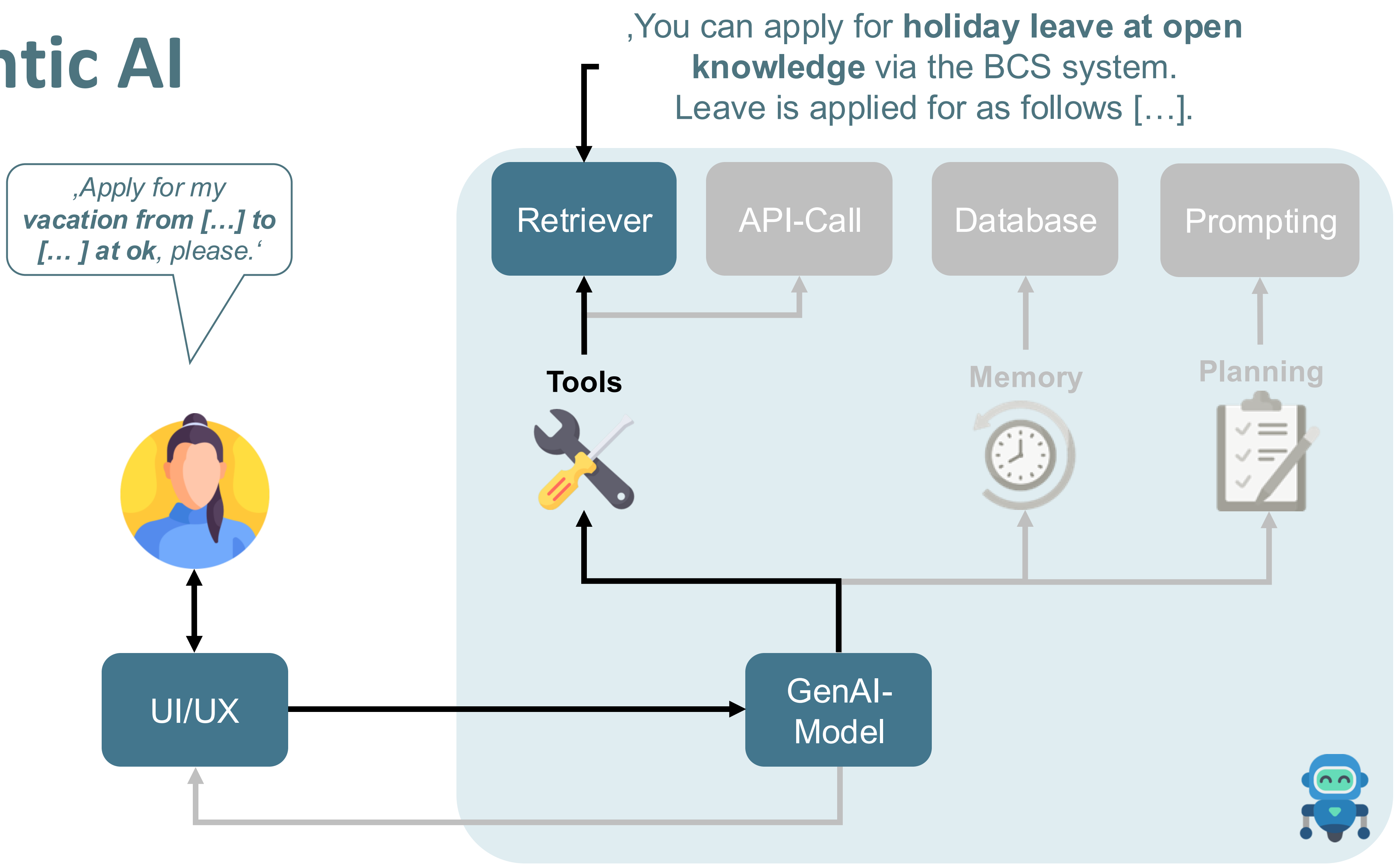


User : {Name: „Tim Wüllner“, ID: „123“}
[…]
‚Apply for my **vacation from […] to […] at ok**, please.‘
Retriever
API-Call
Database
Prompting
Tools
Memory
Planning
UI/UX
GenAI-Model

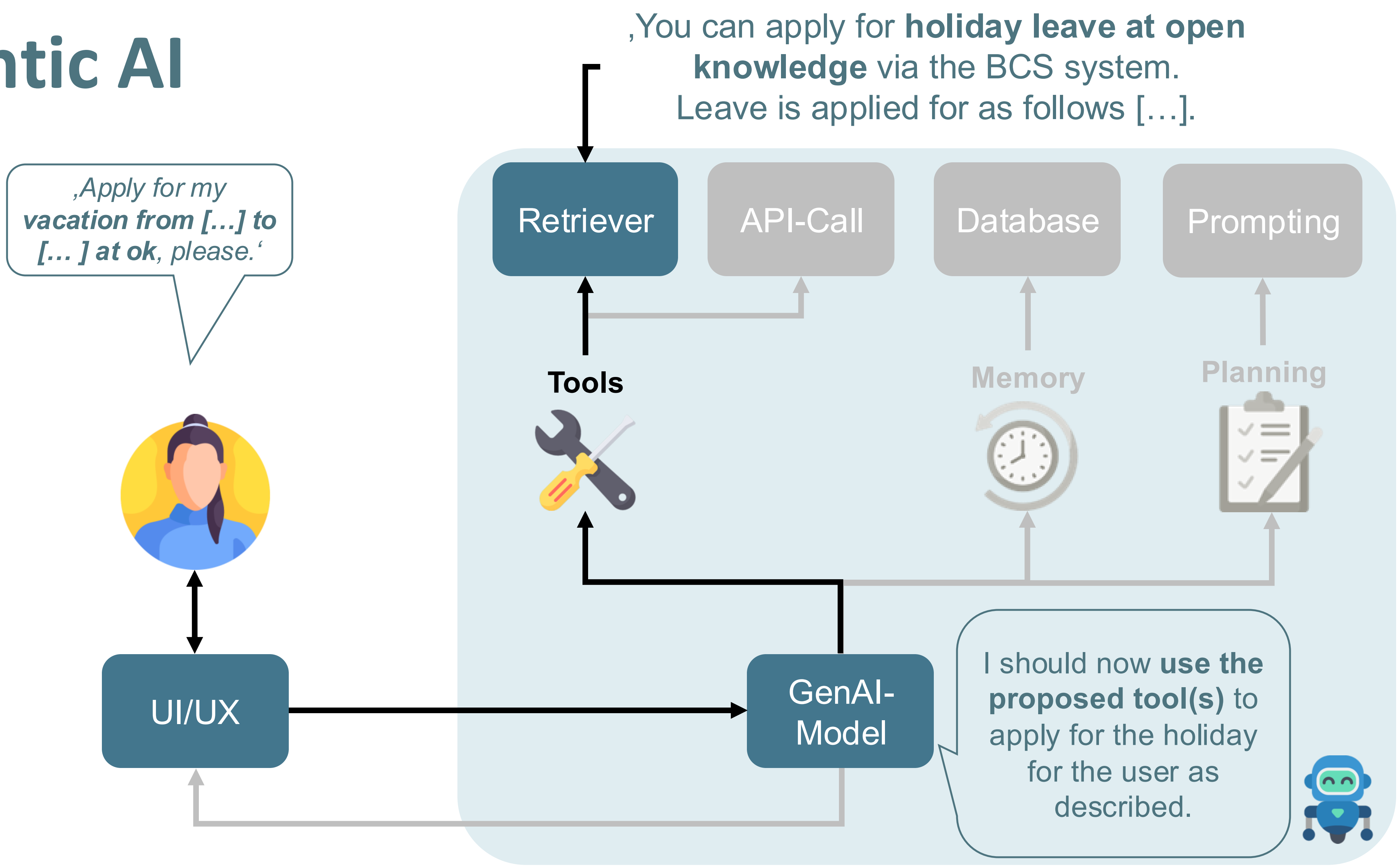
Agentic AI


User : {Name: „Tim Wüllner", ID: „123"}
[…]
‚Apply for my *vacation from […] to […] at ok, please.*'
Retriever
API-Call
Database
Prompting
Tools
Memory
Planning
UI/UX
GenAI-Model
I should now obtain **information on how to apply for leave** at open knowledge.

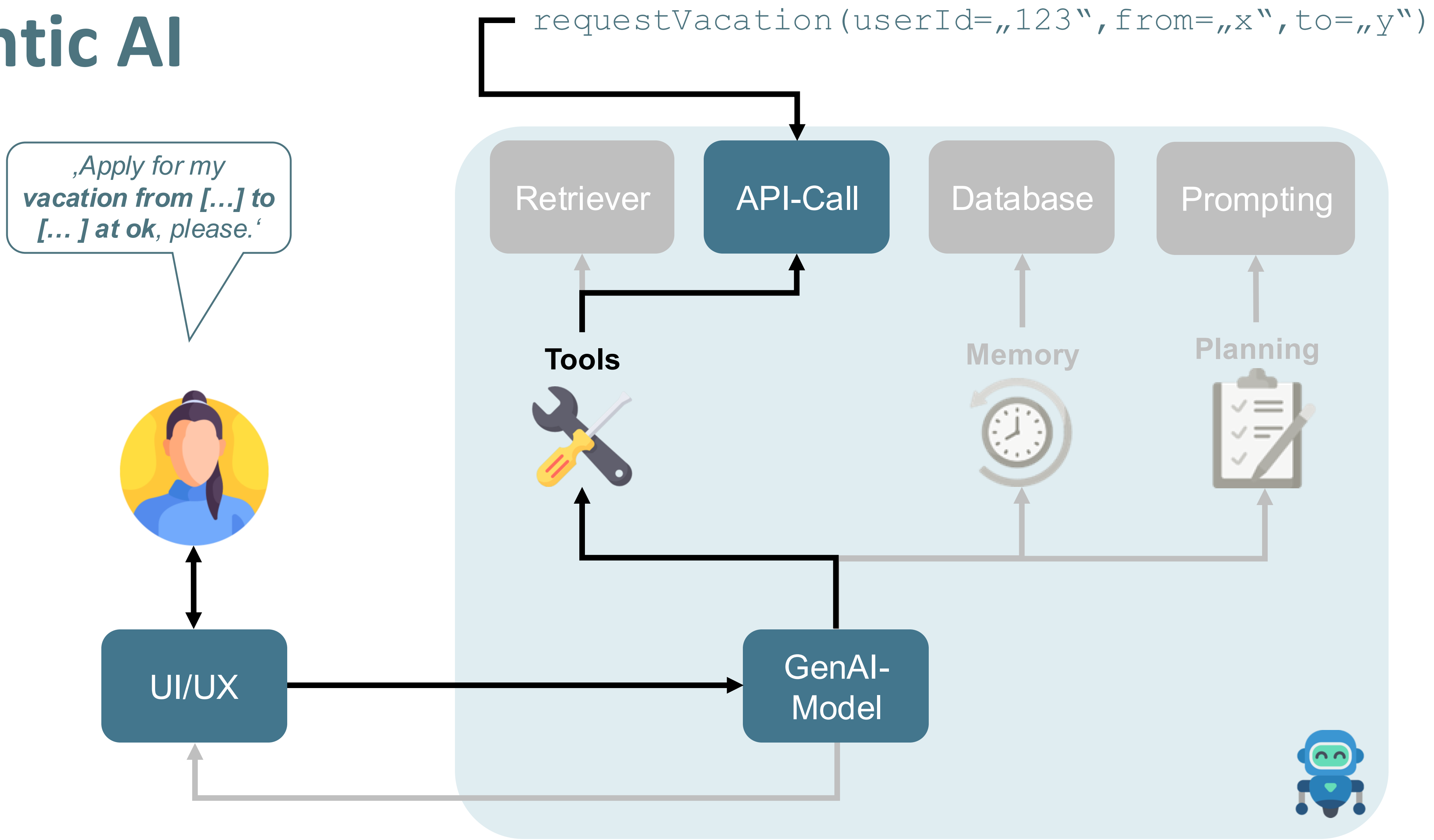
Agentic AI


‚You can apply for **holiday leave at open knowledge** via the BCS system.
Leave is applied for as follows […].
‚Apply for my *vacation from […] to […] at ok*, please.‘
Retriever
API-Call
Database
Prompting
Tools
Memory
Planning
UI/UX
GenAI-Model

Agentic AI

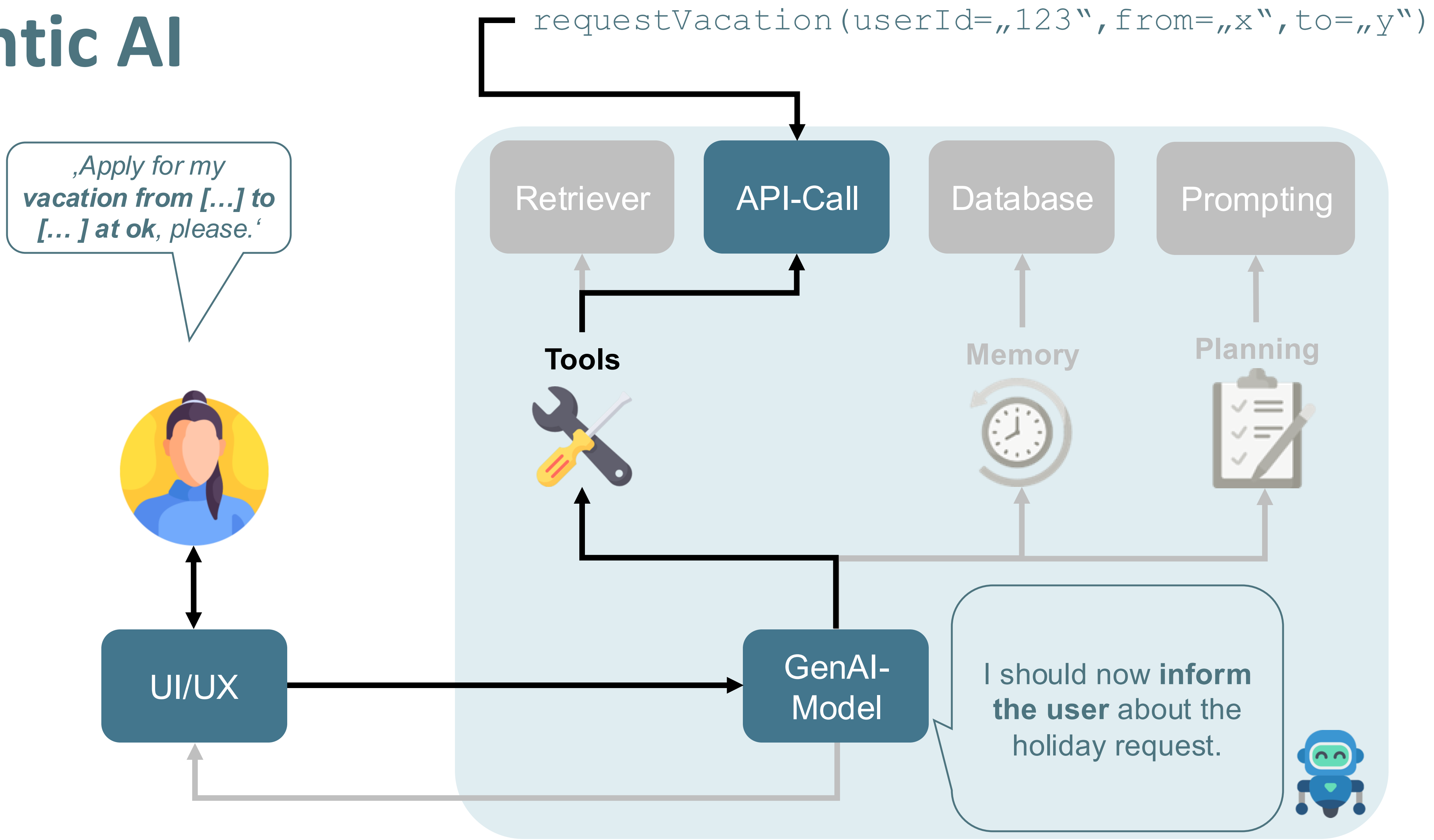

‚You can apply for **holiday leave at open knowledge** via the BCS system.
Leave is applied for as follows […].
‚Apply for my **vacation from […] to […] at ok**, please.‘
Retriever
API-Call
Database
Prompting
Tools
Memory
Planning
UI/UX
GenAI-Model
I should now **use the proposed tool(s)** to apply for the holiday for the user as described.

Agentic AI

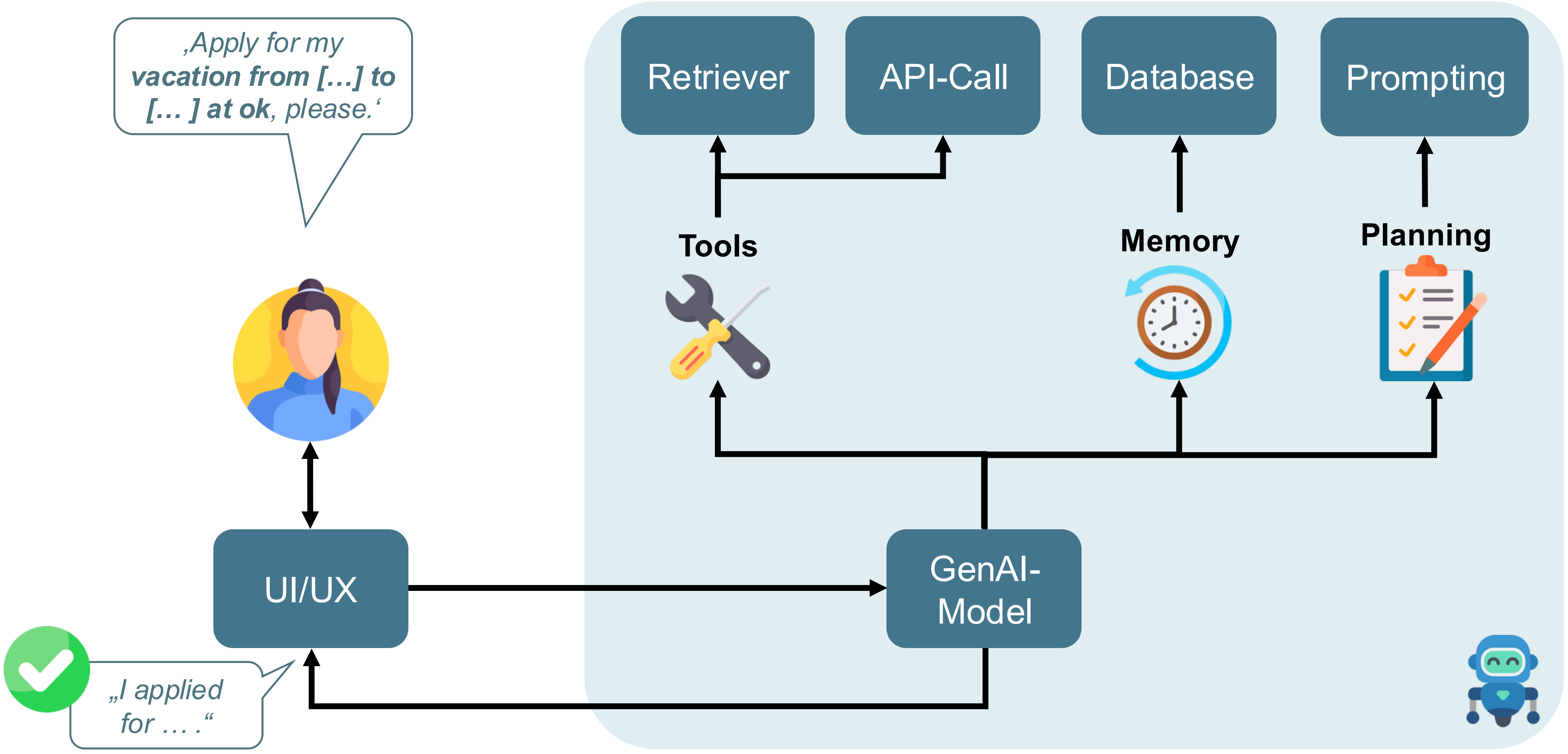


requestVacation(userId=„123“,from=„x“,to=„y“)
‚Apply for my **vacation from […] to […] at ok**, please.‘
Retriever
API-Call
Database
Prompting
Tools
Memory
Planning
UI/UX
GenAI-Model

Agentic AI

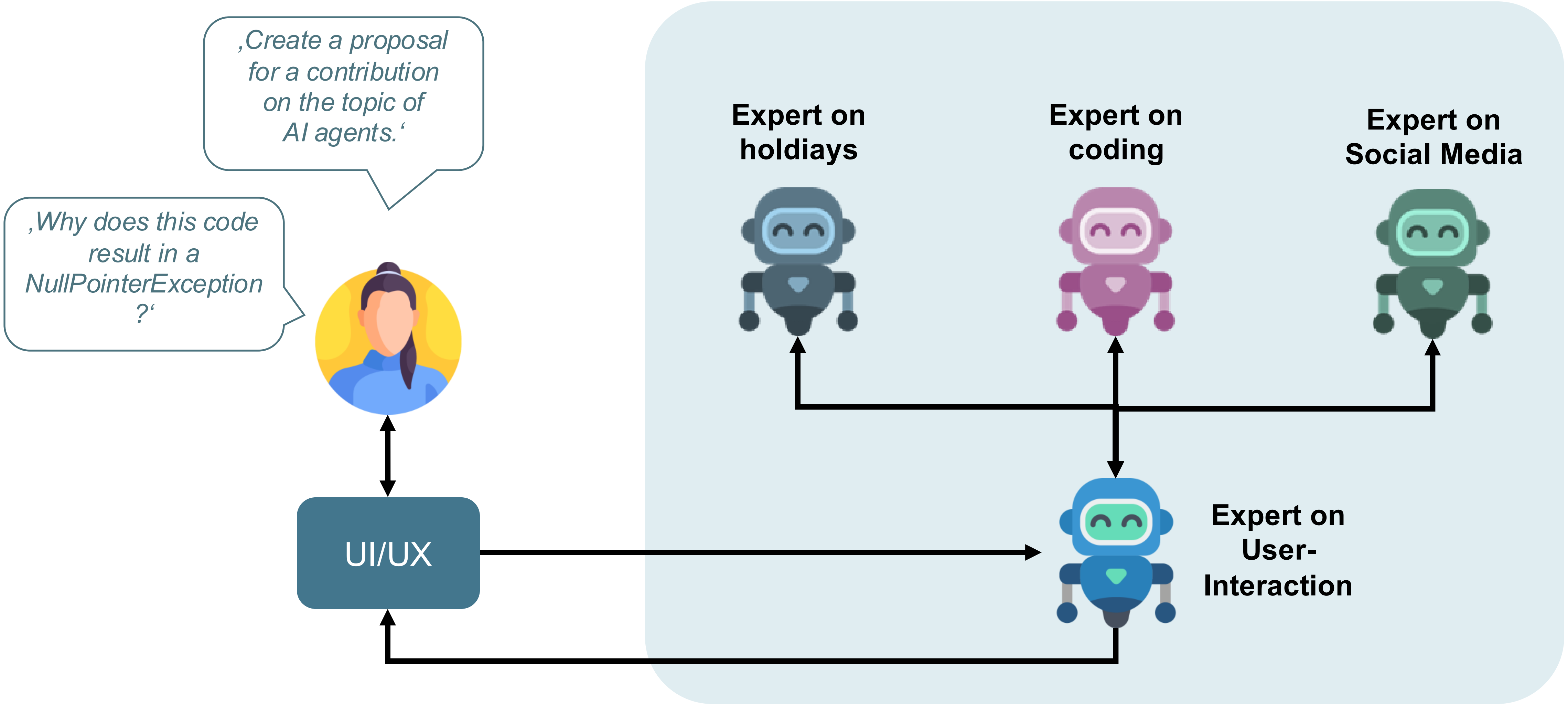

requestVacation(userId=„123“,from=„x“,to=„y“)
‚Apply for my *vacation from […] to […] at ok*, please.‘
Retriever
API-Call
Database
Prompting
Tools
Memory
Planning
UI/UX
GenAI-Model
I should now **inform the user** about the holiday request.

Agentic AI



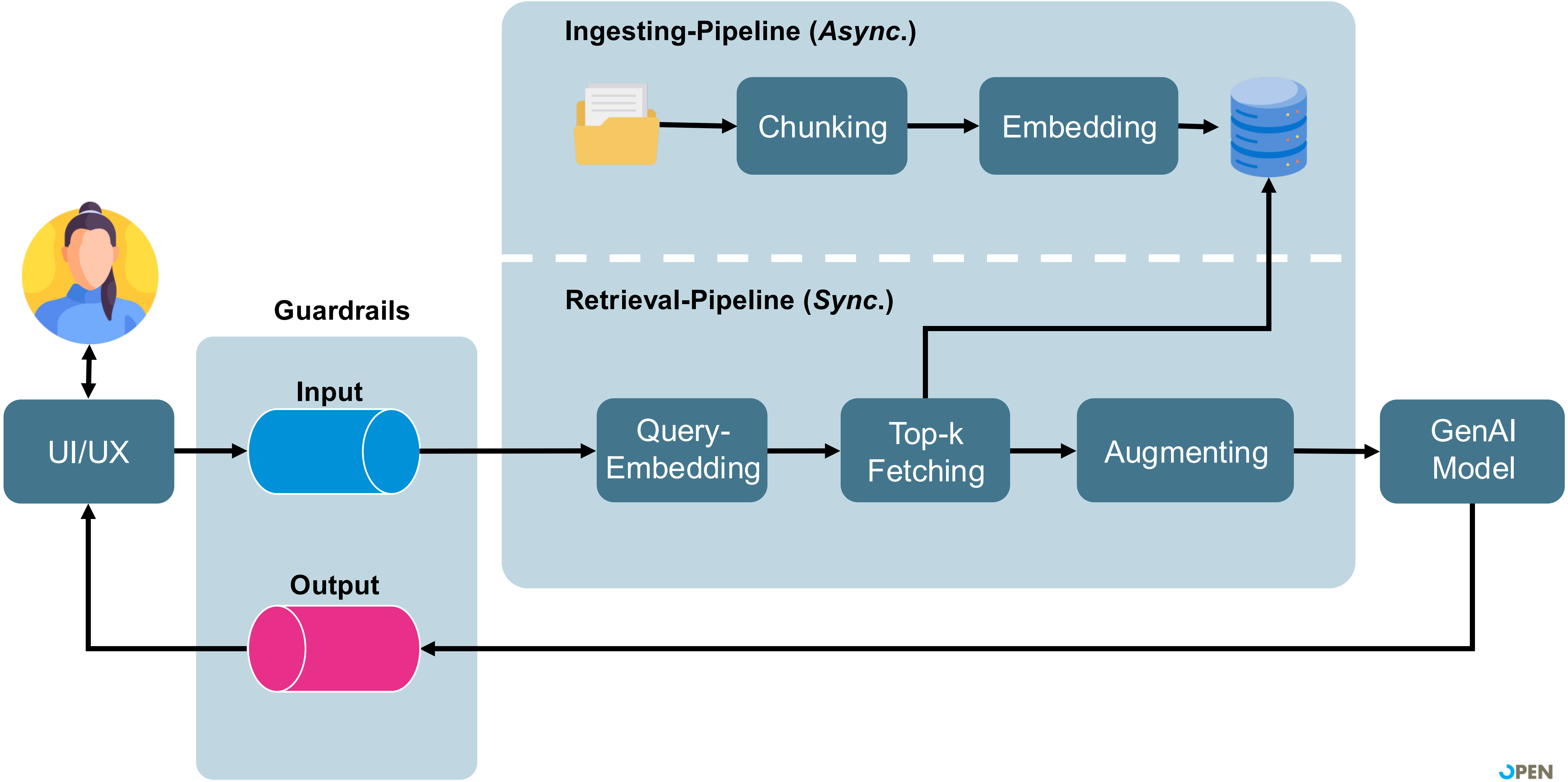
‚Apply for my **vacation from […] to […] at ok**, please.'
Retriever
API-Call
Database
Prompting
Tools
Memory
Planning
UI/UX
GenAI-Model
„I applied for … ."

Agentic AI Multi Agent Systems



Conclusion

What should I take away from the workshop for myself and my team?



OFFENKUNDIGGUT

OPEN KNOW LEDGE

GenAI from prototype to production

Your personal „takeaways“:

- GenAI is powerful. But also expensive.
- Professional prompting is your super power.
- Every model has its own character.
- RAG for your own domain knowledge.
- GenAI is also just software.

Your use case determines the right path!

Time for Questions?

Yes, sure!

#WISSENTEILEN

by **open** knowledge **GmbH**

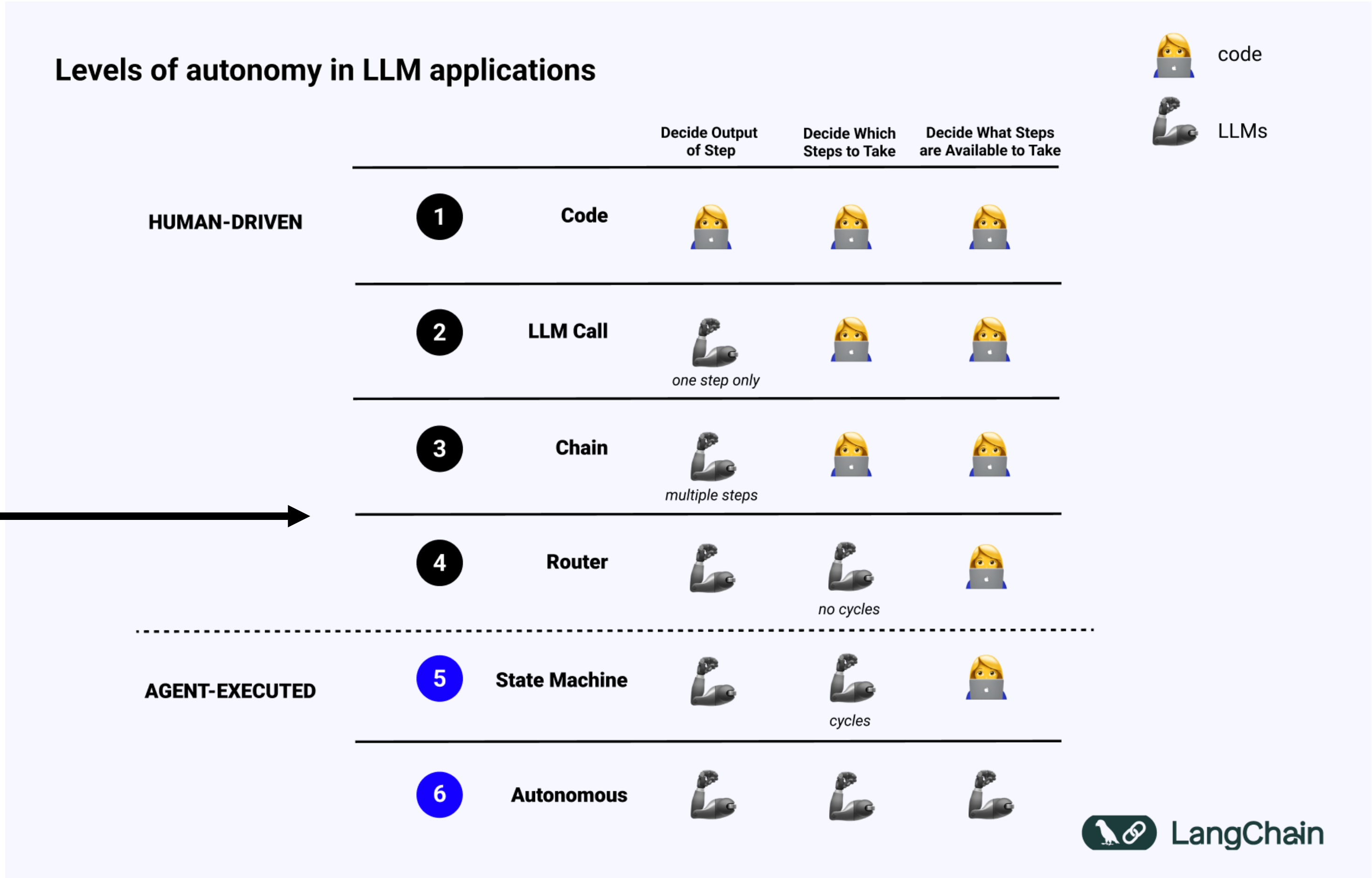
Many Thanks!

Lars Röwekamp, CIO New Technologies

@_**open**Knowledge | @**mobile**Larson

Looking beyond the RAG horizon:
Cognitive Architecture

Wir sind hier



blog.langchain.dev/what-is-a-cognitive-architecture

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